

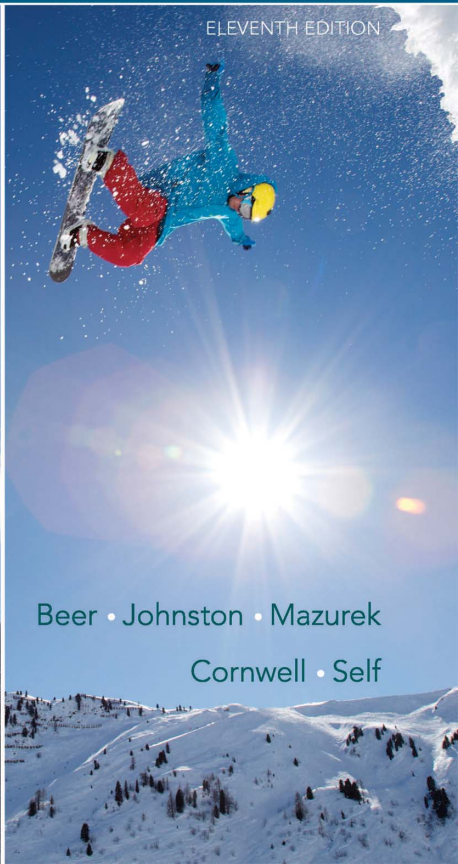
VECTOR MECHANICS for ENGINEERS

STATICS | DYNAMICS



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ELEVENTH EDITION



Beer • Johnston • Mazurek

Cornwell • Self

Eleventh Edition

Vector Mechanics For Engineers

Statics and Dynamics

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VECTOR MECHANICS FOR ENGINEERS: STATICS AND DYNAMICS, ELEVENTH EDITION

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This book is printed on acid-free paper.

1 2 3 4 5 6 7 8 9 0 DOW/DOW 1 0 9 8 7 6 5

ISBN 978-0-07-339824-2

MHID 0-07-339824-1

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Content Licensing Specialist (Image): *Carrie Burger*

Typeface: *10/12 Times LT Std*

Printer: *R. R. Donnelley*

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Library of Congress Cataloging-in-Publication Data

Beer, Ferdinand P. (Ferdinand Pierre), 1915–2003.

Vector mechanics for engineers. Statics and dynamics / Ferdinand P. Beer, Late of Lehigh University, E. Russell Johnston, Jr., Late of University of Connecticut, David F. Mazurek, U.S. Coast Guard Academy, Phillip J. Cornwell, Rose-Hulman Institute of Technology; with the collaboration of Brian P. Self, California Polytechnic State University, San Luis Obispo.—Eleventh edition.

pages cm

Includes index.

ISBN 978-0-07-339824-2

1. Statics. 2. Dynamics. I. Johnston, E. Russell (Elwood Russell), 1925–2010. II. Mazurek, David F. (David Francis) III. Title. IV. Title: Statics and dynamics.

TA350.B3552 2016

620.1'054—dc23

2014041301

The Internet addresses listed in the text were accurate at the time of publication. The inclusion of a website does not indicate an endorsement by the authors or McGraw-Hill Education, and McGraw-Hill Education does not guarantee the accuracy of the information presented at these sites.

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Preface

Objectives

A primary objective in a first course in mechanics is to help develop a student's ability first to analyze problems in a simple and logical manner, and then to apply basic principles to their solutions. A strong conceptual understanding of these basic mechanics principles is essential for successfully solving mechanics problems. We hope this text will help instructors achieve these goals.

General Approach

Vector algebra was introduced at the beginning of the first volume and is used in the presentation of the basic principles of statics, as well as in the solution of many problems, particularly three-dimensional problems. Similarly, the concept of vector differentiation will be introduced early in this volume, and vector analysis will be used throughout the presentation of dynamics. This approach leads to more concise derivations of the fundamental principles of mechanics. It also makes it possible to analyze many problems in kinematics and kinetics which could not be solved by scalar methods. The emphasis in this text, however, remains on the correct understanding of the principles of mechanics and on their application to the solution of engineering problems, and vector analysis is presented chiefly as a convenient tool.[†]

Practical Applications Are Introduced Early. One of the characteristics of the approach used in this book is that mechanics of *particles* is clearly separated from the mechanics of *rigid bodies*. This approach makes it possible to consider simple practical applications at an early stage and to postpone the introduction of the more difficult concepts. For example:

- In *Statics*, the statics of particles is treated first, and the principle of equilibrium of a particle was immediately applied to practical situations involving only concurrent forces. The statics of rigid bodies is considered later, at which time the vector and scalar products of two vectors were introduced and used to define the moment of a force about a point and about an axis.
- In *Dynamics*, the same division is observed. The basic concepts of force, mass, and acceleration, of work and energy, and of impulse and momentum are introduced and first applied to problems involving only particles. Thus, students can familiarize themselves with the three basic methods used in dynamics and learn their respective advantages before facing the difficulties associated with the motion of rigid bodies.

NEW!

The 11th edition has undergone a complete rewrite to modernize and streamline the language throughout the text.

2.2 ADDING FORCES BY COMPONENTS

In Sec. 2.1E, we described how to resolve a force into components. Here we discuss how to add forces by using their components, especially rectangular components. This method is often the most convenient way to add forces and, in practice, is the most common approach. (Note that we can readily extend the properties of vectors established in this section to the rectangular components of any vector quantity, such as velocity or momentum.)

2.2A Rectangular Components of a Force: Unit Vectors

In many problems, it is useful to resolve a force into two components that are perpendicular to each other. Figure 2.14 shows a force F resolved into a component F_x along the x axis and a component F_y along the y axis. The parallelogram drawn to obtain the two components is a rectangle, and F_x and F_y are called **rectangular components**.

The x and y axes are usually chosen to be horizontal and vertical, respectively, as in Fig. 2.14; they may, however, be chosen in any two perpendicular directions, as shown in Fig. 2.15. In determining the

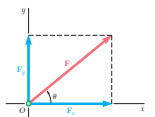


Fig. 2.14 Rectangular components of a force F .

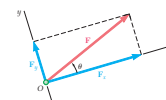


Fig. 2.15 Rectangular components of a force F for axes rotated away from horizontal and vertical.

[†]In a parallel text, *Mechanics for Engineers*, fifth edition, the use of vector algebra is limited to the addition and subtraction of vectors, and vector differentiation is omitted.

17.1 ENERGY METHODS FOR A RIGID BODY

We now use the principle of work and energy to analyze the plane motion of rigid bodies. As we pointed out in Chap. 13, the method of work and energy is particularly well adapted to solving problems involving velocities and displacements. Its main advantage is that the work of forces and the kinetic energy of particles are scalar quantities.

17.1A Principle of Work and Energy

To apply the principle of work and energy to the motion of a rigid body, we again assume that the rigid body is made up of a large number n of particles of mass Δm_i . From Eq. (14.30) of Sec. 14.2B, we have

Principle of work and energy, rigid body

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (17.1)$$

where T_1 , T_2 = the initial and final values of total kinetic energy of particles forming the rigid body

$U_{1 \rightarrow 2}$ = work of all forces acting on various particles of the body

Just as we did in Chap. 13, we can express the work done by nonconservative forces as $U_{1 \rightarrow 2}^{NC}$, and we can define potential energy terms for conservative forces. Then we can express Eq. (17.1) as

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{NC} = T_2 + V_{g_2} + V_{e_2} \quad (17.1')$$

where V_{g_1} and V_{g_2} are the initial and final gravitational potential energy of the center of mass of the rigid body with respect to a reference point or datum, and V_{e_1} and V_{e_2} are the initial and final values of the elastic energy associated with springs in the system.

We obtain the total kinetic energy

$$T = \frac{1}{2} \sum_{i=1}^n \Delta m_i v_i^2 \quad (17.2)$$

by adding positive scalar quantities, so it is itself a positive scalar quantity. You will see later how to determine T for various types of motion of a rigid body.

The expression $U_{1 \rightarrow 2}$ in Eq. (17.1) represents the work of all the forces acting on the various particles of the body whether these forces are internal or external. However, the total work of the internal forces holding together the particles of a rigid body is zero. To see this, consider two particles A and B of a rigid body and the two equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ they exert on each other (Fig. 17.1). Although, in general, small displacements $d\mathbf{r}$ and $d\mathbf{r}'$ of the two particles are different, the components of these displacements along AB must be equal; otherwise, the particles would not remain at the same distance from each other and the body would not be rigid. Therefore, the work of $\mathbf{F}$ is equal in magnitude and

New Concepts Are Introduced in Simple Terms. Since this text is designed for the first course in dynamics, new concepts are presented in simple terms and every step is explained in detail. On the other hand, by discussing the broader aspects of the problems considered, and by stressing methods of general applicability, a definite maturity of approach has been achieved. For example, the concept of potential energy is discussed in the general case of a conservative force. Also, the study of the plane motion of rigid bodies is designed to lead naturally to the study of their general motion in space. This is true in kinematics as well as in kinetics, where the principle of equivalence of external and effective forces is applied directly to the analysis of plane motion, thus facilitating the transition to the study of three-dimensional motion.

Fundamental Principles Are Placed in the Context of Simple Applications. The fact that mechanics is essentially a *deductive* science based on a few fundamental principles is stressed. Derivations have been presented in their logical sequence and with all the rigor warranted at this level. However, the learning process being largely *inductive*, simple applications are considered first. For example:

- The statics of particles precedes the statics of rigid bodies, and problems involving internal forces are postponed until Chap. 6.
- In Chap. 4, equilibrium problems involving only coplanar forces are considered first and solved by ordinary algebra, while problems involving three-dimensional forces and requiring the full use of vector algebra are discussed in the second part of the chapter.
- The kinematics of particles (Chap. 11) precedes the kinematics of rigid bodies (Chap. 15).
- The fundamental principles of the kinetics of rigid bodies are first applied to the solution of two-dimensional problems (Chaps. 16 and 17), which can be more easily visualized by the student, while three-dimensional problems are postponed until Chap. 18.

The Presentation of the Principles of Kinetics Is Unified.

The eleventh edition of *Vector Mechanics for Engineers* retains the unified presentation of the principles of kinetics which characterized the previous ten editions. The concepts of linear and angular momentum are introduced in Chap. 12 so that Newton's second law of motion can be presented not only in its conventional form $\mathbf{F} = m\mathbf{a}$, but also as a law relating, respectively, the sum of the forces acting on a particle and the sum of their moments to the rates of change of the linear and angular momentum of the particle. This makes possible an earlier introduction of the principle of conservation of angular momentum and a more meaningful discussion of the motion of a particle under a central force (Sec. 12.3A). More importantly, this approach can be readily extended to the study of the motion of a system of particles (Chap. 14) and leads to a more concise and unified treatment of the kinetics of rigid bodies in two and three dimensions (Chaps. 16 through 18).

Systematic Problem-Solving Approach. New to this edition of the text, all the sample problems are solved using the steps of **Strategy**, **Modeling**, **Analysis**, and **Reflect & Think**, or the "SMART" approach.

NEW!

Optional Sections Offer Advanced or Specialty Topics.

A large number of optional sections have been included. These sections are indicated by asterisks and thus are easily distinguished from those which form the core of the basic dynamics course. They can be omitted without prejudice to the understanding of the rest of the text.

The topics covered in the optional sections in statics include the reduction of a system of forces of a wrench, applications to hydrostatics, equilibrium of cables, products of inertia and Mohr's circle, the determination of the principal axes and the mass moments of inertia of a body of arbitrary shape, and the method of virtual work. The sections on the inertia properties of three-dimensional bodies are primarily intended for students who will later study in dynamics the three-dimensional motion of rigid bodies.

The topics covered in the optional sections in dynamics include graphical methods for the solution of rectilinear-motion problems, the trajectory of a particle under a central force, the deflection of fluid streams, problems involving jet and rocket propulsion, the kinematics and kinetics of rigid bodies in three dimensions, damped mechanical vibrations, and electrical analogues. These topics will be found of particular interest when dynamics is taught in the junior year.

The material presented in the text and most of the problems require no previous mathematical knowledge beyond algebra, trigonometry, elementary calculus, and the elements of vector algebra presented in Chaps. 2 and 3 of the volume on statics. However, special problems are included, which make use of a more advanced knowledge of calculus, and certain sections, such as Secs. 19.5A and 19.5B on damped vibrations, should be assigned only if students possess the proper mathematical background. In portions of the text using elementary calculus, a greater emphasis is placed on the correct understanding and application of the concepts of differentiation and integration, than on the nimble manipulation of mathematical formulas. In this connection, it should be mentioned that the determination of the centroids of composite areas precedes the calculation of centroids by integration, thus making it possible to establish the concept of moment of area firmly before introducing the use of integration.

Guided Tour

Chapter Introduction. Each chapter begins with a list of learning objectives and an outline that previews chapter topics. An introductory section describes the material to be covered in simple terms, and how it will be applied to the solution of engineering problems.

Chapter Lessons. The body of the text is divided into sections, each consisting of one or more sub-sections, several sample problems, and a large number of end-of-section problems for students to solve. Each section corresponds to a well-defined topic and generally can be covered in one lesson. In a number of cases, however, the instructor will find it desirable to devote more than one lesson to a given topic. *The Instructor's and Solutions Manual* contains suggestions on the coverage of each lesson.

Sample Problems. The Sample Problems are set up in much the same form that students will use when solving assigned problems, and they employ the SMART problem-solving methodology that students are encouraged to use in the solution of their assigned problems. They thus serve the double purpose of reinforcing the text and demonstrating the type of neat and orderly work that students should cultivate in their own solutions. In addition, in-problem references and captions have been added to the sample problem figures for contextual linkage to the step-by-step solution.

Concept Applications. Concept Applications are used within selected theory sections in the Statics volume to amplify certain topics, and they are designed to reinforce the specific material being presented and facilitate its understanding.

Solving Problems on Your Own. A section entitled *Solving Problems on Your Own* is included for each lesson, between the sample problems and the problems to be assigned. The purpose of these sections is to help students organize in their own minds the preceding theory of the text and the solution methods of the sample problems so that they can more successfully solve the homework problems. Also included in these sections are specific suggestions and strategies that will enable the students to more efficiently attack any assigned problems.

Homework Problem Sets. Most of the problems are of a practical nature and should appeal to engineering students. They are primarily designed, however, to illustrate the material presented in the text and to help students understand the principles of mechanics. The problems are grouped according to the portions of material they illustrate and, in general, are arranged in order of increasing difficulty. Problems requiring special attention are indicated by asterisks. Answers to 70 percent of the problems are given at the end of the book. Problems for which the answers are given are set in straight type in the text, while problems for which no answer is given are set in italic and red font color.



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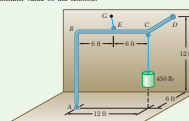
Introduction

The tallest skyscraper in the Western Hemisphere, One World Trade Center is a prominent feature of the New York City skyline. From its foundation to its structural components and mechanical systems, the design and operation of the tower is based on the fundamentals of engineering mechanics.

NEW!

Sample Problem 4.10

A 450-lb load hangs from the corner C of a rigid piece of pipe ABCD that has been bent as shown. The pipe is supported by ball-and-socket joints A and D, which are fastened, respectively, to the floor and to a vertical wall, and by a cable attached at the midpoint E of the portion BC of the pipe and at a point G on the wall. Determine (a) where G should be located if the tension in the cable is to be minimum, (b) the corresponding minimum value of the tension.

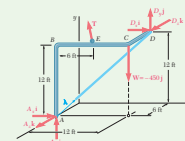


STRATEGY: Draw the free-body diagram of the pipe showing the reactions at A and D. Isolate the unknown tension T and the known weight W by summing moments about the diagonal line AD, and compute values from the equilibrium equations.

MODELING and ANALYSIS:

Free-Body Diagram. The free-body diagram of the pipe includes the load $W = (-450 \text{ lb})\mathbf{j}$, the reactions at A and D, and the force T exerted by the cable (Fig. 1). To eliminate the moments of the forces about the line AD from the computations, take the sum of the moments about the line AD and set it equal to zero. Denote the unit vector along AD by λ , which enables you to write

$$\sum M_{AD} = 0: \lambda \cdot (\lambda \times T) + \lambda \cdot (\lambda \times W) = 0 \quad (1)$$



NEW!

Over 650 of the homework problems in the text are new or revised.

Chapter Review and Summary. Each chapter ends with a review and summary of the material covered in that chapter. Marginal notes are used to help students organize their review work, and cross-references have been included to help them find the portions of material requiring their special attention.

Review Problems. A set of review problems is included at the end of each chapter. These problems provide students further opportunity to apply the most important concepts introduced in the chapter.

Review Problems

- 2.127** Two structural members A and B are bolted to a bracket as shown. Knowing that both members are in compression and that the force is 15 kN in member A and 10 kN in member B , determine by trigonometry the magnitude and direction of the resultant of the forces applied to the bracket by members A and B .

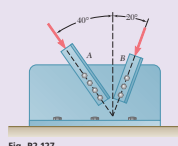


Fig. P2.127

- 2.128** Determine the x and y components of each of the forces shown.

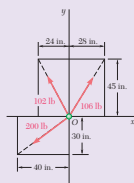


Fig. P2.128

- 2.129** A hoist trolley is subjected to the three forces shown. Knowing that $\alpha = 40^\circ$, determine (a) the required magnitude of the force P if the resultant of the three forces is to be vertical, (b) the corresponding magnitude of the resultant.

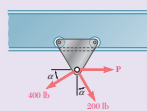


Fig. P2.129

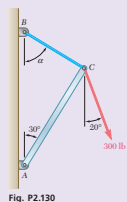


Fig. P2.130

- 2.130** Knowing that $\alpha = 55^\circ$ and that boom AC exerts on pin C a force directed along line AC , determine (a) the magnitude of that force, (b) the tension in cable BC .

Review and Summary

In this chapter, we have studied the effect of forces on particles, i.e., on bodies of such shape and size that we may assume all forces acting on them apply at the same point.

Resultant of Two Forces

Forces are *vector quantities*; they are characterized by a point of application, a magnitude, and a direction, and they add according to the parallelogram law (Fig. 2.30). We can determine the magnitude and direction of the resultant R of two forces P and Q either graphically or by trigonometry using the law of cosines and the law of sines (Sample Prob. 2.1).



Fig. 2.30

Components of a Force

Any given force acting on a particle can be resolved into two or more components, i.e., it can be replaced by two or more forces that have the same effect on the particle. A force F can be resolved into two components P and Q by drawing a parallelogram with F for its diagonal; the components P and Q are then represented by the two adjacent sides of the parallelogram (Fig. 2.31). Again, we can determine the components either graphically or by trigonometry [Sec. 2.1E].

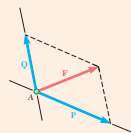


Fig. 2.31

Rectangular Components; Unit Vectors

A force F is resolved into two rectangular components if its components F_x and F_y are perpendicular to each other and are directed along the coordinate axes (Fig. 2.32). Introducing the unit vectors i and j along the x and y axes, respectively, we can write the components and the vector as [Sec. 2.2A]

$$F_x = F_x i \quad F_y = F_y j \quad (2.6)$$

and

$$F = F_x i + F_y j \quad (2.7)$$

where F_x and F_y are the *scalar components* of F . These components, which can be positive or negative, are defined by the relations

$$F_x = F \cos \theta \quad F_y = F \sin \theta \quad (2.8)$$

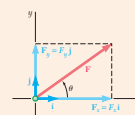


Fig. 2.32

Computer Problems. Accessible through Connect are problem sets for each chapter that are designed to be solved with computational software. Many of these problems are relevant to the design process; they may involve the analysis of a structure for various configurations and loadings of the structure, or the determination of the equilibrium positions of a given mechanism that may require an iterative method of solution. Developing the algorithm required to solve a given mechanics problem will benefit the students in two different ways: (1) it will help them gain a better understanding of the mechanics principles involved; (2) it will provide them with an opportunity to apply their computer skills to the solution of a meaningful engineering problem.

Concept Questions. Educational research has shown that students can often choose appropriate equations and solve algorithmic problems without having a strong conceptual understanding of mechanics principles.[†] To help assess and develop student conceptual understanding, we have included Concept Questions, which are multiple choice problems that require few, if any, calculations. Each possible incorrect answer typically represents a common misconception (e.g., students often think that a vehicle moving in a curved path at constant speed has zero acceleration). Students are encouraged to solve these problems using the principles and techniques discussed in the text and to use these principles to help them develop their intuition. Mastery and discussion of these Concept Questions will deepen students' conceptual understanding and help them to solve dynamics problems.

Free Body and Impulse-Momentum Diagram Practice Problems. Drawing diagrams correctly is a critical step in solving kinetics problems in dynamics. A new type of problem has been added to the text to emphasize the importance of drawing these diagrams. In Chaps. 12 and 16 the Free Body Practice Problems require students to draw a free-body diagram (FBD) showing the applied forces and an equivalent diagram called a “kinetic diagram” (KD) showing $m\mathbf{a}$ or its components and $\bar{I}\alpha$. These diagrams provide students with a pictorial representation of Newton's second law and are critical in helping students to correctly solve kinetic problems. In Chaps. 13 and 17 the Impulse-Momentum Diagram Practice Problems require students to draw diagrams showing the momenta of the bodies before impact, the impulses exerted on the body during impact, and the final momenta of the bodies. The answers to all of these questions can be accessed through Connect.

FREE-BODY PRACTICE PROBLEMS

16.F1 A 6-ft board is placed in a truck with one end resting against a block secured to the floor and the other leaning against a vertical partition. Draw the FBD and KD necessary to determine the maximum allowable acceleration of the truck if the board is to remain in the position shown.

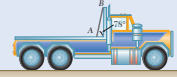


Fig. P16.F1

16.F2 A uniform circular plate of mass 3 kg is attached to two links AC and BD of the same length. Knowing that the plate is released from rest in the position shown, in which lines joining G to A and B are, respectively, horizontal and vertical, draw the FBD and KD for the plate.

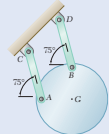


Fig. P16.F2

16.F3 Two uniform disks and two cylinders are assembled as indicated. Disk A weighs 20 lb and disk B weighs 12 lb. Knowing that the system is released from rest, draw the FBD and KD for the whole system.

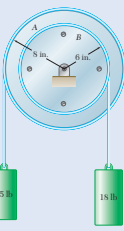


Fig. P16.F3

16.F4 The 400-lb crate shown is lowered by means of two overhead cranes. Knowing the tension in each cable, draw the FBD and KD that can be used to determine the angular acceleration of the crate and the acceleration of the center of gravity.

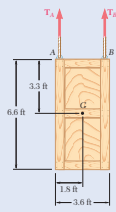


Fig. P16.F4

[†]Hestenes, D., Wells, M., and Swakhamer, G (1992). The force concept inventory. *The Physics Teacher*, 30: 141–158.

Streveler, R. A., Litzinger, T. A., Miller, R. L., and Steif, P. S. (2008). Learning conceptual knowledge in the engineering sciences: Overview and future research directions, *JEE*, 279–294.

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- **Lecture PowerPoint Slides** for each chapter that can be modified. These generally have an introductory application slide, animated worked-out problems that you can do in class with your students, concept questions, and “what-if?” questions at the end of the units.
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Acknowledgments

A special thanks to our colleagues who thoroughly checked the solutions and answers to all problems in this edition and then prepared the solutions for the accompanying *Instructor's and Solutions Manual*, James Widmann of California Polytechnic State University and Amy Mazurek.

The authors thank the many companies that provided photographs for this edition.

We are pleased to acknowledge David Chelton, who carefully reviewed the entire text and provided many helpful suggestions for revising this edition.

The authors also thank the members of the staff at McGraw-Hill for their support and dedication during the preparation of this new edition. We particularly wish to acknowledge the contributions of Global Brand Manager Raghu Srinivasan, Brand Manager Thomas Scaife, Product Developers Robin Reed & Joan Weber, Content Project Manager Jolynn Kilburg, and Program Manager Lora Neyens.

David F. Mazurek
Phillip J. Cornwell
Brian P. Self

The authors gratefully acknowledge the many helpful comments and suggestions offered by focus group attendees and by users of the previous editions of *Vector Mechanics for Engineers*:

George Adams
Northeastern University
William Altenhof
University of Windsor
Sean B. Anderson
Boston University
Manohar Arora
Colorado School of Mines
Gilbert Baladi
Michigan State University
Francois Barthelat
McGill University
Oscar Barton, Jr.
U.S. Naval Academy
M. Asghar Bhatti
University of Iowa
Shaohong Cheng
University of Windsor
Philip Datsis
University of Rhode Island
Timothy A. Doughty
University of Portland

Howard Epstein
University of Connecticut
Asad Esmaeily
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Engineering Department
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Florida Institute of Technology
Jeff Hanson
Texas Tech University
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University of Florida
Shaofan Li
University of California, Berkeley
William R. Murray
Cal Poly State University
Eric Musselman
University of Minnesota, Duluth
Masoud Olia
Wentworth Institute of Technology
Renee K. B. Petersen
Washington State University

Amir G Rezaei
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North Carolina State University
Yixin Shao
McGill University
Muhammad Sharif
The University of Alabama
Anthony Sinclair
University of Toronto
Lizhi Sun
University of California, Irvine
Jeffrey Thomas
Northwestern University
Jiashi Yang
University of Nebraska
Xiangwa Zeng
Case Western Reserve University

List of Symbols

$\mathbf{a}, a$	Acceleration	$\bar{k}$	Centroidal radius of gyration
a	Constant; radius; distance; semimajor axis of ellipse	l	Length
$\bar{\mathbf{a}}, \bar{a}$	Acceleration of mass center	$\mathbf{L}$	Linear momentum
$\mathbf{a}_{B/A}$	Acceleration of B relative to frame in translation with A	L	Length; inductance
$\mathbf{a}_{P/\mathcal{F}}$	Acceleration of P relative to rotating frame $\mathcal{F}$	m	Mass
$\mathbf{a}_c$	Coriolis acceleration	m'	Mass per unit length
$\mathbf{A}, \mathbf{B}, \mathbf{C}, \dots$	Reactions at supports and connections	$\mathbf{M}$	Couple; moment
$A, B, C, \dots$	Points	$\mathbf{M}_O$	Moment about point O
A	Area	$\mathbf{M}_O^R$	Moment resultant about point O
b	Width; distance; semiminor axis of ellipse	M	Magnitude of couple or moment; mass of earth
c	Constant; coefficient of viscous damping	M_{OL}	Moment about axis OL
C	Centroid; instantaneous center of rotation; capacitance	n	Normal direction
d	Distance	$\mathbf{N}$	Normal component of reaction
$\mathbf{e}_n, \mathbf{e}_t$	Unit vectors along normal and tangent	O	Origin of coordinates
$\mathbf{e}_r, \mathbf{e}_\theta$	Unit vectors in radial and transverse directions	$\mathbf{P}$	Force; vector
e	Coefficient of restitution; base of natural logarithms	$\dot{\mathbf{P}}$	Rate of change of vector $\mathbf{P}$ with respect to frame of fixed orientation
E	Total mechanical energy; voltage	q	Mass rate of flow; electric charge
f	Scalar function	$\mathbf{Q}$	Force; vector
f_f	Frequency of forced vibration	$\dot{\mathbf{Q}}$	Rate of change of vector $\mathbf{Q}$ with respect to frame of fixed orientation
f_n	Natural frequency	$(\dot{\mathbf{Q}})_{Oxyz}$	Rate of change of vector $\mathbf{Q}$ with respect to frame $Oxyz$
$\mathbf{F}$	Force; friction force	$\mathbf{r}$	Position vector
g	Acceleration of gravity	$\mathbf{r}_{B/A}$	Position vector of B relative to A
G	Center of gravity; mass center; constant of gravitation	r	Radius; distance; polar coordinate
h	Angular momentum per unit mass	$\mathbf{R}$	Resultant force; resultant vector; reaction
$\mathbf{H}_O$	Angular momentum about point O	R	Radius of earth; resistance
$\dot{\mathbf{H}}_G$	Rate of change of angular momentum $\mathbf{H}_G$ with respect to frame of fixed orientation	$\mathbf{s}$	Position vector
$(\dot{\mathbf{H}}_G)_{Gxyz}$	Rate of change of angular momentum $\mathbf{H}_G$ with respect to rotating frame $Gxyz$	s	Length of arc
$\mathbf{i}, \mathbf{j}, \mathbf{k}$	Unit vectors along coordinate axes	t	Time; thickness; tangential direction
i	Current	$\mathbf{T}$	Force
$I, I_x, \dots$	Moments of inertia	T	Tension; kinetic energy
$\bar{I}$	Centroidal moment of inertia	$\mathbf{u}$	Velocity
$I_{xy}, \dots$	Products of inertia	u	Variable
J	Polar moment of inertia	U	Work
k	Spring constant	U_{1-2}^{NC}	work done by non-conservative forces
k_x, k_y, k_O	Radii of gyration	$\mathbf{v}, v$	Velocity
		v	Speed
		$\bar{\mathbf{v}}, \bar{v}$	Velocity of mass center
		$\mathbf{v}_{B/A}$	Velocity of B relative to frame in translation with A
		$\mathbf{v}_{P/\mathcal{F}}$	Velocity of P relative to rotating frame $\mathcal{F}$

$\mathbf{V}$	Vector product	θ	Angular coordinate; Eulerian angle; angle; polar coordinate
V	Volume; potential energy	μ	Coefficient of friction
w	Load per unit length	ρ	Density; radius of curvature
$\mathbf{W}, W$	Weight; load	τ	Periodic time
x, y, z	Rectangular coordinates; distances	τ_n	Period of free vibration
$\dot{x}, \dot{y}, \dot{z}$	Time derivatives of coordinates x, y, z	ϕ	Angle of friction; Eulerian angle; phase angle; angle
$\bar{x}, \bar{y}, \bar{z}$	Rectangular coordinates of centroid, center of gravity, or mass center	φ	Phase difference
α, α	Angular acceleration	ψ	Eulerian angle
α, β, γ	Angles	ω, ω	Angular velocity
γ	Specific weight	ω_f	Circular frequency of forced vibration
δ	Elongation	ω_n	Natural circular frequency
ε	Eccentricity of conic section or of orbit	Ω	Angular velocity of frame of reference
λ	Unit vector along a line		
η	Efficiency		



1

Introduction

The tallest skyscraper in the Western Hemisphere, One World Trade Center is a prominent feature of the New York City skyline. From its foundation to its structural components and mechanical systems, the design and operation of the tower is based on the fundamentals of engineering mechanics.

Introduction

- 1.1 WHAT IS MECHANICS?
- 1.2 FUNDAMENTAL CONCEPTS AND PRINCIPLES
- 1.3 SYSTEMS OF UNITS
- 1.4 CONVERTING BETWEEN TWO SYSTEMS OF UNITS
- 1.5 METHOD OF SOLVING PROBLEMS
- 1.6 NUMERICAL ACCURACY

Objectives

- **Define** the science of mechanics and examine its fundamental principles.
- **Discuss** and compare the International System of Units and U.S. Customary Units.
- **Discuss** how to approach the solution of mechanics problems, and introduce the SMART problem-solving methodology.
- **Examine** factors that govern numerical accuracy in the solution of a mechanics problem.

1.1 What is Mechanics?

Mechanics is defined as the science that describes and predicts the conditions of rest or motion of bodies under the action of forces. It consists of the mechanics of *rigid bodies*, mechanics of *deformable bodies*, and mechanics of *fluids*.

The mechanics of rigid bodies is subdivided into **statics** and **dynamics**. Statics deals with bodies at rest; dynamics deals with bodies in motion. In this text, we assume bodies are perfectly rigid. In fact, actual structures and machines are never absolutely rigid; they deform under the loads to which they are subjected. However, because these deformations are usually small, they do not appreciably affect the conditions of equilibrium or the motion of the structure under consideration. They are important, though, as far as the resistance of the structure to failure is concerned. Deformations are studied in a course in mechanics of materials, which is part of the mechanics of deformable bodies. The third division of mechanics, the mechanics of fluids, is subdivided into the study of *incompressible fluids* and of *compressible fluids*. An important subdivision of the study of incompressible fluids is *hydraulics*, which deals with applications involving water.

Mechanics is a physical science, since it deals with the study of physical phenomena. However, some teachers associate mechanics with mathematics, whereas many others consider it as an engineering subject. Both these views are justified in part. Mechanics is the foundation of most engineering sciences and is an indispensable prerequisite to their study. However, it does not have the *empiricism* found in some engineering sciences, i.e., it does not rely on experience or observation alone. The rigor of mechanics and the emphasis it places on deductive reasoning makes it resemble mathematics. However, mechanics is not an *abstract* or even a *pure* science; it is an *applied* science.

The purpose of mechanics is to explain and predict physical phenomena and thus to lay the foundations for engineering applications. You need to know statics to determine how much force will be exerted on a point in a bridge design and whether the structure can withstand that force. Determining the force a dam needs to withstand from the water in a river requires statics. You need statics to calculate how much weight a crane can lift, how much force a locomotive needs to pull a freight train, or how

much force a circuit board in a computer can withstand. The concepts of dynamics enable you to analyze the flight characteristics of a jet, design a building to resist earthquakes, and mitigate shock and vibration to passengers inside a vehicle. The concepts of dynamics enable you to calculate how much force you need to send a satellite into orbit, accelerate a 200,000-ton cruise ship, or design a toy truck that doesn't break. You will not learn how to do these things in this course, but the ideas and methods you learn here will be the underlying basis for the engineering applications you will learn in your work.

1.2 Fundamental Concepts and Principles

Although the study of mechanics goes back to the time of Aristotle (384–322 B.C.) and Archimedes (287–212 B.C.), not until Newton (1642–1727) did anyone develop a satisfactory formulation of its fundamental principles. These principles were later modified by d'Alembert, Lagrange, and Hamilton. Their validity remained unchallenged until Einstein formulated his **theory of relativity** (1905). Although its limitations have now been recognized, **newtonian mechanics** still remains the basis of today's engineering sciences.

The basic concepts used in mechanics are *space*, *time*, *mass*, and *force*. These concepts cannot be truly defined; they should be accepted on the basis of our intuition and experience and used as a mental frame of reference for our study of mechanics.

The concept of **space** is associated with the position of a point P . We can define the position of P by providing three lengths measured from a certain reference point, or *origin*, in three given directions. These lengths are known as the *coordinates* of P .

To define an event, it is not sufficient to indicate its position in space. We also need to specify the **time** of the event.

We use the concept of **mass** to characterize and compare bodies on the basis of certain fundamental mechanical experiments. Two bodies of the same mass, for example, are attracted by the earth in the same manner; they also offer the same resistance to a change in translational motion.

A **force** represents the action of one body on another. A force can be exerted by actual contact, like a push or a pull, or at a distance, as in the case of gravitational or magnetic forces. A force is characterized by its *point of application*, its *magnitude*, and its *direction*; a force is represented by a *vector* (Sec. 2.1B).

In newtonian mechanics, space, time, and mass are absolute concepts that are independent of each other. (This is not true in **relativistic mechanics**, where the duration of an event depends upon its position and the mass of a body varies with its velocity.) On the other hand, the concept of force is not independent of the other three. Indeed, one of the fundamental principles of newtonian mechanics listed below is that the resultant force acting on a body is related to the mass of the body and to the manner in which its velocity varies with time.

In this text, you will study the conditions of rest or motion of particles and rigid bodies in terms of the four basic concepts we have introduced. By **particle**, we mean a very small amount of matter, which we

assume occupies a single point in space. A **rigid body** consists of a large number of particles occupying fixed positions with respect to one another. The study of the mechanics of particles is clearly a prerequisite to that of rigid bodies. Besides, we can use the results obtained for a particle directly in a large number of problems dealing with the conditions of rest or motion of actual bodies.

The study of elementary mechanics rests on six fundamental principles, based on experimental evidence.

- **The Parallelogram Law for the Addition of Forces.** Two forces acting on a particle may be replaced by a single force, called their *resultant*, obtained by drawing the diagonal of the parallelogram with sides equal to the given forces (Sec. 2.1A).
- **The Principle of Transmissibility.** The conditions of equilibrium or of motion of a rigid body remain unchanged if a force acting at a given point of the rigid body is replaced by a force of the same magnitude and same direction, but acting at a different point, provided that the two forces have the same line of action (Sec. 3.1B).
- **Newton's Three Laws of Motion.** Formulated by Sir Isaac Newton in the late seventeenth century, these laws can be stated as follows:

FIRST LAW. If the resultant force acting on a particle is zero, the particle remains at rest (if originally at rest) or moves with constant speed in a straight line (if originally in motion) (Sec. 2.3B).

SECOND LAW. If the resultant force acting on a particle is not zero, the particle has an acceleration proportional to the magnitude of the resultant and in the direction of this resultant force.

As you will see in Sec. 12.1, this law can be stated as

$$\mathbf{F} = m\mathbf{a} \quad (1.1)$$

where $\mathbf{F}$, m , and $\mathbf{a}$ represent, respectively, the resultant force acting on the particle, the mass of the particle, and the acceleration of the particle expressed in a consistent system of units.

THIRD LAW. The forces of action and reaction between bodies in contact have the same magnitude, same line of action, and opposite sense (Ch. 6, Introduction).

- **Newton's Law of Gravitation.** Two particles of mass M and m are mutually attracted with equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ of magnitude F (Fig. 1.1), given by the formula

$$F = G \frac{Mm}{r^2} \quad (1.2)$$

where r = the distance between the two particles and G = a universal constant called the *constant of gravitation*. Newton's law of gravitation introduces the idea of an action exerted at a distance and extends the range of application of Newton's third law: the action $\mathbf{F}$ and the reaction $-\mathbf{F}$ in Fig. 1.1 are equal and opposite, and they have the same line of action.

A particular case of great importance is that of the attraction of the earth on a particle located on its surface. The force $\mathbf{F}$ exerted by the earth on the particle is defined as the **weight $\mathbf{W}$** of the particle. Suppose we set

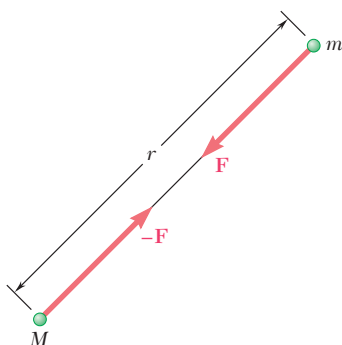


Fig. 1.1 From Newton's law of gravitation, two particles of masses M and m exert forces upon each other of equal magnitude, opposite direction, and the same line of action. This also illustrates Newton's third law of motion.

M equal to the mass of the earth, m equal to the mass of the particle, and r equal to the earth's radius R . Then introducing the constant

$$g = \frac{GM}{R^2} \quad (1.3)$$

we can express the magnitude W of the weight of a particle of mass m as[†]

$$W = mg \quad (1.4)$$

The value of R in formula (1.3) depends upon the elevation of the point considered; it also depends upon its latitude, since the earth is not truly spherical. The value of g therefore varies with the position of the point considered. However, as long as the point actually remains on the earth's surface, it is sufficiently accurate in most engineering computations to assume that g equals 9.81 m/s^2 or 32.2 ft/s^2 .

The principles we have just listed will be introduced in the course of our study of mechanics as they are needed. The statics of particles carried out in Chap. 2 will be based on the parallelogram law of addition and on Newton's first law alone. We introduce the principle of transmissibility in Chap. 3 as we begin the study of the statics of rigid bodies, and we bring in Newton's third law in Chap. 6 as we analyze the forces exerted on each other by the various members forming a structure. We introduce Newton's second law and Newton's law of gravitation in dynamics. We will then show that Newton's first law is a particular case of Newton's second law (Sec. 12.1) and that the principle of transmissibility could be derived from the other principles and thus eliminated (Sec. 16.1D). In the meantime, however, Newton's first and third laws, the parallelogram law of addition, and the principle of transmissibility will provide us with the necessary and sufficient foundation for the entire study of the statics of particles, rigid bodies, and systems of rigid bodies.

As noted earlier, the six fundamental principles listed previously are based on experimental evidence. Except for Newton's first law and the principle of transmissibility, they are independent principles that cannot be derived mathematically from each other or from any other elementary physical principle. On these principles rests most of the intricate structure of newtonian mechanics. For more than two centuries, engineers have solved a tremendous number of problems dealing with the conditions of rest and motion of rigid bodies, deformable bodies, and fluids by applying these fundamental principles. Many of the solutions obtained could be checked experimentally, thus providing a further verification of the principles from which they were derived. Only in the twentieth century has Newton's mechanics found to be at fault, in the study of the motion of atoms and the motion of the planets, where it must be supplemented by the theory of relativity. On the human or engineering scale, however, where velocities are small compared with the speed of light, Newton's mechanics have yet to be disproved.

1.3 Systems of Units

Associated with the four fundamental concepts just discussed are the so-called *kinetic units*, i.e., the units of *length*, *time*, *mass*, and *force*. These units cannot be chosen independently if Eq. (1.1) is to be satisfied.

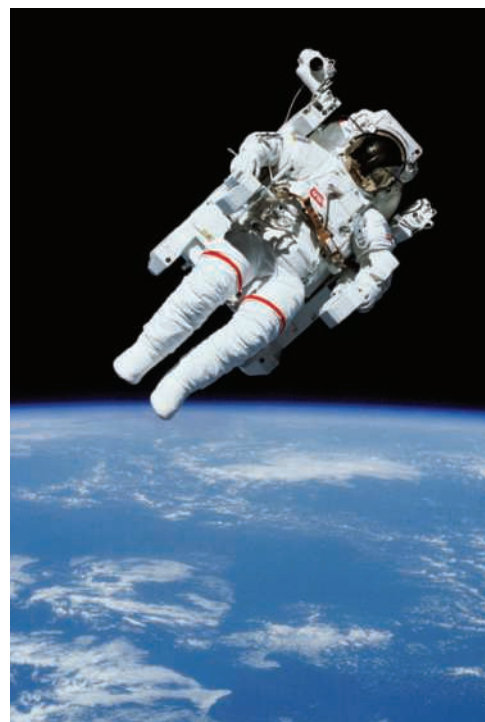


Photo 1.1 When in orbit of the earth, people and objects are said to be *weightless* even though the gravitational force acting is approximately 90% of that experienced on the surface of the earth. This apparent contradiction will be resolved in Chapter 12 when we apply Newton's second law to the motion of particles.

[†]A more accurate definition of the weight $\mathbf{W}$ should take into account the earth's rotation.

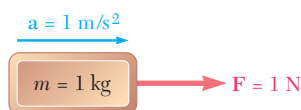


Fig. 1.2 A force of 1 newton applied to a body of mass 1 kg provides an acceleration of 1 m/s^2 .

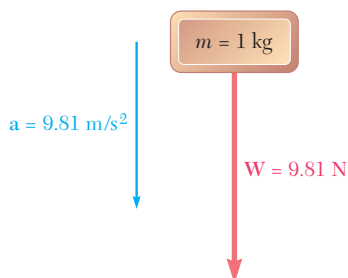


Fig. 1.3 A body of mass 1 kg experiencing an acceleration due to gravity of 9.81 m/s^2 has a weight of 9.81 N.

Three of the units may be defined arbitrarily; we refer to them as **basic units**. The fourth unit, however, must be chosen in accordance with Eq. (1.1) and is referred to as a **derived unit**. Kinetic units selected in this way are said to form a **consistent system of units**.

International System of Units (SI Units).[†] In this system, which will be in universal use after the United States has completed its conversion to SI units, the base units are the units of length, mass, and time, and they are called, respectively, the **meter** (m), the **kilogram** (kg), and the **second** (s). All three are arbitrarily defined. The second was originally chosen to represent $1/86\,400$ of the mean solar day, but it is now defined as the duration of $9\,192\,631\,770$ cycles of the radiation corresponding to the transition between two levels of the fundamental state of the cesium-133 atom. The meter, originally defined as one ten-millionth of the distance from the equator to either pole, is now defined as $1\,650\,763.73$ wavelengths of the orange-red light corresponding to a certain transition in an atom of krypton-86. (The newer definitions are much more precise and with today's modern instrumentation, are easier to verify as a standard.) The kilogram, which is approximately equal to the mass of 0.001 m^3 of water, is defined as the mass of a platinum-iridium standard kept at the International Bureau of Weights and Measures at Sèvres, near Paris, France. The unit of force is a derived unit. It is called the **newton** (N) and is defined as the force that gives an acceleration of 1 m/s^2 to a body of mass 1 kg (Fig. 1.2). From Eq. (1.1), we have

$$1 \text{ N} = (1 \text{ kg})(1 \text{ m/s}^2) = 1 \text{ kg} \cdot \text{m/s}^2 \quad (1.5)$$

The SI units are said to form an *absolute* system of units. This means that the three base units chosen are independent of the location where measurements are made. The meter, the kilogram, and the second may be used anywhere on the earth; they may even be used on another planet and still have the same significance.

The *weight* of a body, or the *force of gravity* exerted on that body, like any other force, should be expressed in newtons. From Eq. (1.4), it follows that the weight of a body of mass 1 kg (Fig. 1.3) is

$$\begin{aligned} W &= mg \\ &= (1 \text{ kg})(9.81 \text{ m/s}^2) \\ &= 9.81 \text{ N} \end{aligned}$$

Multiples and submultiples of the fundamental SI units are denoted through the use of the prefixes defined in Table 1.1. The multiples and submultiples of the units of length, mass, and force most frequently used in engineering are, respectively, the *kilometer* (km) and the *millimeter* (mm); the *megagram*[‡] (Mg) and the *gram* (g); and the *kilonewton* (kN). According to Table 1.1, we have

$$\begin{aligned} 1 \text{ km} &= 1000 \text{ m} & 1 \text{ mm} &= 0.001 \text{ m} \\ 1 \text{ Mg} &= 1000 \text{ kg} & 1 \text{ g} &= 0.001 \text{ kg} \\ 1 \text{ kN} &= 1000 \text{ N} \end{aligned}$$

The conversion of these units into meters, kilograms, and newtons, respectively, can be effected by simply moving the decimal point three places

[†]SI stands for *Système International d'Unités* (French)

[‡]Also known as a *metric ton*.

Table 1.1 SI Prefixes

Multiplication Factor	Prefix [†]	Symbol
1 000 000 000 000 = 10^{12}	tera	T
1 000 000 000 = 10^9	giga	G
1 000 000 = 10^6	mega	M
1 000 = 10^3	kilo	k
100 = 10^2	hecto [‡]	h
10 = 10^1	deka [‡]	da
0.1 = 10^{-1}	deci [‡]	d
0.01 = 10^{-2}	centi [‡]	c
0.001 = 10^{-3}	milli	m
0.000 001 = 10^{-6}	micro	μ
0.000 000 001 = 10^{-9}	nano	n
0.000 000 000 001 = 10^{-12}	pico	p
0.000 000 000 000 001 = 10^{-15}	femto	f
0.000 000 000 000 000 001 = 10^{-18}	atto	a

[†]The first syllable of every prefix is accented, so that the prefix retains its identity. Thus, the preferred pronunciation of kilometer places the accent on the first syllable, not the second.

[‡]The use of these prefixes should be avoided, except for the measurement of areas and volumes and for the nontechnical use of centimeter, as for body and clothing measurements.

to the right or to the left. For example, to convert 3.82 km into meters, move the decimal point three places to the right:

$$3.82 \text{ km} = 3820 \text{ m}$$

Similarly, to convert 47.2 mm into meters, move the decimal point three places to the left:

$$47.2 \text{ mm} = 0.0472 \text{ m}$$

Using engineering notation, you can also write

$$\begin{aligned} 3.82 \text{ km} &= 3.82 \times 10^3 \text{ m} \\ 47.2 \text{ mm} &= 47.2 \times 10^{-3} \text{ m} \end{aligned}$$

The multiples of the unit of time are the *minute* (min) and the *hour* (h). Since 1 min = 60 s and 1 h = 60 min = 3600 s, these multiples cannot be converted as readily as the others.

By using the appropriate multiple or submultiple of a given unit, you can avoid writing very large or very small numbers. For example, it is usually simpler to write 427.2 km rather than 427 200 m and 2.16 mm rather than 0.002 16 m.[†]

Units of Area and Volume. The unit of area is the *square meter* (m^2), which represents the area of a square of side 1 m; the unit of volume is the *cubic meter* (m^3), which is equal to the volume of a cube of side 1 m. In order to avoid exceedingly small or large numerical values when computing areas and volumes, we use systems of subunits obtained by respectively squaring and cubing not only the millimeter, but also two intermediate

[†]Note that when more than four digits appear on either side of the decimal point to express a quantity in SI units—as in 427 000 m or 0.002 16 m—use spaces, never commas, to separate the digits into groups of three. This practice avoids confusion with the comma used in place of a decimal point, which is the convention in many countries.

submultiples of the meter: the *decimeter* (dm) and the *centimeter* (cm). By definition,

$$\begin{aligned}1 \text{ dm} &= 0.1 \text{ m} = 10^{-1} \text{ m} \\1 \text{ cm} &= 0.01 \text{ m} = 10^{-2} \text{ m} \\1 \text{ mm} &= 0.001 \text{ m} = 10^{-3} \text{ m}\end{aligned}$$

Therefore, the submultiples of the unit of area are

$$\begin{aligned}1 \text{ dm}^2 &= (1 \text{ dm})^2 = (10^{-1} \text{ m})^2 = 10^{-2} \text{ m}^2 \\1 \text{ cm}^2 &= (1 \text{ cm})^2 = (10^{-2} \text{ m})^2 = 10^{-4} \text{ m}^2 \\1 \text{ mm}^2 &= (1 \text{ mm})^2 = (10^{-3} \text{ m})^2 = 10^{-6} \text{ m}^2\end{aligned}$$

Similarly, the submultiples of the unit of volume are

$$\begin{aligned}1 \text{ dm}^3 &= (1 \text{ dm})^3 = (10^{-1} \text{ m})^3 = 10^{-3} \text{ m}^3 \\1 \text{ cm}^3 &= (1 \text{ cm})^3 = (10^{-2} \text{ m})^3 = 10^{-6} \text{ m}^3 \\1 \text{ mm}^3 &= (1 \text{ mm})^3 = (10^{-3} \text{ m})^3 = 10^{-9} \text{ m}^3\end{aligned}$$

Note that when measuring the volume of a liquid, the cubic decimeter (dm^3) is usually referred to as a *liter* (L).

Table 1.2 shows other derived SI units used to measure the moment of a force, the work of a force, etc. Although we will introduce these units in later chapters as they are needed, we should note an important rule at

Table 1.2 Principal SI Units Used in Mechanics

Quantity	Unit	Symbol	Formula
Acceleration	Meter per second squared	...	m/s^2
Angle	Radian	rad	$^\circ$
Angular acceleration	Radian per second squared	...	rad/s^2
Angular velocity	Radian per second	...	rad/s
Area	Square meter	...	m^2
Density	Kilogram per cubic meter	...	kg/m^3
Energy	Joule	J	$\text{N}\cdot\text{m}$
Force	Newton	N	$\text{kg}\cdot\text{m/s}^2$
Frequency	Hertz	Hz	s^{-1}
Impulse	Newton-second	...	$\text{kg}\cdot\text{m/s}$
Length	Meter	m	$^\circ$
Mass	Kilogram	kg	$^\circ$
Moment of a force	Newton-meter	...	$\text{N}\cdot\text{m}$
Power	Watt	W	J/s
Pressure	Pascal	Pa	N/m^2
Stress	Pascal	Pa	N/m^2
Time	Second	s	$^\circ$
Velocity	Meter per second	...	m/s
Volume			
Solids	Cubic meter	...	m^3
Liquids	Liter	L	10^{-3} m^3
Work	Joule	J	$\text{N}\cdot\text{m}$

[†]Supplementary unit (1 revolution = 2π rad = 360°).

[‡]Base unit.

this time: When a derived unit is obtained by dividing a base unit by another base unit, you may use a prefix in the numerator of the derived unit, but not in its denominator. For example, the constant k of a spring that stretches 20 mm under a load of 100 N is expressed as

$$k = \frac{100 \text{ N}}{20 \text{ mm}} = \frac{100 \text{ N}}{0.020 \text{ m}} = 5000 \text{ N/m} \text{ or } k = 5 \text{ kN/m}$$

but never as $k = 5 \text{ N/mm}$.

U.S. Customary Units. Most practicing American engineers still commonly use a system in which the base units are those of length, force, and time. These units are, respectively, the *foot* (ft), the *pound* (lb), and the *second* (s). The second is the same as the corresponding SI unit. The foot is defined as 0.3048 m. The pound is defined as the *weight* of a platinum standard, called the *standard pound*, which is kept at the National Institute of Standards and Technology outside Washington D.C., the mass of which is 0.453 592 43 kg. Since the weight of a body depends upon the earth's gravitational attraction, which varies with location, the standard pound should be placed at sea level and at a latitude of 45° to properly define a force of 1 lb. Clearly the U.S. customary units do not form an absolute system of units. Because they depend upon the gravitational attraction of the earth, they form a *gravitational* system of units.

Although the standard pound also serves as the unit of mass in commercial transactions in the United States, it cannot be used that way in engineering computations, because such a unit would not be consistent with the base units defined in the preceding paragraph. Indeed, when acted upon by a force of 1 lb—that is, when subjected to the force of gravity—the standard pound has the acceleration due to gravity, $g = 32.2 \text{ ft/s}^2$ (Fig. 1.4), not the unit acceleration required by Eq. (1.1). The unit of mass consistent with the foot, the pound, and the second is the mass that receives an acceleration of 1 ft/s^2 when a force of 1 lb is applied to it (Fig. 1.5). This unit, sometimes called a *slug*, can be derived from the equation $F = ma$ after substituting 1 lb for F and 1 ft/s^2 for a . We have

$$F = ma \quad 1 \text{ lb} = (1 \text{ slug})(1 \text{ ft/s}^2)$$

This gives us

$$1 \text{ slug} = \frac{1 \text{ lb}}{1 \text{ ft/s}^2} = 1 \text{ lb} \cdot \text{s}^2/\text{ft} \quad (1.6)$$

Comparing Figs. 1.4 and 1.5, we conclude that the slug is a mass 32.2 times larger than the mass of the standard pound.

The fact that, in the U.S. customary system of units, bodies are characterized by their weight in pounds rather than by their mass in slugs is convenient in the study of statics, where we constantly deal with weights and other forces and only seldom deal directly with masses. However, in the study of dynamics, where forces, masses, and accelerations are involved, the mass m of a body is expressed in slugs when its weight W is given in pounds. Recalling Eq. (1.4), we write

$$m = \frac{W}{g} \quad (1.7)$$

where g is the acceleration due to gravity ($g = 32.2 \text{ ft/s}^2$).

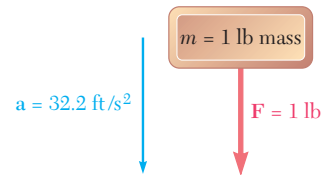


Fig. 1.4 A body of 1 pound mass acted upon by a force of 1 pound has an acceleration of 32.2 ft/s^2 .

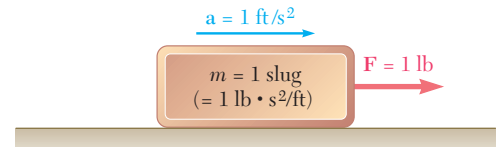


Fig. 1.5 A force of 1 pound applied to a body of mass 1 slug produces an acceleration of 1 ft/s^2 .

Other U.S. customary units frequently encountered in engineering problems are the *mile* (mi), equal to 5280 ft; the *inch* (in.), equal to (1/12) ft; and the *kilopound* (kip), equal to 1000 lb. The *ton* is often used to represent a mass of 2000 lb but, like the pound, must be converted into slugs in engineering computations.

The conversion into feet, pounds, and seconds of quantities expressed in other U.S. customary units is generally more involved and requires greater attention than the corresponding operation in SI units. For example, suppose we are given the magnitude of a velocity $v = 30$ mi/h and want to convert it to ft/s. First we write

$$v = 30 \frac{\text{mi}}{\text{h}}$$

Since we want to get rid of the unit miles and introduce instead the unit feet, we should multiply the right-hand member of the equation by an expression containing miles in the denominator and feet in the numerator. However, since we do not want to change the value of the right-hand side of the equation, the expression used should have a value equal to unity. The quotient (5280 ft)/(1 mi) is such an expression. Operating in a similar way to transform the unit hour into seconds, we have

$$v = \left(30 \frac{\text{mi}}{\text{h}}\right) \left(\frac{5280 \text{ ft}}{1 \text{ mi}}\right) \left(\frac{1 \text{ h}}{3600 \text{ s}}\right)$$

Carrying out the numerical computations and canceling out units that appear in both the numerator and the denominator, we obtain

$$v = 44 \frac{\text{ft}}{\text{s}} = 44 \text{ ft/s}$$

1.4 Converting Between Two Systems of Units

In many situations, an engineer might need to convert into SI units a numerical result obtained in U.S. customary units or vice versa. Because the unit of time is the same in both systems, only two kinetic base units need be converted. Thus, since all other kinetic units can be derived from these base units, only two conversion factors need be remembered.

Units of Length. By definition, the U.S. customary unit of length is

$$1 \text{ ft} = 0.3048 \text{ m} \quad (1.8)$$

It follows that

$$1 \text{ mi} = 5280 \text{ ft} = 5280(0.3048 \text{ m}) = 1609 \text{ m}$$

or

$$1 \text{ mi} = 1.609 \text{ km} \quad (1.9)$$

Also,

$$1 \text{ in.} = \frac{1}{12} \text{ ft} = \frac{1}{12}(0.3048 \text{ m}) = 0.0254 \text{ m}$$

or

$$1 \text{ in.} = 25.4 \text{ mm} \quad (1.10)$$

Units of Force. Recall that the U.S. customary unit of force (pound) is defined as the weight of the standard pound (of mass 0.4536 kg) at sea level and at a latitude of 45° (where $g = 9.807 \text{ m/s}^2$). Then, using Eq. (1.4), we write

$$\begin{aligned} W &= mg \\ 1 \text{ lb} &= (0.4536 \text{ kg})(9.807 \text{ m/s}^2) = 4.448 \text{ kg}\cdot\text{m/s}^2 \end{aligned}$$

From Eq. (1.5), this reduces to

$$1 \text{ lb} = 4.448 \text{ N} \quad (1.11)$$

Units of Mass. The U.S. customary unit of mass (slug) is a derived unit. Thus, using Eqs. (1.6), (1.8), and (1.11), we have

$$1 \text{ slug} = 1 \text{ lb}\cdot\text{s}^2/\text{ft} = \frac{1 \text{ lb}}{1 \text{ ft/s}^2} = \frac{4.448 \text{ N}}{0.3048 \text{ m/s}^2} = 14.59 \text{ N}\cdot\text{s}^2/\text{m}$$

Again, from Eq. (1.5),

$$1 \text{ slug} = 1 \text{ lb}\cdot\text{s}^2/\text{ft} = 14.59 \text{ kg} \quad (1.12)$$

Although it cannot be used as a consistent unit of mass, recall that the mass of the standard pound is, by definition,

$$1 \text{ pound mass} = 0.4536 \text{ kg} \quad (1.13)$$

We can use this constant to determine the *mass* in SI units (kilograms) of a body that has been characterized by its *weight* in U.S. customary units (pounds).

To convert a derived U.S. customary unit into SI units, simply multiply or divide by the appropriate conversion factors. For example, to convert the moment of a force that is measured as $M = 47 \text{ lb}\cdot\text{in.}$ into SI units, use formulas (1.10) and (1.11) and write

$$\begin{aligned} M &= 47 \text{ lb}\cdot\text{in.} = 47(4.448 \text{ N})(25.4 \text{ mm}) \\ &= 5310 \text{ N}\cdot\text{mm} = 5.31 \text{ N}\cdot\text{m} \end{aligned}$$

You can also use conversion factors to convert a numerical result obtained in SI units into U.S. customary units. For example, if the moment of a force is measured as $M = 40 \text{ N}\cdot\text{m}$, follow the procedure at the end of Sec. 1.3 to write

$$M = 40 \text{ N}\cdot\text{m} = (40 \text{ N}\cdot\text{m})\left(\frac{1 \text{ lb}}{4.448 \text{ N}}\right)\left(\frac{1 \text{ ft}}{0.3048 \text{ m}}\right)$$

Carrying out the numerical computations and canceling out units that appear in both the numerator and the denominator, you obtain

$$M = 29.5 \text{ lb}\cdot\text{ft}$$

The U.S. customary units most frequently used in mechanics are listed in Table 1.3 with their SI equivalents.

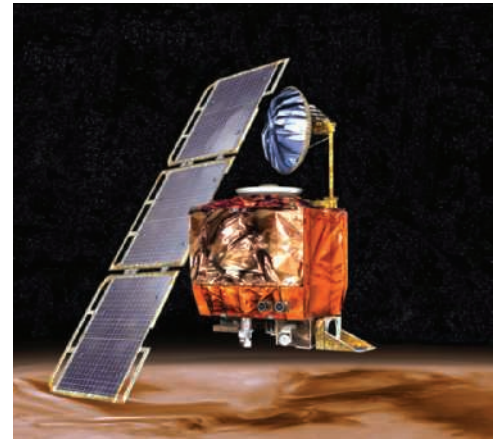


Photo 1.2 In 1999, The Mars Climate Orbiter entered orbit around Mars at too low an altitude and disintegrated. Investigation showed that the software on board the probe interpreted force instructions in newtons, but the software at mission control on the earth was generating those instructions in terms of pounds.

Table 1.3 U.S. Customary Units and Their SI Equivalents

Quantity	U.S. Customary Unit	SI Equivalent
Acceleration	ft/s ²	0.3048 m/s ²
	in./s ²	0.0254 m/s ²
Area	ft ²	0.0929 m ²
	in ²	645.2 mm ²
Energy	ft·lb	1.356 J
Force	kip	4.448 kN
	lb	4.448 N
	oz	0.2780 N
Impulse	lb·s	4.448 N·s
Length	ft	0.3048 m
	in.	25.40 mm
	mi	1.609 km
Mass	oz mass	28.35 g
	lb mass	0.4536 kg
	slug	14.59 kg
	ton	907.2 kg
Moment of a force	lb·ft	1.356 N·m
	lb·in.	0.1130 N·m
Moment of inertia		
	Of an area	in ⁴ 0.4162 × 10 ⁶ mm ⁴
Of a mass	lb·ft·s ²	1.356 kg·m ²
Momentum	lb·s	4.448 kg·m/s
Power	ft·lb/s	1.356 W
	hp	745.7 W
Pressure or stress	lb/ft ²	47.88 Pa
	lb/in ² (psi)	6.895 kPa
Velocity	ft/s	0.3048 m/s
	in./s	0.0254 m/s
	mi/h (mph)	0.4470 m/s
	mi/h (mph)	1.609 km/h
Volume	ft ³	0.02832 m ³
	in ³	16.39 cm ³
Liquids	gal	3.785 L
	qt	0.9464 L
Work	ft·lb	1.356 J

1.5 Method of Solving Problems

You should approach a problem in mechanics as you would approach an actual engineering situation. By drawing on your own experience and intuition about physical behavior, you will find it easier to understand and formulate the problem. Once you have clearly stated and understood the problem, however, there is no place in its solution for arbitrary methodologies.

The solution must be based on the six fundamental principles stated in Sec. 1.2 or on theorems derived from them.

Every step you take in the solution must be justified on this basis. Strict rules must be followed, which lead to the solution in an almost automatic fashion, leaving no room for your intuition or “feeling.” After you have obtained an answer, you should check it. Here again, you may call upon

your common sense and personal experience. If you are not completely satisfied with the result, you should carefully check your formulation of the problem, the validity of the methods used for its solution, and the accuracy of your computations.

In general, you can usually solve problems in several different ways; there is no one approach that works best for everybody. However, we have found that students often find it helpful to have a general set of guidelines to use for framing problems and planning solutions. In the Sample Problems throughout this text, we use a four-step method for approaching problems, which we refer to as the SMART methodology: **S**trategy, **M**odeling, **A**nalysis, and **R**eflect and **T**hink.

1. **Strategy.** The statement of a problem should be clear and precise, and it should contain the given data and indicate what information is required. The first step in solving the problem is to decide what concepts you have learned that apply to the given situation and to connect the data to the required information. It is often useful to work backward from the information you are trying to find: Ask yourself what quantities you need to know to obtain the answer, and if some of these quantities are unknown, how can you find them from the given data.
2. **Modeling.** The first step in modeling is to define the system; that is, clearly define what you are setting aside for analysis. After you have selected a system, draw a neat sketch showing all quantities involved with a separate diagram for each body in the problem. For equilibrium problems, indicate clearly the forces acting on each body along with any relevant geometrical data, such as lengths and angles. (These diagrams are known as **free-body diagrams** and are described in detail in Sec. 2.3C and the beginning of Ch. 4.)
3. **Analysis.** After you have drawn the appropriate diagrams, use the fundamental principles of mechanics listed in Sec. 1.2 to write equations expressing the conditions of rest or motion of the bodies considered. Each equation should be clearly related to one of the free-body diagrams and should be numbered. If you do not have enough equations to solve for the unknowns, try selecting another system, or reexamine your strategy to see if you can apply other principles to the problem. Once you have obtained enough equations, you can find a numerical solution by following the usual rules of algebra, neatly recording each step and the intermediate results. Alternatively, you can solve the resulting equations with your calculator or a computer. (For multipart problems, it is sometimes convenient to present the Modeling and Analysis steps together, but they are both essential parts of the overall process.)
4. **Reflect and Think.** After you have obtained the answer, check it carefully. Does it make sense in the context of the original problem? For instance, the problem may ask for the force at a given point of a structure. If your answer is negative, what does that mean for the force at the point?

You can often detect mistakes in *reasoning* by checking the units. For example, to determine the moment of a force of 50 N about a point 0.60 m from its line of action, we write (Sec. 3.3A)

$$M = Fd = (30 \text{ N})(0.60 \text{ m}) = 30 \text{ N}\cdot\text{m}$$

The unit N·m obtained by multiplying newtons by meters is the correct unit for the moment of a force; if you had obtained another unit, you would know that some mistake had been made.

You can often detect errors in *computation* by substituting the numerical answer into an equation that was not used in the solution and verifying that the equation is satisfied. The importance of correct computations in engineering cannot be overemphasized.

1.6 Numerical Accuracy

The accuracy of the solution to a problem depends upon two items: (1) the accuracy of the given data and (2) the accuracy of the computations performed. The solution cannot be more accurate than the less accurate of these two items.

For example, suppose the loading of a bridge is known to be 75 000 lb with a possible error of 100 lb either way. The relative error that measures the degree of accuracy of the data is

$$\frac{100 \text{ lb}}{75\,000 \text{ lb}} = 0.0013 = 0.13\%$$

In computing the reaction at one of the bridge supports, it would be meaningless to record it as 14 322 lb. The accuracy of the solution cannot be greater than 0.13%, no matter how precise the computations are, and the possible error in the answer may be as large as $(0.13/100)(14\,322 \text{ lb}) \approx 20 \text{ lb}$. The answer should be properly recorded as $14\,320 \pm 20 \text{ lb}$.

In engineering problems, the data are seldom known with an accuracy greater than 0.2%. It is therefore seldom justified to write answers with an accuracy greater than 0.2%. A practical rule is to use four figures to record numbers beginning with a “1” and three figures in all other cases. Unless otherwise indicated, you should assume the data given in a problem are known with a comparable degree of accuracy. A force of 40 lb, for example, should be read as 40.0 lb, and a force of 15 lb should be read as 15.00 lb.

Electronic calculators are widely used by practicing engineers and engineering students. The speed and accuracy of these calculators facilitate the numerical computations in the solution of many problems. However, you should not record more significant figures than can be justified merely because you can obtain them easily. As noted previously, an accuracy greater than 0.2% is seldom necessary or meaningful in the solution of practical engineering problems.



2

Statics of Particles

Many engineering problems can be solved by considering the equilibrium of a "particle." In the case of this beam that is being hoisted into position, a relation between the tensions in the various cables involved can be obtained by considering the equilibrium of the hook to which the cables are attached.

Introduction

2.1 ADDITION OF PLANAR FORCES

2.1A Force on a Particle: Resultant of Two Forces

2.1B Vectors

2.1C Addition of Vectors

2.1D Resultant of Several Concurrent Forces

2.1E Resolution of a Force into Components

2.2 ADDING FORCES BY COMPONENTS

2.2A Rectangular Components of a Force: Unit Vectors

2.2B Addition of Forces by Summing X and Y Components

2.3 FORCES AND EQUILIBRIUM IN A PLANE

2.3A Equilibrium of a Particle

2.3B Newton's First Law of Motion

2.3C Problems Involving the Equilibrium of a Particle: Free-Body Diagrams

2.4 ADDING FORCES IN SPACE

2.4A Rectangular Components of a Force in Space

2.4B Force Defined by Its Magnitude and Two Points on Its Line of Action

2.4C Addition of Concurrent Forces in Space

2.5 FORCES AND EQUILIBRIUM IN SPACE

Objectives

- **Describe** force as a vector quantity.
- **Examine** vector operations useful for the analysis of forces.
- **Determine** the resultant of multiple forces acting on a particle.
- **Resolve** forces into components.
- **Add** forces that have been resolved into rectangular components.
- **Introduce** the concept of the free-body diagram.
- **Use** free-body diagrams to assist in the analysis of planar and spatial particle equilibrium problems.

Introduction

In this chapter, you will study the effect of forces acting on particles. By the word “particle” we do not mean only tiny bits of matter, like an atom or an electron. Instead, we mean that the sizes and shapes of the bodies under consideration do not significantly affect the solutions of the problems. Another way of saying this is that we assume all forces acting on a given body act at the same point. This does not mean the object must be tiny—if you were modeling the mechanics of the Milky Way galaxy, for example, you could treat the Sun and the entire Solar System as just a particle.

Our first step is to explain how to replace two or more forces acting on a given particle by a single force having the same effect as the original forces. This single equivalent force is called the *resultant* of the original forces. After this step, we will derive the relations among the various forces acting on a particle in a state of *equilibrium*. We will use these relations to determine some of the forces acting on the particle.

The first part of this chapter deals with forces contained in a single plane. Because two lines determine a plane, this situation arises any time we can reduce the problem to one of a particle subjected to two forces that support a third force, such as a crate suspended from two chains or a traffic light held in place by two cables. In the second part of this chapter, we examine the more general case of forces in three-dimensional space.

2.1 ADDITION OF PLANAR FORCES

Many important practical situations in engineering involve forces in the same plane. These include forces acting on a pulley, projectile motion, and an object in equilibrium on a flat surface. We will examine this situation first before looking at the added complications of forces acting in three-dimensional space.

2.1A Force on a Particle: Resultant of Two Forces

A force represents the action of one body on another. It is generally characterized by its **point of application**, its **magnitude**, and its **direction**. Forces acting on a given particle, however, have the same point of application. Thus, each force considered in this chapter is completely defined by its magnitude and direction.

The magnitude of a force is characterized by a certain number of units. As indicated in Chap. 1, the SI units used by engineers to measure the magnitude of a force are the newton (N) and its multiple the kilonewton (kN), which is equal to 1000 N. The U.S. customary units used for the same purpose are the pound (lb) and its multiple the kilopound (kip), which is equal to 1000 lb. We saw in Chapter 1 that a force of 445 N is equivalent to a force of 100 lb or that a force of 100 N equals a force of about 22.5 lb.

We define the direction of a force by its **line of action** and the **sense** of the force. The line of action is the infinite straight line along which the force acts; it is characterized by the angle it forms with some fixed axis (Fig. 2.1). The force itself is represented by a segment of that line; through the use of an appropriate scale, we can choose the length of this segment to represent the magnitude of the force. We indicate the sense of the force by an arrowhead. It is important in defining a force to indicate its sense. Two forces having the same magnitude and the same line of action but a different sense, such as the forces shown in Fig. 2.1*a* and *b*, have directly opposite effects on a particle.

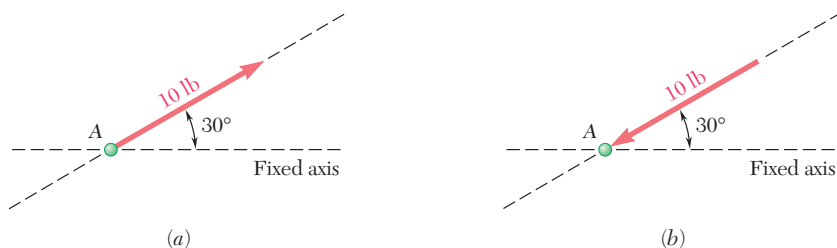


Fig. 2.1 The line of action of a force makes an angle with a given fixed axis. (a) The sense of the 10-lb force is away from particle A; (b) the sense of the 10-lb force is toward particle A.

Experimental evidence shows that two forces **P** and **Q** acting on a particle A (Fig. 2.2*a*) can be replaced by a single force **R** that has the same effect on the particle (Fig. 2.2*c*). This force is called the **resultant** of the forces **P** and **Q**. We can obtain **R**, as shown in Fig. 2.2*b*, by constructing a parallelogram, using **P** and **Q** as two adjacent sides. **The diagonal that passes through A represents the resultant.** This method for finding the resultant is known as the **parallelogram law** for the addition of two forces. This law is based on experimental evidence; it cannot be proved or derived mathematically.

2.1B Vectors

We have just seen that forces do not obey the rules of addition defined in ordinary arithmetic or algebra. For example, two forces acting at a right angle to each other, one of 4 lb and the other of 3 lb, add up to a force of

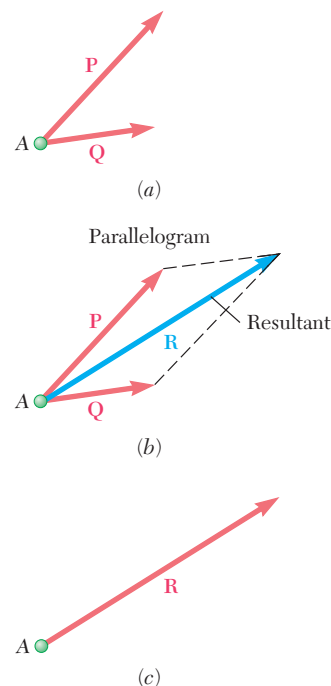


Fig. 2.2 (a) Two forces **P** and **Q** act on particle A. (b) Draw a parallelogram with **P** and **Q** as the adjacent sides and label the diagonal that passes through A as **R**. (c) **R** is the resultant of the two forces **P** and **Q** and is equivalent to their sum.



Photo 2.1 In its purest form, a tug-of-war pits two opposite and almost-equal forces against each other. Whichever team can generate the larger force, wins. As you can see, a competitive tug-of-war can be quite intense.

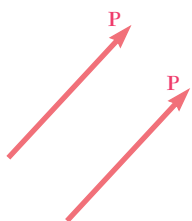


Fig. 2.3 Equal vectors have the same magnitude and the same direction, even if they have different points of application.

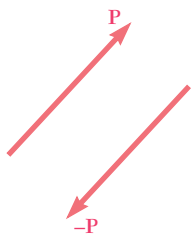


Fig. 2.4 The negative vector of a given vector has the same magnitude but the opposite direction of the given vector.

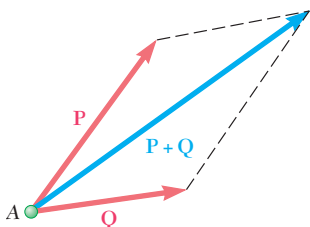


Fig. 2.5 Using the parallelogram law to add two vectors.

5 lb acting at an angle between them, *not* to a force of 7 lb. Forces are not the only quantities that follow the parallelogram law of addition. As you will see later, *displacements*, *velocities*, *accelerations*, and *momenta* are other physical quantities possessing magnitude and direction that add according to the parallelogram law. All of these quantities can be represented mathematically by **vectors**. Those physical quantities that have magnitude but not direction, such as *volume*, *mass*, or *energy*, are represented by plain numbers often called **scalars** to distinguish them from vectors.

Vectors are defined as **mathematical expressions possessing magnitude and direction, which add according to the parallelogram law**. Vectors are represented by arrows in diagrams and are distinguished from scalar quantities in this text through the use of boldface type (**P**). In longhand writing, a vector may be denoted by drawing a short arrow above the letter used to represent it ($\vec{P}$). The magnitude of a vector defines the length of the arrow used to represent it. In this text, we use italic type to denote the magnitude of a vector. Thus, the magnitude of the vector **P** is denoted by *P*.

A vector used to represent a force acting on a given particle has a well-defined point of application—namely, the particle itself. Such a vector is said to be a *fixed*, or *bound*, vector and cannot be moved without modifying the conditions of the problem. Other physical quantities, however, such as couples (see Chap. 3), are represented by vectors that may be freely moved in space; these vectors are called *free* vectors. Still other physical quantities, such as forces acting on a rigid body (see Chap. 3), are represented by vectors that can be moved along their lines of action; they are known as *sliding* vectors.

Two vectors that have the same magnitude and the same direction are said to be *equal*, whether or not they also have the same point of application (Fig. 2.3); equal vectors may be denoted by the same letter.

The *negative vector* of a given vector **P** is defined as a vector having the same magnitude as **P** and a direction opposite to that of **P** (Fig. 2.4); the negative of the vector **P** is denoted by $-\mathbf{P}$. The vectors **P** and $-\mathbf{P}$ are commonly referred to as **equal and opposite** vectors. Clearly, we have

$$\mathbf{P} + (-\mathbf{P}) = 0$$

2.1C Addition of Vectors

By definition, vectors add according to the parallelogram law. Thus, we obtain the sum of two vectors **P** and **Q** by attaching the two vectors to the same point *A* and constructing a parallelogram, using **P** and **Q** as two adjacent sides (Fig. 2.5). The diagonal that passes through *A* represents the sum of the vectors **P** and **Q**, denoted by $\mathbf{P} + \mathbf{Q}$. The fact that the sign $+$ is used for both vector and scalar addition should not cause any confusion if vector and scalar quantities are always carefully distinguished. Note that the magnitude of the vector $\mathbf{P} + \mathbf{Q}$ is *not*, in general, equal to the sum $P + Q$ of the magnitudes of the vectors **P** and **Q**.

Since the parallelogram constructed on the vectors **P** and **Q** does not depend upon the order in which **P** and **Q** are selected, we conclude that the addition of two vectors is *commutative*, and we write

$$\mathbf{P} + \mathbf{Q} = \mathbf{Q} + \mathbf{P} \quad (2.1)$$

From the parallelogram law, we can derive an alternative method for determining the sum of two vectors, known as the **triangle rule**. Consider Fig. 2.5, where the sum of the vectors $\mathbf{P}$ and $\mathbf{Q}$ has been determined by the parallelogram law. Since the side of the parallelogram opposite $\mathbf{Q}$ is equal to $\mathbf{Q}$ in magnitude and direction, we could draw only half of the parallelogram (Fig. 2.6a). The sum of the two vectors thus can be found by **arranging $\mathbf{P}$ and $\mathbf{Q}$ in tip-to-tail fashion and then connecting the tail of $\mathbf{P}$ with the tip of $\mathbf{Q}$** . If we draw the other half of the parallelogram, as in Fig. 2.6b, we obtain the same result, confirming that vector addition is commutative.

We define *subtraction* of a vector as the addition of the corresponding negative vector. Thus, we determine the vector $\mathbf{P} - \mathbf{Q}$, representing the difference between the vectors $\mathbf{P}$ and $\mathbf{Q}$, by adding to $\mathbf{P}$ the negative vector $-\mathbf{Q}$ (Fig. 2.7). We write

$$\mathbf{P} - \mathbf{Q} = \mathbf{P} + (-\mathbf{Q}) \quad (2.2)$$

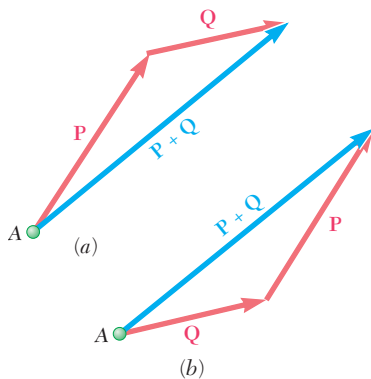


Fig. 2.6 The triangle rule of vector addition. (a) Adding vector $\mathbf{Q}$ to vector $\mathbf{P}$ equals (b) adding vector $\mathbf{P}$ to vector $\mathbf{Q}$.

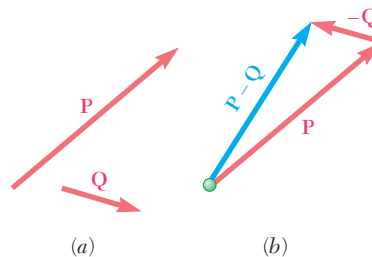


Fig. 2.7 Vector subtraction: Subtracting vector $\mathbf{Q}$ from vector $\mathbf{P}$ is the same as adding vector $-\mathbf{Q}$ to vector $\mathbf{P}$.

Here again we should observe that, although we use the same sign to denote both vector and scalar subtraction, we avoid confusion by taking care to distinguish between vector and scalar quantities.

We now consider the *sum of three or more vectors*. The sum of three vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ is, by *definition*, obtained by first adding the vectors $\mathbf{P}$ and $\mathbf{Q}$ and then adding the vector $\mathbf{S}$ to the vector $\mathbf{P} + \mathbf{Q}$. We write

$$\mathbf{P} + \mathbf{Q} + \mathbf{S} = (\mathbf{P} + \mathbf{Q}) + \mathbf{S} \quad (2.3)$$

Similarly, we obtain the sum of four vectors by adding the fourth vector to the sum of the first three. It follows that we can obtain the sum of any number of vectors by applying the parallelogram law repeatedly to successive pairs of vectors until all of the given vectors are replaced by a single vector.

If the given vectors are *coplanar*, i.e., if they are contained in the same plane, we can obtain their sum graphically. For this case, repeated application of the triangle rule is simpler than applying the parallelogram law. In Fig. 2.8a, we find the sum of three vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ in this manner. The triangle rule is first applied to obtain the sum $\mathbf{P} + \mathbf{Q}$ of the

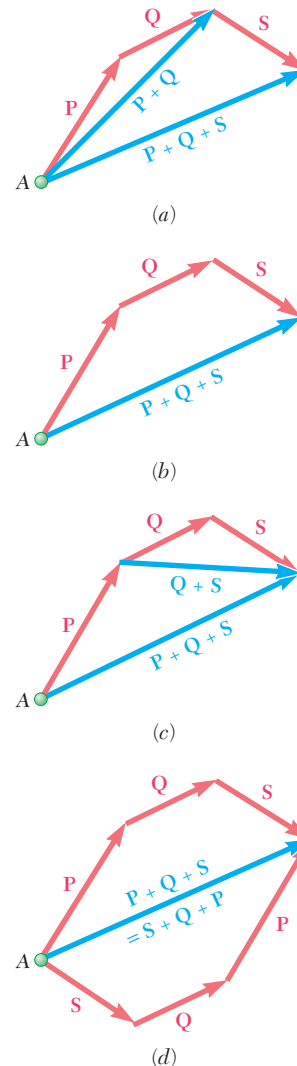


Fig. 2.8 Graphical addition of vectors. (a) Applying the triangle rule twice to add three vectors; (b) the vectors can be added in one step by the polygon rule; (c) vector addition is associative; (d) the order of addition is immaterial.

vectors $\mathbf{P}$ and $\mathbf{Q}$; we apply it again to obtain the sum of the vectors $\mathbf{P} + \mathbf{Q}$ and $\mathbf{S}$. However, we could have omitted determining the vector $\mathbf{P} + \mathbf{Q}$ and obtain the sum of the three vectors directly, as shown in Fig. 2.8b, by **arranging the given vectors in tip-to-tail fashion and connecting the tail of the first vector with the tip of the last one**. This is known as the **polygon rule** for the addition of vectors.

The result would be unchanged if, as shown in Fig. 2.8c, we had replaced the vectors $\mathbf{Q}$ and $\mathbf{S}$ by their sum $\mathbf{Q} + \mathbf{S}$. We may thus write

$$\mathbf{P} + \mathbf{Q} + \mathbf{S} = (\mathbf{P} + \mathbf{Q}) + \mathbf{S} = \mathbf{P} + (\mathbf{Q} + \mathbf{S}) \quad (2.4)$$

which expresses the fact that vector addition is *associative*. Recalling that vector addition also has been shown to be commutative in the case of two vectors, we can write

$$\begin{aligned} \mathbf{P} + \mathbf{Q} + \mathbf{S} &= (\mathbf{P} + \mathbf{Q}) + \mathbf{S} = \mathbf{S} + (\mathbf{P} + \mathbf{Q}) \\ &= \mathbf{S} + (\mathbf{Q} + \mathbf{P}) = \mathbf{S} + \mathbf{Q} + \mathbf{P} \end{aligned} \quad (2.5)$$

This expression, as well as others we can obtain in the same way, shows that the order in which several vectors are added together is immaterial (Fig. 2.8d).

Product of a Scalar and a Vector. It is convenient to denote the sum $\mathbf{P} + \mathbf{P}$ by $2\mathbf{P}$, the sum $\mathbf{P} + \mathbf{P} + \mathbf{P}$ by $3\mathbf{P}$, and, in general, the sum of n equal vectors $\mathbf{P}$ by the product $n\mathbf{P}$. Therefore, we define the product $n\mathbf{P}$ of a positive integer n and a vector $\mathbf{P}$ as a vector having the same direction as $\mathbf{P}$ and the magnitude nP . Extending this definition to include all scalars and recalling the definition of a negative vector given earlier, we define the product $k\mathbf{P}$ of a scalar k and a vector $\mathbf{P}$ as a vector having the same direction as $\mathbf{P}$ (if k is positive) or a direction opposite to that of $\mathbf{P}$ (if k is negative) and a magnitude equal to the product of P and the absolute value of k (Fig. 2.9).

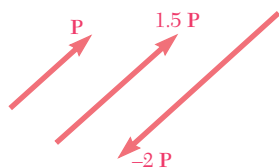


Fig. 2.9 Multiplying a vector by a scalar changes the vector's magnitude, but not its direction (unless the scalar is negative, in which case the direction is reversed).

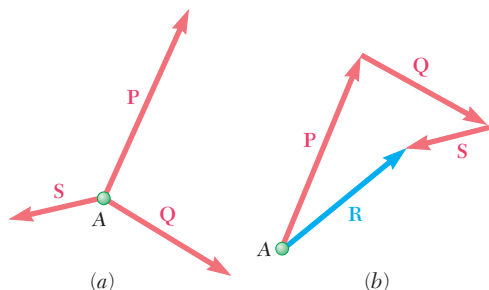


Fig. 2.10 Concurrent forces can be added by the polygon rule.

2.1D Resultant of Several Concurrent Forces

Consider a particle A acted upon by several coplanar forces, i.e., by several forces contained in the same plane (Fig. 2.10a). Since the forces all pass through A , they are also said to be *concurrent*. We can add the vectors representing the forces acting on A by the polygon rule (Fig. 2.10b). Since the use of the polygon rule is equivalent to the repeated application of the parallelogram law, the vector $\mathbf{R}$ obtained in this way represents the resultant of the given concurrent forces. That is, the single force $\mathbf{R}$ has the same effect on the particle A as the given forces. As before, the order in which we add the vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ representing the given forces is immaterial.

2.1E Resolution of a Force into Components

We have seen that two or more forces acting on a particle may be replaced by a single force that has the same effect on the particle. Conversely, a single

force $\mathbf{F}$ acting on a particle may be replaced by two or more forces that, together, have the same effect on the particle. These forces are called **components** of the original force $\mathbf{F}$, and the process of substituting them for $\mathbf{F}$ is called **resolving the force $\mathbf{F}$ into components**.

Clearly, each force $\mathbf{F}$ can be resolved into an infinite number of possible sets of components. Sets of *two components* $\mathbf{P}$ and $\mathbf{Q}$ are the most important as far as practical applications are concerned. However, even then, the number of ways in which a given force $\mathbf{F}$ may be resolved into two components is unlimited (Fig. 2.11).

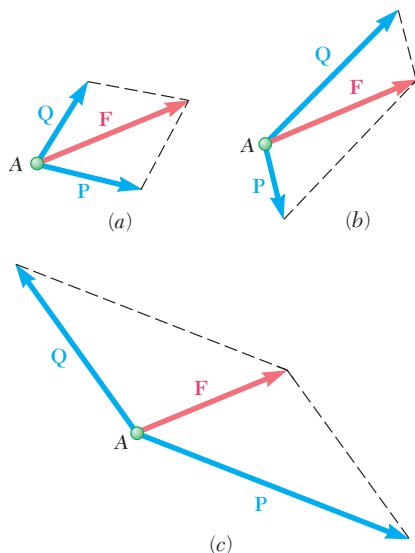


Fig. 2.11 Three possible sets of components for a given force vector $\mathbf{F}$.

In many practical problems, we start with a given vector $\mathbf{F}$ and want to determine a useful set of components. Two cases are of particular interest:

1. **One of the Two Components, $\mathbf{P}$, Is Known.** We obtain the second component, $\mathbf{Q}$, by applying the triangle rule and joining the tip of $\mathbf{P}$ to the tip of $\mathbf{F}$ (Fig. 2.12). We can determine the magnitude and direction of $\mathbf{Q}$ graphically or by trigonometry. Once we have determined $\mathbf{Q}$, both components $\mathbf{P}$ and $\mathbf{Q}$ should be applied at A .
2. **The Line of Action of Each Component Is Known.** We obtain the magnitude and sense of the components by applying the parallelogram law and drawing lines through the tip of $\mathbf{F}$ that are parallel to the given lines of action (Fig. 2.13). This process leads to two well-defined components, $\mathbf{P}$ and $\mathbf{Q}$, which can be determined graphically or computed trigonometrically by applying the law of sines.

You will encounter many similar cases; for example, you might know the direction of one component while the magnitude of the other component is to be as small as possible (see Sample Prob. 2.2). In all cases, you need to draw the appropriate triangle or parallelogram that satisfies the given conditions.

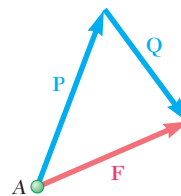


Fig. 2.12 When component $\mathbf{P}$ is known, use the triangle rule to find component $\mathbf{Q}$.

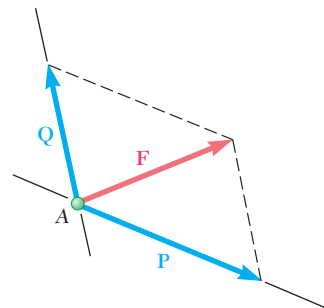


Fig. 2.13 When the lines of action are known, use the parallelogram rule to determine components $\mathbf{P}$ and $\mathbf{Q}$.

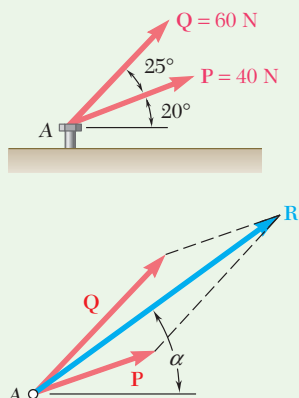


Fig. 1 Parallelogram law applied to add forces **P** and **Q**.

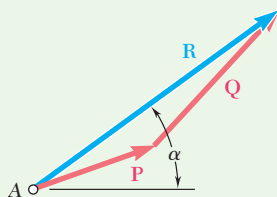


Fig. 2 Triangle rule applied to add forces **P** and **Q**.

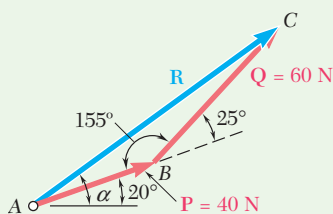


Fig. 3 Geometry of triangle rule applied to add forces **P** and **Q**.

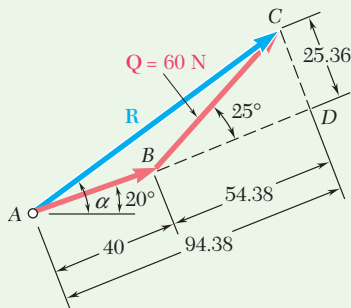


Fig. 4 Alternative geometry of triangle rule applied to add forces **P** and **Q**.

Sample Problem 2.1

Two forces **P** and **Q** act on a bolt **A**. Determine their resultant.

STRATEGY: Two lines determine a plane, so this is a problem of two coplanar forces. You can solve the problem graphically or by trigonometry.

MODELING: For a graphical solution, you can use the parallelogram rule or the triangle rule for addition of vectors. For a trigonometric solution, you can use the law of cosines and law of sines or use a right-triangle approach.

ANALYSIS:

Graphical Solution. Draw to scale a parallelogram with sides equal to **P** and **Q** (Fig. 1). Measure the magnitude and direction of the resultant. They are

$$R = 98 \text{ N} \quad \alpha = 35^\circ \quad \mathbf{R} = 98 \text{ N} \nearrow 35^\circ \quad \blacktriangleleft$$

You can also use the triangle rule. Draw forces **P** and **Q** in tip-to-tail fashion (Fig. 2). Again measure the magnitude and direction of the resultant. The answers should be the same.

$$R = 98 \text{ N} \quad \alpha = 35^\circ \quad \mathbf{R} = 98 \text{ N} \nearrow 35^\circ \quad \blacktriangleleft$$

Trigonometric Solution. Using the triangle rule again, you know two sides and the included angle (Fig. 3). Apply the law of cosines.

$$\begin{aligned} R^2 &= P^2 + Q^2 - 2PQ \cos B \\ R^2 &= (40 \text{ N})^2 + (60 \text{ N})^2 - 2(40 \text{ N})(60 \text{ N}) \cos 155^\circ \\ R &= 97.73 \text{ N} \end{aligned}$$

Now apply the law of sines:

$$\frac{\sin A}{Q} = \frac{\sin B}{R} \quad \frac{\sin A}{60 \text{ N}} = \frac{\sin 155^\circ}{97.73 \text{ N}} \quad (1)$$

Solving Eq. (1) for $\sin A$, you obtain

$$\sin A = \frac{(60 \text{ N}) \sin 155^\circ}{97.73 \text{ N}}$$

Using a calculator, compute this quotient, and then obtain its arc sine:

$$A = 15.04^\circ \quad \alpha = 20^\circ + A = 35.04^\circ$$

Use three significant figures to record the answer (cf. Sec. 1.6):

$$\mathbf{R} = 97.7 \text{ N} \nearrow 35.0^\circ \quad \blacktriangleleft$$

Alternative Trigonometric Solution. Construct the right triangle **BCD** (Fig. 4) and compute

$$\begin{aligned} CD &= (60 \text{ N}) \sin 25^\circ = 25.36 \text{ N} \\ BD &= (60 \text{ N}) \cos 25^\circ = 54.38 \text{ N} \end{aligned}$$

Then, using triangle ACD , you have

$$\tan A = \frac{25.36 \text{ N}}{94.38 \text{ N}} \quad A = 15.04^\circ$$

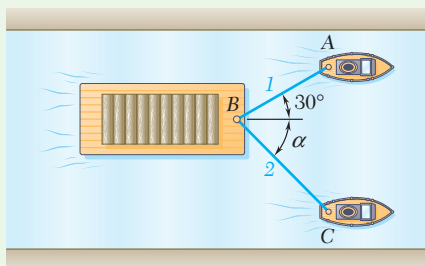
$$R = \frac{25.36}{\sin A} \quad R = 97.73 \text{ N}$$

Again,

$$\alpha = 20^\circ + A = 35.04^\circ \quad \mathbf{R = 97.7 \text{ N} } \angle 35.0^\circ \quad \blacktriangleleft$$

REFLECT and THINK: An analytical solution using trigonometry provides for greater accuracy. However, it is helpful to use a graphical solution as a check.

Sample Problem 2.2



Two tugboats are pulling a barge. If the resultant of the forces exerted by the tugboats is a 5000-lb force directed along the axis of the barge, determine (a) the tension in each of the ropes, given that $\alpha = 45^\circ$, (b) the value of α for which the tension in rope 2 is minimum.

STRATEGY: This is a problem of two coplanar forces. You can solve the first part either graphically or analytically. In the second part, a graphical approach readily shows the necessary direction for rope 2, and you can use an analytical approach to complete the solution.

MODELING: You can use the parallelogram law or the triangle rule to solve part (a). For part (b), use a variation of the triangle rule.

ANALYSIS: a. Tension for $\alpha = 45^\circ$.

Graphical Solution. Use the parallelogram law. The resultant (the diagonal of the parallelogram) is equal to 5000 lb and is directed to the right. Draw the sides parallel to the ropes (Fig. 1). If the drawing is done to scale, you should measure

$$T_1 = 3700 \text{ lb} \quad T_2 = 2600 \text{ lb} \quad \blacktriangleleft$$

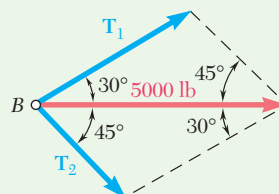


Fig. 1 Parallelogram law applied to add forces T_1 and T_2 .

(continued)

Trigonometric Solution. Use the triangle rule. Note that the triangle in Fig. 2 represents half of the parallelogram shown in Fig. 1. Using the law of sines,

$$\frac{T_1}{\sin 45^\circ} = \frac{T_2}{\sin 30^\circ} = \frac{5000 \text{ lb}}{\sin 105^\circ}$$

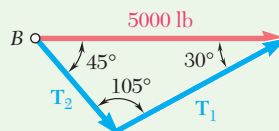


Fig. 2 Triangle rule applied to add forces T_1 and T_2 .

With a calculator, compute and store the value of the last quotient. Multiply this value successively by $\sin 45^\circ$ and $\sin 30^\circ$, obtaining

$$T_1 = 3660 \text{ lb} \quad T_2 = 2590 \text{ lb} \quad \blacktriangleleft$$

b. Value of α for Minimum T_2 . To determine the value of α for which the tension in rope 2 is minimum, use the triangle rule again. In Fig. 3, line $1-1'$ is the known direction of T_1 . Several possible directions of T_2 are shown by the lines $2-2'$. The minimum value of T_2 occurs when T_1 and T_2 are perpendicular (Fig. 4). Thus, the minimum value of T_2 is

$$T_2 = (5000 \text{ lb}) \sin 30^\circ = 2500 \text{ lb}$$

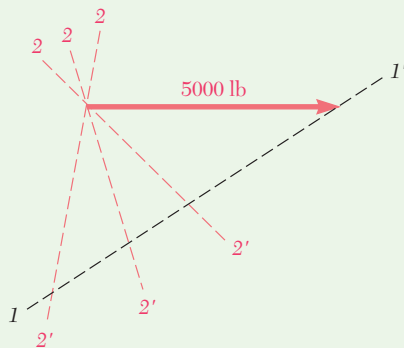


Fig. 3 Determination of direction of minimum T_2 .

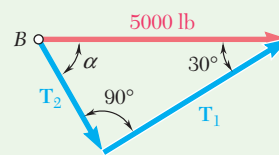


Fig. 4 Triangle rule applied for minimum T_2 .

Corresponding values of T_1 and α are

$$T_1 = (5000 \text{ lb}) \cos 30^\circ = 4330 \text{ lb}$$

$$\alpha = 90^\circ - 30^\circ \quad \alpha = 60^\circ \quad \blacktriangleleft$$

REFLECT and THINK: Part (a) is a straightforward application of resolving a vector into components. The key to part (b) is recognizing that the minimum value of T_2 occurs when T_1 and T_2 are perpendicular.

SOLVING PROBLEMS ON YOUR OWN

The preceding sections were devoted to adding vectors by using the parallelogram law, triangle rule, and polygon rule with application to forces.

We presented two sample problems. In Sample Prob. 2.1, we used the parallelogram law to determine the resultant of two forces of known magnitude and direction. In Sample Prob. 2.2, we used it to resolve a given force into two components of known direction.

You will now be asked to solve problems on your own. Some may resemble one of the sample problems; others may not. What all problems and sample problems in this section have in common is that they can be solved by direct application of the parallelogram law.

Your solution of a given problem should consist of the following steps:

1. Identify which forces are the applied forces and which is the resultant. It is often helpful to write the vector equation that shows how the forces are related. For example, in Sample Prob. 2.1 you could write

$$\mathbf{R} = \mathbf{P} + \mathbf{Q}$$

You may want to keep this relation in mind as you formulate the next part of the solution.

2. Draw a parallelogram with the applied forces as two adjacent sides and the resultant as the included diagonal (Fig. 2.2). Alternatively, you can use the **triangle rule** with the applied forces drawn in tip-to-tail fashion and the resultant extending from the tail of the first vector to the tip of the second (Fig. 2.6).

3. Indicate all dimensions. Using one of the triangles of the parallelogram or the triangle constructed according to the triangle rule, indicate all dimensions—whether sides or angles—and determine the unknown dimensions either graphically or by trigonometry.

4. Recall the laws of trigonometry. If you use trigonometry, remember that the law of cosines should be applied first if two sides and the included angle are known [Sample Prob. 2.1], and the law of sines should be applied first if one side and all angles are known [Sample Prob. 2.2].

If you have had prior exposure to mechanics, you might be tempted to ignore the solution techniques of this lesson in favor of resolving the forces into rectangular components. The component method is important and is considered in the next section, but use of the parallelogram law simplifies the solution of many problems and should be mastered first.

Problems

- 2.1** Two forces are applied as shown to a hook. Determine graphically the magnitude and direction of their resultant using (a) the parallelogram law, (b) the triangle rule.

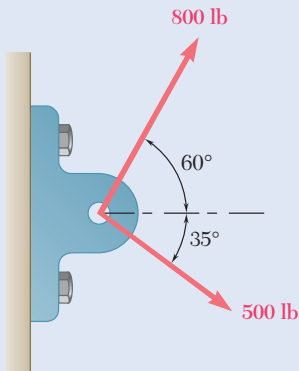


Fig. P2.2

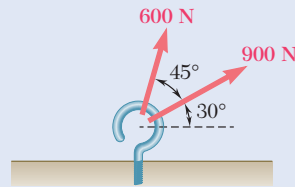


Fig. P2.1

- 2.2** Two forces are applied as shown to a bracket support. Determine graphically the magnitude and direction of their resultant using (a) the parallelogram law, (b) the triangle rule.

- 2.3** Two structural members B and C are bolted to bracket A . Knowing that both members are in tension and that $P = 10$ kN and $Q = 15$ kN, determine graphically the magnitude and direction of the resultant force exerted on the bracket using (a) the parallelogram law, (b) the triangle rule.

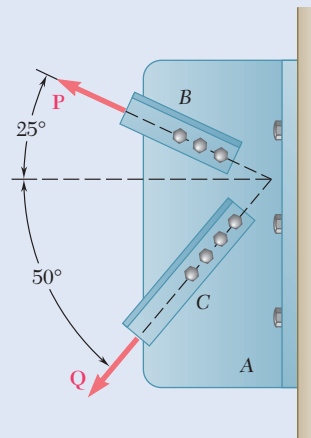


Fig. P2.3 and P2.4

- 2.4** Two structural members B and C are bolted to bracket A . Knowing that both members are in tension and that $P = 6$ kips and $Q = 4$ kips, determine graphically the magnitude and direction of the resultant force exerted on the bracket using (a) the parallelogram law, (b) the triangle rule.

- 2.5** A stake is being pulled out of the ground by means of two ropes as shown. Knowing that $\alpha = 30^\circ$, determine by trigonometry (a) the magnitude of the force $\mathbf{P}$ so that the resultant force exerted on the stake is vertical, (b) the corresponding magnitude of the resultant.

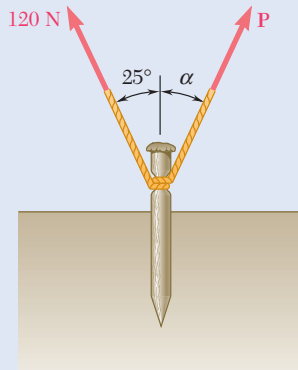


Fig. P2.5

- 2.6** A telephone cable is clamped at A to the pole AB . Knowing that the tension in the left-hand portion of the cable is $T_1 = 800$ lb, determine by trigonometry (a) the required tension T_2 in the right-hand portion if the resultant $\mathbf{R}$ of the forces exerted by the cable at A is to be vertical, (b) the corresponding magnitude of $\mathbf{R}$.

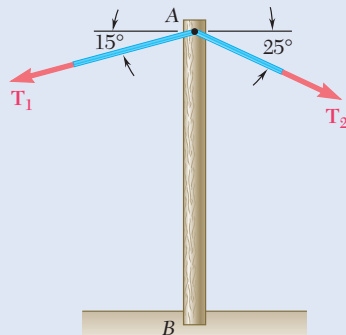


Fig. P2.6 and P2.7

- 2.7** A telephone cable is clamped at A to the pole AB . Knowing that the tension in the right-hand portion of the cable is $T_2 = 1000$ lb, determine by trigonometry (a) the required tension T_1 in the left-hand portion if the resultant $\mathbf{R}$ of the forces exerted by the cable at A is to be vertical, (b) the corresponding magnitude of $\mathbf{R}$.
- 2.8** A disabled automobile is pulled by means of two ropes as shown. The tension in rope AB is 2.2 kN, and the angle α is 25° . Knowing that the resultant of the two forces applied at A is directed along the axis of the automobile, determine by trigonometry (a) the tension in rope AC , (b) the magnitude of the resultant of the two forces applied at A .
- 2.9** A disabled automobile is pulled by means of two ropes as shown. Knowing that the tension in rope AB is 3 kN, determine by trigonometry the tension in rope AC and the value of α so that the resultant force exerted at A is a 4.8 -kN force directed along the axis of the automobile.

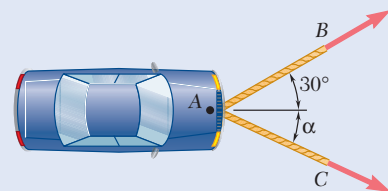


Fig. P2.8 and P2.9

- 2.10** Two forces are applied as shown to a hook support. Knowing that the magnitude of $\mathbf{P}$ is 35 N, determine by trigonometry (a) the required angle α if the resultant $\mathbf{R}$ of the two forces applied to the support is to be horizontal, (b) the corresponding magnitude of $\mathbf{R}$.

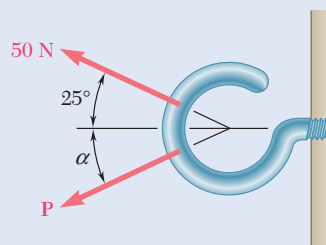


Fig. P2.10

- 2.11** A steel tank is to be positioned in an excavation. Knowing that $\alpha = 20^\circ$, determine by trigonometry (a) the required magnitude of the force $\mathbf{P}$ if the resultant $\mathbf{R}$ of the two forces applied at A is to be vertical, (b) the corresponding magnitude of $\mathbf{R}$.

- 2.12** A steel tank is to be positioned in an excavation. Knowing that the magnitude of $\mathbf{P}$ is 500 lb, determine by trigonometry (a) the required angle α if the resultant $\mathbf{R}$ of the two forces applied at A is to be vertical, (b) the corresponding magnitude of $\mathbf{R}$.

- 2.13** A steel tank is to be positioned in an excavation. Determine by trigonometry (a) the magnitude and direction of the smallest force $\mathbf{P}$ for which the resultant $\mathbf{R}$ of the two forces applied at A is vertical, (b) the corresponding magnitude of $\mathbf{R}$.

- 2.14** For the hook support of Prob. 2.10, determine by trigonometry (a) the magnitude and direction of the smallest force $\mathbf{P}$ for which the resultant $\mathbf{R}$ of the two forces applied to the support is horizontal, (b) the corresponding magnitude of $\mathbf{R}$.

- 2.15** For the hook support shown, determine by trigonometry the magnitude and direction of the resultant of the two forces applied to the support.

- 2.16** Solve Prob. 2.1 by trigonometry.

- 2.17** Solve Prob. 2.4 by trigonometry.

- 2.18** For the stake of Prob. 2.5, knowing that the tension in one rope is 120 N, determine by trigonometry the magnitude and direction of the force $\mathbf{P}$ so that the resultant is a vertical force of 160 N.

- 2.19** Two forces $\mathbf{P}$ and $\mathbf{Q}$ are applied to the lid of a storage bin as shown. Knowing that $P = 48$ N and $Q = 60$ N, determine by trigonometry the magnitude and direction of the resultant of the two forces.

- 2.20** Two forces $\mathbf{P}$ and $\mathbf{Q}$ are applied to the lid of a storage bin as shown. Knowing that $P = 60$ N and $Q = 48$ N, determine by trigonometry the magnitude and direction of the resultant of the two forces.

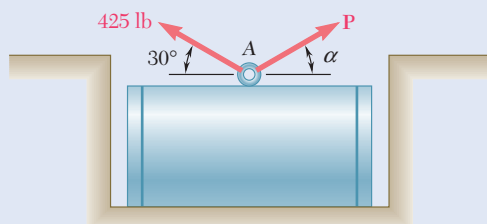


Fig. P2.11, P2.12 and P2.13

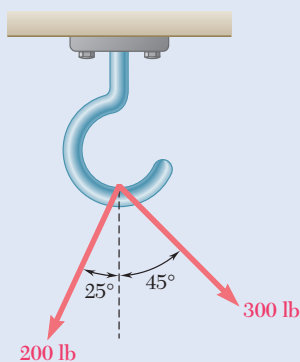


Fig. P2.15

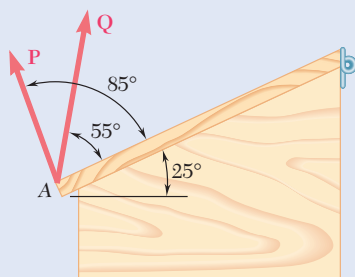


Fig. P2.19 and P2.20

2.2 ADDING FORCES BY COMPONENTS

In Sec. 2.1E, we described how to resolve a force into components. Here we discuss how to add forces by using their components, especially rectangular components. This method is often the most convenient way to add forces and, in practice, is the most common approach. (Note that we can readily extend the properties of vectors established in this section to the rectangular components of any vector quantity, such as velocity or momentum.)

2.2A Rectangular Components of a Force: Unit Vectors

In many problems, it is useful to resolve a force into two components that are perpendicular to each other. Figure 2.14 shows a force $\mathbf{F}$ resolved into a component $\mathbf{F}_x$ along the x axis and a component $\mathbf{F}_y$ along the y axis. The parallelogram drawn to obtain the two components is a rectangle, and $\mathbf{F}_x$ and $\mathbf{F}_y$ are called **rectangular components**.

The x and y axes are usually chosen to be horizontal and vertical, respectively, as in Fig. 2.14; they may, however, be chosen in any two perpendicular directions, as shown in Fig. 2.15. In determining the

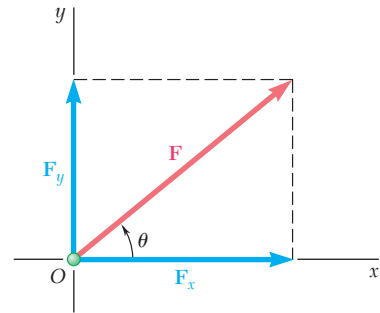


Fig. 2.14 Rectangular components of a force $\mathbf{F}$.

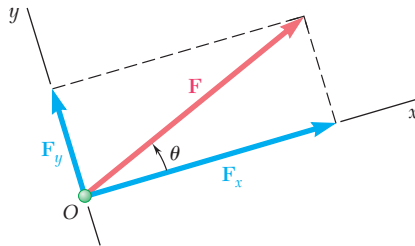


Fig. 2.15 Rectangular components of a force $\mathbf{F}$ for axes rotated away from horizontal and vertical.

rectangular components of a force, you should think of the construction lines shown in Figs. 2.14 and 2.15 as being *parallel* to the x and y axes, rather than *perpendicular* to these axes. This practice will help avoid mistakes in determining *oblique* components, as in Sec. 2.1E.

Force in Terms of Unit Vectors. To simplify working with rectangular components, we introduce two vectors of unit magnitude, directed respectively along the positive x and y axes. These vectors are called **unit vectors** and are denoted by $\mathbf{i}$ and $\mathbf{j}$, respectively (Fig. 2.16). Recalling the

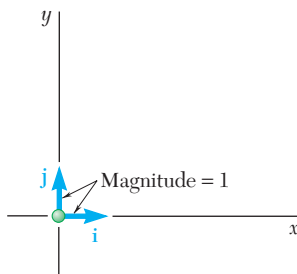


Fig. 2.16 Unit vectors along the x and y axes.

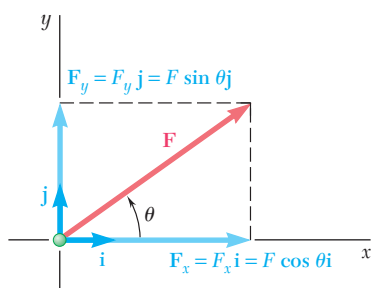


Fig. 2.17 Expressing the components of $\mathbf{F}$ in terms of unit vectors with scalar multipliers.

definition of the product of a scalar and a vector given in Sec. 2.1C, note that we can obtain the rectangular components $\mathbf{F}_x$ and $\mathbf{F}_y$ of a force $\mathbf{F}$ by multiplying respectively the unit vectors $\mathbf{i}$ and $\mathbf{j}$ by appropriate scalars (Fig. 2.17). We have

$$\mathbf{F}_x = F_x \mathbf{i} \quad \mathbf{F}_y = F_y \mathbf{j} \quad (2.6)$$

and

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} \quad (2.7)$$

The scalars F_x and F_y may be positive or negative, depending upon the sense of $\mathbf{F}_x$ and of $\mathbf{F}_y$, but their absolute values are equal to the magnitudes of the component forces $\mathbf{F}_x$ and $\mathbf{F}_y$, respectively. The scalars F_x and F_y are called the **scalar components** of the force $\mathbf{F}$, whereas the actual component forces $\mathbf{F}_x$ and $\mathbf{F}_y$ should be referred to as the **vector components** of $\mathbf{F}$. However, when there exists no possibility of confusion, we may refer to the vector as well as the scalar components of $\mathbf{F}$ as simply the **components** of $\mathbf{F}$. Note that the scalar component F_x is positive when the vector component $\mathbf{F}_x$ has the same sense as the unit vector $\mathbf{i}$ (i.e., the same sense as the positive x axis) and is negative when $\mathbf{F}_x$ has the opposite sense. A similar conclusion holds for the sign of the scalar component F_y .

Scalar Components. Denoting by F the magnitude of the force $\mathbf{F}$ and by θ the angle between $\mathbf{F}$ and the x axis, which is measured counter-clockwise from the positive x axis (Fig. 2.17), we may express the scalar components of $\mathbf{F}$ as

$$F_x = F \cos \theta \quad F_y = F \sin \theta \quad (2.8)$$

These relations hold for any value of the angle θ from 0° to 360° , and they define the signs as well as the absolute values of the scalar components F_x and F_y .

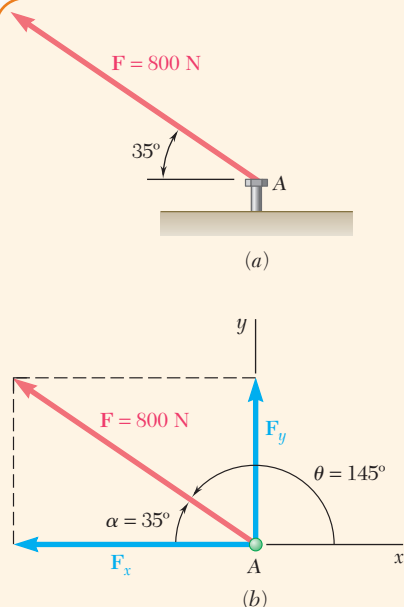


Fig. 2.18 (a) Force $\mathbf{F}$ exerted on a bolt; (b) rectangular components of $\mathbf{F}$.

Concept Application 2.1

A force of 800 N is exerted on a bolt A as shown in Fig. 2.18a. Determine the horizontal and vertical components of the force.

Solution

In order to obtain the correct sign for the scalar components F_x and F_y , we could substitute the value $180^\circ - 35^\circ = 145^\circ$ for θ in Eqs. (2.8). However, it is often more practical to determine by inspection the signs of F_x and F_y (Fig. 2.18b) and then use the trigonometric functions of the angle $\alpha = 35^\circ$. Therefore,

$$F_x = -F \cos \alpha = -(800 \text{ N}) \cos 35^\circ = -655 \text{ N}$$

$$F_y = +F \sin \alpha = +(800 \text{ N}) \sin 35^\circ = +459 \text{ N}$$

The vector components of $\mathbf{F}$ are thus

$$\mathbf{F}_x = -(655 \text{ N})\mathbf{i} \quad \mathbf{F}_y = +(459 \text{ N})\mathbf{j}$$

and we may write $\mathbf{F}$ in the form

$$\mathbf{F} = -(655 \text{ N})\mathbf{i} + (459 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

Concept Application 2.2

A man pulls with a force of 300 N on a rope attached to the top of a building, as shown in Fig. 2.19*a*. What are the horizontal and vertical components of the force exerted by the rope at point A?

Solution

You can see from Fig. 2.19*b* that

$$F_x = +(300 \text{ N}) \cos \alpha \quad F_y = -(300 \text{ N}) \sin \alpha$$

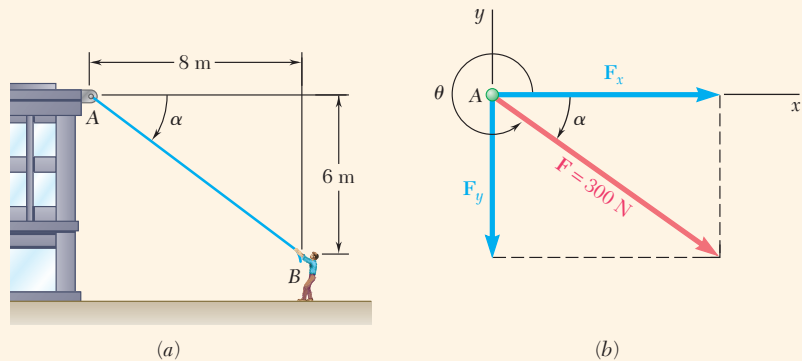


Fig. 2.19 (a) A man pulls on a rope attached to a building; (b) components of the rope's force $\mathbf{F}$.

Observing that $AB = 10 \text{ m}$, we find from Fig. 2.19*a*

$$\cos \alpha = \frac{8 \text{ m}}{AB} = \frac{8 \text{ m}}{10 \text{ m}} = \frac{4}{5} \quad \sin \alpha = \frac{6 \text{ m}}{AB} = \frac{6 \text{ m}}{10 \text{ m}} = \frac{3}{5}$$

We thus obtain

$$F_x = +(300 \text{ N}) \frac{4}{5} = +240 \text{ N} \quad F_y = -(300 \text{ N}) \frac{3}{5} = -180 \text{ N}$$

This gives us a total force of

$$\mathbf{F} = (240 \text{ N})\mathbf{i} - (180 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

Direction of a Force. When a force $\mathbf{F}$ is defined by its rectangular components F_x and F_y (see Fig. 2.17), we can find the angle θ defining its direction from

$$\tan \theta = \frac{F_y}{F_x} \quad (2.9)$$

We can obtain the magnitude F of the force by applying the Pythagorean theorem,

$$F = \sqrt{F_x^2 + F_y^2} \quad (2.10)$$

or by solving for F from one of the Eqs. (2.8).

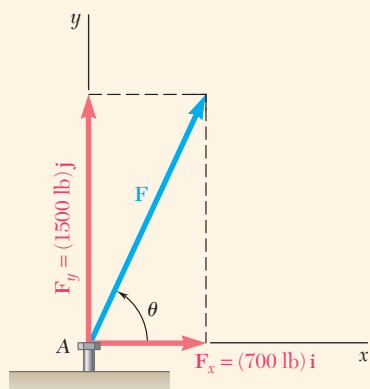


Fig. 2.20 Components of a force F exerted on a bolt.

Concept Application 2.3

A force $\mathbf{F} = (700 \text{ lb})\mathbf{i} + (1500 \text{ lb})\mathbf{j}$ is applied to a bolt A . Determine the magnitude of the force and the angle θ it forms with the horizontal.

Solution

First draw a diagram showing the two rectangular components of the force and the angle θ (Fig. 2.20). From Eq. (2.9), you obtain

$$\tan \theta = \frac{F_y}{F_x} = \frac{1500 \text{ lb}}{700 \text{ lb}}$$

Using a calculator, enter 1500 lb and divide by 700 lb; computing the arc tangent of the quotient gives you $\theta = 65.0^\circ$. Solve the second of Eqs. (2.8) for F to get

$$F = \frac{F_y}{\sin \theta} = \frac{1500 \text{ lb}}{\sin 65.0^\circ} = 1655 \text{ lb}$$

The last calculation is easier if you store the value of F_y when originally entered; you may then recall it and divide it by $\sin \theta$.

2.2B Addition of Forces by Summing X and Y Components

We described in Sec. 2.1A how to add forces according to the parallelogram law. From this law, we derived two other methods that are more readily applicable to the graphical solution of problems: the triangle rule for the addition of two forces and the polygon rule for the addition of three or more forces. We also explained that the force triangle used to define the resultant of two forces could be used to obtain a trigonometric solution.

However, when we need to add three or more forces, we cannot obtain any practical trigonometric solution from the force polygon that defines the resultant of the forces. In this case, the best approach is to obtain an analytic solution of the problem by resolving each force into two rectangular components.

Consider, for instance, three forces $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ acting on a particle A (Fig. 2.21a). Their resultant $\mathbf{R}$ is defined by the relation

$$\mathbf{R} = \mathbf{P} + \mathbf{Q} + \mathbf{S} \quad (2.11)$$

Resolving each force into its rectangular components, we have

$$\begin{aligned} R_x \mathbf{i} + R_y \mathbf{j} &= P_x \mathbf{i} + P_y \mathbf{j} + Q_x \mathbf{i} + Q_y \mathbf{j} + S_x \mathbf{i} + S_y \mathbf{j} \\ &= (P_x + Q_x + S_x) \mathbf{i} + (P_y + Q_y + S_y) \mathbf{j} \end{aligned}$$

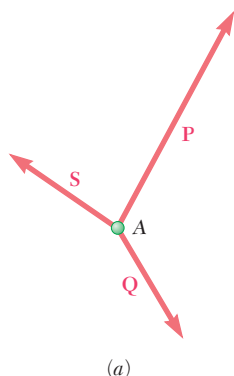


Fig. 2.21 (a) Three forces acting on a particle.

From this equation, we can see that

$$R_x = P_x + Q_x + S_x \quad R_y = P_y + Q_y + S_y \quad (2.12)$$

or for short,

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad (2.13)$$

We thus conclude that **when several forces are acting on a particle, we obtain the scalar components R_x and R_y of the resultant $\mathbf{R}$ by adding algebraically the corresponding scalar components of the given forces.** (Clearly, this result also applies to the addition of other vector quantities, such as velocities, accelerations, or momenta.)

In practice, determining the resultant $\mathbf{R}$ is carried out in three steps, as illustrated in Fig. 2.21.

1. Resolve the given forces (Fig. 2.21a) into their x and y components (Fig. 2.21b).

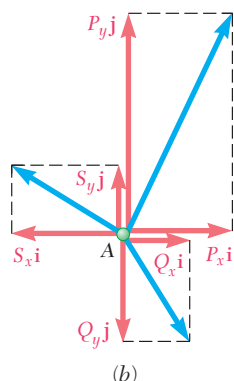


Fig. 2.21 (b) Rectangular components of each force.

2. Add these components to obtain the x and y components of $\mathbf{R}$ (Fig. 2.21c).

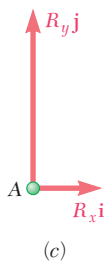


Fig. 2.21 (c) Summation of the components.

3. Apply the parallelogram law to determine the resultant $\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j}$ (Fig. 2.21d).

The procedure just described is most efficiently carried out if you arrange the computations in a table (see Sample Problem 2.3). Although this is the only practical analytic method for adding three or more forces, it is also often preferred to the trigonometric solution in the case of adding two forces.

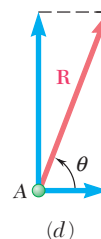


Fig. 2.21 (d) Determining the resultant from its components.

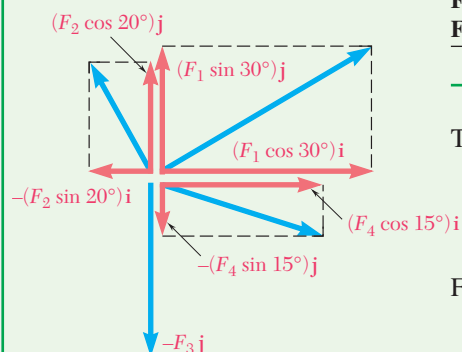
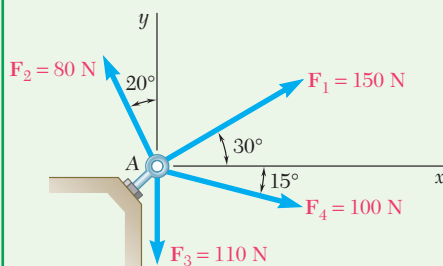


Fig. 1 Rectangular components of each force.

Sample Problem 2.3

Four forces act on bolt A as shown. Determine the resultant of the forces on the bolt.

STRATEGY: The simplest way to approach a problem of adding four forces is to resolve the forces into components.

MODELING: As we mentioned, solving this kind of problem is usually easier if you arrange the components of each force in a table. In the table below, we entered the x and y components of each force as determined by trigonometry (Fig. 1). According to the convention adopted in this section, the scalar number representing a force component is positive if the force component has the same sense as the corresponding coordinate axis. Thus, x components acting to the right and y components acting upward are represented by positive numbers.

ANALYSIS:

Force	Magnitude, N	x Component, N	y Component, N
F_1	150	+129.9	+75.0
F_2	80	-27.4	+75.2
F_3	110	0	-110.0
F_4	100	+96.6	-25.9
		$R_x = +199.1$	$R_y = +14.3$

Thus, the resultant $\mathbf{R}$ of the four forces is

$$\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j} \quad \mathbf{R} = (199.1 \text{ N})\mathbf{i} + (14.3 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

You can now determine the magnitude and direction of the resultant. From the triangle shown in Fig. 2, you have

$$\tan \alpha = \frac{R_y}{R_x} = \frac{14.3 \text{ N}}{199.1 \text{ N}} \quad \alpha = 4.1^\circ$$

$$R = \frac{14.3 \text{ N}}{\sin \alpha} = 199.6 \text{ N} \quad \mathbf{R} = 199.6 \text{ N} \nearrow 4.1^\circ \quad \blacktriangleleft$$

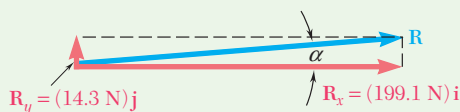


Fig. 2 Resultant of the given force system.

REFLECT and THINK: Arranging data in a table not only helps you keep track of the calculations, but also makes things simpler for using a calculator on similar computations.

SOLVING PROBLEMS ON YOUR OWN

You saw in the preceding lesson that we can determine the resultant of two forces either graphically or from the trigonometry of an oblique triangle.

A. When three or more forces are involved, the best way to determine their resultant $\mathbf{R}$ is by first resolving each force into **rectangular components**. You may encounter either of two cases, depending upon the way in which each of the given forces is defined.

Case 1. The force $\mathbf{F}$ is defined by its magnitude F and the angle α it forms with the x axis. Obtain the x and y components of the force by multiplying F by $\cos \alpha$ and $\sin \alpha$, respectively [Concept Application 2.1].

Case 2. The force $\mathbf{F}$ is defined by its magnitude F and the coordinates of two points A and B on its line of action (Fig. 2.19). Find the angle α that $\mathbf{F}$ forms with the x axis by trigonometry, and then use the process of Case 1. However, you can also find the components of $\mathbf{F}$ directly from proportions among the various dimensions involved without actually determining α [Concept Application 2.2].

B. Rectangular components of the resultant. Obtain the components R_x and R_y of the resultant by adding the corresponding components of the given forces algebraically [Sample Prob. 2.3].

You can express the resultant in vectorial form using the unit vectors $\mathbf{i}$ and $\mathbf{j}$, which are directed along the x and y axes, respectively:

$$\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j}$$

Alternatively, you can determine the *magnitude and direction* of the resultant by solving the right triangle of sides R_x and R_y for R and for the angle that $\mathbf{R}$ forms with the x axis.

Problems

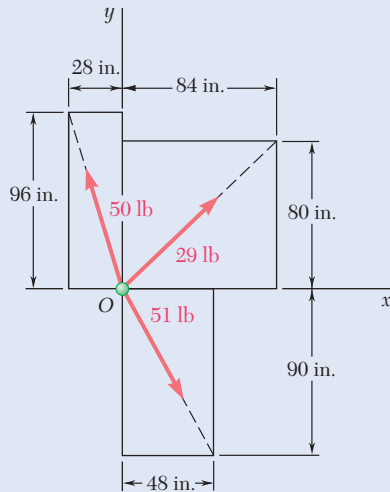


Fig. P2.21

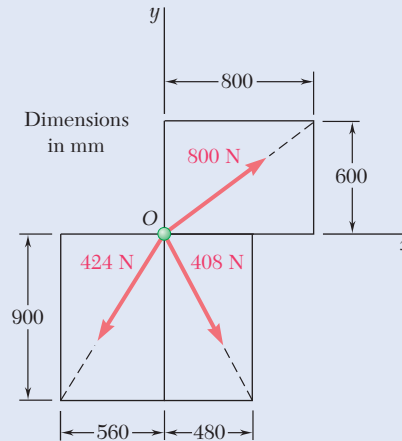


Fig. P2.22

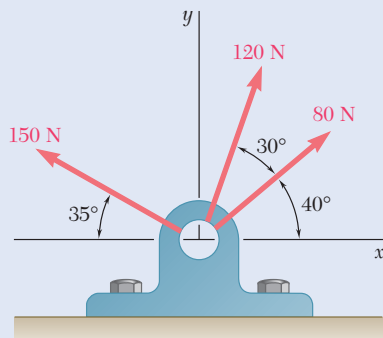


Fig. P2.23

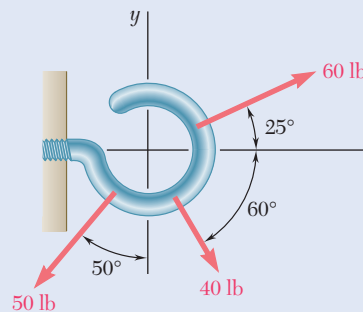


Fig. P2.24

- 2.25** Member BC exerts on member AC a force $\mathbf{P}$ directed along line BC . Knowing that $\mathbf{P}$ must have a 325-N horizontal component, determine (a) the magnitude of the force $\mathbf{P}$, (b) its vertical component.

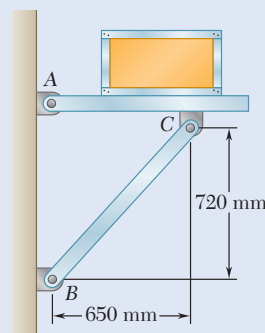


Fig. P2.25

- 2.26** Member BD exerts on member ABC a force $\mathbf{P}$ directed along line BD . Knowing that $\mathbf{P}$ must have a 300-lb horizontal component, determine (a) the magnitude of the force $\mathbf{P}$, (b) its vertical component.
- 2.27** The hydraulic cylinder BC exerts on member AB a force $\mathbf{P}$ directed along line BC . Knowing that $\mathbf{P}$ must have a 600-N component perpendicular to member AB , determine (a) the magnitude of the force $\mathbf{P}$, (b) its component along line AB .

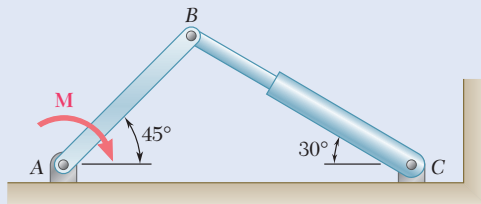


Fig. P2.27

- 2.28** Cable AC exerts on beam AB a force $\mathbf{P}$ directed along line AC . Knowing that $\mathbf{P}$ must have a 350-lb vertical component, determine (a) the magnitude of the force $\mathbf{P}$, (b) its horizontal component.
- 2.29** The hydraulic cylinder BD exerts on member ABC a force $\mathbf{P}$ directed along line BD . Knowing that $\mathbf{P}$ must have a 750-N component perpendicular to member ABC , determine (a) the magnitude of the force $\mathbf{P}$, (b) its component parallel to ABC .

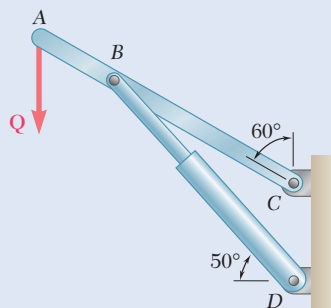


Fig. P2.29

- 2.30** The guy wire BD exerts on the telephone pole AC a force $\mathbf{P}$ directed along BD . Knowing that $\mathbf{P}$ must have a 720-N component perpendicular to the pole AC , determine (a) the magnitude of the force $\mathbf{P}$, (b) its component along line AC .
- 2.31** Determine the resultant of the three forces of Prob. 2.21.
- 2.32** Determine the resultant of the three forces of Prob. 2.23.
- 2.33** Determine the resultant of the three forces of Prob. 2.24.
- 2.34** Determine the resultant of the three forces of Prob. 2.22.

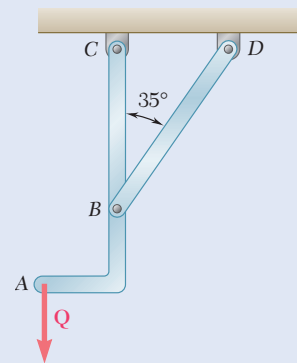


Fig. P2.26

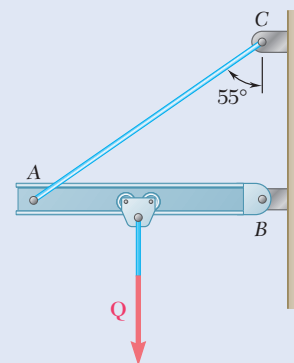


Fig. P2.28

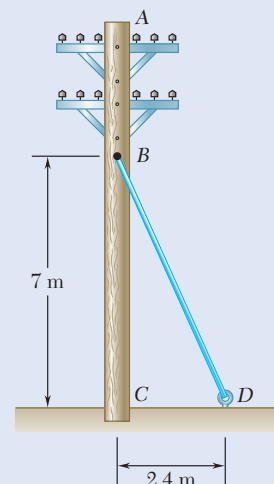


Fig. P2.30

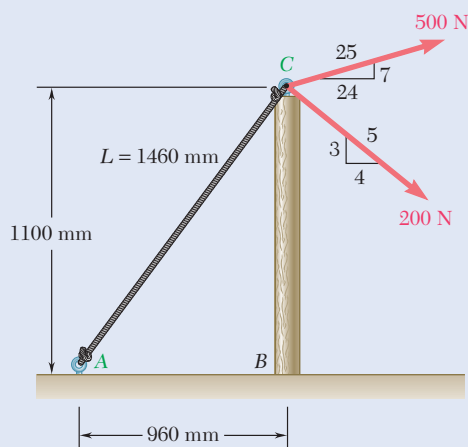


Fig. P2.36

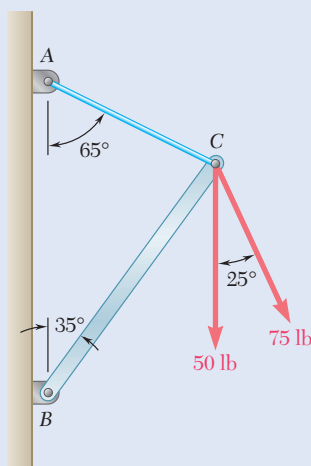


Fig. P2.41

2.35 Knowing that $\alpha = 35^\circ$, determine the resultant of the three forces shown.

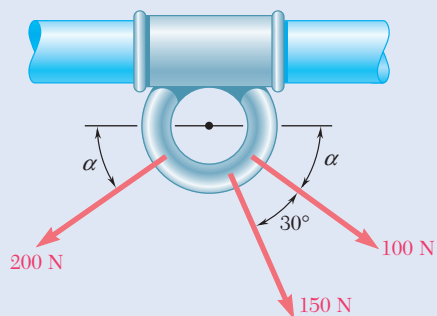


Fig. P2.35

2.36 Knowing that the tension in rope AC is 365 N, determine the resultant of the three forces exerted at point C of post BC.

2.37 Knowing that $\alpha = 40^\circ$, determine the resultant of the three forces shown.

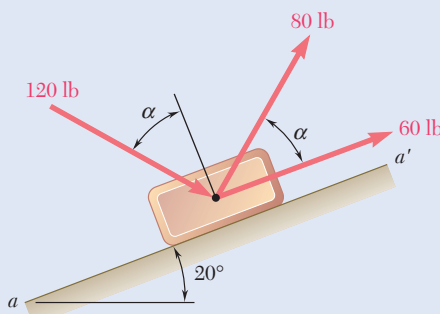


Fig. P2.37 and P2.38

2.38 Knowing that $\alpha = 75^\circ$, determine the resultant of the three forces shown.

2.39 For the collar of Prob. 2.35, determine (a) the required value of α if the resultant of the three forces shown is to be vertical, (b) the corresponding magnitude of the resultant.

2.40 For the post of Prob. 2.36, determine (a) the required tension in rope AC if the resultant of the three forces exerted at point C is to be horizontal, (b) the corresponding magnitude of the resultant.

2.41 Determine (a) the required tension in cable AC, knowing that the resultant of the three forces exerted at point C of boom BC must be directed along BC, (b) the corresponding magnitude of the resultant.

2.42 For the block of Probs. 2.37 and 2.38, determine (a) the required value of α if the resultant of the three forces shown is to be parallel to the incline, (b) the corresponding magnitude of the resultant.

2.3 FORCES AND EQUILIBRIUM IN A PLANE

Now that we have seen how to add forces, we can proceed to one of the key concepts in this course: the equilibrium of a particle. The connection between equilibrium and the sum of forces is very direct: a particle can be in equilibrium only when the sum of the forces acting on it is zero.

2.3A Equilibrium of a Particle

In the preceding sections, we discussed methods for determining the resultant of several forces acting on a particle. Although it has not occurred in any of the problems considered so far, it is quite possible for the resultant to be zero. In such a case, the net effect of the given forces is zero, and the particle is said to be in **equilibrium**. We thus have the definition:

When the resultant of all the forces acting on a particle is zero, the particle is in equilibrium.

A particle acted upon by two forces is in equilibrium if the two forces have the same magnitude and the same line of action but opposite sense. The resultant of the two forces is then zero, as shown in Fig. 2.22.

Another case of equilibrium of a particle is represented in Fig. 2.23a, where four forces are shown acting on particle A. In Fig. 2.23b, we use the polygon rule to determine the resultant of the given forces. Starting from point O with $\mathbf{F}_1$ and arranging the forces in tip-to-tail fashion, we find that the tip of $\mathbf{F}_4$ coincides with the starting point O. Thus, the resultant $\mathbf{R}$ of the given system of forces is zero, and the particle is in equilibrium.

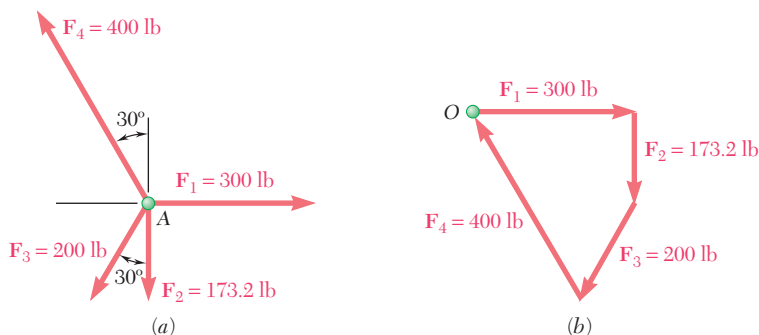


Fig. 2.23 (a) Four forces acting on particle A; (b) using the polygon law to find the resultant of the forces in (a), which is zero because the particle is in equilibrium.

The closed polygon drawn in Fig. 2.23b provides a *graphical* expression of the equilibrium of A. To express *algebraically* the conditions for the equilibrium of a particle, we write

Equilibrium of a particle

$$\mathbf{R} = \Sigma \mathbf{F} = 0$$

(2.14)



Photo 2.2 Forces acting on the carabiner include the weight of the girl and her harness, and the force exerted by the pulley attachment. Treating the carabiner as a particle, it is in equilibrium because the resultant of all forces acting on it is zero.

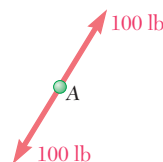


Fig. 2.22 When a particle is in equilibrium, the resultant of all forces acting on the particle is zero.

Resolving each force $\mathbf{F}$ into rectangular components, we have

$$\Sigma (F_x \mathbf{i} + F_y \mathbf{j}) = 0 \quad \text{or} \quad (\Sigma F_x) \mathbf{i} + (\Sigma F_y) \mathbf{j} = 0$$

We conclude that the necessary and sufficient conditions for the equilibrium of a particle are

Equilibrium of a particle
(scalar equations)

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad (2.15)$$

Returning to the particle shown in Fig. 2.23, we can check that the equilibrium conditions are satisfied. We have

$$\begin{aligned} \Sigma F_x &= 300 \text{ lb} - (200 \text{ lb}) \sin 30^\circ - (400 \text{ lb}) \sin 30^\circ \\ &= 300 \text{ lb} - 100 \text{ lb} - 200 \text{ lb} = 0 \\ \Sigma F_y &= -173.2 \text{ lb} - (200 \text{ lb}) \cos 30^\circ + (400 \text{ lb}) \cos 30^\circ \\ &= -173.2 \text{ lb} - 173.2 \text{ lb} + 346.4 \text{ lb} = 0 \end{aligned}$$

2.3B Newton's First Law of Motion

As we discussed in Section 1.2, Sir Isaac Newton formulated three fundamental laws upon which the science of mechanics is based. The first of these laws can be stated as:

If the resultant force acting on a particle is zero, the particle will remain at rest (if originally at rest) or will move with constant speed in a straight line (if originally in motion).

From this law and from the definition of equilibrium just presented, we can see that a particle in equilibrium is either at rest or moving in a straight line with constant speed. If a particle does not behave in either of these ways, it is not in equilibrium, and the resultant force on it is not zero. In the following section, we consider various problems concerning the equilibrium of a particle.

Note that most of statics involves using Newton's first law to analyze an equilibrium situation. In practice, this means designing a bridge or a building that remains stable and does not fall over. It also means understanding the forces that might act to disturb equilibrium, such as a strong wind or a flood of water. The basic idea is pretty simple, but the applications can be quite complicated.

2.3C Free-Body Diagrams and Problem Solving

In practice, a problem in engineering mechanics is derived from an actual physical situation. A sketch showing the physical conditions of the problem is known as a **space diagram**.

The methods of analysis discussed in the preceding sections apply to a system of forces acting on a particle. A large number of problems involving actual structures, however, can be reduced to problems concerning the equilibrium of a particle. The method is to choose a significant particle and draw a separate diagram showing this particle and all the

forces acting on it. Such a diagram is called a **free-body diagram**. (The name derives from the fact that when drawing the chosen body, or particle, it is “free” from all other bodies in the actual situation.)

As an example, consider the 75-kg crate shown in the space diagram of Fig. 2.24a. This crate was lying between two buildings, and is now being lifted onto a truck, which will remove it. The crate is supported by a vertical cable that is joined at A to two ropes, which pass over pulleys attached to the buildings at B and C . We want to determine the tension in each of the ropes AB and AC .

In order to solve this problem, we first draw a free-body diagram showing a particle in equilibrium. Since we are interested in the rope tensions, the free-body diagram should include at least one of these tensions or, if possible, both tensions. You can see that point A is a good free body for this problem. The free-body diagram of point A is shown in Fig. 2.24b. It shows point A and the forces exerted on A by the vertical cable and the two ropes. The force exerted by the cable is directed downward, and its magnitude is equal to the weight W of the crate. Recalling Eq. (1.4), we write

$$W = mg = (75 \text{ kg})(9.81 \text{ m/s}^2) = 736 \text{ N}$$

and indicate this value in the free-body diagram. The forces exerted by the two ropes are not known. Since they are respectively equal in magnitude to the tensions in rope AB and rope AC , we denote them by $\mathbf{T}_{AB}$ and $\mathbf{T}_{AC}$ and draw them away from A in the directions shown in the space diagram. No other detail is included in the free-body diagram.

Since point A is in equilibrium, the three forces acting on it must form a closed triangle when drawn in tip-to-tail fashion. We have drawn this **force triangle** in Fig. 2.24c. The values T_{AB} and T_{AC} of the tensions in the ropes may be found graphically if the triangle is drawn to scale, or they may be found by trigonometry. If we choose trigonometry, we use the law of sines:

$$\frac{T_{AB}}{\sin 60^\circ} = \frac{T_{AC}}{\sin 40^\circ} = \frac{736 \text{ N}}{\sin 80^\circ}$$

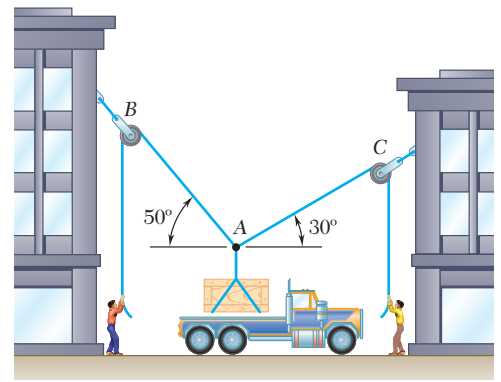
$$T_{AB} = 647 \text{ N} \quad T_{AC} = 480 \text{ N}$$

When a particle is in equilibrium under three forces, you can solve the problem by drawing a force triangle. When a particle is in equilibrium under more than three forces, you can solve the problem graphically by drawing a force polygon. If you need an analytic solution, you should solve the **equations of equilibrium** given in Sec. 2.3A:

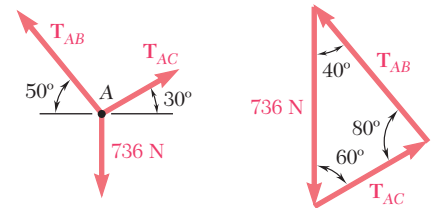
$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad (2.15)$$

These equations can be solved for no more than *two unknowns*. Similarly, the force triangle used in the case of equilibrium under three forces can be solved for only two unknowns.

The most common types of problems are those in which the two unknowns represent (1) the two components (or the magnitude and direction) of a single force or (2) the magnitudes of two forces, each of known direction. Problems involving the determination of the maximum or minimum value of the magnitude of a force are also encountered (see Probs. 2.57 through 2.61).



(a) Space diagram



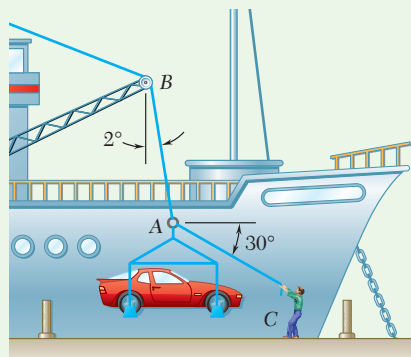
(b) Free-body diagram

(c) Force triangle

Fig. 2.24 (a) The space diagram shows the physical situation of the problem; (b) the free-body diagram shows one central particle and the forces acting on it; (c) the force triangle can be solved with the law of sines. Note that the forces form a closed triangle because the particle is in equilibrium and the resultant force is zero.



Photo 2.3 As illustrated in Fig. 2.24, it is possible to determine the tensions in the cables supporting the shaft shown by treating the hook as a particle and then applying the equations of equilibrium to the forces acting on the hook.



Sample Problem 2.4

In a ship-unloading operation, a 3500-lb automobile is supported by a cable. A worker ties a rope to the cable at A and pulls on it in order to center the automobile over its intended position on the dock. At the moment illustrated, the automobile is stationary, the angle between the cable and the vertical is 2° , and the angle between the rope and the horizontal is 30° . What are the tensions in the rope and cable?

STRATEGY: This is a problem of equilibrium under three coplanar forces. You can treat point A as a particle and solve the problem using a force triangle.

MODELING and ANALYSIS:

Free-Body Diagram. Choose point A as the particle and draw the complete free-body diagram (Fig. 1). T_{AB} is the tension in the cable AB, and T_{AC} is the tension in the rope.

Equilibrium Condition. Since only three forces act on point A, draw a force triangle to express that it is in equilibrium (Fig. 2). Using the law of sines,

$$\frac{T_{AB}}{\sin 120^\circ} = \frac{T_{AC}}{\sin 2^\circ} = \frac{3500 \text{ lb}}{\sin 58^\circ}$$

With a calculator, compute and store the value of the last quotient. Multiplying this value successively by $\sin 120^\circ$ and $\sin 2^\circ$, you obtain

$$T_{AB} = 3570 \text{ lb} \quad T_{AC} = 144 \text{ lb} \quad \blacktriangleleft$$

REFLECT and THINK: This is a common problem of knowing one force in a three-force equilibrium problem and calculating the other forces from the given geometry. This basic type of problem will occur often as part of more complicated situations in this text.

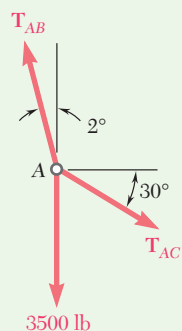


Fig. 1 Free-body diagram of particle A.

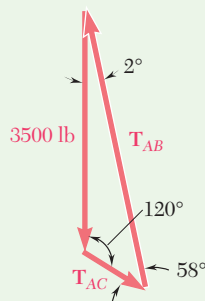


Fig. 2 Force triangle of the forces acting on particle A.

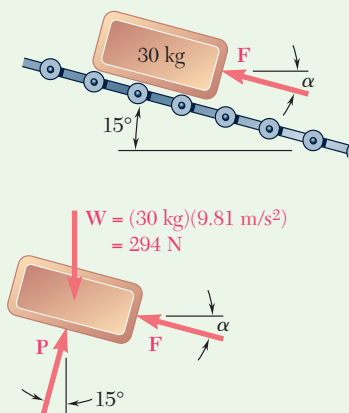


Fig. 1 Free-body diagram of package, treated as a particle.

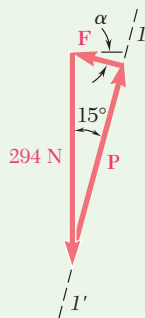


Fig. 2 Force triangle of the forces acting on package.

Sample Problem 2.5

Determine the magnitude and direction of the smallest force $\mathbf{F}$ that maintains the 30-kg package shown in equilibrium. Note that the force exerted by the rollers on the package is perpendicular to the incline.

STRATEGY: This is an equilibrium problem with three coplanar forces that you can solve with a force triangle. The new wrinkle is to determine a minimum force. You can approach this part of the solution in a way similar to Sample Problem 2.2.

MODELING and ANALYSIS:

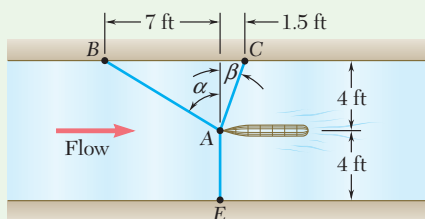
Free-Body Diagram. Choose the package as a free body, assuming that it can be treated as a particle. Then draw the corresponding free-body diagram (Fig. 1).

Equilibrium Condition. Since only three forces act on the free body, draw a force triangle to express that it is in equilibrium (Fig. 2). Line $l-l'$ represents the known direction of $\mathbf{P}$. In order to obtain the minimum value of the force $\mathbf{F}$, choose the direction of $\mathbf{F}$ to be perpendicular to that of $\mathbf{P}$. From the geometry of this triangle,

$$F = (294 \text{ N}) \sin 15^\circ = 76.1 \text{ N} \quad \alpha = 15^\circ$$

$$\mathbf{F} = 76.1 \text{ N} \searrow 15^\circ \quad \blacktriangleleft$$

REFLECT and THINK: Determining maximum and minimum forces to maintain equilibrium is a common practical problem. Here the force needed is about 25% of the weight of the package, which seems reasonable for an incline of 15° .



Sample Problem 2.6

For a new sailboat, a designer wants to determine the drag force that may be expected at a given speed. To do so, she places a model of the proposed hull in a test channel and uses three cables to keep its bow on the center-line of the channel. Dynamometer readings indicate that for a given speed, the tension is 40 lb in cable AB and 60 lb in cable AE . Determine the drag force exerted on the hull and the tension in cable AC .

STRATEGY: The cables all connect at point A , so you can treat that as a particle in equilibrium. Because four forces act at A (tensions in three cables and the drag force), you should use the equilibrium conditions and sum forces by components to solve for the unknown forces.

(continued)

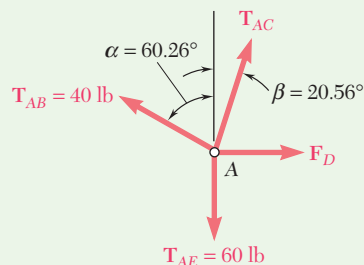


Fig. 1 Free-body diagram of particle A.

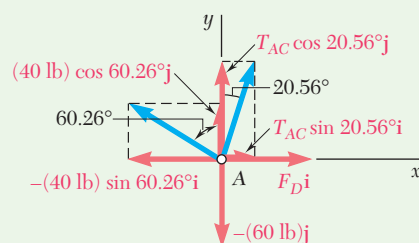


Fig. 2 Rectangular components of forces acting on particle A.

MODELING and ANALYSIS:

Determining the Angles. First, determine the angles α and β defining the direction of cables AB and AC:

$$\tan \alpha = \frac{7 \text{ ft}}{4 \text{ ft}} = 1.75 \quad \tan \beta = \frac{1.5 \text{ ft}}{4 \text{ ft}} = 0.375$$

$$\alpha = 60.26^\circ \quad \beta = 20.56^\circ$$

Free-Body Diagram. Choosing point A as a free body, draw the free-body diagram (Fig. 1). It includes the forces exerted by the three cables on the hull, as well as the drag force $\mathbf{F}_D$ exerted by the flow.

Equilibrium Condition. Because point A is in equilibrium, the resultant of all forces is zero:

$$\mathbf{R} = \mathbf{T}_{AB} + \mathbf{T}_{AC} + \mathbf{T}_{AE} + \mathbf{F}_D = 0 \quad (1)$$

Because more than three forces are involved, resolve the forces into x and y components (Fig. 2):

$$\begin{aligned} \mathbf{T}_{AB} &= -(40 \text{ lb}) \sin 60.26^\circ \mathbf{i} + (40 \text{ lb}) \cos 60.26^\circ \mathbf{j} \\ &= -(34.73 \text{ lb}) \mathbf{i} + (19.84 \text{ lb}) \mathbf{j} \end{aligned}$$

$$\begin{aligned} \mathbf{T}_{AC} &= T_{AC} \sin 20.56^\circ \mathbf{i} + T_{AC} \cos 20.56^\circ \mathbf{j} \\ &= 0.3512 T_{AC} \mathbf{i} + 0.9363 T_{AC} \mathbf{j} \end{aligned}$$

$$\mathbf{T}_{AE} = -(60 \text{ lb}) \mathbf{j}$$

$$\mathbf{F}_D = F_D \mathbf{i}$$

Substituting these expressions into Eq. (1) and factoring the unit vectors $\mathbf{i}$ and $\mathbf{j}$, you have

$$(-34.73 \text{ lb} + 0.3512 T_{AC} + F_D) \mathbf{i} + (19.84 \text{ lb} + 0.9363 T_{AC} - 60 \text{ lb}) \mathbf{j} = 0$$

This equation is satisfied if, and only if, the coefficients of $\mathbf{i}$ and $\mathbf{j}$ are each equal to zero. You obtain the following two equilibrium equations, which express, respectively, that the sum of the x components and the sum of the y components of the given forces must be zero.

$$(\Sigma F_x = 0:) \quad -34.73 \text{ lb} + 0.3512 T_{AC} + F_D = 0 \quad (2)$$

$$(\Sigma F_y = 0:) \quad 19.84 \text{ lb} + 0.9363 T_{AC} - 60 \text{ lb} = 0 \quad (3)$$

From Eq. (3), you find

$$T_{AC} = +42.9 \text{ lb} \quad \blacktriangleleft$$

Substituting this value into Eq. (2) yields

$$F_D = +19.66 \text{ lb} \quad \blacktriangleleft$$

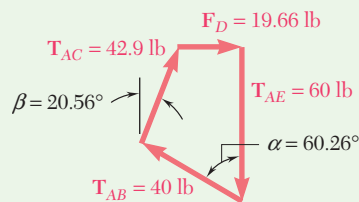


Fig. 3 Force polygon of forces acting on particle A.

REFLECT and THINK: In drawing the free-body diagram, you assumed a sense for each unknown force. A positive sign in the answer indicates that the assumed sense is correct. You can draw the complete force polygon (Fig. 3) to check the results.

SOLVING PROBLEMS ON YOUR OWN

When a particle is in **equilibrium**, the resultant of the forces acting on the particle must be zero. Expressing this fact in the case of a particle under *coplanar forces* provides you with two relations among these forces. As in the preceding sample problems, you can use these relations to determine two unknowns—such as the magnitude and direction of one force or the magnitudes of two forces.

Drawing a clear and accurate free-body diagram is a must in the solution of any equilibrium problem. This diagram shows the particle and all of the forces acting on it. Indicate in your free-body diagram the magnitudes of known forces, as well as any angle or dimensions that define the direction of a force. Any unknown magnitude or angle should be denoted by an appropriate symbol. Nothing else should be included in the free-body diagram. Skipping this step might save you pencil and paper, but it is very likely to lead you to a wrong solution.

Case 1. If the free-body diagram involves only **three forces**, the rest of the solution is best carried out by drawing these forces in tip-to-tail fashion to form a **force triangle**. You can solve this triangle graphically or by trigonometry for no more than two unknowns [Sample Probs. 2.4 and 2.5].

Case 2. If the free-body diagram indicates **more than three forces**, it is most practical to use an *analytic solution*. Select x and y axes and resolve each of the forces into x and y components. Setting the sum of the x components and the sum of the y components of all the forces to zero, you obtain two equations that you can solve for no more than two unknowns [Sample Prob. 2.6].

We strongly recommend that, when using an analytic solution, you write the equations of equilibrium in the same form as Eqs. (2) and (3) of Sample Prob. 2.6. The practice adopted by some students of initially placing the unknowns on the left side of the equation and the known quantities on the right side may lead to confusion in assigning the appropriate sign to each term.

Regardless of the method used to solve a two-dimensional equilibrium problem, you can determine at most two unknowns. If a two-dimensional problem involves more than two unknowns, you must obtain one or more additional relations from the information contained in the problem statement.

Problems

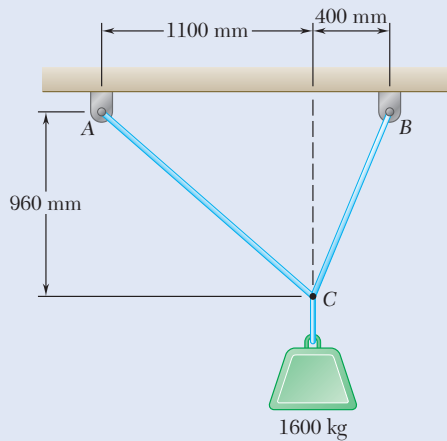


Fig. P2.F1

FREE-BODY PRACTICE PROBLEMS

- 2.F1** Two cables are tied together at C and loaded as shown. Draw the free-body diagram needed to determine the tension in AC and BC .
- 2.F2** Two forces of magnitude $T_A = 8$ kips and $T_B = 15$ kips are applied as shown to a welded connection. Knowing that the connection is in equilibrium, draw the free-body diagram needed to determine the magnitudes of the forces T_C and T_D .

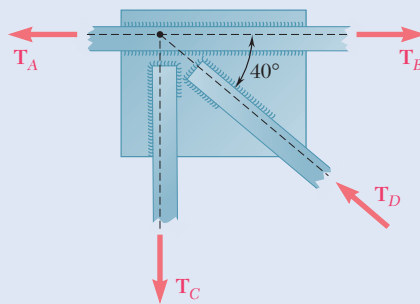


Fig. P2.F2

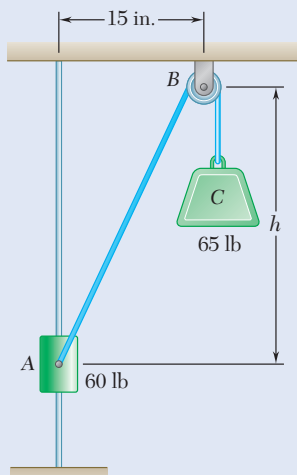


Fig. P2.F3

- 2.F3** The 60-lb collar A can slide on a frictionless vertical rod and is connected as shown to a 65-lb counterweight C . Draw the free-body diagram needed to determine the value of h for which the system is in equilibrium.
- 2.F4** A chairlift has been stopped in the position shown. Knowing that each chair weighs 250 N and that the skier in chair E weighs 765 N, draw the free-body diagrams needed to determine the weight of the skier in chair F .

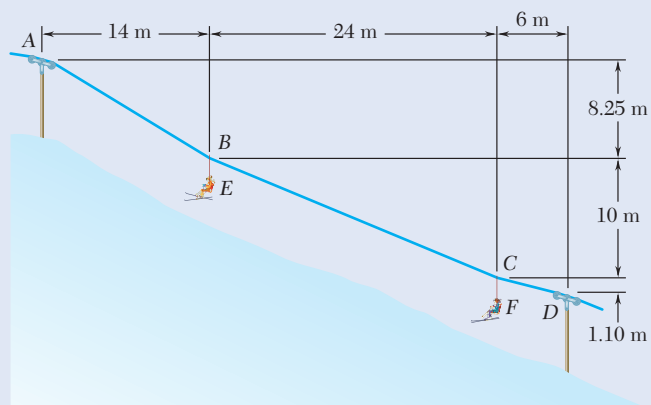


Fig. P2.F4

END-OF-SECTION PROBLEMS

- 2.43** Two cables are tied together at C and are loaded as shown. Determine the tension (a) in cable AC , (b) in cable BC .

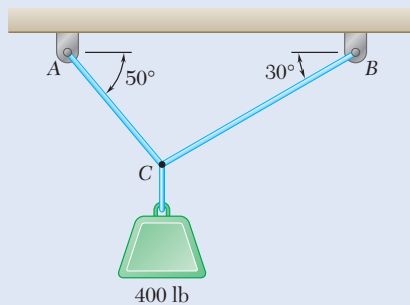


Fig. P2.43

- 2.44** Two cables are tied together at C and are loaded as shown. Knowing that $\alpha = 30^\circ$, determine the tension (a) in cable AC , (b) in cable BC .

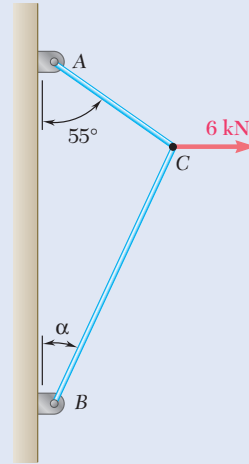


Fig. P2.44

- 2.45** Two cables are tied together at C and loaded as shown. Determine the tension (a) in cable AC , (b) in cable BC .

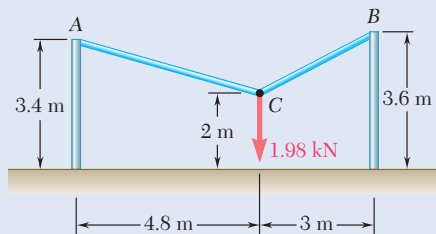


Fig. P2.45

- 2.46** Two cables are tied together at C and are loaded as shown. Knowing that $P = 500$ N and $\alpha = 60^\circ$, determine the tension in (a) in cable AC , (b) in cable BC .

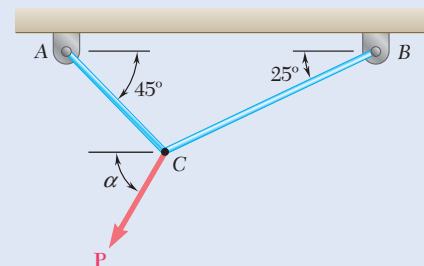


Fig. P2.46

- 2.47** Two cables are tied together at C and are loaded as shown. Determine the tension (a) in cable AC , (b) in cable BC .

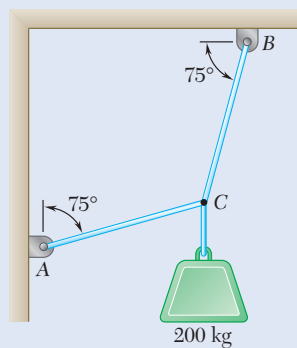


Fig. P2.47

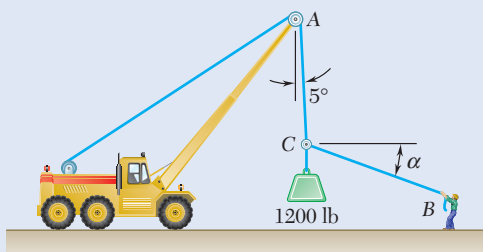


Fig. P2.48

2.48 Knowing that $\alpha = 20^\circ$, determine the tension (a) in cable AC, (b) in rope BC.

2.49 Two cables are tied together at C and are loaded as shown. Knowing that $P = 300$ N, determine the tension in cables AC and BC.

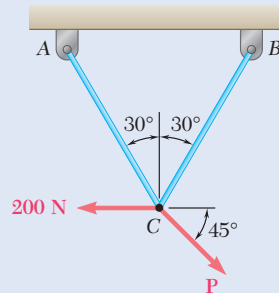


Fig. P2.49 and P2.50

2.50 Two cables are tied together at C and are loaded as shown. Determine the range of values of P for which both cables remain taut.

2.51 Two forces P and Q are applied as shown to an aircraft connection. Knowing that the connection is in equilibrium and that $P = 500$ lb and $Q = 650$ lb, determine the magnitudes of the forces exerted on rods A and B.

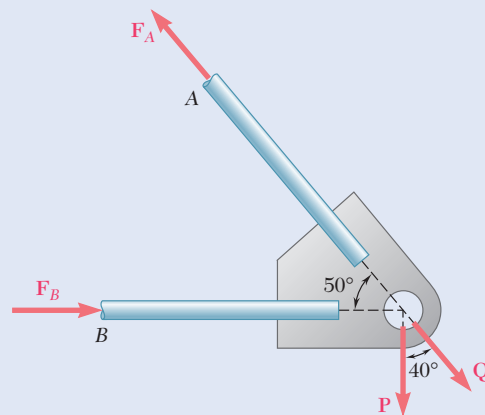


Fig. P2.51 and P2.52

2.52 Two forces P and Q are applied as shown to an aircraft connection. Knowing that the connection is in equilibrium and that the magnitudes of the forces exerted on rods A and B are $F_A = 750$ lb and $F_B = 400$ lb, determine the magnitudes of P and Q .

2.53 A welded connection is in equilibrium under the action of the four forces shown. Knowing that $F_A = 8$ kN and $F_B = 16$ kN, determine the magnitudes of the other two forces.

2.54 A welded connection is in equilibrium under the action of the four forces shown. Knowing that $F_A = 5$ kN and $F_D = 6$ kN, determine the magnitudes of the other two forces.

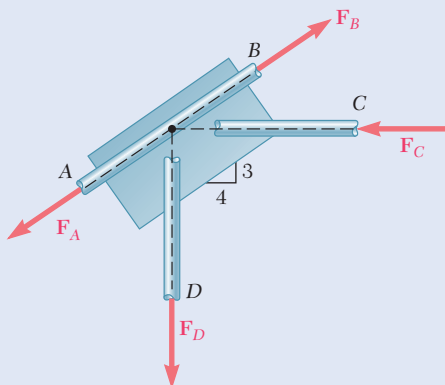


Fig. P2.53 and P2.54

2.55 A sailor is being rescued using a boatswain's chair that is suspended from a pulley that can roll freely on the support cable ACB and is pulled at a constant speed by cable CD . Knowing that $\alpha = 30^\circ$ and $\beta = 10^\circ$ and that the combined weight of the boatswain's chair and the sailor is 200 lb, determine the tension (a) in the support cable ACB , (b) in the traction cable CD .

2.56 A sailor is being rescued using a boatswain's chair that is suspended from a pulley that can roll freely on the support cable ACB and is pulled at a constant speed by cable CD . Knowing that $\alpha = 25^\circ$ and $\beta = 15^\circ$ and that the tension in cable CD is 20 lb, determine (a) the combined weight of the boatswain's chair and the sailor, (b) the tension in the support cable ACB .

2.57 For the cables of Prob. 2.44, find the value of α for which the tension is as small as possible (a) in cable BC , (b) in both cables simultaneously. In each case determine the tension in each cable.

2.58 For the cables of Prob. 2.46, it is known that the maximum allowable tension is 600 N in cable AC and 750 N in cable BC . Determine (a) the maximum force P that can be applied at C , (b) the corresponding value of α .

2.59 For the situation described in Fig. P2.48, determine (a) the value of α for which the tension in rope BC is as small as possible, (b) the corresponding value of the tension.

2.60 Two cables tied together at C are loaded as shown. Determine the range of values of Q for which the tension will not exceed 60 lb in either cable.

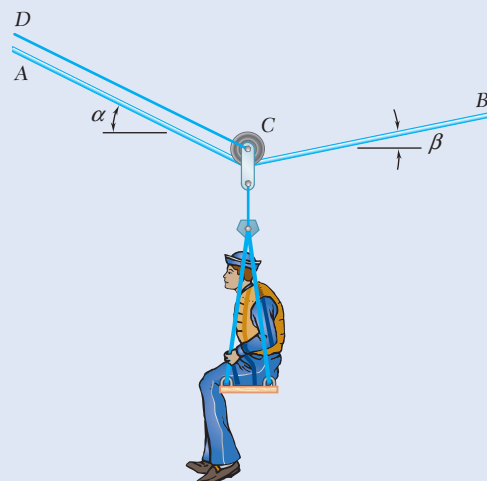


Fig. P2.55 and P2.56

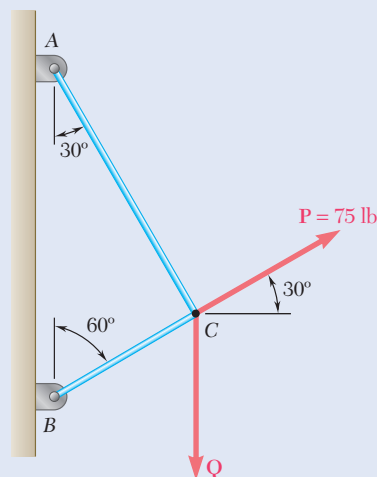


Fig. P2.60

2.61 A movable bin and its contents have a combined weight of 2.8 kN. Determine the shortest chain sling ACB that can be used to lift the loaded bin if the tension in the chain is not to exceed 5 kN.

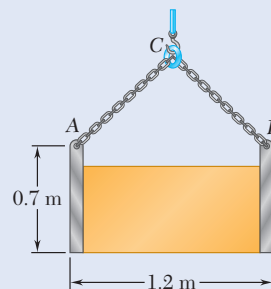


Fig. P2.61

- 2.62** For $W = 800 \text{ N}$, $P = 200 \text{ N}$, and $d = 600 \text{ mm}$, determine the value of h consistent with equilibrium.

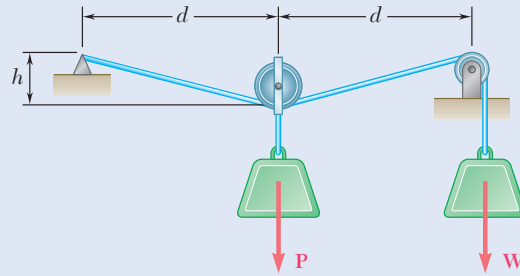


Fig. P2.62

- 2.63** Collar A is connected as shown to a 50-lb load and can slide on a frictionless horizontal rod. Determine the magnitude of the force $\mathbf{P}$ required to maintain the equilibrium of the collar when (a) $x = 4.5 \text{ in.}$, (b) $x = 15 \text{ in.}$

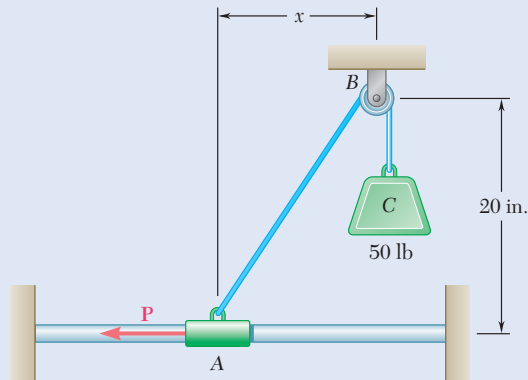


Fig. P2.63 and P2.64

- 2.64** Collar A is connected as shown to a 50-lb load and can slide on a frictionless horizontal rod. Determine the distance x for which the collar is in equilibrium when $P = 48 \text{ lb}$.
- 2.65** Three forces are applied to a bracket as shown. The directions of the two 150-N forces may vary, but the angle between these forces is always 50° . Determine the range of values of α for which the magnitude of the resultant of the forces acting at A is less than 600 N.

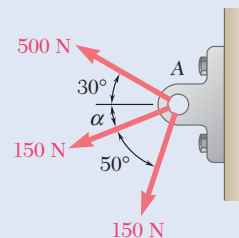


Fig. P2.65

- 2.66** A 200-kg crate is to be supported by the rope-and-pulley arrangement shown. Determine the magnitude and direction of the force $\mathbf{P}$ that must be exerted on the free end of the rope to maintain equilibrium. (*Hint:* The tension in the rope is the same on each side of a simple pulley. This can be proved by the methods of Chap. 4.)

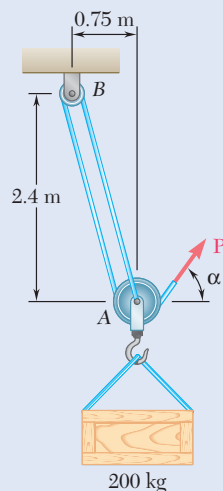


Fig. P2.66

- 2.67** A 600-lb crate is supported by several rope-and-pulley arrangements as shown. Determine for each arrangement the tension in the rope. (See the hint for Prob. 2.66.)

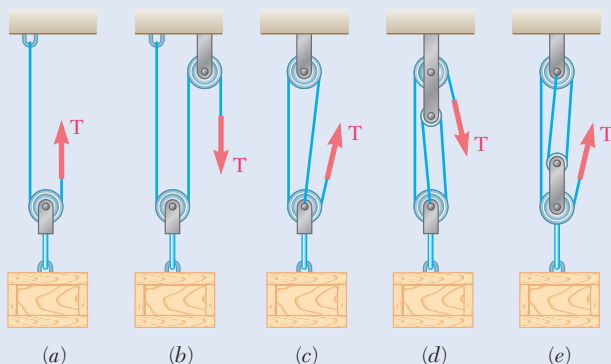


Fig. P2.67

- 2.68** Solve parts *b* and *d* of Prob. 2.67, assuming that the free end of the rope is attached to the crate.
- 2.69** A load $\mathbf{Q}$ is applied to pulley C , which can roll on the cable ACB . The pulley is held in the position shown by a second cable CAD , which passes over the pulley A and supports a load $\mathbf{P}$. Knowing that $P = 750 \text{ N}$, determine (a) the tension in cable ACB , (b) the magnitude of load $\mathbf{Q}$.
- 2.70** An 1800-N load $\mathbf{Q}$ is applied to pulley C , which can roll on the cable ACB . The pulley is held in the position shown by a second cable CAD , which passes over the pulley A and supports a load $\mathbf{P}$. Determine (a) the tension in cable ACB , (b) the magnitude of load $\mathbf{P}$.

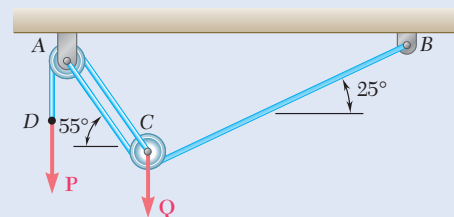


Fig. P2.69 and P2.70

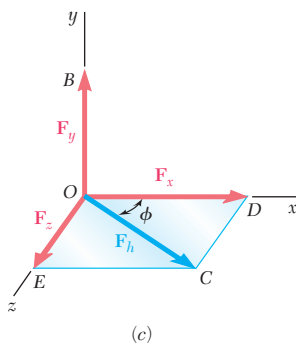
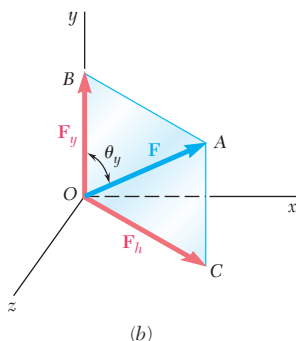
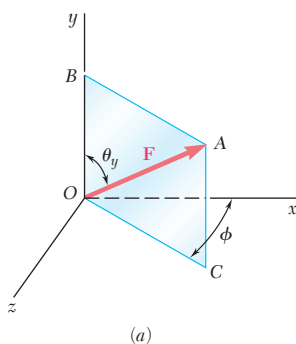


Fig. 2.25 (a) A force $\mathbf{F}$ in an xyz coordinate system; (b) components of $\mathbf{F}$ along the y axis and in the xz plane; (c) components of $\mathbf{F}$ along the three rectangular axes.

2.4 ADDING FORCES IN SPACE

The problems considered in the first part of this chapter involved only two dimensions; they were formulated and solved in a single plane. In the last part of this chapter, we discuss problems involving the three dimensions of space.

2.4A Rectangular Components of a Force in Space

Consider a force $\mathbf{F}$ acting at the origin O of the system of rectangular coordinates x , y , and z . To define the direction of $\mathbf{F}$, we draw the vertical plane $OBAC$ containing $\mathbf{F}$ (Fig. 2.25a). This plane passes through the vertical y axis; its orientation is defined by the angle ϕ it forms with the xy plane. The direction of $\mathbf{F}$ within the plane is defined by the angle θ_y that $\mathbf{F}$ forms with the y axis. We can resolve the force $\mathbf{F}$ into a vertical component $\mathbf{F}_y$ and a horizontal component $\mathbf{F}_h$; this operation, shown in Fig. 2.25b, is carried out in plane $OBAC$ according to the rules developed earlier. The corresponding scalar components are

$$F_y = F \cos \theta_y \quad F_h = F \sin \theta_y \quad (2.16)$$

However, we can also resolve $\mathbf{F}_h$ into two rectangular components $\mathbf{F}_x$ and $\mathbf{F}_z$ along the x and z axes, respectively. This operation, shown in Fig. 2.25c, is carried out in the xz plane. We obtain the following expressions for the corresponding scalar components:

$$\begin{aligned} F_x &= F_h \cos \phi = F \sin \theta_y \cos \phi \\ F_z &= F_h \sin \phi = F \sin \theta_y \sin \phi \end{aligned} \quad (2.17)$$

The given force $\mathbf{F}$ thus has been resolved into three rectangular vector components $\mathbf{F}_x$, $\mathbf{F}_y$, $\mathbf{F}_z$, which are directed along the three coordinate axes.

We can now apply the Pythagorean theorem to the triangles OAB and OCD of Fig. 2.25:

$$\begin{aligned} F^2 &= (OA)^2 = (OB)^2 + (BA)^2 = F_y^2 + F_h^2 \\ F_h^2 &= (OC)^2 = (OD)^2 + (DC)^2 = F_x^2 + F_z^2 \end{aligned}$$

Eliminating F_h^2 from these two equations and solving for F , we obtain the following relation between the magnitude of $\mathbf{F}$ and its rectangular scalar components:

Magnitude of a force in space

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} \quad (2.18)$$

The relationship between the force $\mathbf{F}$ and its three components $\mathbf{F}_x$, $\mathbf{F}_y$, and $\mathbf{F}_z$ is more easily visualized if we draw a “box” having $\mathbf{F}_x$, $\mathbf{F}_y$, and $\mathbf{F}_z$ for edges, as shown in Fig. 2.26. The force $\mathbf{F}$ is then represented by the main diagonal OA of this box. Figure 2.26b shows the right triangle

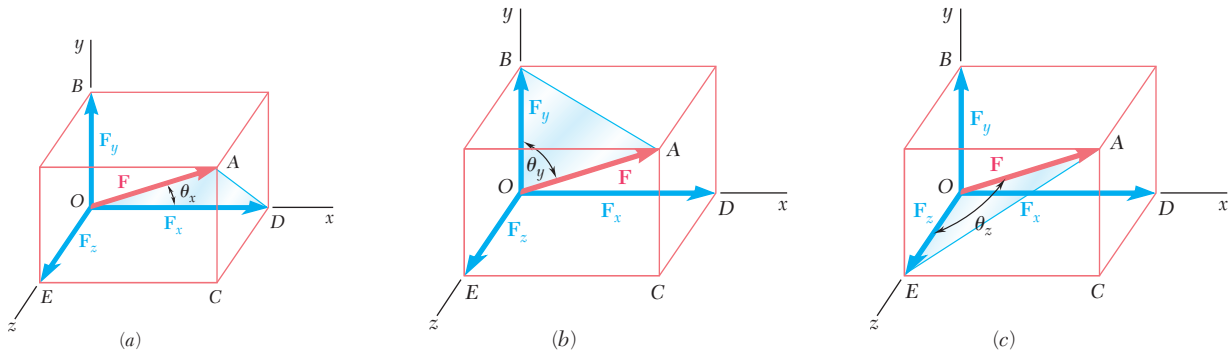


Fig. 2.26 (a) Force $\mathbf{F}$ in a three-dimensional box, showing its angle with the x axis; (b) force $\mathbf{F}$ and its angle with the y axis; (c) force $\mathbf{F}$ and its angle with the z axis.

OAB used to derive the first of the formulas (2.16): $F_y = F \cos \theta_y$. In Fig. 2.26a and c, two other right triangles have also been drawn: OAD and OAE . These triangles occupy positions in the box comparable with that of triangle OAB . Denoting by θ_x and θ_z , respectively, the angles that $\mathbf{F}$ forms with the x and z axes, we can derive two formulas similar to $F_y = F \cos \theta_y$. We thus write

Scalar components of a force $\mathbf{F}$

$$F_x = F \cos \theta_x \quad F_y = F \cos \theta_y \quad F_z = F \cos \theta_z \quad (2.19)$$

The three angles θ_x , θ_y , and θ_z define the direction of the force $\mathbf{F}$; they are more commonly used for this purpose than the angles θ_y and ϕ introduced at the beginning of this section. The cosines of θ_x , θ_y , and θ_z are known as the **direction cosines** of the force $\mathbf{F}$.

Introducing the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, which are directed respectively along the x , y , and z axes (Fig. 2.27), we can express $\mathbf{F}$ in the form

Vector expression of a force $\mathbf{F}$

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k} \quad (2.20)$$

where the scalar components F_x , F_y , and F_z are defined by the relations in Eq. (2.19).

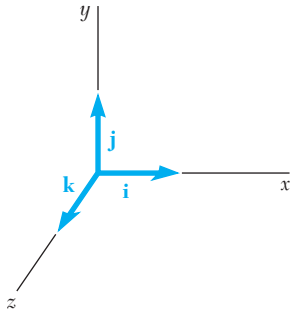


Fig. 2.27 The three unit vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ lie along the three coordinate axes x , y , z , respectively.

Concept Application 2.4

A force of 500 N forms angles of 60° , 45° , and 120° , respectively, with the x , y , and z axes. Find the components F_x , F_y , and F_z of the force and express the force in terms of unit vectors.

Solution

Substitute $F = 500$ N, $\theta_x = 60^\circ$, $\theta_y = 45^\circ$, and $\theta_z = 120^\circ$ into formulas (2.19). The scalar components of $\mathbf{F}$ are then

$$F_x = (500 \text{ N}) \cos 60^\circ = +250 \text{ N}$$

$$F_y = (500 \text{ N}) \cos 45^\circ = +354 \text{ N}$$

$$F_z = (500 \text{ N}) \cos 120^\circ = -250 \text{ N}$$

Carrying these values into Eq. (2.20), you have

$$\mathbf{F} = (250 \text{ N})\mathbf{i} + (354 \text{ N})\mathbf{j} - (250 \text{ N})\mathbf{k}$$

As in the case of two-dimensional problems, a plus sign indicates that the component has the same sense as the corresponding axis, and a minus sign indicates that it has the opposite sense.

The angle a force $\mathbf{F}$ forms with an axis should be measured from the positive side of the axis and is always between 0 and 180° . An angle θ_x smaller than 90° (acute) indicates that $\mathbf{F}$ (assumed attached to O) is on the same side of the yz plane as the positive x axis; $\cos \theta_x$ and F_x are then positive. An angle θ_x larger than 90° (obtuse) indicates that $\mathbf{F}$ is on the other side of the yz plane; $\cos \theta_x$ and F_x are then negative. In Concept Application 2.4, the angles θ_x and θ_y are acute and θ_z is obtuse; consequently, F_x and F_y are positive and F_z is negative.

Substituting into Eq. (2.20) the expressions obtained for F_x , F_y , and F_z in Eq. (2.19), we have

$$\mathbf{F} = F(\cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k}) \quad (2.21)$$

This equation shows that the force $\mathbf{F}$ can be expressed as the product of the scalar F and the vector

$$\boldsymbol{\lambda} = \cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k} \quad (2.22)$$

Clearly, the vector $\boldsymbol{\lambda}$ is a vector whose magnitude is equal to 1 and whose direction is the same as that of $\mathbf{F}$ (Fig. 2.33). The vector $\boldsymbol{\lambda}$ is referred to as the **unit vector along the line of action** of $\mathbf{F}$. It follows from Eq. (2.22) that the components of the unit vector $\boldsymbol{\lambda}$ are respectively equal to the direction cosines of the line of action of $\mathbf{F}$:

$$\lambda_x = \cos \theta_x \quad \lambda_y = \cos \theta_y \quad \lambda_z = \cos \theta_z \quad (2.23)$$

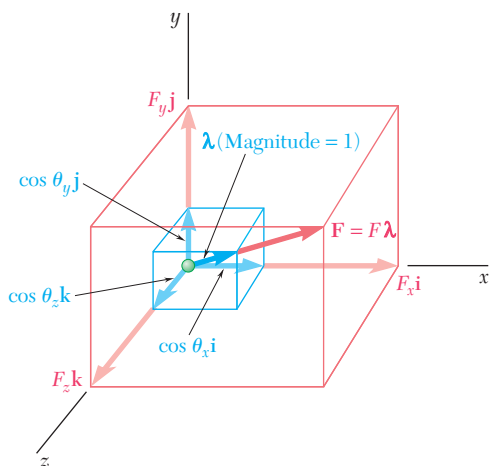


Fig. 2.28 Force $\mathbf{F}$ can be expressed as the product of its magnitude F and a unit vector $\boldsymbol{\lambda}$ in the direction of $\mathbf{F}$. Also shown are the components of $\mathbf{F}$ and its unit vector.

Note that the values of the three angles θ_x , θ_y , and θ_z are not independent. Recalling that the sum of the squares of the components of a vector is equal to the square of its magnitude, we can write

$$\lambda_x^2 + \lambda_y^2 + \lambda_z^2 = 1$$

Substituting for λ_x , λ_y , and λ_z from Eq. (2.23), we obtain

**Relationship among
direction cosines**

$$\cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z = 1 \quad (2.24)$$

In Concept Application 2.4, for instance, once the values $\theta_x = 60^\circ$ and $\theta_y = 45^\circ$ have been selected, the value of θ_z *must* be equal to 60° or 120° in order to satisfy the identity in Eq. (2.24).

When the components F_x , F_y , and F_z of a force $\mathbf{F}$ are given, we can obtain the magnitude F of the force from Eq. (2.18). We can then solve relations in Eq. (2.19) for the direction cosines as

$$\cos\theta_x = \frac{F_x}{F} \quad \cos\theta_y = \frac{F_y}{F} \quad \cos\theta_z = \frac{F_z}{F} \quad (2.25)$$

From the direction cosines, we can find the angles θ_x , θ_y , and θ_z characterizing the direction of $\mathbf{F}$.

Concept Application 2.5

A force $\mathbf{F}$ has the components $F_x = 20$ lb, $F_y = -30$ lb, and $F_z = 60$ lb. Determine its magnitude F and the angles θ_x , θ_y , and θ_z it forms with the coordinate axes.

Solution

You can obtain the magnitude of $\mathbf{F}$ from formula (2.18):

$$\begin{aligned} F &= \sqrt{F_x^2 + F_y^2 + F_z^2} \\ &= \sqrt{(20 \text{ lb})^2 + (-30 \text{ lb})^2 + (60 \text{ lb})^2} \\ &= \sqrt{4900} \text{ lb} = 70 \text{ lb} \end{aligned}$$

Substituting the values of the components and magnitude of $\mathbf{F}$ into Eqs. (2.25), the direction cosines are

$$\cos\theta_x = \frac{F_x}{F} = \frac{20 \text{ lb}}{70 \text{ lb}} \quad \cos\theta_y = \frac{F_y}{F} = \frac{-30 \text{ lb}}{70 \text{ lb}} \quad \cos\theta_z = \frac{F_z}{F} = \frac{60 \text{ lb}}{70 \text{ lb}}$$

Calculating each quotient and its arc cosine gives you

$$\theta_x = 73.4^\circ \quad \theta_y = 115.4^\circ \quad \theta_z = 31.0^\circ$$

These computations can be carried out easily with a calculator.

2.4B Force Defined by its Magnitude and Two Points on its Line of Action

In many applications, the direction of a force $\mathbf{F}$ is defined by the coordinates of two points, $M(x_1, y_1, z_1)$ and $N(x_2, y_2, z_2)$, located on its line of action (Fig. 2.29). Consider the vector $\overrightarrow{MN}$ joining M and N and of the same

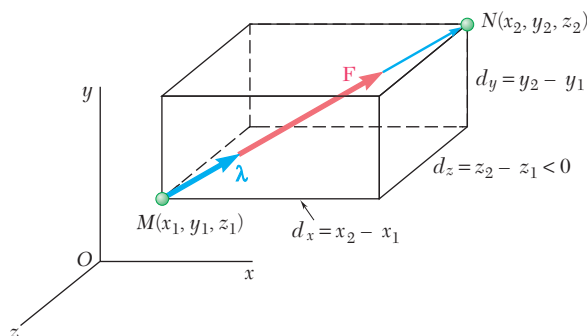


Fig. 2.29 A case where the line of action of force $\mathbf{F}$ is determined by the two points M and N . We can calculate the components of $\mathbf{F}$ and its direction cosines from the vector $\overrightarrow{MN}$.

sense as a force $\mathbf{F}$. Denoting its scalar components by d_x , d_y , and d_z , respectively, we write

$$\overrightarrow{MN} = d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k} \quad (2.26)$$

We can obtain a unit vector $\boldsymbol{\lambda}$ along the line of action of $\mathbf{F}$ (i.e., along the line MN) by dividing the vector $\overrightarrow{MN}$ by its magnitude MN . Substituting for $\overrightarrow{MN}$ from Eq. (2.26) and observing that MN is equal to the distance d from M to N , we have

$$\boldsymbol{\lambda} = \frac{\overrightarrow{MN}}{MN} = \frac{1}{d}(d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k}) \quad (2.27)$$

Recalling that $\mathbf{F}$ is equal to the product of F and $\boldsymbol{\lambda}$, we have

$$\mathbf{F} = F\boldsymbol{\lambda} = \frac{F}{d}(d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k}) \quad (2.28)$$

It follows that the scalar components of $\mathbf{F}$ are, respectively,

**Scalar components
of force $\mathbf{F}$**

$$F_x = \frac{Fd_x}{d} \quad F_y = \frac{Fd_y}{d} \quad F_z = \frac{Fd_z}{d} \quad (2.29)$$

The relations in Eq. (2.29) considerably simplify the determination of the components of a force $\mathbf{F}$ of given magnitude F when the line of action of $\mathbf{F}$ is defined by two points M and N . The calculation consists of

first subtracting the coordinates of M from those of N , then determining the components of the vector $\overrightarrow{MN}$ and the distance d from M to N . Thus,

$$d_x = x_2 - x_1 \quad d_y = y_2 - y_1 \quad d_z = z_2 - z_1$$

$$d = \sqrt{d_x^2 + d_y^2 + d_z^2}$$

Substituting for F and for d_x , d_y , d_z , and d into the relations in Eq. (2.29), we obtain the components F_x , F_y , and F_z of the force.

We can then obtain the angles θ_x , θ_y , and θ_z that $\mathbf{F}$ forms with the coordinate axes from Eqs. (2.25). Comparing Eqs. (2.22) and (2.27), we can write

**Direction cosines
of force $\mathbf{F}$**

$$\cos \theta_x = \frac{d_x}{d} \quad \cos \theta_y = \frac{d_y}{d} \quad \cos \theta_z = \frac{d_z}{d} \quad (2.30)$$

In other words, we can determine the angles θ_x , θ_y , and θ_z directly from the components and the magnitude of the vector $\overrightarrow{MN}$.

2.4C Addition of Concurrent Forces in Space

We can determine the resultant $\mathbf{R}$ of two or more forces in space by summing their rectangular components. Graphical or trigonometric methods are generally not practical in the case of forces in space.

The method followed here is similar to that used in Sec. 2.2B with coplanar forces. Setting

$$\mathbf{R} = \Sigma \mathbf{F}$$

we resolve each force into its rectangular components:

$$\begin{aligned} R_x \mathbf{i} + R_y \mathbf{j} + R_z \mathbf{k} &= \Sigma (F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}) \\ &= (\Sigma F_x) \mathbf{i} + (\Sigma F_y) \mathbf{j} + (\Sigma F_z) \mathbf{k} \end{aligned}$$

From this equation, it follows that

**Rectangular components
of the resultant**

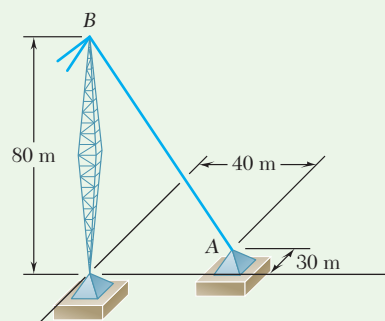
$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad R_z = \Sigma F_z \quad (2.31)$$

The magnitude of the resultant and the angles θ_x , θ_y , and θ_z that the resultant forms with the coordinate axes are obtained using the method discussed earlier in this section. We end up with

**Resultant of concurrent
forces in space**

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2} \quad (2.32)$$

$$\cos \theta_x = \frac{R_x}{R} \quad \cos \theta_y = \frac{R_y}{R} \quad \cos \theta_z = \frac{R_z}{R} \quad (2.33)$$



Sample Problem 2.7

A tower guy wire is anchored by means of a bolt at A . The tension in the wire is 2500 N. Determine (a) the components F_x , F_y , and F_z of the force acting on the bolt and (b) the angles θ_x , θ_y , and θ_z defining the direction of the force.

STRATEGY: From the given distances, we can determine the length of the wire and the direction of a unit vector along it. From that, we can find the components of the tension and the angles defining its direction.

MODELING and ANALYSIS:

a. Components of the Force. The line of action of the force acting on the bolt passes through points A and B , and the force is directed from A to B . The components of the vector $\overrightarrow{AB}$, which has the same direction as the force, are

$$d_x = -40 \text{ m} \quad d_y = +80 \text{ m} \quad d_z = +30 \text{ m}$$

The total distance from A to B is

$$AB = d = \sqrt{d_x^2 + d_y^2 + d_z^2} = 94.3 \text{ m}$$

Denoting the unit vectors along the coordinate axes by $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, you have

$$\overrightarrow{AB} = -(40 \text{ m})\mathbf{i} + (80 \text{ m})\mathbf{j} + (30 \text{ m})\mathbf{k}$$

Introducing the unit vector $\boldsymbol{\lambda} = \overrightarrow{AB}/AB$ (Fig. 1), you can express $\mathbf{F}$ in terms of $\overrightarrow{AB}$ as

$$\mathbf{F} = F\boldsymbol{\lambda} = F \frac{\overrightarrow{AB}}{AB} = \frac{2500 \text{ N}}{94.3 \text{ m}} \overrightarrow{AB}$$

Substituting the expression for $\overrightarrow{AB}$ gives you

$$\begin{aligned} \mathbf{F} &= \frac{2500 \text{ N}}{94.3 \text{ m}} [-(40 \text{ m})\mathbf{i} + (80 \text{ m})\mathbf{j} + (30 \text{ m})\mathbf{k}] \\ &= -(1060 \text{ N})\mathbf{i} + (2120 \text{ N})\mathbf{j} + (795 \text{ N})\mathbf{k} \end{aligned}$$

The components of $\mathbf{F}$, therefore, are

$$F_x = -1060 \text{ N} \quad F_y = +2120 \text{ N} \quad F_z = +795 \text{ N} \quad \blacktriangleleft$$

b. Direction of the Force. Using Eqs. (2.25), you can write the direction cosines directly (Fig. 2):

$$\begin{aligned} \cos \theta_x &= \frac{F_x}{F} = \frac{-1060 \text{ N}}{2500 \text{ N}} & \cos \theta_y &= \frac{F_y}{F} = \frac{+2120 \text{ N}}{2500 \text{ N}} \\ \cos \theta_z &= \frac{F_z}{F} = \frac{+795 \text{ N}}{2500 \text{ N}} \end{aligned}$$

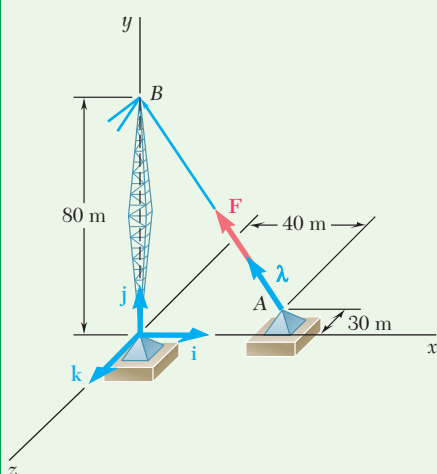


Fig. 1 Cable force acting on bolt at A , and its unit vector.

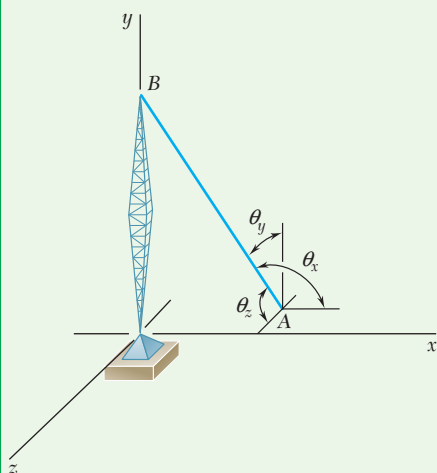


Fig. 2 Direction angles for cable AB .

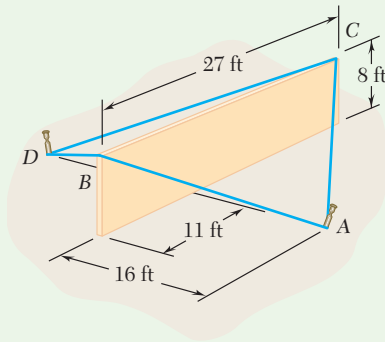
Calculating each quotient and its arc cosine, you obtain

$$\theta_x = 115.1^\circ \quad \theta_y = 32.0^\circ \quad \theta_z = 71.5^\circ \quad \blacktriangleleft$$

(Note. You could have obtained this same result by using the components and magnitude of the vector $\overrightarrow{AB}$ rather than those of the force $\mathbf{F}$.)

REFLECT and THINK: It makes sense that, for a given geometry, only a certain set of components and angles characterize a given resultant force. The methods in this section allow you to translate back and forth between forces and geometry.

Sample Problem 2.8



A wall section of precast concrete is temporarily held in place by the cables shown. If the tension is 840 lb in cable AB and 1200 lb in cable AC, determine the magnitude and direction of the resultant of the forces exerted by cables AB and AC on stake A.

STRATEGY: This is a problem in adding concurrent forces in space. The simplest approach is to first resolve the forces into components and to then sum the components and find the resultant.

MODELING and ANALYSIS:

Components of the Forces. First resolve the force exerted by each cable on stake A into x , y , and z components. To do this, determine the components and magnitude of the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, measuring them from A toward the wall section (Fig. 1). Denoting the unit vectors along the coordinate axes by $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, these vectors are

$$\begin{aligned} \overrightarrow{AB} &= -(16 \text{ ft})\mathbf{i} + (8 \text{ ft})\mathbf{j} + (11 \text{ ft})\mathbf{k} & AB &= 21 \text{ ft} \\ \overrightarrow{AC} &= -(16 \text{ ft})\mathbf{i} + (8 \text{ ft})\mathbf{j} - (16 \text{ ft})\mathbf{k} & AC &= 24 \text{ ft} \end{aligned}$$

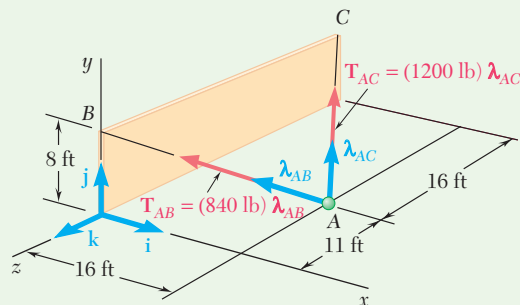


Fig. 1 Cable forces acting on stake at A, and their unit vectors.

Denoting by λ_{AB} the unit vector along AB , the tension in AB is

$$\mathbf{T}_{AB} = T_{AB}\lambda_{AB} = T_{AB}\frac{\overrightarrow{AB}}{AB} = \frac{840 \text{ lb}}{21 \text{ ft}} \overrightarrow{AB}$$

Substituting the expression found for $\overrightarrow{AB}$, the tension becomes

$$\mathbf{T}_{AB} = \frac{840 \text{ lb}}{21 \text{ ft}} [-(16 \text{ ft})\mathbf{i} + (8 \text{ ft})\mathbf{j} + (11 \text{ ft})\mathbf{k}]$$

$$\mathbf{T}_{AB} = -(640 \text{ lb})\mathbf{i} + (320 \text{ lb})\mathbf{j} + (440 \text{ lb})\mathbf{k}$$

Similarly, denoting by λ_{AC} the unit vector along AC , the tension in AC is

$$\mathbf{T}_{AC} = T_{AC}\lambda_{AC} = T_{AC}\frac{\overrightarrow{AC}}{AC} = \frac{1200 \text{ lb}}{24 \text{ ft}} \overrightarrow{AC}$$

$$\mathbf{T}_{AC} = -(800 \text{ lb})\mathbf{i} + (400 \text{ lb})\mathbf{j} - (800 \text{ lb})\mathbf{k}$$

Resultant of the Forces. The resultant $\mathbf{R}$ of the forces exerted by the two cables is

$$\mathbf{R} = \mathbf{T}_{AB} + \mathbf{T}_{AC} = -(1440 \text{ lb})\mathbf{i} + (720 \text{ lb})\mathbf{j} - (360 \text{ lb})\mathbf{k}$$

You can now determine the magnitude and direction of the resultant as

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2} = \sqrt{(-1440)^2 + (720)^2 + (-360)^2}$$

$$R = 1650 \text{ lb} \quad \blacktriangleleft$$

The direction cosines come from Eqs. (2.33):

$$\begin{aligned} \cos \theta_x &= \frac{R_x}{R} = \frac{-1440 \text{ lb}}{1650 \text{ lb}} & \cos \theta_y &= \frac{R_y}{R} = \frac{+720 \text{ lb}}{1650 \text{ lb}} \\ \cos \theta_z &= \frac{R_z}{R} = \frac{-360 \text{ lb}}{1650 \text{ lb}} \end{aligned}$$

Calculating each quotient and its arc cosine, the angles are

$$\theta_x = 150.8^\circ \quad \theta_y = 64.1^\circ \quad \theta_z = 102.6^\circ \quad \blacktriangleleft$$

REFLECT and THINK: Based on visual examination of the cable forces, you might have anticipated that θ_x for the resultant should be obtuse and θ_y should be acute. The outcome of θ_z was not as apparent.

SOLVING PROBLEMS ON YOUR OWN

In this section, we saw that we can define a **force in space** by its magnitude and direction or by the three rectangular components F_x , F_y , and F_z .

A. When a force is defined by its magnitude and direction, you can find its rectangular components F_x , F_y , and F_z as follows.

Case 1. If the direction of the force $\mathbf{F}$ is defined by the angles θ_y and ϕ shown in Fig. 2.25, projections of $\mathbf{F}$ through these angles or their complements will yield the components of $\mathbf{F}$ [Eqs. (2.17)]. Note that to find the x and z components of $\mathbf{F}$, first project $\mathbf{F}$ onto the horizontal plane; the projection $\mathbf{F}_h$ obtained in this way is then resolved into the components $\mathbf{F}_x$ and $\mathbf{F}_z$ (Fig. 2.25c).

Case 2. If the direction of the force $\mathbf{F}$ is defined by the angles θ_x , θ_y , and θ_z that $\mathbf{F}$ forms with the coordinate axes, you can obtain each component by multiplying the magnitude F of the force by the cosine of the corresponding angle [Concept Application 2.4]:

$$F_x = F \cos \theta_x \qquad F_y = F \cos \theta_y \qquad F_z = F \cos \theta_z$$

Case 3. If the direction of the force $\mathbf{F}$ is defined by two points M and N located on its line of action (Fig. 2.29), first express the vector $\overrightarrow{MN}$ drawn from M to N in terms of its components d_x , d_y , and d_z and the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$:

$$\overrightarrow{MN} = d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k}$$

Then determine the unit vector $\boldsymbol{\lambda}$ along the line of action of $\mathbf{F}$ by dividing the vector $\overrightarrow{MN}$ by its magnitude MN . Multiplying $\boldsymbol{\lambda}$ by the magnitude of $\mathbf{F}$ gives you the desired expression for $\mathbf{F}$ in terms of its rectangular components [Sample Prob. 2.7]:

$$\mathbf{F} = F\boldsymbol{\lambda} = \frac{F}{d}(d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k})$$

It is helpful to use a consistent and meaningful system of notation when determining the rectangular components of a force. The method used in this text is illustrated in Sample Prob. 2.8, where the force $\mathbf{T}_{AB}$ acts from stake A toward point B . Note that the subscripts have been ordered to agree with the direction of the force. We recommend that you adopt the same notation, as it will help you identify point 1 (the first subscript) and point 2 (the second subscript).

When calculating the vector defining the line of action of a force, you might think of its scalar components as the number of steps you must take in each coordinate direction to go from point 1 to point 2. It is essential that you always remember to assign the correct sign to each of the components.

(continued)

B. When a force is defined by its rectangular components F_x , F_y , and F_z , you can obtain its magnitude F from

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

You can determine the direction cosines of the line of action of $\mathbf{F}$ by dividing the components of the force by F :

$$\cos \theta_x = \frac{F_x}{F} \quad \cos \theta_y = \frac{F_y}{F} \quad \cos \theta_z = \frac{F_z}{F}$$

From the direction cosines, you can obtain the angles θ_x , θ_y , and θ_z that $\mathbf{F}$ forms with the coordinate axes [Concept Application 2.5].

C. To determine the resultant R of two or more forces in three-dimensional space, first determine the rectangular components of each force by one of the procedures described previously. Adding these components will yield the components R_x , R_y , and R_z of the resultant. You can then obtain the magnitude and direction of the resultant as indicated previously for a force $\mathbf{F}$ [Sample Prob. 2.8].

Problems

END-OF-SECTION PROBLEMS

- 2.71** Determine (a) the x , y , and z components of the 600-N force, (b) the angles θ_x , θ_y , and θ_z that the force forms with the coordinate axes.
- 2.72** Determine (a) the x , y , and z components of the 450-N force, (b) the angles θ_x , θ_y , and θ_z that the force forms with the coordinate axes.
- 2.73** A gun is aimed at a point A located 35° east of north. Knowing that the barrel of the gun forms an angle of 40° with the horizontal and that the maximum recoil force is 400 N, determine (a) the x , y , and z components of that force, (b) the values of the angles θ_x , θ_y , and θ_z defining the direction of the recoil force. (Assume that the x , y , and z axes are directed, respectively, east, up, and south.)
- 2.74** Solve Prob. 2.73 assuming that point A is located 15° north of west and that the barrel of the gun forms an angle of 25° with the horizontal.
- 2.75** The angle between spring AB and the post DA is 30° . Knowing that the tension in the spring is 50 lb, determine (a) the x , y , and z components of the force exerted on the circular plate at B , (b) the angles θ_x , θ_y , and θ_z defining the direction of the force at B .
- 2.76** The angle between spring AC and the post DA is 30° . Knowing that the tension in the spring is 40 lb, determine (a) the x , y , and z components of the force exerted on the circular plate at C , (b) the angles θ_x , θ_y , and θ_z defining the direction of the force at C .
- 2.77** Cable AB is 65 ft long, and the tension in that cable is 3900 lb. Determine (a) the x , y , and z components of the force exerted by the cable on the anchor B , (b) the angles θ_x , θ_y , and θ_z defining the direction of that force.
- 2.78** Cable AC is 70 ft long, and the tension in that cable is 5250 lb. Determine (a) the x , y , and z components of the force exerted by the cable on the anchor C , (b) the angles θ_x , θ_y , and θ_z defining the direction of that force.
- 2.79** Determine the magnitude and direction of the force $\mathbf{F} = (240 \text{ N})\mathbf{i} - (270 \text{ N})\mathbf{j} + (680 \text{ N})\mathbf{k}$.
- 2.80** Determine the magnitude and direction of the force $\mathbf{F} = (320 \text{ N})\mathbf{i} + (400 \text{ N})\mathbf{j} - (250 \text{ N})\mathbf{k}$.
- 2.81** A force acts at the origin of a coordinate system in a direction defined by the angles $\theta_x = 69.3^\circ$ and $\theta_z = 57.9^\circ$. Knowing that the y component of the force is -174.0 lb , determine (a) the angle θ_y , (b) the other components and the magnitude of the force.

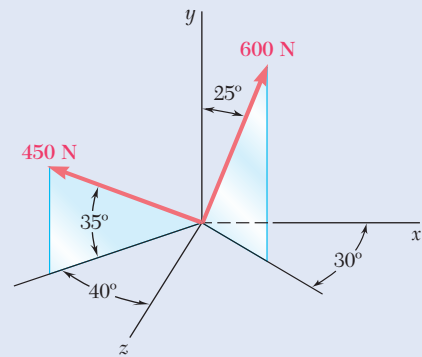


Fig. P2.71 and P2.72

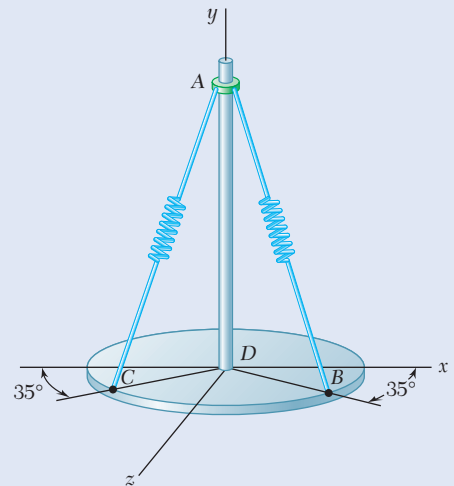


Fig. P2.75 and P2.76

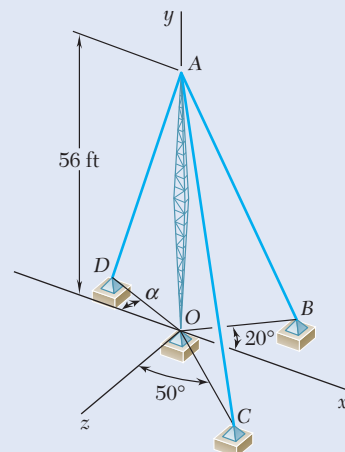


Fig. P2.77 and P2.78

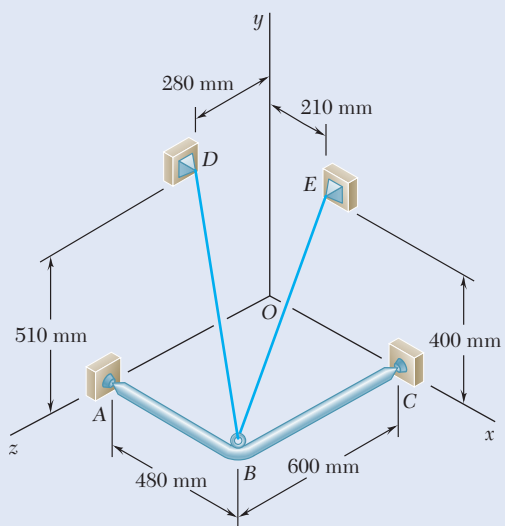


Fig. P2.85

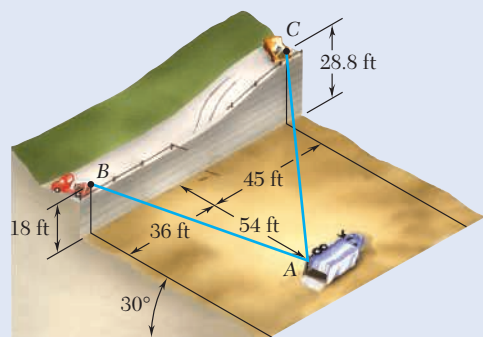


Fig. P2.87 and P2.88

2.82 A force acts at the origin of a coordinate system in a direction defined by the angles $\theta_x = 70.9^\circ$ and $\theta_y = 144.9^\circ$. Knowing that the z component of the force is -52.0 lb, determine (a) the angle θ_z , (b) the other components and the magnitude of the force.

2.83 A force $\mathbf{F}$ of magnitude 210 N acts at the origin of a coordinate system. Knowing that $F_x = 80$ N, $\theta_z = 151.2^\circ$, and $F_y < 0$, determine (a) the components F_y and F_z , (b) the angles θ_x and θ_y .

2.84 A force $\mathbf{F}$ of magnitude 1200 N acts at the origin of a coordinate system. Knowing that $\theta_x = 65^\circ$, $\theta_y = 40^\circ$, and $F_z > 0$, determine (a) the components of the force, (b) the angle θ_z .

2.85 A frame ABC is supported in part by cable DBE that passes through a frictionless ring at B . Knowing that the tension in the cable is 385 N, determine the components of the force exerted by the cable on the support at D .

2.86 For the frame and cable of Prob. 2.85, determine the components of the force exerted by the cable on the support at E .

2.87 In order to move a wrecked truck, two cables are attached at A and pulled by winches B and C as shown. Knowing that the tension in cable AB is 2 kips, determine the components of the force exerted at A by the cable.

2.88 In order to move a wrecked truck, two cables are attached at A and pulled by winches B and C as shown. Knowing that the tension in cable AC is 1.5 kips, determine the components of the force exerted at A by the cable.

2.89 A rectangular plate is supported by three cables as shown. Knowing that the tension in cable AB is 408 N, determine the components of the force exerted on the plate at B .

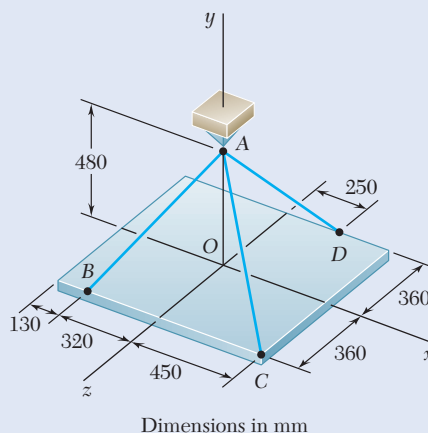


Fig. P2.89 and P2.90

2.90 A rectangular plate is supported by three cables as shown. Knowing that the tension in cable AD is 429 N, determine the components of the force exerted on the plate at D .

- 2.91** Find the magnitude and direction of the resultant of the two forces shown knowing that $P = 300$ N and $Q = 400$ N.
- 2.92** Find the magnitude and direction of the resultant of the two forces shown knowing that $P = 400$ N and $Q = 300$ N.
- 2.93** Knowing that the tension is 425 lb in cable AB and 510 lb in cable AC , determine the magnitude and direction of the resultant of the forces exerted at A by the two cables.

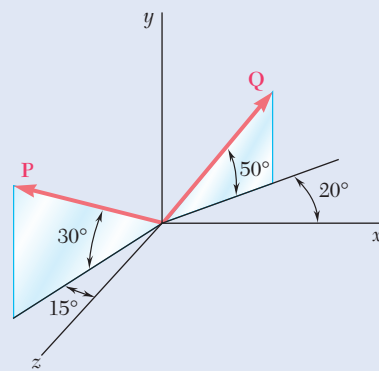


Fig. P2.91 and P2.92

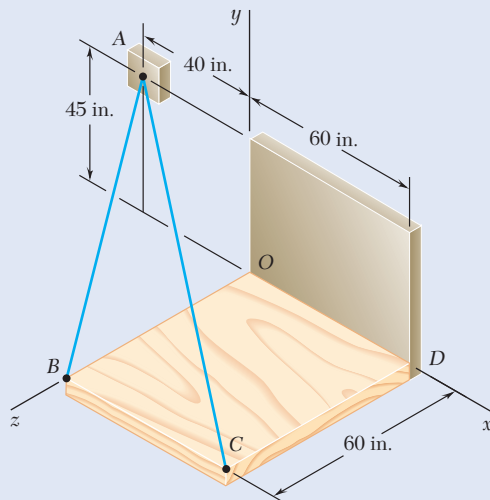


Fig. P2.93 and P2.94

- 2.94** Knowing that the tension is 510 lb in cable AB and 425 lb in cable AC , determine the magnitude and direction of the resultant of the forces exerted at A by the two cables.
- 2.95** For the frame of Prob. 2.85, determine the magnitude and direction of the resultant of the forces exerted by the cable at B knowing that the tension in the cable is 385 N.
- 2.96** For the plate of Prob. 2.89, determine the tensions in cables AB and AD knowing that the tension in cable AC is 54 N and that the resultant of the forces exerted by the three cables at A must be vertical.
- 2.97** The boom OA carries a load P and is supported by two cables as shown. Knowing that the tension in cable AB is 183 lb and that the resultant of the load P and of the forces exerted at A by the two cables must be directed along OA , determine the tension in cable AC .
- 2.98** For the boom and loading of Prob. 2.97, determine the magnitude of the load P .

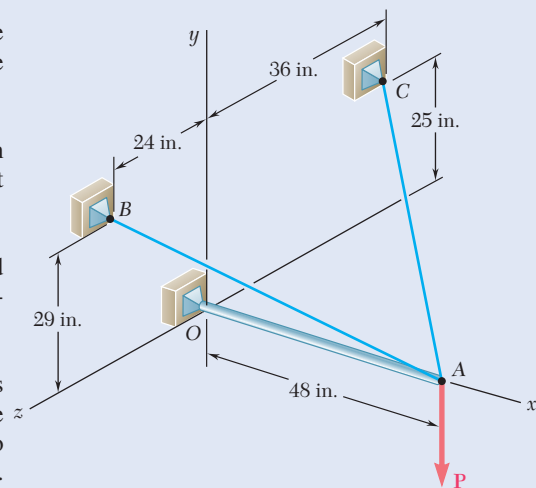


Fig. P2.97

2.5 FORCES AND EQUILIBRIUM IN SPACE

According to the definition given in Sec. 2.3, a particle A is in equilibrium if the resultant of all the forces acting on A is zero. The components R_x , R_y , and R_z of the resultant of forces in space are given by equations (2.31); when the components of the resultant are zero, we have

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma F_z = 0 \quad (2.34)$$

Equations (2.34) represent the necessary and sufficient conditions for the equilibrium of a particle in space. We can use them to solve problems dealing with the equilibrium of a particle involving no more than three unknowns.

The first step in solving three-dimensional equilibrium problems is to draw a free-body diagram showing the particle in equilibrium and *all* of the forces acting on it. You can then write the equations of equilibrium (2.34) and solve them for three unknowns. In the more common types of problems, these unknowns will represent (1) the three components of a single force or (2) the magnitude of three forces, each of known direction.

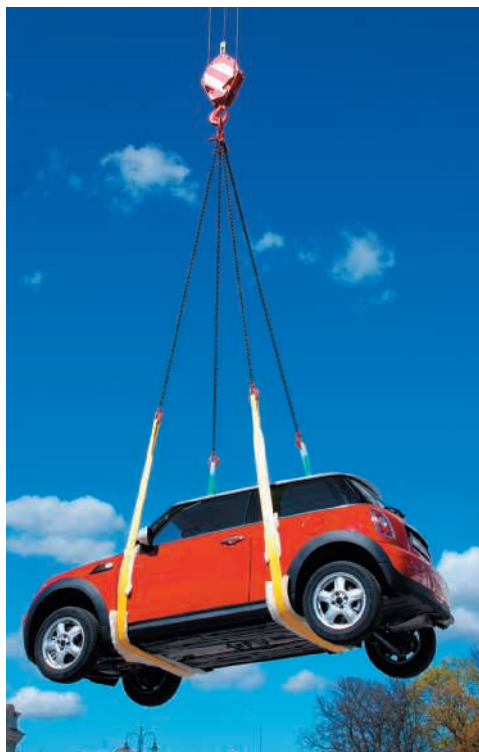
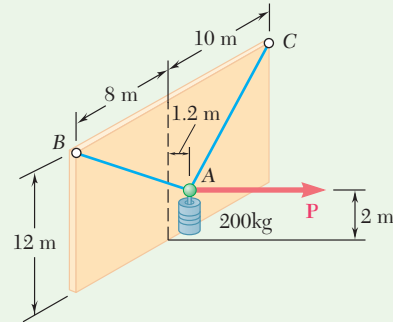


Photo 2.4 Although we cannot determine the tension in the four cables supporting the car by using the three equations (2.34), we can obtain a relation among the tensions by analyzing the equilibrium of the hook.

Sample Problem 2.9

A 200-kg cylinder is hung by means of two cables AB and AC that are attached to the top of a vertical wall. A horizontal force $\mathbf{P}$ perpendicular to the wall holds the cylinder in the position shown. Determine the magnitude of $\mathbf{P}$ and the tension in each cable.



STRATEGY: Connection point A is acted upon by four forces, including the weight of the cylinder. You can use the given geometry to express the force components of the cables and then apply equilibrium conditions to calculate the tensions.

MODELING and ANALYSIS:

Free-Body Diagram. Choose point A as a free body; this point is subjected to four forces, three of which are of unknown magnitude. Introducing the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, resolve each force into rectangular components (Fig. 1):

$$\mathbf{P} = P\mathbf{i}$$

$$\mathbf{W} = -mg\mathbf{j} = -(200 \text{ kg})(9.81 \text{ m/s}^2)\mathbf{j} = -(1962 \text{ N})\mathbf{j} \quad (1)$$

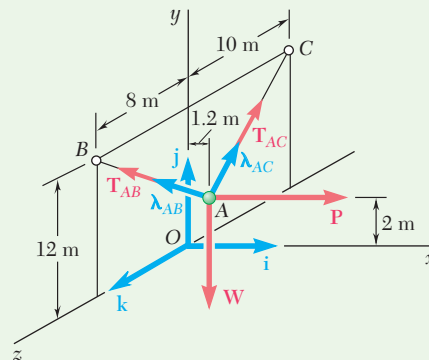


Fig. 1 Free-body diagram of particle A .

For $\mathbf{T}_{AB}$ and $\mathbf{T}_{AC}$, it is first necessary to determine the components and magnitudes of the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Denoting the unit vector along AB by $\boldsymbol{\lambda}_{AB}$, you can write $\mathbf{T}_{AB}$ as

$$\begin{aligned}\overrightarrow{AB} &= -(1.2 \text{ m})\mathbf{i} + (10 \text{ m})\mathbf{j} + (8 \text{ m})\mathbf{k} & AB &= 12.862 \text{ m} \\ \boldsymbol{\lambda}_{AB} &= \frac{\overrightarrow{AB}}{12.862 \text{ m}} = -0.09330\mathbf{i} + 0.7775\mathbf{j} + 0.6220\mathbf{k} \\ \mathbf{T}_{AB} &= T_{AB}\boldsymbol{\lambda}_{AB} = -0.09330T_{AB}\mathbf{i} + 0.7775T_{AB}\mathbf{j} + 0.6220T_{AB}\mathbf{k} \quad (2)\end{aligned}$$

Similarly, denoting the unit vector along AC by $\boldsymbol{\lambda}_{AC}$, you have for $\mathbf{T}_{AC}$

$$\begin{aligned}\overrightarrow{AC} &= -(1.2 \text{ m})\mathbf{i} + (10 \text{ m})\mathbf{j} - (10 \text{ m})\mathbf{k} & AC &= 14.193 \text{ m} \\ \boldsymbol{\lambda}_{AC} &= \frac{\overrightarrow{AC}}{14.193 \text{ m}} = -0.08455\mathbf{i} + 0.7046\mathbf{j} - 0.7046\mathbf{k} \\ \mathbf{T}_{AC} &= T_{AC}\boldsymbol{\lambda}_{AC} = -0.08455T_{AC}\mathbf{i} + 0.7046T_{AC}\mathbf{j} - 0.7046T_{AC}\mathbf{k} \quad (3)\end{aligned}$$

Equilibrium Condition. Since A is in equilibrium, you must have

$$\Sigma \mathbf{F} = 0: \quad \mathbf{T}_{AB} + \mathbf{T}_{AC} + \mathbf{P} + \mathbf{W} = 0$$

or substituting from Eqs. (1), (2), and (3) for the forces and factoring $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, you have

$$\begin{aligned} &(-0.09330T_{AB} - 0.08455T_{AC} + P)\mathbf{i} \\ &+ (0.7775T_{AB} + 0.7046T_{AC} - 1962 \text{ N})\mathbf{j} \\ &+ (0.6220T_{AB} - 0.7046T_{AC})\mathbf{k} = 0 \end{aligned}$$

Setting the coefficients of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ equal to zero, you can write three scalar equations, which express that the sums of the x , y , and z components of the forces are respectively equal to zero.

$$\begin{aligned}(\Sigma F_x = 0:) & \quad -0.09330T_{AB} - 0.08455T_{AC} + P = 0 \\ (\Sigma F_y = 0:) & \quad +0.7775T_{AB} + 0.7046T_{AC} - 1962 \text{ N} = 0 \\ (\Sigma F_z = 0:) & \quad +0.6220T_{AB} - 0.7046T_{AC} = 0\end{aligned}$$

Solving these equations, you obtain

$$P = 235 \text{ N} \quad T_{AB} = 1402 \text{ N} \quad T_{AC} = 1238 \text{ N} \quad \blacktriangleleft$$

REFLECT and THINK: The solution of the three unknown forces yielded positive results, which is completely consistent with the physical situation of this problem. Conversely, if one of the cable force results had been negative, thereby reflecting compression instead of tension, you should recognize that the solution is in error.

SOLVING PROBLEMS ON YOUR OWN

We saw earlier that when a particle is in **equilibrium**, the resultant of the forces acting on the particle must be zero. In the case of the equilibrium of a particle in three-dimensional space, this equilibrium condition provides you with three relations among the forces acting on the particle. These relations may be used to determine three unknowns—usually the magnitudes of three forces.

The solution usually consists of the following steps:

1. Draw a free-body diagram of the particle. This diagram shows the particle and all the forces acting on it. Indicate on the diagram the magnitudes of known forces, as well as any angles or dimensions that define the direction of a force. Any unknown magnitude or angle should be denoted by an appropriate symbol. Nothing else should be included in the free-body diagram.

2. Resolve each force into rectangular components. Following the method used earlier, determine for each force $\mathbf{F}$ the unit vector $\boldsymbol{\lambda}$ defining the direction of that force, and express $\mathbf{F}$ as the product of its magnitude F and $\boldsymbol{\lambda}$. You will obtain an expression of the form

$$\mathbf{F} = F\boldsymbol{\lambda} = \frac{F}{d}(d_x\mathbf{i} + d_y\mathbf{j} + d_z\mathbf{k})$$

where d , d_x , d_y , and d_z are dimensions obtained from the free-body diagram of the particle. If you know the magnitude as well as the direction of the force, then F is known and the expression obtained for $\mathbf{F}$ is well defined; otherwise F is one of the three unknowns that should be determined.

3. Set the resultant, or sum, of the forces exerted on the particle equal to zero. You will obtain a vector equation consisting of terms containing the unit vectors $\mathbf{i}$, $\mathbf{j}$, or $\mathbf{k}$. Group the terms containing the same unit vector and factor that vector. For the vector equation to be satisfied, you must set the coefficient of each of the unit vectors equal to zero. This yields three scalar equations that you can solve for no more than three unknowns [Sample Prob. 2.9].

Problems

FREE-BODY PRACTICE PROBLEMS

2.F5 Three cables are used to tether a balloon as shown. Knowing that the tension in cable AC is 444 N , draw the free-body diagram needed to determine the vertical force $\mathbf{P}$ exerted by the balloon at A .

2.F6 A container of mass $m = 120\text{ kg}$ is supported by three cables as shown. Draw the free-body diagram needed to determine the tension in each cable.

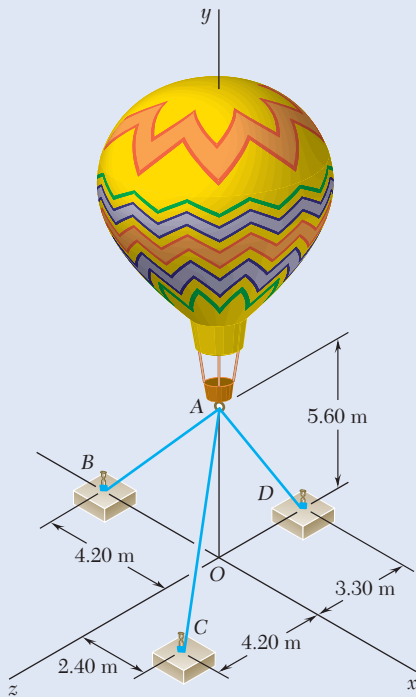


Fig. P2.F5

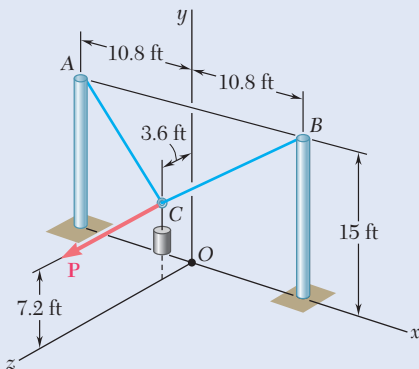


Fig. P2.F7

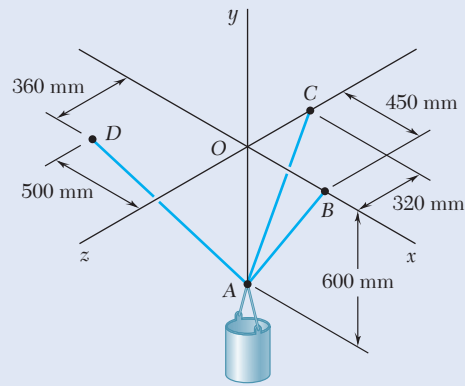


Fig. P2.F6

2.F7 A 150-lb cylinder is supported by two cables AC and BC that are attached to the top of vertical posts. A horizontal force $\mathbf{P}$, which is perpendicular to the plane containing the posts, holds the cylinder in the position shown. Draw the free-body diagram needed to determine the magnitude of $\mathbf{P}$ and the force in each cable.

2.F8 A transmission tower is held by three guy wires attached to a pin at A and anchored by bolts at B , C , and D . Knowing that the tension in wire AB is 630 lb , draw the free-body diagram needed to determine the vertical force $\mathbf{P}$ exerted by the tower on the pin at A .

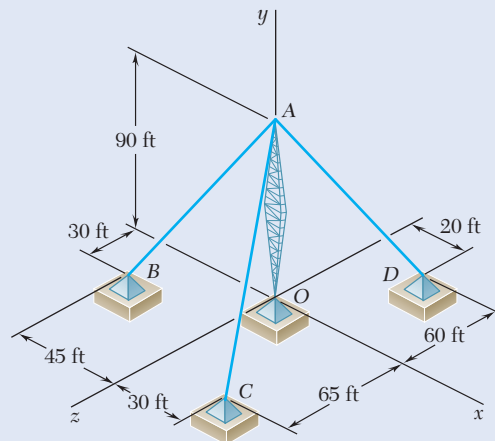


Fig. P2.F8

END-OF-SECTION PROBLEMS

2.99 A container is supported by three cables that are attached to a ceiling as shown. Determine the weight W of the container knowing that the tension in cable AB is 6 kN.

2.100 A container is supported by three cables that are attached to a ceiling as shown. Determine the weight W of the container, knowing that the tension in cable AD is 4.3 kN.

2.101 Three cables are used to tether a balloon as shown. Determine the vertical force $\mathbf{P}$ exerted by the balloon at A knowing that the tension in cable AD is 481 N.

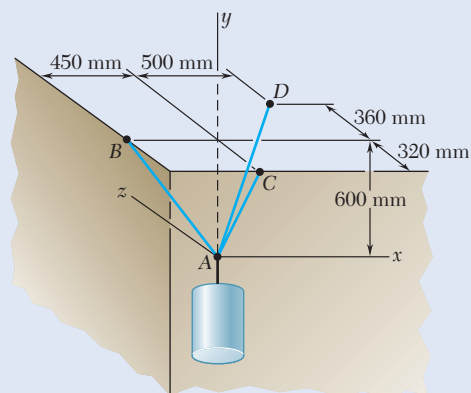


Fig. P2.99 and P2.100

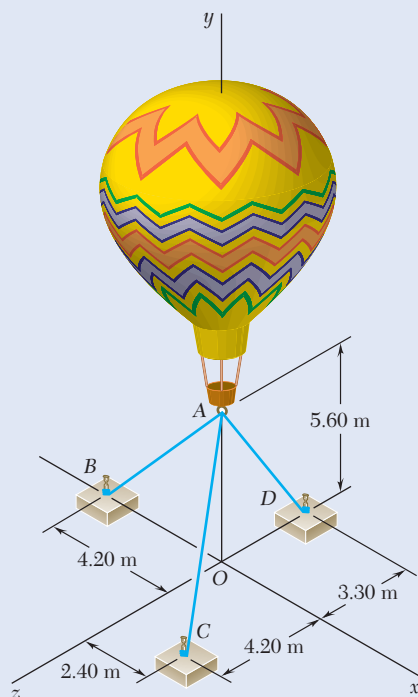


Fig. P2.101 and P2.102

2.102 Three cables are used to tether a balloon as shown. Knowing that the balloon exerts an 800-N vertical force at A , determine the tension in each cable.

2.103 A 36-lb triangular plate is supported by three wires as shown. Determine the tension in each wire, knowing that $a = 6$ in.

2.104 Solve Prob. 2.103, assuming that $a = 8$ in.

2.105 A crate is supported by three cables as shown. Determine the weight of the crate knowing that the tension in cable AC is 544 lb.

2.106 A 1600-lb crate is supported by three cables as shown. Determine the tension in each cable.

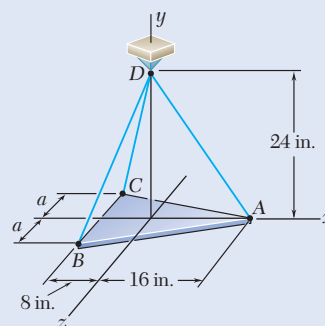


Fig. P2.103

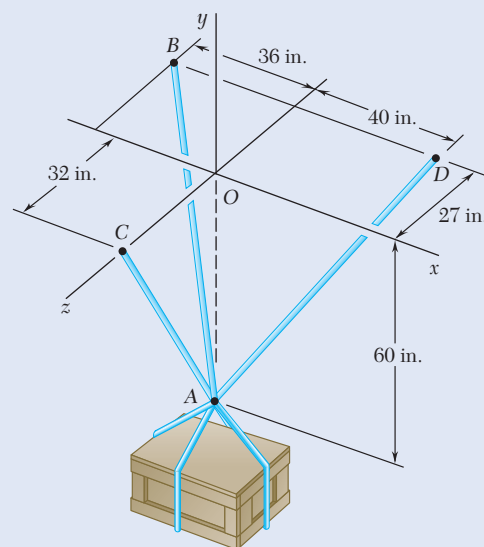
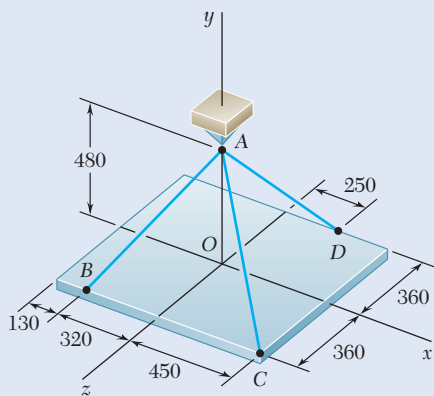


Fig. P2.105 and P2.106

- 2.107** Three cables are connected at A , where the forces $\mathbf{P}$ and $\mathbf{Q}$ are applied as shown. Knowing that $Q = 0$, find the value of P for which the tension in cable AD is 305 N.



Dimensions in mm

Fig. P2.109 and P2.110

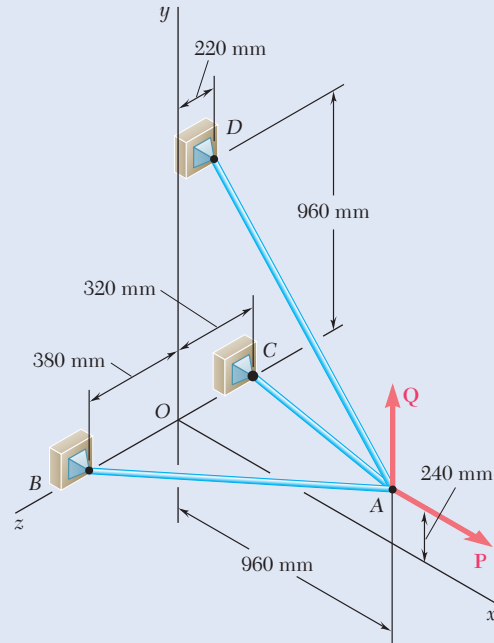


Fig. P2.107 and P2.108

- 2.108** Three cables are connected at A , where the forces $\mathbf{P}$ and $\mathbf{Q}$ are applied as shown. Knowing that $P = 1200$ N, determine the values of Q for which cable AD is taut.

- 2.109** A rectangular plate is supported by three cables as shown. Knowing that the tension in cable AC is 60 N, determine the weight of the plate.

- 2.110** A rectangular plate is supported by three cables as shown. Knowing that the tension in cable AD is 520 N, determine the weight of the plate.

- 2.111** A transmission tower is held by three guy wires attached to a pin at A and anchored by bolts at B , C , and D . If the tension in wire AB is 840 lb, determine the vertical force $\mathbf{P}$ exerted by the tower on the pin at A .

- 2.112** A transmission tower is held by three guy wires attached to a pin at A and anchored by bolts at B , C , and D . If the tension in wire AC is 590 lb, determine the vertical force $\mathbf{P}$ exerted by the tower on the pin at A .

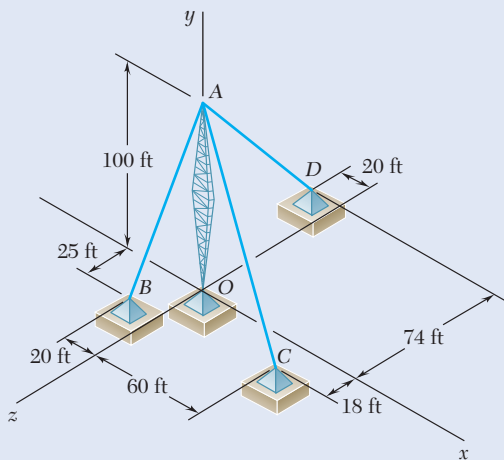


Fig. P2.111 and P2.112

- 2.113** In trying to move across a slippery icy surface, a 175-lb man uses two ropes AB and AC . Knowing that the force exerted on the man by the icy surface is perpendicular to that surface, determine the tension in each rope.

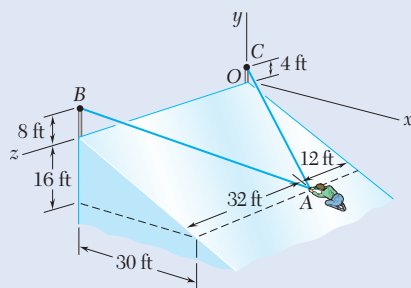


Fig. P2.113

- 2.114** Solve Prob. 2.113 assuming that a friend is helping the man at A by pulling on him with a force $\mathbf{P} = -(45 \text{ lb})\mathbf{k}$.
- 2.115** For the rectangular plate of Probs. 2.109 and 2.110, determine the tension in each of the three cables knowing that the weight of the plate is 792 N.
- 2.116** For the cable system of Probs. 2.107 and 2.108, determine the tension in each cable knowing that $P = 2880 \text{ N}$ and $Q = 0$.
- 2.117** For the cable system of Probs. 2.107 and 2.108, determine the tension in each cable knowing that $P = 2880 \text{ N}$ and $Q = 576 \text{ N}$.
- 2.118** For the cable system of Probs. 2.107 and 2.108, determine the tension in each cable knowing that $P = 2880 \text{ N}$ and $Q = -576 \text{ N}$ (Q is directed downward).
- 2.119** For the transmission tower of Probs. 2.111 and 2.112, determine the tension in each guy wire knowing that the tower exerts on the pin at A an upward vertical force of 1800 lb.
- 2.120** Three wires are connected at point D , which is located 18 in. below the T-shaped pipe support ABC . Determine the tension in each wire when a 180-lb cylinder is suspended from point D as shown.

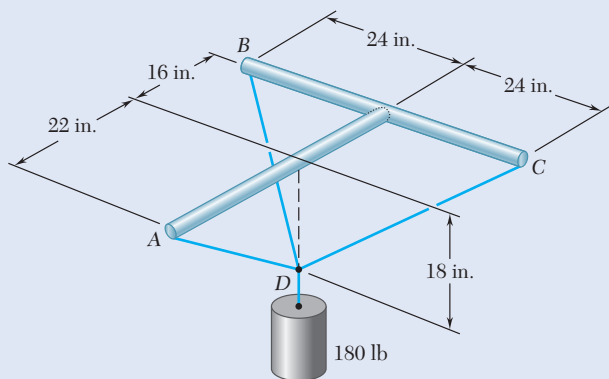


Fig. P2.120

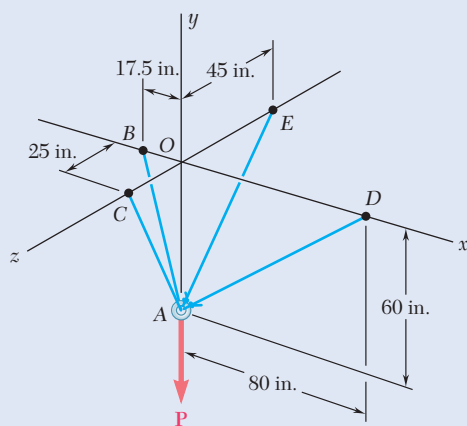


Fig. P2.123

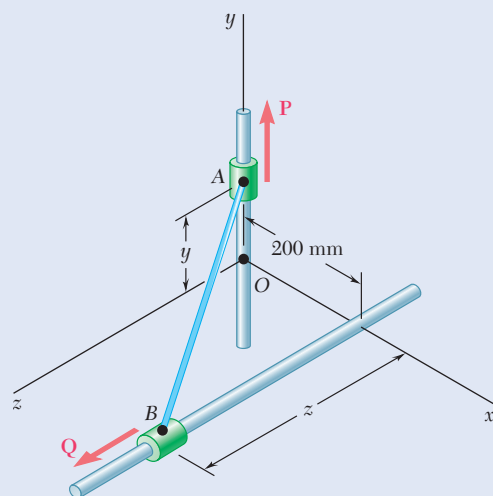


Fig. P2.125

- 2.121** A container of weight W is suspended from ring A , to which cables AC and AE are attached. A force $\mathbf{P}$ is applied to the end F of a third cable that passes over a pulley at B and through ring A and that is attached to a support at D . Knowing that $W = 1000$ N, determine the magnitude of $\mathbf{P}$. (Hint: The tension is the same in all portions of cable $FBAD$.)

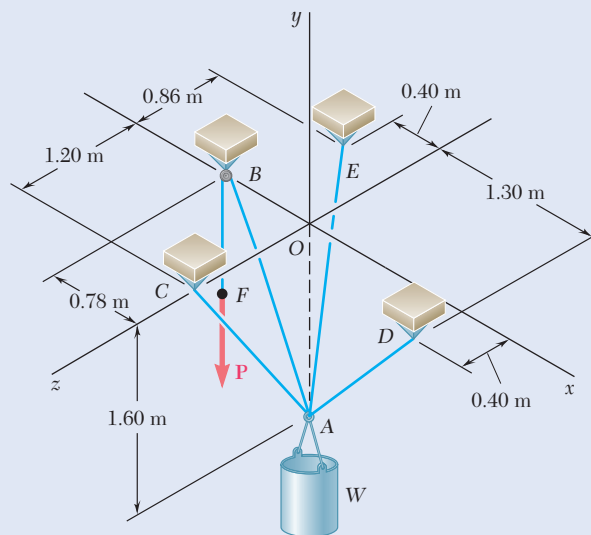


Fig. P2.121

- 2.122** Knowing that the tension in cable AC of the system described in Prob. 2.121 is 150 N, determine (a) the magnitude of the force $\mathbf{P}$, (b) the weight W of the container.
- 2.123** Cable BAC passes through a frictionless ring A and is attached to fixed supports at B and C , while cables AD and AE are both tied to the ring and are attached, respectively, to supports at D and E . Knowing that a 200-lb vertical load $\mathbf{P}$ is applied to ring A , determine the tension in each of the three cables.
- 2.124** Knowing that the tension in cable AE of Prob. 2.123 is 75 lb, determine (a) the magnitude of the load $\mathbf{P}$, (b) the tension in cables BAC and AD .
- 2.125** Collars A and B are connected by a 525-mm-long wire and can slide freely on frictionless rods. If a force $\mathbf{P} = (341 \text{ N})\mathbf{j}$ is applied to collar A , determine (a) the tension in the wire when $y = 155$ mm, (b) the magnitude of the force $\mathbf{Q}$ required to maintain the equilibrium of the system.
- 2.126** Solve Prob. 2.125 assuming that $y = 275$ mm.

Review and Summary

In this chapter, we have studied the effect of forces on particles, i.e., on bodies of such shape and size that we may assume all forces acting on them apply at the same point.

Resultant of Two Forces

Forces are *vector quantities*; they are characterized by a point of application, a magnitude, and a direction, and they add according to the parallelogram law (Fig. 2.30). We can determine the magnitude and direction of the resultant $\mathbf{R}$ of two forces $\mathbf{P}$ and $\mathbf{Q}$ either graphically or by trigonometry using the law of cosines and the law of sines [Sample Prob. 2.1].

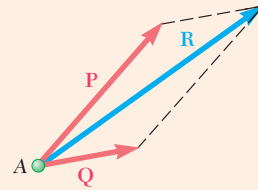


Fig. 2.30

Components of a Force

Any given force acting on a particle can be resolved into two or more components, i.e., it can be replaced by two or more forces that have the same effect on the particle. A force $\mathbf{F}$ can be resolved into two components $\mathbf{P}$ and $\mathbf{Q}$ by drawing a parallelogram with $\mathbf{F}$ for its diagonal; the components $\mathbf{P}$ and $\mathbf{Q}$ are then represented by the two adjacent sides of the parallelogram (Fig. 2.31). Again, we can determine the components either graphically or by trigonometry [Sec. 2.1E].

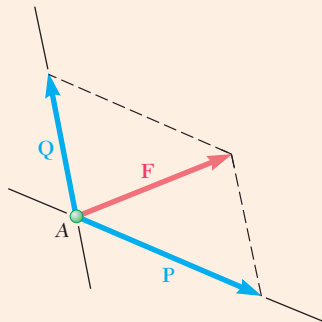


Fig. 2.31

Rectangular Components; Unit Vectors

A force $\mathbf{F}$ is resolved into two rectangular components if its components $\mathbf{F}_x$ and $\mathbf{F}_y$ are perpendicular to each other and are directed along the coordinate axes (Fig. 2.32). Introducing the unit vectors $\mathbf{i}$ and $\mathbf{j}$ along the x and y axes, respectively, we can write the components and the vector as [Sec. 2.2A]

$$\mathbf{F}_x = F_x \mathbf{i} \quad \mathbf{F}_y = F_y \mathbf{j} \quad (2.6)$$

and

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} \quad (2.7)$$

where F_x and F_y are the *scalar components* of $\mathbf{F}$. These components, which can be positive or negative, are defined by the relations

$$F_x = F \cos \theta \quad F_y = F \sin \theta \quad (2.8)$$

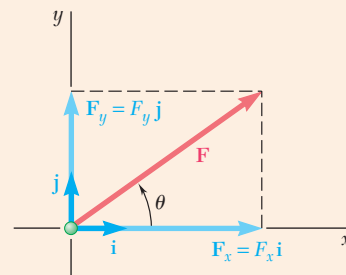


Fig. 2.32

When the rectangular components F_x and F_y of a force $\mathbf{F}$ are given, we can obtain the angle θ defining the direction of the force from

$$\tan \theta = \frac{F_y}{F_x} \quad (2.9)$$

We can obtain the magnitude F of the force by solving one of the equations (2.8) for F or by applying the Pythagorean theorem:

$$F = \sqrt{F_x^2 + F_y^2} \quad (2.10)$$

Resultant of Several Coplanar Forces

When three or more coplanar forces act on a particle, we can obtain the rectangular components of their resultant $\mathbf{R}$ by adding the corresponding components of the given forces algebraically [Sec. 2.2B]:

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad (2.13)$$

The magnitude and direction of $\mathbf{R}$ then can be determined from relations similar to Eqs. (2.9) and (2.10) [Sample Prob. 2.3].

Forces in Space

A force $\mathbf{F}$ in three-dimensional space can be resolved into rectangular components F_x , F_y , and F_z [Sec. 2.4A]. Denoting by θ_x , θ_y , and θ_z , respectively, the angles that $\mathbf{F}$ forms with the x , y , and z axes (Fig. 2.33), we have

$$F_x = F \cos \theta_x \quad F_y = F \cos \theta_y \quad F_z = F \cos \theta_z \quad (2.19)$$

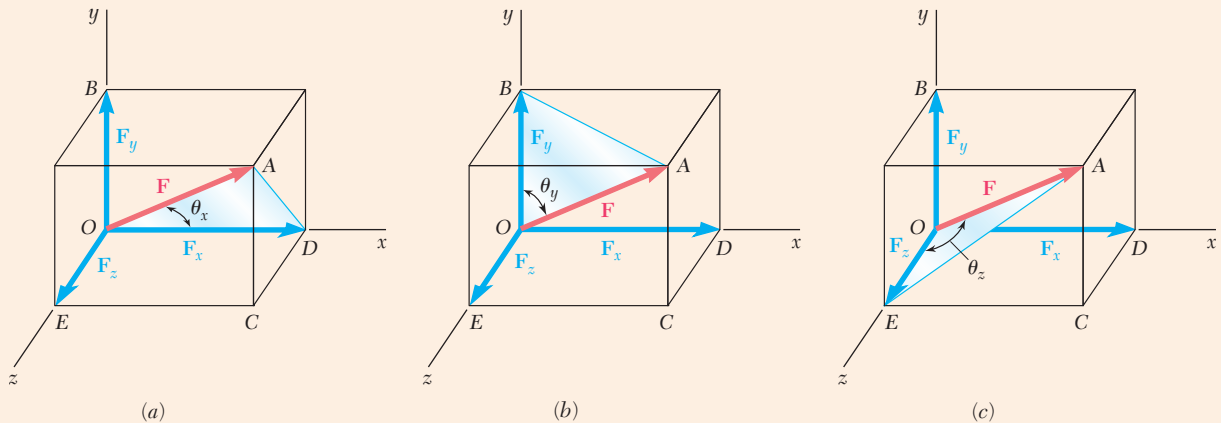


Fig. 2.33

Direction Cosines

The cosines of θ_x , θ_y , and θ_z are known as the *direction cosines* of the force $\mathbf{F}$. Introducing the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ along the coordinate axes, we can write $\mathbf{F}$ as

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k} \quad (2.20)$$

or

$$\mathbf{F} = F(\cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k}) \quad (2.21)$$

This last equation shows (Fig. 2.34) that $\mathbf{F}$ is the product of its magnitude F and the unit vector expressed by

$$\boldsymbol{\lambda} = \cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k}$$

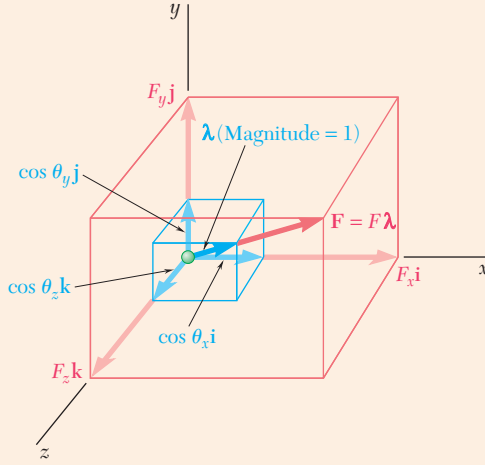


Fig. 2.34

Since the magnitude of $\boldsymbol{\lambda}$ is equal to unity, we must have

$$\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1 \quad (2.24)$$

When we are given the rectangular components F_x , F_y , and F_z of a force $\mathbf{F}$, we can find the magnitude F of the force by

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} \quad (2.18)$$

and the direction cosines of $\mathbf{F}$ are obtained from Eqs. (2.19). We have

$$\cos \theta_x = \frac{F_x}{F} \quad \cos \theta_y = \frac{F_y}{F} \quad \cos \theta_z = \frac{F_z}{F} \quad (2.25)$$

When a force $\mathbf{F}$ is defined in three-dimensional space by its magnitude F and two points M and N on its line of action [Sec. 2.4B], we can obtain its rectangular components by first expressing the vector $\overrightarrow{MN}$ joining points M and N in terms of its components d_x , d_y , and d_z (Fig. 2.35):

$$\overrightarrow{MN} = d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k} \quad (2.26)$$

We next determine the unit vector $\boldsymbol{\lambda}$ along the line of action of $\mathbf{F}$ by dividing $\overrightarrow{MN}$ by its magnitude $MN = d$:

$$\boldsymbol{\lambda} = \frac{\overrightarrow{MN}}{MN} = \frac{1}{d} (d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k}) \quad (2.27)$$

Recalling that $\mathbf{F}$ is equal to the product of F and $\boldsymbol{\lambda}$, we have

$$\mathbf{F} = F \boldsymbol{\lambda} = \frac{F}{d} (d_x \mathbf{i} + d_y \mathbf{j} + d_z \mathbf{k}) \quad (2.28)$$

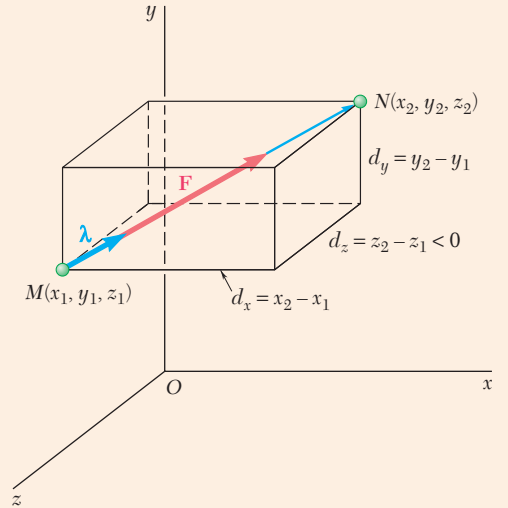


Fig. 2.35

From this equation it follows [Sample Probs. 2.7 and 2.8] that the scalar components of $\mathbf{F}$ are, respectively,

$$F_x = \frac{Fd_x}{d} \quad F_y = \frac{Fd_y}{d} \quad F_z = \frac{Fd_z}{d} \quad (2.29)$$

Resultant of Forces in Space

When two or more forces act on a particle in three-dimensional space, we can obtain the rectangular components of their resultant $\mathbf{R}$ by adding the corresponding components of the given forces algebraically [Sec. 2.4C]. We have

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad R_z = \Sigma F_z \quad (2.31)$$

We can then determine the magnitude and direction of $\mathbf{R}$ from relations similar to Eqs. (2.18) and (2.25) [Sample Prob. 2.8].

Equilibrium of a Particle

A particle is said to be in equilibrium when the resultant of all the forces acting on it is zero [Sec. 2.3A]. The particle remains at rest (if originally at rest) or moves with constant speed in a straight line (if originally in motion) [Sec. 2.3B].

Free-Body Diagram

To solve a problem involving a particle in equilibrium, first draw a free-body diagram of the particle showing all of the forces acting on it [Sec. 2.3C]. If only three coplanar forces act on the particle, you can draw a force triangle to express that the particle is in equilibrium. Using graphical methods of trigonometry, you can solve this triangle for no more than two unknowns [Sample Prob. 2.4]. If more than three coplanar forces are involved, you should use the equations of equilibrium:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad (2.15)$$

These equations can be solved for no more than two unknowns [Sample Prob. 2.6].

Equilibrium in Space

When a particle is in equilibrium in three-dimensional space [Sec. 2.5], use the three equations of equilibrium:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma F_z = 0 \quad (2.34)$$

These equations can be solved for no more than three unknowns [Sample Prob. 2.9].

Review Problems

2.127 Two structural members A and B are bolted to a bracket as shown. Knowing that both members are in compression and that the force is 15 kN in member A and 10 kN in member B , determine by trigonometry the magnitude and direction of the resultant of the forces applied to the bracket by members A and B .

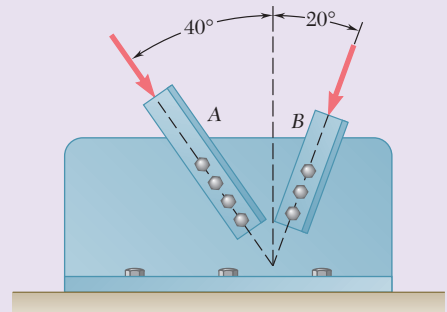


Fig. P2.127

2.128 Determine the x and y components of each of the forces shown.

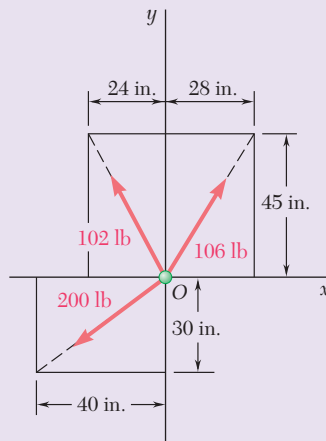


Fig. P2.128

2.129 A hoist trolley is subjected to the three forces shown. Knowing that $\alpha = 40^\circ$, determine (a) the required magnitude of the force P if the resultant of the three forces is to be vertical, (b) the corresponding magnitude of the resultant.

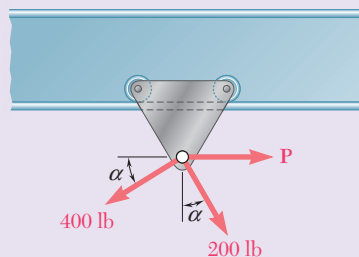


Fig. P2.129

2.130 Knowing that $\alpha = 55^\circ$ and that boom AC exerts on pin C a force directed along line AC , determine (a) the magnitude of that force, (b) the tension in cable BC .

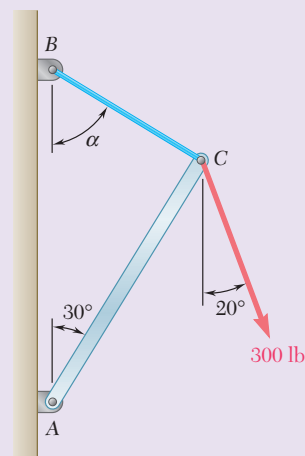


Fig. P2.130

- 2.131** Two cables are tied together at C and loaded as shown. Knowing that $P = 360$ N, determine the tension (a) in cable AC , (b) in cable BC .

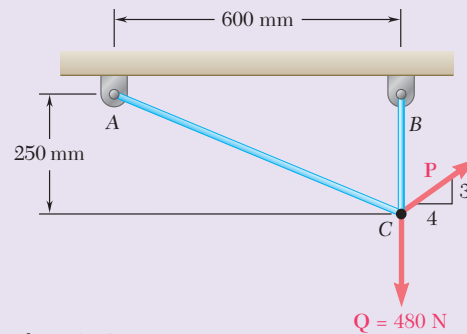


Fig. P2.131

- 2.132** Two cables tied together at C are loaded as shown. Knowing that the maximum allowable tension in each cable is 800 N, determine (a) the magnitude of the largest force P that can be applied at C , (b) the corresponding value of α .

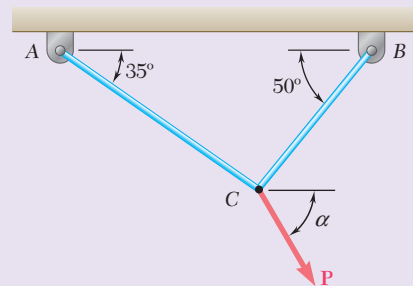


Fig. P2.132

- 2.133** The end of the coaxial cable AE is attached to the pole AB , which is strengthened by the guy wires AC and AD . Knowing that the tension in wire AC is 120 lb, determine (a) the components of the force exerted by this wire on the pole, (b) the angles θ_x , θ_y , and θ_z that the force forms with the coordinate axes.

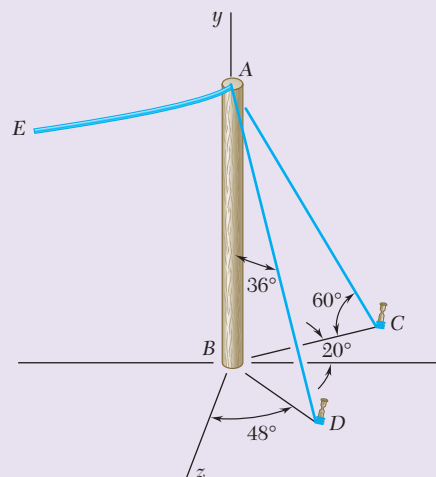


Fig. P2.133

- 2.134** Knowing that the tension in cable AC is 2130 N, determine the components of the force exerted on the plate at C .

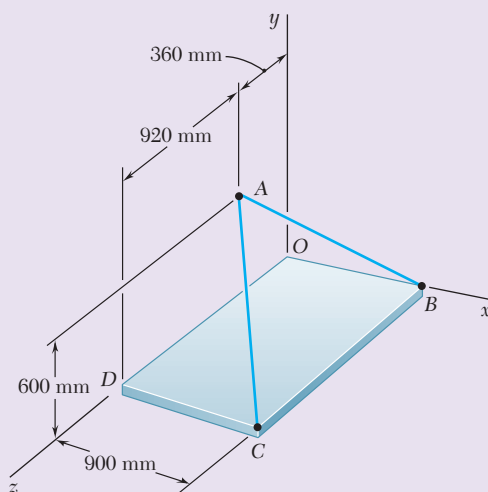


Fig. P2.134

2.135 Find the magnitude and direction of the resultant of the two forces shown knowing that $P = 600 \text{ N}$ and $Q = 450 \text{ N}$.

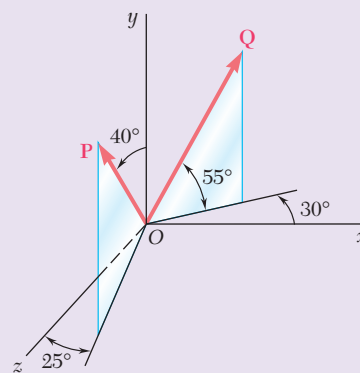


Fig. P2.135

2.136 A container of weight W is suspended from ring A . Cable BAC passes through the ring and is attached to fixed supports at B and C . Two forces $\mathbf{P} = P\mathbf{i}$ and $\mathbf{Q} = Q\mathbf{k}$ are applied to the ring to maintain the container in the position shown. Knowing that $W = 376 \text{ N}$, determine P and Q . (*Hint: The tension is the same in both portions of cable BAC .*)

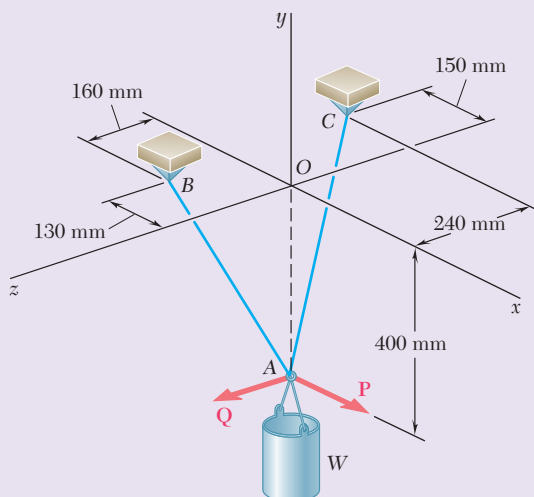


Fig. P2.136

2.137 Collars A and B are connected by a 25-in.-long wire and can slide freely on frictionless rods. If a 60-lb force $\mathbf{Q}$ is applied to collar B as shown, determine (a) the tension in the wire when $x = 9 \text{ in.}$, (b) the corresponding magnitude of the force $\mathbf{P}$ required to maintain the equilibrium of the system.

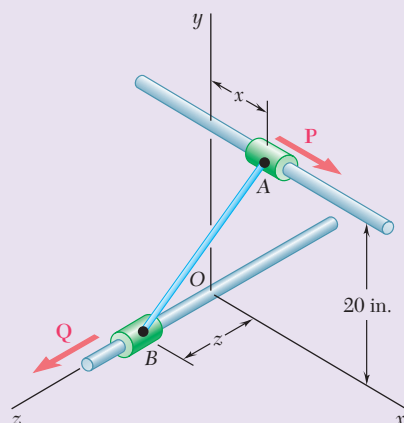


Fig. P2.137 and P2.138

2.138 Collars A and B are connected by a 25-in.-long wire and can slide freely on frictionless rods. Determine the distances x and z for which the equilibrium of the system is maintained when $P = 120 \text{ lb}$ and $Q = 60 \text{ lb}$.



3

Rigid Bodies: Equivalent Systems of Forces

Four tugboats work together to free the oil tanker Coastal Eagle Point that ran aground while attempting to navigate a channel in Tampa Bay. It will be shown in this chapter that the forces exerted on the ship by the tugboats could be replaced by an equivalent force exerted by a single, more powerful, tugboat.

Introduction

3.1 FORCES AND MOMENTS

3.1A External and Internal Forces

3.1B Principle of Transmissibility: Equivalent Forces

3.1C Vector Products

3.1D Rectangular Components of Vector Products

3.1E Moment of a Force about a Point

3.1F Rectangular Components of the Moment of a Force

3.2 MOMENT OF A FORCE ABOUT AN AXIS

3.2A Scalar Products

3.2B Mixed Triple Products

3.2C Moment of a Force about a Given Axis

3.3 COUPLES AND FORCE- COUPLE SYSTEMS

3.3A Moment of a Couple

3.3B Equivalent Couples

3.3C Addition of Couples

3.3D Couple Vectors

3.3E Resolution of a Given Force into a Force at O and a Couple

3.4 SIMPLIFYING SYSTEMS OF FORCES

3.4A Reducing a System of Forces to a Force-Couple System

3.4B Equivalent and Equipollent Systems of Forces

3.4C Further Reduction of a System of Forces

*3.4D Reduction of a System of Forces to a Wrench

Review and Summary

Objectives

- **Discuss** the principle of transmissibility that enables a force to be treated as a sliding vector.
- **Define** the moment of a force about a point.
- **Examine** vector and scalar products, useful in analysis involving moments.
- **Apply** Varignon's Theorem to simplify certain moment analyses.
- **Define** the mixed triple product and use it to determine the moment of a force about an axis.
- **Define** the moment of a couple, and consider the particular properties of couples.
- **Resolve** a given force into an equivalent force-couple system at another point.
- **Reduce** a system of forces into an equivalent force-couple system.
- **Examine** circumstances where a system of forces can be reduced to a single force.
- **Define** a wrench and consider how any general system of forces can be reduced to a wrench.

Introduction

In Chapter 2, we assumed that each of the bodies considered could be treated as a single particle. Such a view, however, is not always possible. In general, a body should be treated as a combination of a large number of particles. In this case, we need to consider the size of the body as well as the fact that forces act on different parts of the body and thus have different points of application.

Most of the bodies considered in elementary mechanics are assumed to be rigid. We define a **rigid body** as one that does not deform. Actual structures and machines are never absolutely rigid and deform under the loads to which they are subjected. However, these deformations are usually small and do not appreciably affect the conditions of equilibrium or the motion of the structure under consideration. They are important, though, as far as the resistance of the structure to failure is concerned and are considered in the study of mechanics of materials.

In this chapter, you will study the effect of forces exerted on a rigid body, and you will learn how to replace a given system of forces by a simpler equivalent system. This analysis rests on the fundamental assumption that the effect of a given force on a rigid body remains unchanged if that force is moved along its line of action (*principle of transmissibility*). It follows that forces acting on a rigid body can be represented by *sliding vectors*, as indicated earlier in Sec. 2.1B.

Two important concepts associated with the effect of a force on a rigid body are the *moment of a force about a point* (Sec. 3.1E) and the

moment of a force about an axis (Sec. 3.2C). The determination of these quantities involves computing vector products and scalar products of two vectors, so in this chapter, we introduce the fundamentals of vector algebra and apply them to the solution of problems involving forces acting on rigid bodies.

Another concept introduced in this chapter is that of a *couple*, i.e., the combination of two forces that have the same magnitude, parallel lines of action, and opposite sense (Sec. 3.3A). As you will see, we can replace any system of forces acting on a rigid body by an equivalent system consisting of one force acting at a given point and one couple. This basic combination is called a *force-couple system*. In the case of concurrent, coplanar, or parallel forces, we can further reduce the equivalent force-couple system to a single force, called the *resultant* of the system, or to a single couple, called the *resultant couple* of the system.

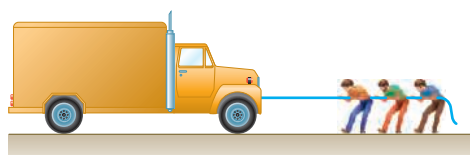
3.1 FORCES AND MOMENTS

The basic definition of a force does not change if the force acts on a point or on a rigid body. However, the effects of the force can be very different, depending on factors such as the point of application or line of action of that force. As a result, calculations involving forces acting on a rigid body are generally more complicated than situations involving forces acting on a point. We begin by examining some general classifications of forces acting on rigid bodies.

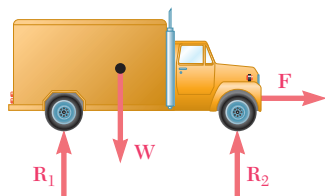
3.1A External and Internal Forces

Forces acting on rigid bodies can be separated into two groups: (1) *external forces* and (2) *internal forces*.

1. **External forces** are exerted by other bodies on the rigid body under consideration. They are entirely responsible for the external behavior of the rigid body, either causing it to move or ensuring that it remains at rest. We shall be concerned only with external forces in this chapter and in Chaps. 4 and 5.
2. **Internal forces** hold together the particles forming the rigid body. If the rigid body is structurally composed of several parts, the forces holding the component parts together are also defined as internal forces. We will consider internal forces in Chaps. 6 and 7.



(a)



(b)

Fig. 3.1 (a) Three people pulling on a truck with a rope; (b) free-body diagram of the truck, shown as a rigid body instead of a particle.

As an example of external forces, consider the forces acting on a disabled truck that three people are pulling forward by means of a rope attached to the front bumper (Fig. 3.1a). The external forces acting on the truck are shown in a **free-body diagram** (Fig. 3.1b). Note that this free-body diagram shows the entire object, not just a particle representing the object. Let us first consider the **weight** of the truck. Although it embodies the effect of the earth's pull on each of the particles forming the truck, the weight can be represented by the single force **W**. The **point of application** of this force—that is, the point at which the force acts—is defined as the **center of gravity** of the truck. (In Chap. 5, we will show how to determine the location of centers of gravity.) The weight **W** tends to make the truck move vertically downward. In fact, it would actually cause the truck to

move downward, i.e., to fall, if it were not for the presence of the ground. The ground opposes the downward motion of the truck by means of the reactions $\mathbf{R}_1$ and $\mathbf{R}_2$. These forces are exerted *by* the ground *on* the truck and must therefore be included among the external forces acting on the truck.

The people pulling on the rope exert the force $\mathbf{F}$. The point of application of $\mathbf{F}$ is on the front bumper. The force $\mathbf{F}$ tends to make the truck move forward in a straight line and does actually make it move, since no external force opposes this motion. (We are ignoring rolling resistance here for simplicity.) This forward motion of the truck, during which each straight line keeps its original orientation (the floor of the truck remains horizontal, and the walls remain vertical), is known as a **translation**. Other forces might cause the truck to move differently. For example, the force exerted by a jack placed under the front axle would cause the truck to pivot about its rear axle. Such a motion is a **rotation**. We conclude, therefore, that each *external force* acting on a *rigid body* can, if unopposed, impart to the rigid body a motion of translation or rotation, or both.

3.1B Principle of Transmissibility: Equivalent Forces

The **principle of transmissibility** states that the conditions of equilibrium or motion of a rigid body remain unchanged if a force $\mathbf{F}$ acting at a given point of the rigid body is replaced by a force $\mathbf{F}'$ of the same magnitude and same direction, but acting at a different point, *provided that the two forces have the same line of action* (Fig. 3.2). The two forces $\mathbf{F}$ and $\mathbf{F}'$ have the same effect on the rigid body and are said to be **equivalent forces**. This principle, which states that the action of a force may be *transmitted* along its line of action, is based on experimental evidence. It *cannot* be derived from the properties established so far in this text and therefore must be accepted as an experimental law. (You will see in Sec. 16.1D that we *can* derive the principle of transmissibility from the study of the dynamics of rigid bodies, but this study requires the use of Newton's second and third laws and of several other concepts as well.) Therefore, our study of the statics of rigid bodies is based on the three principles introduced so far: the parallelogram law of vector addition, Newton's first law, and the principle of transmissibility.

We indicated in Chap. 2 that we could represent the forces acting on a particle by vectors. These vectors had a well-defined point of application—namely, the particle itself—and were therefore fixed, or bound, vectors. In the case of forces acting on a rigid body, however, the point of application of the force does not matter, as long as the line of action remains unchanged. Thus, forces acting on a rigid body must be represented by a different kind of vector, known as a **sliding vector**, since forces are allowed to slide along their lines of action. Note that all of the properties we derive in the following sections for the forces acting on a rigid body are valid more generally for any system of sliding vectors. In order to keep our presentation more intuitive, however, we will carry it out in terms of physical forces rather than in terms of mathematical sliding vectors.

Returning to the example of the truck, we first observe that the line of action of the force $\mathbf{F}$ is a horizontal line passing through both the front

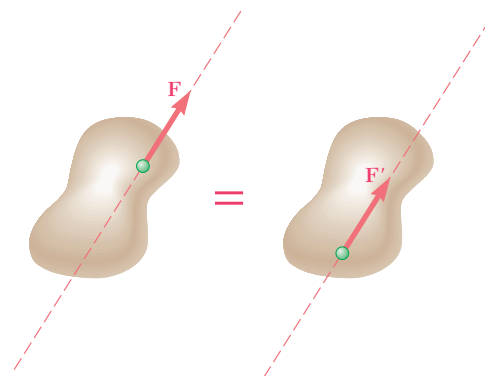


Fig. 3.2 Two forces $\mathbf{F}$ and $\mathbf{F}'$ are equivalent if they have the same magnitude and direction and the same line of action, even if they act at different points.

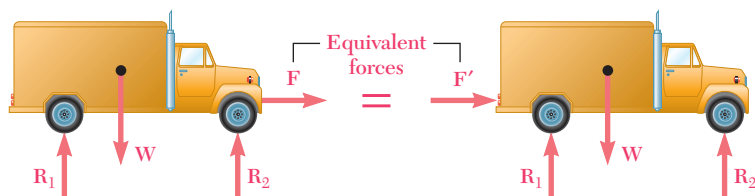


Fig. 3.3 Force $\mathbf{F}'$ is equivalent to force $\mathbf{F}$, so the motion of the truck is the same whether you pull it or push it.

and rear bumpers of the truck (Fig. 3.3). Using the principle of transmissibility, we can therefore replace $\mathbf{F}$ by an *equivalent force* $\mathbf{F}'$ acting on the rear bumper. In other words, the conditions of motion are unaffected, and all of the other external forces acting on the truck ($\mathbf{W}$, $\mathbf{R}_1$, $\mathbf{R}_2$) remain unchanged if the people push on the rear bumper instead of pulling on the front bumper.

The principle of transmissibility and the concept of equivalent forces have limitations. Consider, for example, a short bar AB acted upon by equal and opposite axial forces $\mathbf{P}_1$ and $\mathbf{P}_2$, as shown in Fig. 3.4a. According

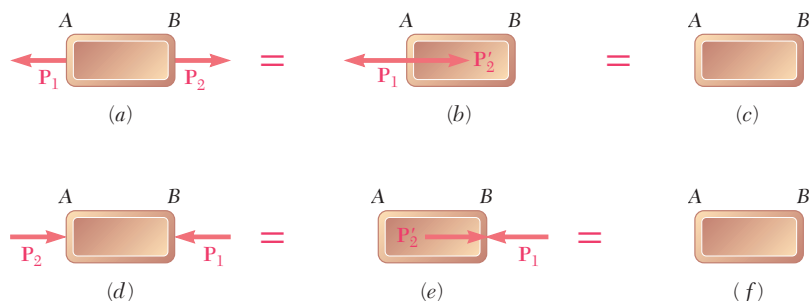


Fig. 3.4 (a–c) A set of equivalent forces acting on bar AB ; (d–f) another set of equivalent forces acting on bar AB . Both sets produce the same external effect (equilibrium in this case) but different internal forces and deformations.

to the principle of transmissibility, we can replace force $\mathbf{P}_2$ by a force $\mathbf{P}'_2$ having the same magnitude, the same direction, and the same line of action but acting at A instead of B (Fig. 3.4b). The forces $\mathbf{P}_1$ and $\mathbf{P}'_2$ acting on the same particle can be added according to the rules of Chap. 2, and since these forces are equal and opposite, their sum is equal to zero. Thus, in terms of the external behavior of the bar, the original system of forces shown in Fig. 3.4a is equivalent to no force at all (Fig. 3.4c).

Consider now the two equal and opposite forces $\mathbf{P}_1$ and $\mathbf{P}_2$ acting on the bar AB as shown in Fig. 3.4d. We can replace the force $\mathbf{P}_2$ by a force $\mathbf{P}'_2$ having the same magnitude, the same direction, and the same line of action but acting at B instead of at A (Fig. 3.4e). We can add forces $\mathbf{P}_1$ and $\mathbf{P}'_2$, and their sum is again zero (Fig. 3.4f). From the point of view of the mechanics of rigid bodies, the systems shown in Fig. 3.4a and d are thus equivalent. However, the *internal forces* and *deformations* produced by the two systems are clearly different. The bar of Fig. 3.4a is in *tension* and, if not absolutely rigid, increases in length slightly; the bar of Fig. 3.4d is in *compression* and, if not absolutely rigid, decreases in length slightly. Thus, although we can use the principle of transmissibility to determine the

conditions of motion or equilibrium of rigid bodies and to compute the external forces acting on these bodies, it should be avoided, or at least used with care, in determining internal forces and deformations.

3.1C Vector Products

In order to gain a better understanding of the effect of a force on a rigid body, we need to introduce a new concept, the *moment of a force about a point*. However, this concept is more clearly understood and is applied more effectively if we first add to the mathematical tools at our disposal the vector product of two vectors.

The **vector product** of two vectors $\mathbf{P}$ and $\mathbf{Q}$ is defined as the vector $\mathbf{V}$ that satisfies the following conditions.

1. The line of action of $\mathbf{V}$ is perpendicular to the plane containing $\mathbf{P}$ and $\mathbf{Q}$ (Fig. 3.5a).
2. The magnitude of $\mathbf{V}$ is the product of the magnitudes of $\mathbf{P}$ and $\mathbf{Q}$ and of the sine of the angle θ formed by $\mathbf{P}$ and $\mathbf{Q}$ (the measure of which is always 180° or less). We thus have

Magnitude of a vector product

$$V = PQ \sin \theta \quad (3.1)$$

3. The direction of $\mathbf{V}$ is obtained from the **right-hand rule**. Close your right hand and hold it so that your fingers are curled in the same sense as the rotation through θ that brings the vector $\mathbf{P}$ in line with the vector $\mathbf{Q}$. Your thumb then indicates the direction of the vector $\mathbf{V}$ (Fig. 3.5b). Note that if $\mathbf{P}$ and $\mathbf{Q}$ do not have a common point of application, you should first redraw them from the same point. The three vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{V}$ —taken in that order—are said to form a *right-handed triad*.[†]

As stated previously, the vector $\mathbf{V}$ satisfying these three conditions (which define it uniquely) is referred to as the *vector product* of $\mathbf{P}$ and $\mathbf{Q}$. It is represented by the mathematical expression

Vector product

$$\mathbf{V} = \mathbf{P} \times \mathbf{Q} \quad (3.2)$$

Because of this notation, the vector product of two vectors $\mathbf{P}$ and $\mathbf{Q}$ is also referred to as the *cross product* of $\mathbf{P}$ and $\mathbf{Q}$.

It follows from Eq. (3.1) that if the vectors $\mathbf{P}$ and $\mathbf{Q}$ have either the same direction or opposite directions, their vector product is zero. In the general case when the angle θ formed by the two vectors is neither 0° nor 180° , Eq. (3.1) has a simple geometric interpretation: The magnitude V of the vector product of $\mathbf{P}$ and $\mathbf{Q}$ is equal to the area of the parallelogram that has $\mathbf{P}$ and $\mathbf{Q}$ for sides (Fig. 3.6). The vector product $\mathbf{P} \times \mathbf{Q}$ is

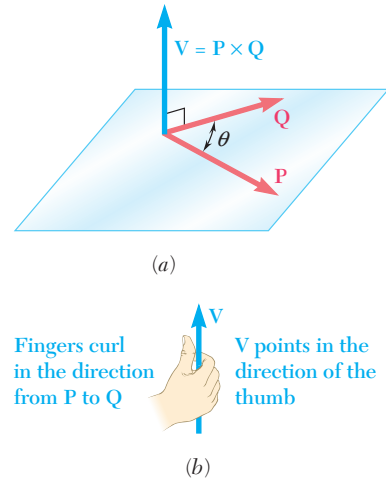


Fig. 3.5 (a) The vector product $\mathbf{V}$ has the magnitude $PQ \sin \theta$ and is perpendicular to the plane of $\mathbf{P}$ and $\mathbf{Q}$; (b) you can determine the direction of $\mathbf{V}$ by using the right-hand rule.

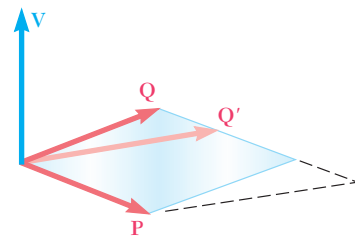


Fig. 3.6 The magnitude of the vector product $\mathbf{V}$ equals the area of the parallelogram formed by $\mathbf{P}$ and $\mathbf{Q}$. If you change $\mathbf{Q}$ to $\mathbf{Q}'$ in such a way that the parallelogram changes shape but $\mathbf{P}$ and the area are still the same, then the magnitude of $\mathbf{V}$ remains the same.

[†]Note that the x , y , and z axes used in Chap. 2 form a right-handed system of orthogonal axes and that the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ defined in Sec. 2.4A form a right-handed orthogonal triad.

therefore unchanged if we replace $\mathbf{Q}$ by a vector $\mathbf{Q}'$ that is coplanar with $\mathbf{P}$ and $\mathbf{Q}$ such that the line joining the tips of $\mathbf{Q}$ and $\mathbf{Q}'$ is parallel to $\mathbf{P}$:

$$\mathbf{V} = \mathbf{P} \times \mathbf{Q} = \mathbf{P} \times \mathbf{Q}' \quad (3.3)$$

From the third condition used to define the vector product $\mathbf{V}$ of $\mathbf{P}$ and $\mathbf{Q}$ —namely, that $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{V}$ must form a right-handed triad—it follows that vector products *are not commutative*; i.e., $\mathbf{Q} \times \mathbf{P}$ is not equal to $\mathbf{P} \times \mathbf{Q}$. Indeed, we can easily check that $\mathbf{Q} \times \mathbf{P}$ is represented by the vector $-\mathbf{V}$, which is equal and opposite to $\mathbf{V}$:

$$\mathbf{Q} \times \mathbf{P} = -(\mathbf{P} \times \mathbf{Q}) \quad (3.4)$$

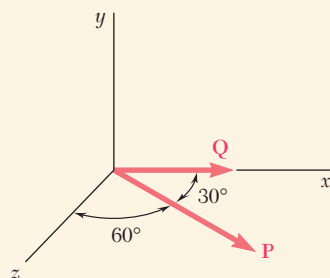


Fig. 3.7 Two vectors $\mathbf{P}$ and $\mathbf{Q}$ with angle between them.

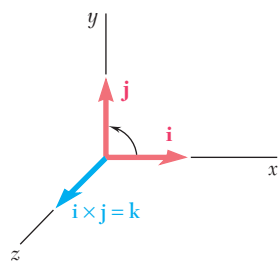
Concept Application 3.1

Let us compute the vector product $\mathbf{V} = \mathbf{P} \times \mathbf{Q}$, where the vector $\mathbf{P}$ is of magnitude 6 and lies in the xz plane at an angle of 30° with the x axis, and where the vector $\mathbf{Q}$ is of magnitude 4 and lies along the x axis (Fig. 3.7).

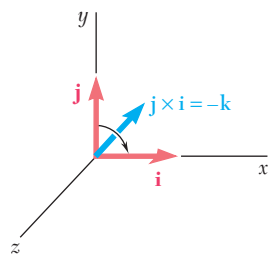
Solution

It follows immediately from the definition of the vector product that the vector $\mathbf{V}$ must lie along the y axis, directed upward, with the magnitude

$$V = PQ \sin \theta = (6)(4) \sin 30^\circ = 12 \quad \blacksquare$$



(a)



(b)

Fig. 3.8 (a) The vector product of the $\mathbf{i}$ and $\mathbf{j}$ unit vectors is the $\mathbf{k}$ unit vector; (b) the vector product of the $\mathbf{j}$ and $\mathbf{i}$ unit vectors is the $-\mathbf{k}$ unit vector.

We saw that the commutative property does not apply to vector products. However, it can be demonstrated that the *distributive* property

$$\mathbf{P} \times (\mathbf{Q}_1 + \mathbf{Q}_2) = \mathbf{P} \times \mathbf{Q}_1 + \mathbf{P} \times \mathbf{Q}_2 \quad (3.5)$$

does hold.

A third property, the associative property, does not apply to vector products; we have in general

$$(\mathbf{P} \times \mathbf{Q}) \times \mathbf{S} \neq \mathbf{P} \times (\mathbf{Q} \times \mathbf{S}) \quad (3.6)$$

3.1D Rectangular Components of Vector Products

Before we turn back to forces acting on rigid bodies, let's look at a more convenient way to express vector products using rectangular components. To do this, we use the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ that were defined in Chap. 2.

Consider first the vector product $\mathbf{i} \times \mathbf{j}$ (Fig. 3.8a). Since both vectors have a magnitude equal to 1 and since they are at a right angle to each other, their vector product is also a unit vector. This unit vector must be $\mathbf{k}$, since the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are mutually perpendicular and form a

right-handed triad. Similarly, it follows from the right-hand rule given in Sec. 3.1C that the product $\mathbf{j} \times \mathbf{i}$ is equal to $-\mathbf{k}$ (Fig. 3.8*b*). Finally, note that the vector product of a unit vector with itself, such as $\mathbf{i} \times \mathbf{i}$, is equal to zero, since both vectors have the same direction. Thus, we can list the vector products of all the various possible pairs of unit vectors:

$$\begin{array}{lll} \mathbf{i} \times \mathbf{i} = \mathbf{0} & \mathbf{j} \times \mathbf{i} = -\mathbf{k} & \mathbf{k} \times \mathbf{i} = \mathbf{j} \\ \mathbf{i} \times \mathbf{j} = \mathbf{k} & \mathbf{j} \times \mathbf{j} = \mathbf{0} & \mathbf{k} \times \mathbf{j} = -\mathbf{i} \\ \mathbf{i} \times \mathbf{k} = -\mathbf{j} & \mathbf{j} \times \mathbf{k} = \mathbf{i} & \mathbf{k} \times \mathbf{k} = \mathbf{0} \end{array} \quad (3.7)$$

We can determine the sign of the vector product of two unit vectors simply by arranging them in a circle and reading them in the order of the multiplication (Fig. 3.9). The product is positive if they follow each other in counterclockwise order and is negative if they follow each other in clockwise order.

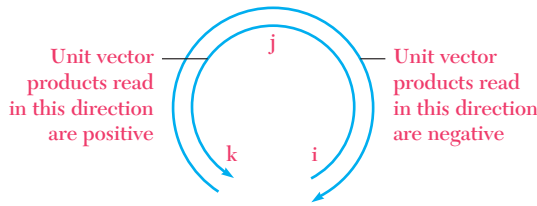


Fig. 3.9 Arrange the three letters $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ in a counterclockwise circle. You can use the order of letters for the three unit vectors in a vector product to determine its sign.

We can now easily express the vector product $\mathbf{V}$ of two given vectors $\mathbf{P}$ and $\mathbf{Q}$ in terms of the rectangular components of these vectors. Resolving $\mathbf{P}$ and $\mathbf{Q}$ into components, we first write

$$\mathbf{V} = \mathbf{P} \times \mathbf{Q} = (P_x\mathbf{i} + P_y\mathbf{j} + P_z\mathbf{k}) \times (Q_x\mathbf{i} + Q_y\mathbf{j} + Q_z\mathbf{k})$$

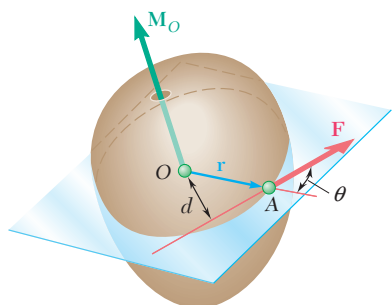
Making use of the distributive property, we express $\mathbf{V}$ as the sum of vector products, such as $P_x\mathbf{i} \times Q_y\mathbf{j}$. We find that each of the expressions obtained is equal to the vector product of two unit vectors, such as $\mathbf{i} \times \mathbf{j}$, multiplied by the product of two scalars, such as P_xQ_y . Recalling the identities of Eq. (3.7) and factoring out $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, we obtain

$$\mathbf{V} = (P_yQ_z - P_zQ_y)\mathbf{i} + (P_zQ_x - P_xQ_z)\mathbf{j} + (P_xQ_y - P_yQ_x)\mathbf{k} \quad (3.8)$$

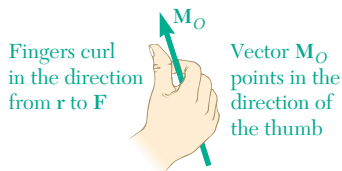
Thus, the rectangular components of the vector product $\mathbf{V}$ are

Rectangular components of a vector product

$$\begin{array}{l} V_x = P_yQ_z - P_zQ_y \\ V_y = P_zQ_x - P_xQ_z \\ V_z = P_xQ_y - P_yQ_x \end{array} \quad (3.9)$$



(a)



(b)

Fig. 3.10 Moment of a force about a point. (a) The moment $\mathbf{M}_O$ is the vector product of the position vector $\mathbf{r}$ and the force $\mathbf{F}$; (b) a right-hand rule indicates the sense of $\mathbf{M}_O$.

Returning to Eq. (3.8), notice that the right-hand side represents the expansion of a determinant. Thus, we can express the vector product $\mathbf{V}$ in the following form, which is more easily memorized:[†]

Rectangular components of a vector product (determinant form)

$$\mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix} \quad (3.10)$$

3.1E Moment of a Force about a Point

We are now ready to consider a force $\mathbf{F}$ acting on a rigid body (Fig. 3.10a). As we know, the force $\mathbf{F}$ is represented by a vector that defines its magnitude and direction. However, the effect of the force on the rigid body depends also upon its point of application A . The position of A can be conveniently defined by the vector $\mathbf{r}$ that joins the fixed reference point O with A ; this vector is known as the *position vector* of A . The position vector $\mathbf{r}$ and the force $\mathbf{F}$ define the plane shown in Fig. 3.10a.

We define the **moment of $\mathbf{F}$ about O** as the vector product of $\mathbf{r}$ and $\mathbf{F}$:

Moment of a force about a point O

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (3.11)$$

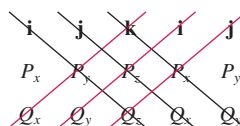
According to the definition of the vector product given in Sec. 3.1C, the moment $\mathbf{M}_O$ must be perpendicular to the plane containing O and force $\mathbf{F}$. The sense of $\mathbf{M}_O$ is defined by the sense of the rotation that will bring vector $\mathbf{r}$ in line with vector $\mathbf{F}$; this rotation is observed as *counterclockwise* by an observer located at the tip of $\mathbf{M}_O$. Another way of defining the sense of $\mathbf{M}_O$ is furnished by a variation of the right-hand rule: Close your right hand and hold it so that your fingers curl in the sense of the rotation that $\mathbf{F}$ would impart to the rigid body about a fixed axis directed along the line of action of $\mathbf{M}_O$. Then your thumb indicates the sense of the moment $\mathbf{M}_O$ (Fig. 3.10b).

Finally, denoting by θ the angle between the lines of action of the position vector $\mathbf{r}$ and the force $\mathbf{F}$, we find that the magnitude of the moment of $\mathbf{F}$ about O is

Magnitude of the moment of a force

$$M_O = rF \sin \theta = Fd \quad (3.12)$$

[†]Any determinant consisting of three rows and three columns can be evaluated by repeating the first and second columns and forming products along each diagonal line. The sum of the products obtained along the red lines is then subtracted from the sum of the products obtained along the black lines.



where d represents the perpendicular distance from O to the line of action of $\mathbf{F}$ (see Fig. 3.10). Experimentally, the tendency of a force $\mathbf{F}$ to make a rigid body rotate about a fixed axis perpendicular to the force depends upon the distance of $\mathbf{F}$ from that axis, as well as upon the magnitude of $\mathbf{F}$. For example, a child's breath can exert enough force to make a toy propeller spin (Fig. 3.11a), but a wind turbine requires the force of a substantial wind to rotate the blades and generate electrical power (Fig. 3.11b). However, the perpendicular distance between the rotation point and the line of action of the force (often called the *moment arm*) is just as important. If you want to apply a small moment to turn a nut on a pipe without breaking it, you might use a small pipe wrench that gives you a small moment



(a) Small force



(b) Large force



(c) Small moment arm



(d) Large moment arm

Fig. 3.11 (a, b) The moment of a force depends on the magnitude of the force; (c, d) it also depends on the length of the moment arm.

arm (Fig. 3.11c). But if you need a larger moment, you could use a large wrench with a long moment arm (Fig. 3.11d). Therefore,

The magnitude of M_O measures the tendency of the force F to make the rigid body rotate about a fixed axis directed along M_O .

In the SI system of units, where a force is expressed in newtons (N) and a distance in meters (m), the moment of a force is expressed in newton-meters (N·m). In the U.S. customary system of units, where a force is expressed in pounds and a distance in feet or inches, the moment of a force is expressed in lb·ft or lb·in.

Note that although the moment M_O of a force about a point depends upon the magnitude, the line of action, and the sense of the force, it does *not* depend upon the actual position of the point of application of the force along its line of action. Conversely, the moment M_O of a force F does not characterize the position of the point of application of F .

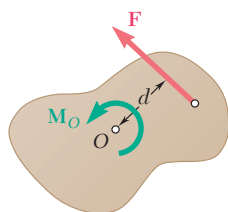
However, as we will see shortly, the moment M_O of a force F of a given magnitude and direction *completely defines the line of action of F* . Indeed, the line of action of F must lie in a plane through O perpendicular to the moment M_O ; its distance d from O must be equal to the quotient M_O/F of the magnitudes of M_O and F ; and the sense of M_O determines whether the line of action of F occurs on one side or the other of the point O .

Recall from Sec. 3.1B that the principle of transmissibility states that two forces F and F' are equivalent (i.e., have the same effect on a rigid body) if they have the same magnitude, same direction, and same line of action. We can now restate this principle:

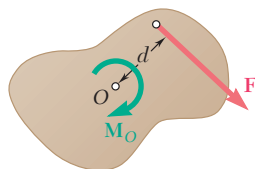
Two forces F and F' are equivalent if, and only if, they are equal (i.e., have the same magnitude and same direction) and have equal moments about a given point O .

The necessary and sufficient conditions for two forces F and F' to be equivalent are thus

$$\mathbf{F} = \mathbf{F}' \quad \text{and} \quad M_O = M'_O \quad (3.13)$$



(a) $M_O = +Fd$



(b) $M_O = -Fd$

Fig. 3.12 (a) A moment that tends to produce a counterclockwise rotation is positive; (b) a moment that tends to produce a clockwise rotation is negative.

We should observe that if the relations of Eqs. (3.13) hold for a given point O , they hold for any other point.

Two-Dimensional Problems. Many applications in statics deal with two-dimensional structures. Such structures have length and breadth but only negligible depth. Often, they are subjected to forces contained in the plane of the structure. We can easily represent two-dimensional structures and the forces acting on them on a sheet of paper or on a blackboard. Their analysis is therefore considerably simpler than that of three-dimensional structures and forces.

Consider, for example, a rigid slab acted upon by a force F in the plane of the slab (Fig. 3.12). The moment of F about a point O , which is chosen in the plane of the figure, is represented by a vector M_O perpendicular to that plane and of magnitude Fd . In the case of Fig. 3.12a, the vector M_O points *out of* the page, whereas in the case of Fig. 3.12b, it points *into* the page. As we look at the figure, we observe in the first case

that $\mathbf{F}$ tends to rotate the slab counterclockwise and in the second case that it tends to rotate the slab clockwise. Therefore, it is natural to refer to the sense of the moment of $\mathbf{F}$ about O in Fig. 3.12a as counterclockwise $\nearrow$, and in Fig. 3.12b as clockwise $\searrow$.

Since the moment of a force $\mathbf{F}$ acting in the plane of the figure must be perpendicular to that plane, we need only specify the *magnitude* and the *sense* of the moment of $\mathbf{F}$ about O . We do this by assigning to the magnitude M_O of the moment a positive or negative sign according to whether the vector $\mathbf{M}_O$ points out of or into the page.

3.1F Rectangular Components of the Moment of a Force

We can use the distributive property of vector products to determine the moment of the resultant of several *concurrent forces*. If several forces $\mathbf{F}_1, \mathbf{F}_2, \dots$ are applied at the same point A (Fig. 3.13) and if we denote by $\mathbf{r}$ the position vector of A , it follows immediately from Eq. (3.5) that

$$\mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2 + \cdots) = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2 + \cdots \quad (3.14)$$

In words,

The moment about a given point O of the resultant of several concurrent forces is equal to the sum of the moments of the various forces about the same point O .

This property, which was originally established by the French mathematician Pierre Varignon (1654–1722) long before the introduction of vector algebra, is known as **Varignon's theorem**.

The relation in Eq. (3.14) makes it possible to replace the direct determination of the moment of a force $\mathbf{F}$ by determining the moments of two or more component forces. As you will see shortly, $\mathbf{F}$ is generally resolved into components parallel to the coordinate axes. However, it may be more expeditious in some instances to resolve $\mathbf{F}$ into components that are not parallel to the coordinate axes (see Sample Prob. 3.3).

In general, determining the moment of a force in space is considerably simplified if the force and the position vector of its point of application are resolved into rectangular x, y , and z components. Consider, for example, the moment $\mathbf{M}_O$ about O of a force $\mathbf{F}$ whose components are F_x, F_y , and F_z and that is applied at a point A with coordinates x, y , and z (Fig. 3.14). Since the components of the position vector $\mathbf{r}$ are respectively equal to the coordinates x, y , and z of the point A , we can write $\mathbf{r}$ and $\mathbf{F}$ as

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (3.15)$$

$$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k} \quad (3.16)$$

Substituting for $\mathbf{r}$ and $\mathbf{F}$ from Eqs. (3.15) and (3.16) into

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (3.11)$$

and recalling Eqs. (3.8) and (3.9), we can write the moment $\mathbf{M}_O$ of $\mathbf{F}$ about O in the form

$$\mathbf{M}_O = M_x\mathbf{i} + M_y\mathbf{j} + M_z\mathbf{k} \quad (3.17)$$

where the components M_x, M_y , and M_z are defined by the relations

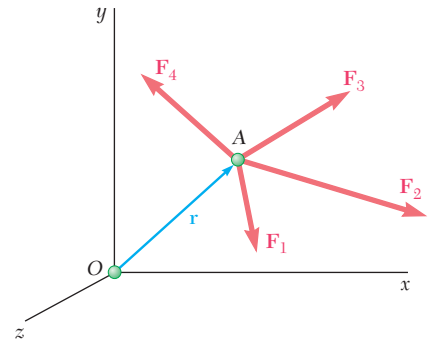


Fig. 3.13 Varignon's theorem says that the moment about point O of the resultant of these four forces equals the sum of the moments about point O of the individual forces.

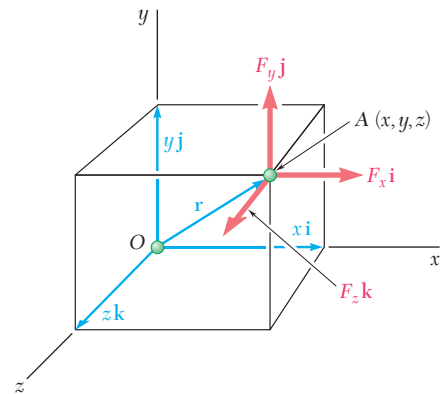


Fig. 3.14 The moment $\mathbf{M}_O$ about point O of a force $\mathbf{F}$ applied at point A is the vector product of the position vector $\mathbf{r}$ and the force $\mathbf{F}$, which can both be expressed in rectangular components.

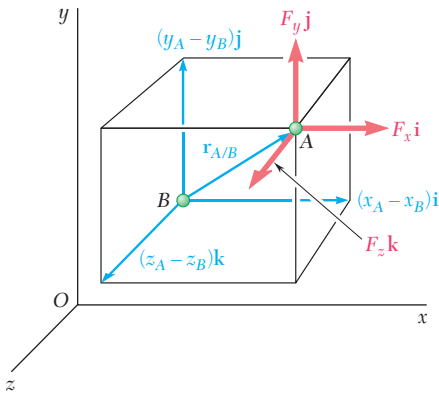


Fig. 3.15 The moment $\mathbf{M}_B$ about the point B of a force $\mathbf{F}$ applied at point A is the vector product of the position vector $\mathbf{r}_{A/B}$ and force $\mathbf{F}$.

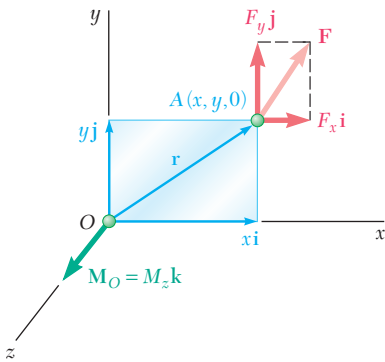


Fig. 3.16 In a two-dimensional problem, the moment $\mathbf{M}_O$ of a force $\mathbf{F}$ applied at A in the xy plane reduces to the z component of the vector product of $\mathbf{r}$ with $\mathbf{F}$.

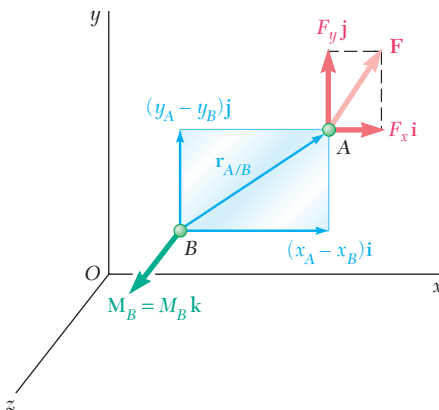


Fig. 3.17 In a two-dimensional problem, the moment $\mathbf{M}_B$ about a point B of a force $\mathbf{F}$ applied at A in the xy plane reduces to the z component of the vector product of $\mathbf{r}_{A/B}$ with $\mathbf{F}$.

Rectangular components of a moment

$$\begin{aligned} M_x &= yF_z - zF_y \\ M_y &= zF_x - xF_z \\ M_z &= xF_y - yF_x \end{aligned} \quad (3.18)$$

As you will see in Sec. 3.2C, the scalar components M_x , M_y , and M_z of the moment $\mathbf{M}_O$ measure the tendency of the force $\mathbf{F}$ to impart to a rigid body a rotation about the x , y , and z axes, respectively. Substituting from Eq. (3.18) into Eq. (3.17), we can also write $\mathbf{M}_O$ in the form of the determinant, as

$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix} \quad (3.19)$$

To compute the moment $\mathbf{M}_B$ about an arbitrary point B of a force $\mathbf{F}$ applied at A (Fig. 3.15), we must replace the position vector $\mathbf{r}$ in Eq. (3.11) by a vector drawn from B to A . This vector is the *position vector of A relative to B* , denoted by $\mathbf{r}_{A/B}$. Observing that $\mathbf{r}_{A/B}$ can be obtained by subtracting $\mathbf{r}_B$ from $\mathbf{r}_A$, we write

$$\mathbf{M}_B = \mathbf{r}_{A/B} \times \mathbf{F} = (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F} \quad (3.20)$$

or using the determinant form,

$$\mathbf{M}_B = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_{A/B} & y_{A/B} & z_{A/B} \\ F_x & F_y & F_z \end{vmatrix} \quad (3.21)$$

where $x_{A/B}$, $y_{A/B}$, and $z_{A/B}$ denote the components of the vector $\mathbf{r}_{A/B}$:

$$x_{A/B} = x_A - x_B \quad y_{A/B} = y_A - y_B \quad z_{A/B} = z_A - z_B$$

In the case of two-dimensional problems, we can assume without loss of generality that the force $\mathbf{F}$ lies in the xy plane (Fig. 3.16). Setting $z = 0$ and $F_z = 0$ in Eq. (3.19), we obtain

$$\mathbf{M}_O = (xF_y - yF_x)\mathbf{k}$$

We can verify that the moment of $\mathbf{F}$ about O is perpendicular to the plane of the figure and that it is completely defined by the scalar

$$M_O = M_z = xF_y - yF_x \quad (3.22)$$

As noted earlier, a positive value for M_O indicates that the vector $\mathbf{M}_O$ points out of the paper (the force $\mathbf{F}$ tends to rotate the body counterclockwise about O), and a negative value indicates that the vector $\mathbf{M}_O$ points into the paper (the force $\mathbf{F}$ tends to rotate the body clockwise about O).

To compute the moment about $B(x_B, y_B)$ of a force lying in the xy plane and applied at $A(x_A, y_A)$ (Fig. 3.17), we set $z_{A/B} = 0$ and $F_z = 0$ in Eq. (3.21) and note that the vector $\mathbf{M}_B$ is perpendicular to the xy plane and is defined in magnitude and sense by the scalar

$$M_B = (x_A - x_B)F_y - (y_A - y_B)F_x \quad (3.23)$$

Sample Problem 3.1

A 100-lb vertical force is applied to the end of a lever, which is attached to a shaft at O . Determine (a) the moment of the 100-lb force about O ; (b) the horizontal force applied at A that creates the same moment about O ; (c) the smallest force applied at A that creates the same moment about O ; (d) how far from the shaft a 240-lb vertical force must act to create the same moment about O ; (e) whether any one of the forces obtained in parts b, c, or d is equivalent to the original force.

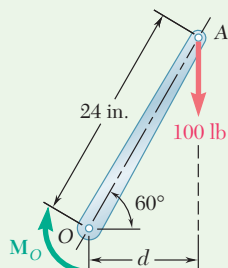


Fig. 1 Determination of the moment of the 100-lb force about O using perpendicular distance d .

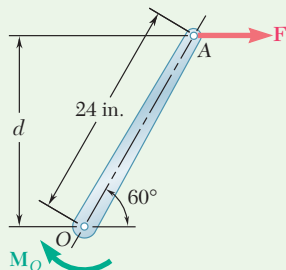


Fig. 2 Determination of horizontal force at A that creates same moment about O .

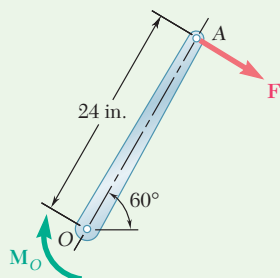
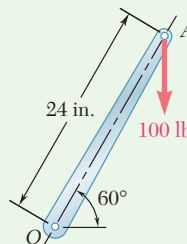


Fig. 3 Determination of smallest force at A that creates same moment about O .



STRATEGY: The calculations asked for all involve variations on the basic defining equation of a moment, $M_O = Fd$.

MODELING and ANALYSIS:

a. Moment about O . The perpendicular distance from O to the line of action of the 100-lb force (Fig. 1) is

$$d = (24 \text{ in.}) \cos 60^\circ = 12 \text{ in.}$$

The magnitude of the moment about O of the 100-lb force is

$$M_O = Fd = (100 \text{ lb})(12 \text{ in.}) = 1200 \text{ lb}\cdot\text{in.}$$

Since the force tends to rotate the lever clockwise about O , represent the moment by a vector $\mathbf{M}_O$ perpendicular to the plane of the figure and pointing *into* the paper. You can express this fact with the notation

$$\mathbf{M}_O = 1200 \text{ lb}\cdot\text{in.} \downarrow \blacktriangleleft$$

b. Horizontal Force. In this case, you have (Fig. 2)

$$d = (24 \text{ in.}) \sin 60^\circ = 20.8 \text{ in.}$$

Since the moment about O must be 1200 lb·in., you obtain

$$\begin{aligned} M_O &= Fd \\ 1200 \text{ lb}\cdot\text{in.} &= F(20.8 \text{ in.}) \\ F &= 57.7 \text{ lb} \quad \mathbf{F} = 57.7 \text{ lb} \rightarrow \blacktriangleleft \end{aligned}$$

c. Smallest Force. Since $M_O = Fd$, the smallest value of F occurs when d is maximum. Choose the force perpendicular to OA and note that $d = 24 \text{ in.}$ (Fig. 3); thus

$$\begin{aligned} M_O &= Fd \\ 1200 \text{ lb}\cdot\text{in.} &= F(24 \text{ in.}) \\ F &= 50 \text{ lb} \quad \mathbf{F} = 50 \text{ lb} \searrow 30^\circ \blacktriangleleft \end{aligned}$$

(continued)

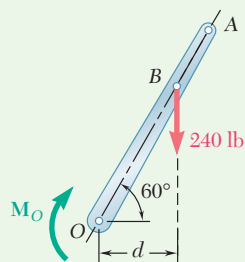


Fig. 4 Position of vertical 240-lb force that creates same moment about O .

d. 240-lb Vertical Force. In this case (Fig. 4), $M_O = Fd$ yields

$$1200 \text{ lb}\cdot\text{in.} = (240 \text{ lb})d \quad d = 5 \text{ in.}$$

but

$$OB \cos 60^\circ = d$$

so

$$OB = 10 \text{ in.} \quad \blacktriangleleft$$

e. None of the forces considered in parts *b*, *c*, or *d* is equivalent to the original 100-lb force. Although they have the same moment about O , they have different x and y components. In other words, although each force tends to rotate the shaft in the same direction, each causes the lever to pull on the shaft in a different way.

REFLECT and THINK: Various combinations of force and lever arm can produce equivalent moments, but the system of force and moment produces a different overall effect in each case.

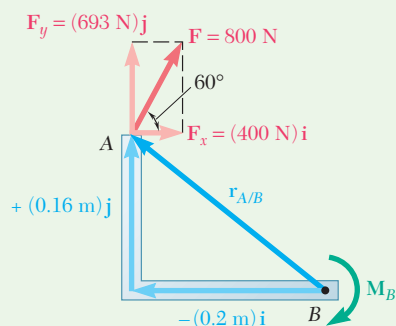
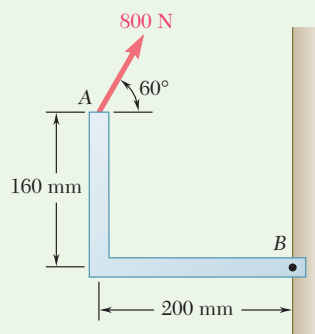


Fig. 1 The moment $\mathbf{M}_B$ is determined from the vector product of position vector $\mathbf{r}_{A/B}$ and force vector $\mathbf{F}$.

Sample Problem 3.2

A force of 800 N acts on a bracket as shown. Determine the moment of the force about B .

STRATEGY: You can resolve both the force and the position vector from B to A into rectangular components and then use a vector approach to complete the solution.

MODELING and ANALYSIS: Obtain the moment $\mathbf{M}_B$ of the force $\mathbf{F}$ about B by forming the vector product

$$\mathbf{M}_B = \mathbf{r}_{A/B} \times \mathbf{F}$$

where $\mathbf{r}_{A/B}$ is the vector drawn from B to A (Fig. 1). Resolving $\mathbf{r}_{A/B}$ and $\mathbf{F}$ into rectangular components, you have

$$\begin{aligned} \mathbf{r}_{A/B} &= -(0.2 \text{ m})\mathbf{i} + (0.16 \text{ m})\mathbf{j} \\ \mathbf{F} &= (800 \text{ N}) \cos 60^\circ \mathbf{i} + (800 \text{ N}) \sin 60^\circ \mathbf{j} \\ &= (400 \text{ N})\mathbf{i} + (693 \text{ N})\mathbf{j} \end{aligned}$$

Recalling the relations in Eq. (3.7) for the cross products of unit vectors (Sec. 3.5), you obtain

$$\begin{aligned} \mathbf{M}_B &= \mathbf{r}_{A/B} \times \mathbf{F} = [-(0.2 \text{ m})\mathbf{i} + (0.16 \text{ m})\mathbf{j}] \times [(400 \text{ N})\mathbf{i} + (693 \text{ N})\mathbf{j}] \\ &= -(138.6 \text{ N}\cdot\text{m})\mathbf{k} - (64.0 \text{ N}\cdot\text{m})\mathbf{k} \\ &= -(202.6 \text{ N}\cdot\text{m})\mathbf{k} \end{aligned}$$

$$\mathbf{M}_B = 203 \text{ N}\cdot\text{m} \quad \blacktriangleleft$$

The moment $\mathbf{M}_B$ is a vector perpendicular to the plane of the figure and pointing *into* the page.

(continued)

REFLECT and THINK: We can also use a scalar approach to solve this problem using the components for the force $\mathbf{F}$ and the position vector $\mathbf{r}_{A/B}$. Following the right-hand rule for assigning signs, we have

$$+\curvearrowright M_B = \Sigma M_B = \Sigma Fd = -(400 \text{ N})(0.16 \text{ m}) - (693 \text{ N})(0.2 \text{ m}) = -202.6 \text{ N}\cdot\text{m}$$

$$\mathbf{M}_B = 203 \text{ N}\cdot\text{m} \downarrow \blacktriangleleft$$

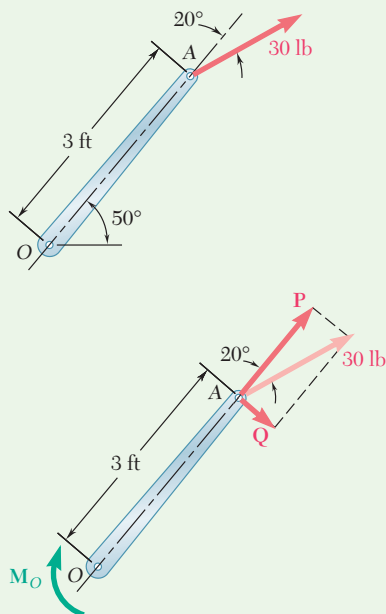


Fig. 1 30-lb force at A resolved into components $\mathbf{P}$ and $\mathbf{Q}$ to simplify the determination of the moment $\mathbf{M}_O$.

Sample Problem 3.3

A 30-lb force acts on the end of the 3-ft lever as shown. Determine the moment of the force about O .

STRATEGY: Resolving the force into components that are perpendicular and parallel to the axis of the lever greatly simplifies the moment calculation.

MODELING and ANALYSIS: Replace the force by two components: one component $\mathbf{P}$ in the direction of OA and one component $\mathbf{Q}$ perpendicular to OA (Fig. 1). Since O is on the line of action of $\mathbf{P}$, the moment of $\mathbf{P}$ about O is zero. Thus, the moment of the 30-lb force reduces to the moment of $\mathbf{Q}$, which is clockwise and can be represented by a negative scalar.

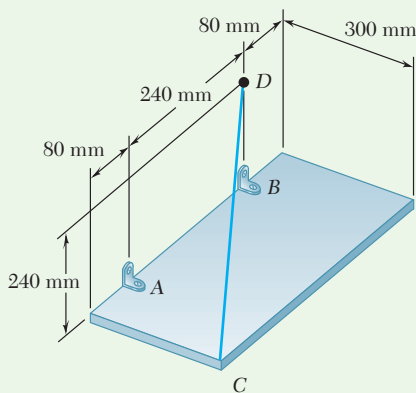
$$Q = (30 \text{ lb}) \sin 20^\circ = 10.26 \text{ lb}$$

$$M_O = -Q(3 \text{ ft}) = -(10.26 \text{ lb})(3 \text{ ft}) = -30.8 \text{ lb}\cdot\text{ft}$$

Since the value obtained for the scalar M_O is negative, the moment $\mathbf{M}_O$ points *into* the page. You can write it as

$$\mathbf{M}_O = 30.8 \text{ lb}\cdot\text{ft} \downarrow \blacktriangleleft$$

REFLECT and THINK: Always be alert for simplifications that can reduce the amount of computation.



Sample Problem 3.4

A rectangular plate is supported by brackets at A and B and by a wire CD . If the tension in the wire is 200 N, determine the moment about A of the force exerted by the wire on point C .

STRATEGY: The solution requires resolving the tension in the wire and the position vector from A to C into rectangular components. You will need a unit vector approach to determine the force components.

MODELING and ANALYSIS: Obtain the moment $\mathbf{M}_A$ about A of the force $\mathbf{F}$ exerted by the wire on point C by forming the vector product

$$\mathbf{M}_A = \mathbf{r}_{C/A} \times \mathbf{F} \quad (1)$$

(continued)

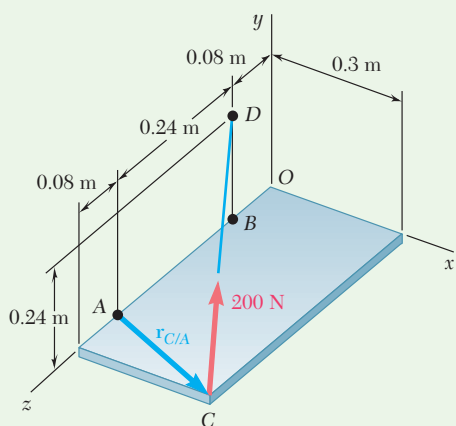


Fig. 1 The moment $\mathbf{M}_A$ is determined from position vector $\mathbf{r}_{C/A}$ and force vector $\mathbf{F}$.

where $\mathbf{r}_{C/A}$ is the vector from A to C

$$\mathbf{r}_{C/A} = \overrightarrow{AC} = (0.3 \text{ m})\mathbf{i} + (0.08 \text{ m})\mathbf{k} \quad (2)$$

and $\mathbf{F}$ is the 200-N force directed along CD (Fig. 1). Introducing the unit vector

$$\boldsymbol{\lambda} = \overrightarrow{CD}/CD,$$

you can express $\mathbf{F}$ as

$$\mathbf{F} = F\boldsymbol{\lambda} = (200 \text{ N}) \frac{\overrightarrow{CD}}{CD} \quad (3)$$

Resolving the vector $\overrightarrow{CD}$ into rectangular components, you have

$$\overrightarrow{CD} = -(0.3 \text{ m})\mathbf{i} + (0.24 \text{ m})\mathbf{j} - (0.32 \text{ m})\mathbf{k} \quad CD = 0.50 \text{ m}$$

Substituting into (3) gives you

$$\begin{aligned} \mathbf{F} &= \frac{200 \text{ N}}{0.50 \text{ m}} [-(0.3 \text{ m})\mathbf{i} + (0.24 \text{ m})\mathbf{j} - (0.32 \text{ m})\mathbf{k}] \\ &= -(120 \text{ N})\mathbf{i} + (96 \text{ N})\mathbf{j} - (128 \text{ N})\mathbf{k} \end{aligned} \quad (4)$$

Substituting for $\mathbf{r}_{C/A}$ and $\mathbf{F}$ from (2) and (4) into (1) and recalling the relations in Eq. (3.7) of Sec. 3.1D, you obtain (Fig. 2)

$$\begin{aligned} \mathbf{M}_A &= \mathbf{r}_{C/A} \times \mathbf{F} = (0.3\mathbf{i} + 0.08\mathbf{k}) \times (-120\mathbf{i} + 96\mathbf{j} - 128\mathbf{k}) \\ &= (0.3)(96)\mathbf{k} + (0.3)(-128)(-\mathbf{j}) + (0.08)(-120)\mathbf{j} + (0.08)(96)(-\mathbf{i}) \end{aligned}$$

$$\mathbf{M}_A = -(7.68 \text{ N}\cdot\text{m})\mathbf{i} + (28.8 \text{ N}\cdot\text{m})\mathbf{j} + (28.8 \text{ N}\cdot\text{m})\mathbf{k} \quad \blacktriangleleft$$

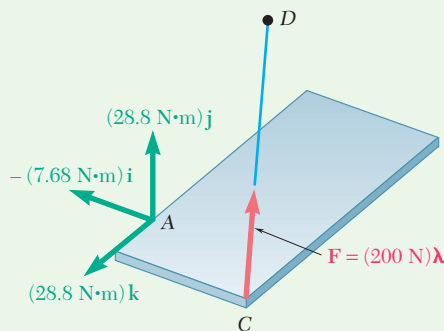


Fig. 2 Components of moment $\mathbf{M}_A$ applied at A.

Alternative Solution. As indicated in Sec. 3.1F, you can also express the moment $\mathbf{M}_A$ in the form of a determinant:

$$\mathbf{M}_A = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_C - x_A & y_C - y_A & z_C - z_A \\ F_x & F_y & F_z \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.3 & 0 & 0.08 \\ -120 & 96 & -128 \end{vmatrix}$$

$$\mathbf{M}_A = -(7.68 \text{ N}\cdot\text{m})\mathbf{i} + (28.8 \text{ N}\cdot\text{m})\mathbf{j} + (28.8 \text{ N}\cdot\text{m})\mathbf{k} \quad \blacktriangleleft$$

REFLECT and THINK: Two-dimensional problems often are solved easily using a scalar approach, but the versatility of a vector analysis is quite apparent in a three-dimensional problem such as this.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced the *vector product* or *cross product* of two vectors. In the following problems, you will use the vector product to compute the *moment of a force about a point* and also to determine the *perpendicular distance* from a point to a line.

We defined the moment of the force $\mathbf{F}$ about the point O of a rigid body as

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (3.11)$$

where $\mathbf{r}$ is the position vector *from O to any point* on the line of action of $\mathbf{F}$. Since the vector product is not commutative, it is absolutely necessary when computing such a product that you place the vectors in the proper order and that each vector have the correct sense. The moment $\mathbf{M}_O$ is important because its magnitude is a measure of the tendency of the force $\mathbf{F}$ to cause the rigid body to rotate about an axis directed along $\mathbf{M}_O$.

1. Computing the moment $\mathbf{M}_O$ of a force in two dimensions. You can use one of the following procedures:

- Use Eq. (3.12), $M_O = Fd$, which expresses the magnitude of the moment as the product of the magnitude of $\mathbf{F}$ and the *perpendicular distance* d from O to the line of action of $\mathbf{F}$ [Sample Prob. 3.1].
- Express $\mathbf{r}$ and $\mathbf{F}$ in component form and formally evaluate the vector product $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$ [Sample Prob. 3.2].
- Resolve $\mathbf{F}$ into components respectively parallel and perpendicular to the position vector $\mathbf{r}$. Only the perpendicular component contributes to the moment of $\mathbf{F}$ [Sample Prob. 3.3].
- Use Eq. (3.22), $M_O = M_z = xF_y - yF_x$. When applying this method, the simplest approach is to treat the scalar components of $\mathbf{r}$ and $\mathbf{F}$ as positive and then to assign, by observation, the proper sign to the moment produced by each force component [Sample Prob. 3.2].

2. Computing the moment $\mathbf{M}_O$ of a force $\mathbf{F}$ in three dimensions. Following the method of Sample Prob. 3.4, the first step in the calculation is to select the most convenient (simplest) position vector $\mathbf{r}$. You should next express $\mathbf{F}$ in terms of its rectangular components. The final step is to evaluate the vector product $\mathbf{r} \times \mathbf{F}$ to determine the moment. In most three-dimensional problems you will find it easiest to calculate the vector product using a determinant.

3. Determining the perpendicular distance d from a point A to a given line. First assume that a force $\mathbf{F}$ of known magnitude F lies along the given line. Next determine its moment about A by forming the vector product $\mathbf{M}_A = \mathbf{r} \times \mathbf{F}$, and calculate this product as indicated above. Then compute its magnitude M_A . Finally, substitute the values of F and M_A into the equation $M_A = Fd$ and solve for d .

Problems

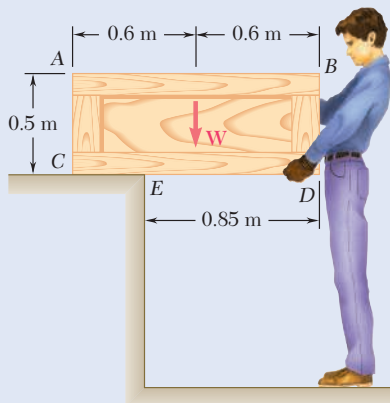


Fig. P3.1 and P3.2

3.1 A crate of mass 80 kg is held in the position shown. Determine (a) the moment produced by the weight W of the crate about E , (b) the smallest force applied at B that creates a moment of equal magnitude and opposite sense about E .

3.2 A crate of mass 80 kg is held in the position shown. Determine (a) the moment produced by the weight W of the crate about E , (b) the smallest force applied at A that creates a moment of equal magnitude and opposite sense about E , (c) the magnitude, sense, and point of application on the bottom of the crate of the smallest vertical force that creates a moment of equal magnitude and opposite sense about E .

3.3 It is known that a vertical force of 200 lb is required to remove the nail at C from the board. As the nail first starts moving, determine (a) the moment about B of the force exerted on the nail, (b) the magnitude of the force P that creates the same moment about B if $\alpha = 10^\circ$, and (c) the smallest force P that creates the same moment about B .

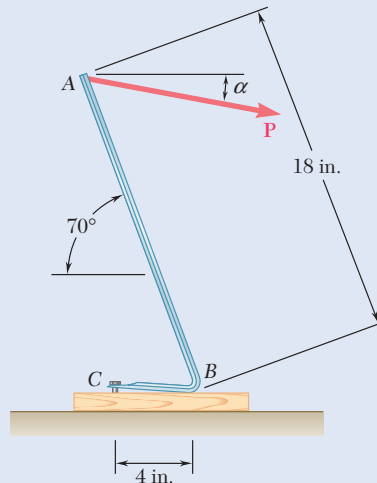


Fig. P3.3

3.4 A 300-N force is applied at A as shown. Determine (a) the moment of the 300-N force about D , (b) the smallest force applied at B that creates the same moment about D .

3.5 A 300-N force is applied at A as shown. Determine (a) the moment of the 300-N force about D , (b) the magnitude and sense of the horizontal force applied at C that creates the same moment about D , (c) the smallest force applied at C that creates the same moment about D .

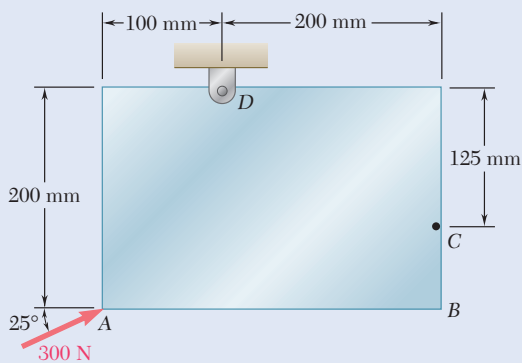


Fig. P3.4 and P3.5

3.6 A 20-lb force is applied to the control rod AB as shown. Knowing that the length of the rod is 9 in. and that $\alpha = 25^\circ$, determine the moment of the force about point B by resolving the force into horizontal and vertical components.

3.7 A 20-lb force is applied to the control rod AB as shown. Knowing that the length of the rod is 9 in. and that $\alpha = 25^\circ$, determine the moment of the force about point B by resolving the force into components along AB and in a direction perpendicular to AB .

3.8 A 20-lb force is applied to the control rod AB as shown. Knowing that the length of the rod is 9 in. and that the moment of the force about B is 120 lb-in. clockwise, determine the value of α .

3.9 Rod AB is held in place by the cord AC . Knowing that the tension in the cord is 1350 N and that $c = 360$ mm, determine the moment about B of the force exerted by the cord at point A by resolving that force into horizontal and vertical components applied (a) at point A , (b) at point C .

3.10 Rod AB is held in place by the cord AC . Knowing that $c = 840$ mm and that the moment about B of the force exerted by the cord at point A is 756 N-m, determine the tension in the cord.

3.11 and 3.12 The tailgate of a car is supported by the hydraulic lift BC . If the lift exerts a 125-lb force directed along its centerline on the ball and socket at B , determine the moment of the force about A .

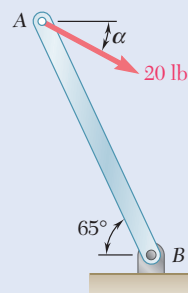


Fig. P3.6 through P3.8

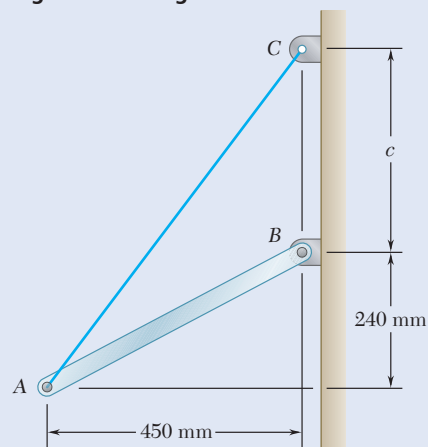


Fig. P3.9 and P3.10

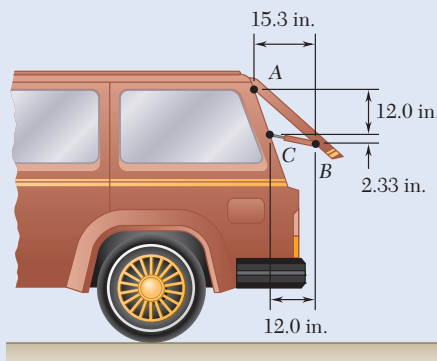


Fig. P3.11

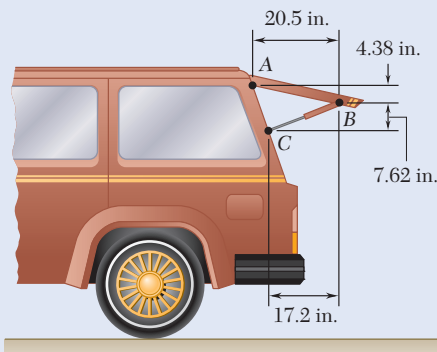


Fig. P3.12

3.13 and 3.14 It is known that the connecting rod AB exerts on the crank BC a 2.5-kN force directed down and to the left along the centerline of AB . Determine the moment of the force about C .

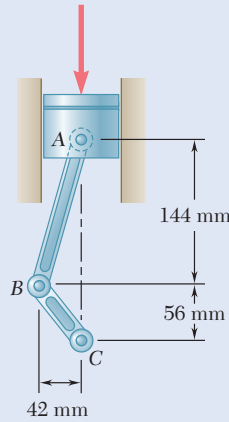


Fig. P3.13

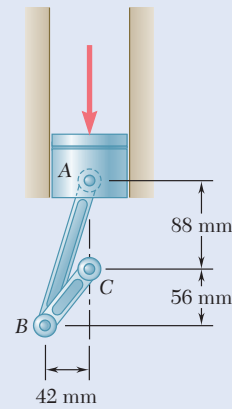


Fig. P3.14

3.15 Form the vector products $\mathbf{B} \times \mathbf{C}$ and $\mathbf{B}' \times \mathbf{C}$, where $B = B'$, and use the results obtained to prove the identity

$$\sin \alpha \cos \beta = \frac{1}{2} \sin (\alpha + \beta) + \frac{1}{2} \sin (\alpha - \beta).$$

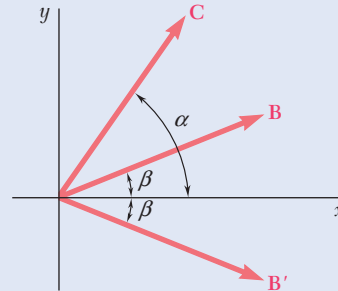


Fig. P3.15

3.16 The vectors $\mathbf{P}$ and $\mathbf{Q}$ are two adjacent sides of a parallelogram. Determine the area of the parallelogram when (a) $\mathbf{P} = -8\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$ and $\mathbf{Q} = 3\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$, (b) $\mathbf{P} = 7\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{Q} = -3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$.

3.17 A plane contains the vectors $\mathbf{A}$ and $\mathbf{B}$. Determine the unit vector normal to the plane when $\mathbf{A}$ and $\mathbf{B}$ are equal to, respectively, (a) $2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$ and $5\mathbf{i} - 8\mathbf{j} - 6\mathbf{k}$, (b) $4\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$ and $-3\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}$.

3.18 A line passes through the points (12 m, 8 m) and (-3 m, -5 m). Determine the perpendicular distance d from the line to the origin O of the system of coordinates.

3.19 Determine the moment about the origin O of the force $\mathbf{F} = 4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ that acts at a point A . Assume that the position vector of A is (a) $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$, (b) $\mathbf{r} = -8\mathbf{i} + 6\mathbf{j} - 10\mathbf{k}$, (c) $\mathbf{r} = 8\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$.

3.20 Determine the moment about the origin O of the force $\mathbf{F} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ that acts at a point A . Assume that the position vector of A is (a) $\mathbf{r} = 3\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$, (b) $\mathbf{r} = \mathbf{i} - 4\mathbf{j} - 2\mathbf{k}$, (c) $\mathbf{r} = 4\mathbf{i} + 6\mathbf{j} - 8\mathbf{k}$.

3.21 Before the trunk of a large tree is felled, cables AB and BC are attached as shown. Knowing that the tensions in cables AB and BC are 555 N and 660 N, respectively, determine the moment about O of the resultant force exerted on the tree by the cables at B .

3.22 The 12-ft boom AB has a fixed end A . A steel cable is stretched from the free end B of the boom to a point C located on the vertical wall. If the tension in the cable is 380 lb, determine the moment about A of the force exerted by the cable at B .

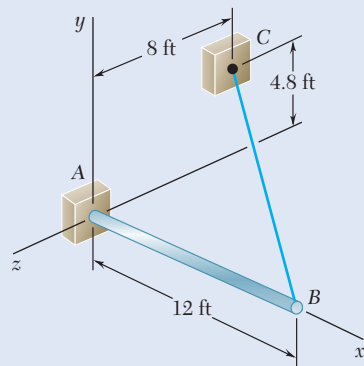


Fig. P3.22

3.23 A 200-N force is applied as shown to the bracket ABC . Determine the moment of the force about A .

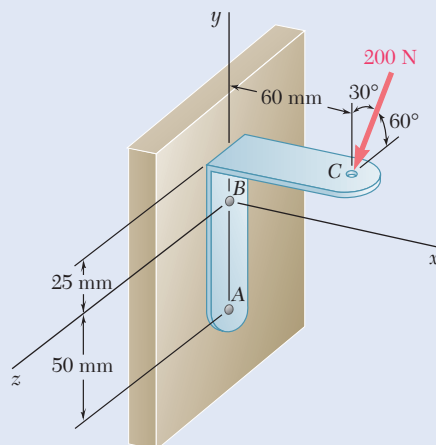


Fig. P3.23

3.24 The wire AE is stretched between the corners A and E of a bent plate. Knowing that the tension in the wire is 435 N, determine the moment about O of the force exerted by the wire (a) on corner A , (b) on corner E .

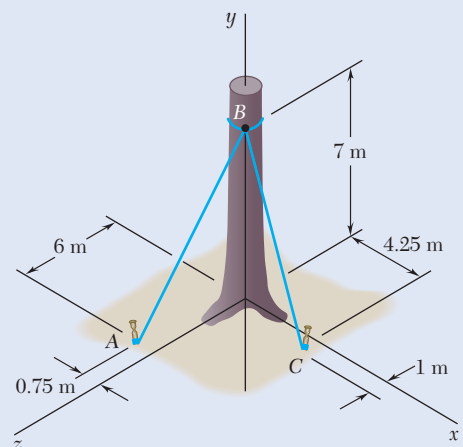


Fig. P3.21

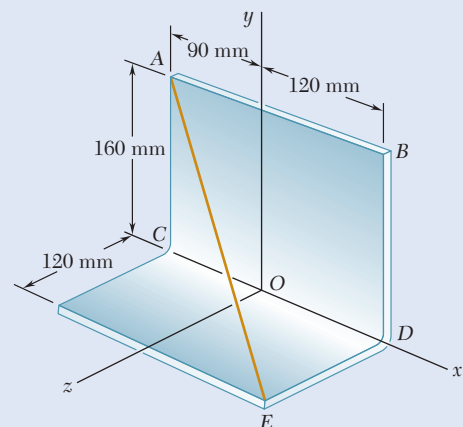


Fig. P3.24

- 3.25** A 6-ft-long fishing rod AB is securely anchored in the sand of a beach. After a fish takes the bait, the resulting force in the line is 6 lb. Determine the moment about A of the force exerted by the line at B .

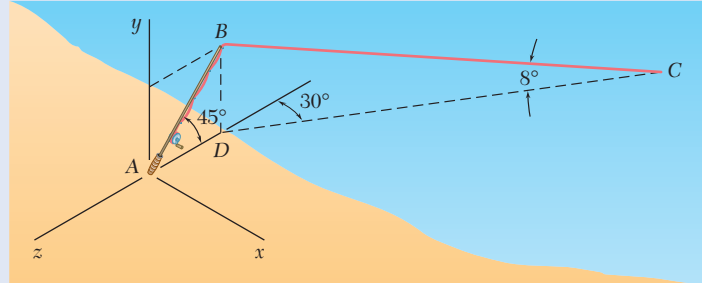


Fig. P3.25

- 3.26** A precast concrete wall section is temporarily held by two cables as shown. Knowing that the tension in cable BD is 900 N, determine the moment about point O of the force exerted by the cable at B .

- 3.27** In Prob. 3.22, determine the perpendicular distance from point A to cable BC .

- 3.28** In Prob. 3.24, determine the perpendicular distance from point O to wire AE .

- 3.29** In Prob. 3.24, determine the perpendicular distance from point B to wire AE .

- 3.30** In Prob. 3.25, determine the perpendicular distance from point A to a line drawn through points B and C .

- 3.31** In Prob. 3.25, determine the perpendicular distance from point D to a line drawn through points B and C .

- 3.32** In Prob. 3.26, determine the perpendicular distance from point O to cable BD .

- 3.33** In Prob. 3.26, determine the perpendicular distance from point C to cable BD .

- 3.34** Determine the value of a that minimizes the perpendicular distance from point C to a section of pipeline that passes through points A and B .

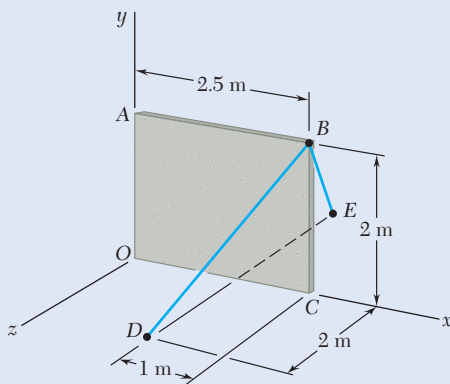


Fig. P3.26

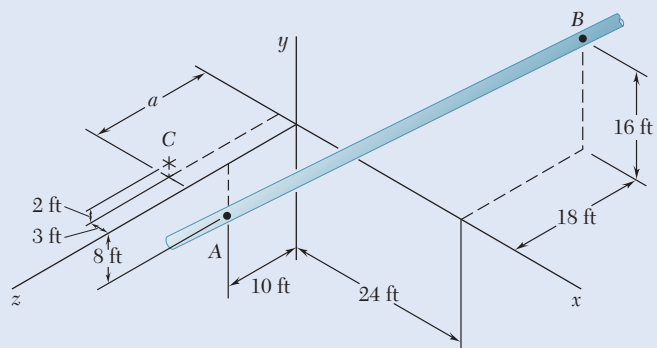


Fig. P3.34

3.2 MOMENT OF A FORCE ABOUT AN AXIS

We want to extend the idea of the moment about a point to the often useful concept of the moment about an axis. However, first we need to introduce another tool of vector mathematics. We have seen that the vector product multiplies two vectors together and produces a new vector. Here we examine the scalar product, which multiplies two vectors together and produces a scalar quantity.

3.2A Scalar Products

The **scalar product** of two vectors $\mathbf{P}$ and $\mathbf{Q}$ is defined as the product of the magnitudes of $\mathbf{P}$ and $\mathbf{Q}$ and of the cosine of the angle θ formed between them (Fig. 3.18). The scalar product of $\mathbf{P}$ and $\mathbf{Q}$ is denoted by $\mathbf{P} \cdot \mathbf{Q}$.

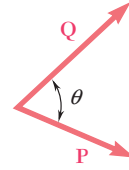


Fig. 3.18 Two vectors $\mathbf{P}$ and $\mathbf{Q}$ and the angle θ between them.

Scalar product $\mathbf{P} \cdot \mathbf{Q} = PQ \cos \theta$ (3.24)

Note that this expression is not a vector but a *scalar*, which explains the name *scalar product*. Because of the notation used, $\mathbf{P} \cdot \mathbf{Q}$ is also referred to as the *dot product* of the vectors $\mathbf{P}$ and $\mathbf{Q}$.

It follows from its very definition that the scalar product of two vectors is *commutative*, i.e., that

$$\mathbf{P} \cdot \mathbf{Q} = \mathbf{Q} \cdot \mathbf{P} \quad (3.25)$$

It can also be proven that the scalar product is *distributive*, as shown by

$$\mathbf{P} \cdot (\mathbf{Q}_1 + \mathbf{Q}_2) = \mathbf{P} \cdot \mathbf{Q}_1 + \mathbf{P} \cdot \mathbf{Q}_2 \quad (3.26)$$

As far as the associative property is concerned, this property cannot apply to scalar products. Indeed, $(\mathbf{P} \cdot \mathbf{Q}) \cdot \mathbf{S}$ has no meaning, because $\mathbf{P} \cdot \mathbf{Q}$ is not a vector but a scalar.

We can also express the scalar product of two vectors $\mathbf{P}$ and $\mathbf{Q}$ in terms of their rectangular components. Resolving $\mathbf{P}$ and $\mathbf{Q}$ into components, we first write

$$\mathbf{P} \cdot \mathbf{Q} = (P_x \mathbf{i} + P_y \mathbf{j} + P_z \mathbf{k}) \cdot (Q_x \mathbf{i} + Q_y \mathbf{j} + Q_z \mathbf{k})$$

Making use of the distributive property, we express $\mathbf{P} \cdot \mathbf{Q}$ as the sum of scalar products, such as $P_x \mathbf{i} \cdot Q_x \mathbf{i}$ and $P_x \mathbf{i} \cdot Q_y \mathbf{j}$. However, from the definition of the scalar product, it follows that the scalar products of the unit vectors are either zero or one.

$$\begin{array}{lll} \mathbf{i} \cdot \mathbf{i} = 1 & \mathbf{j} \cdot \mathbf{j} = 1 & \mathbf{k} \cdot \mathbf{k} = 1 \\ \mathbf{i} \cdot \mathbf{j} = 0 & \mathbf{j} \cdot \mathbf{k} = 0 & \mathbf{k} \cdot \mathbf{i} = 0 \end{array} \quad (3.27)$$

Thus, the expression for $\mathbf{P} \cdot \mathbf{Q}$ reduces to

Scalar product

$$\mathbf{P} \cdot \mathbf{Q} = P_x Q_x + P_y Q_y + P_z Q_z \quad (3.28)$$

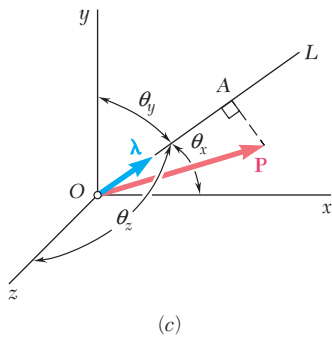
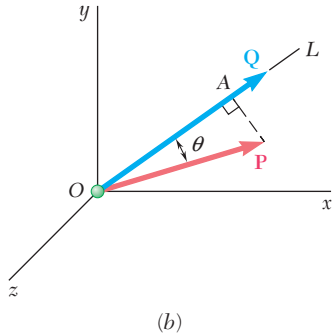
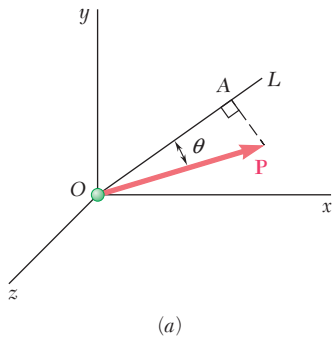


Fig. 3.19 (a) The projection of vector $\mathbf{P}$ at an angle θ to a line OL ; (b) the projection of $\mathbf{P}$ and a vector $\mathbf{Q}$ along OL ; (c) the projection of $\mathbf{P}$, a unit vector $\boldsymbol{\lambda}$ along OL , and the angles of OL with the coordinate axes.

In the particular case when $\mathbf{P}$ and $\mathbf{Q}$ are equal, we note that

$$\mathbf{P} \cdot \mathbf{P} = P_x^2 + P_y^2 + P_z^2 = P^2 \quad (3.29)$$

Applications of the Scalar Product

- 1. Angle formed by two given vectors.** Let two vectors be given in terms of their components:

$$\mathbf{P} = P_x \mathbf{i} + P_y \mathbf{j} + P_z \mathbf{k}$$

$$\mathbf{Q} = Q_x \mathbf{i} + Q_y \mathbf{j} + Q_z \mathbf{k}$$

To determine the angle formed by the two vectors, we equate the expressions obtained in Eqs. (3.24) and (3.28) for their scalar product,

$$PQ \cos \theta = P_x Q_x + P_y Q_y + P_z Q_z$$

Solving for $\cos \theta$, we have

$$\cos \theta = \frac{P_x Q_x + P_y Q_y + P_z Q_z}{PQ} \quad (3.30)$$

- 2. Projection of a vector on a given axis.** Consider a vector $\mathbf{P}$ forming an angle θ with an axis, or directed line, OL (Fig. 3.19a). We define the *projection of $\mathbf{P}$ on the axis OL* as the scalar

$$P_{OL} = P \cos \theta \quad (3.31)$$

The projection P_{OL} is equal in absolute value to the length of the segment OA . It is positive if OA has the same sense as the axis OL —that is, if θ is acute—and negative otherwise. If $\mathbf{P}$ and OL are at a right angle, the projection of $\mathbf{P}$ on OL is zero.

Now consider a vector $\mathbf{Q}$ directed along OL and of the same sense as OL (Fig. 3.19b). We can express the scalar product of $\mathbf{P}$ and $\mathbf{Q}$ as

$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \theta = P_{OL} Q \quad (3.32)$$

from which it follows that

$$P_{OL} = \frac{\mathbf{P} \cdot \mathbf{Q}}{Q} = \frac{P_x Q_x + P_y Q_y + P_z Q_z}{Q} \quad (3.33)$$

In the particular case when the vector selected along OL is the unit vector $\boldsymbol{\lambda}$ (Fig. 3.19c), we have

$$P_{OL} = \mathbf{P} \cdot \boldsymbol{\lambda} \quad (3.34)$$

Recall from Sec. 2.4A that the components of $\boldsymbol{\lambda}$ along the coordinate axes are respectively equal to the direction cosines of OL . Resolving $\mathbf{P}$ and $\boldsymbol{\lambda}$ into rectangular components, we can express the projection of $\mathbf{P}$ on OL as

$$P_{OL} = P_x \cos \theta_x + P_y \cos \theta_y + P_z \cos \theta_z \quad (3.35)$$

where θ_x , θ_y , and θ_z denote the angles that the axis OL forms with the coordinate axes.

3.2B Mixed Triple Products

We have now seen both forms of multiplying two vectors together: the vector product and the scalar product. Here we define the **mixed triple product** of the three vectors $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ as the scalar expression

Mixed triple product

$$\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) \quad (3.36)$$

This is obtained by forming the scalar product of $\mathbf{S}$ with the vector product of $\mathbf{P}$ and $\mathbf{Q}$. [In Chapter 15, we will introduce another kind of triple product, called the vector triple product, $\mathbf{S} \times (\mathbf{P} \times \mathbf{Q})$.]

The mixed triple product of $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ has a simple geometrical interpretation (Fig. 3.20a). Recall from Sec. 3.4 that the vector $\mathbf{P} \times \mathbf{Q}$ is perpendicular to the plane containing $\mathbf{P}$ and $\mathbf{Q}$ and that its magnitude is equal to the area of the parallelogram that has $\mathbf{P}$ and $\mathbf{Q}$ for sides. Also, Eq. (3.32) indicates that we can obtain the scalar product of $\mathbf{S}$ and $\mathbf{P} \times \mathbf{Q}$ by multiplying the magnitude of $\mathbf{P} \times \mathbf{Q}$ (i.e., the area of the parallelogram defined by $\mathbf{P}$ and $\mathbf{Q}$) by the projection of $\mathbf{S}$ on the vector $\mathbf{P} \times \mathbf{Q}$ (i.e., by the projection of $\mathbf{S}$ on the normal to the plane containing the parallelogram). The mixed triple product is thus equal, in absolute value, to the volume of the parallelepiped having the vectors $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ for sides (Fig. 3.20b). The sign of the mixed triple product is positive if $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ form a right-handed triad and negative if they form a left-handed triad. [That is, $\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q})$ is negative if the rotation that brings $\mathbf{P}$ into line with $\mathbf{Q}$ is observed as clockwise from the tip of $\mathbf{S}$]. The mixed triple product is zero if $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ are coplanar.

Since the parallelepiped defined in this way is independent of the order in which the three vectors are taken, the six mixed triple products that can be formed with $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ all have the same absolute value, although not the same sign. It is easily shown that

$$\begin{aligned} \mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) &= \mathbf{P} \cdot (\mathbf{Q} \times \mathbf{S}) = \mathbf{Q} \cdot (\mathbf{S} \times \mathbf{P}) \\ &= -\mathbf{S} \cdot (\mathbf{Q} \times \mathbf{P}) = -\mathbf{P} \cdot (\mathbf{S} \times \mathbf{Q}) = -\mathbf{Q} \cdot (\mathbf{P} \times \mathbf{S}) \end{aligned} \quad (3.37)$$

Arranging the letters representing the three vectors counterclockwise in a circle (Fig. 3.21), we observe that the sign of the mixed triple product remains unchanged if the vectors are permuted in such a way that they still read in counterclockwise order. Such a permutation is said to be a *circular permutation*. It also follows from Eq. (3.37) and from the commutative property of scalar products that the mixed triple product of $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ can be defined equally well as $\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q})$ or $(\mathbf{S} \times \mathbf{P}) \cdot \mathbf{Q}$.

We can also express the mixed triple product of the vectors $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ in terms of the rectangular components of these vectors. Denoting $\mathbf{P} \times \mathbf{Q}$ by $\mathbf{V}$ and using formula (3.28) to express the scalar product of $\mathbf{S}$ and $\mathbf{V}$, we have

$$\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) = \mathbf{S} \cdot \mathbf{V} = S_x V_x + S_y V_y + S_z V_z$$

Substituting from the relations in Eq. (3.9) for the components of $\mathbf{V}$, we obtain

$$\begin{aligned} \mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) &= S_x(P_y Q_z - P_z Q_y) + S_y(P_z Q_x - P_x Q_z) \\ &\quad + S_z(P_x Q_y - P_y Q_x) \end{aligned} \quad (3.38)$$

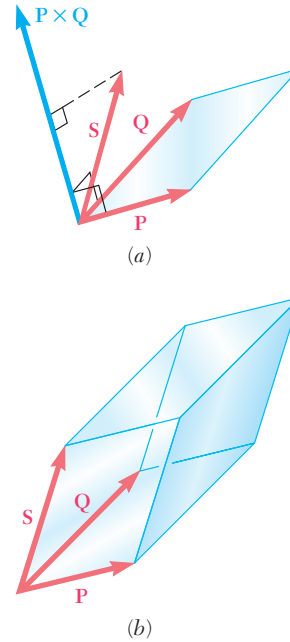


Fig. 3.20 (a) The mixed triple product is equal to the magnitude of the cross product of two vectors multiplied by the projection of the third vector onto that cross product; (b) the result equals the volume of the parallelepiped formed by the three vectors.

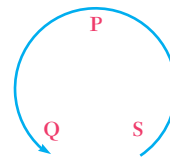


Fig. 3.21 Counterclockwise arrangement for determining the sign of the mixed triple product of three vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$.

We can write this expression in a more compact form if we observe that it represents the expansion of a determinant:

**Mixed triple product,
determinant form**

$$\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) = \begin{vmatrix} S_x & S_y & S_z \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix} \quad (3.39)$$

By applying the rules governing the permutation of rows in a determinant, we could easily verify the relations in Eq. (3.37), which we derived earlier from geometrical considerations.

3.2C Moment of a Force about a Given Axis

Now that we have the necessary mathematical tools, we can introduce the concept of moment of a force about an axis. Consider again a force $\mathbf{F}$ acting on a rigid body and the moment $\mathbf{M}_O$ of that force about O (Fig. 3.22). Let OL be an axis through O .

We define the moment M_{OL} of $\mathbf{F}$ about OL as the projection OC of the moment $\mathbf{M}_O$ onto the axis OL .

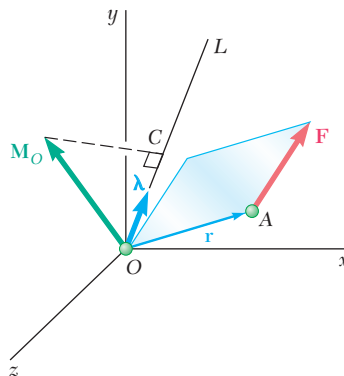


Fig. 3.22 The moment $\mathbf{M}_{OL}$ of a force $\mathbf{F}$ about the axis OL is the projection on OL of the moment $\mathbf{M}_O$. The calculation involves the unit vector λ along OL and the position vector $\mathbf{r}$ from O to A , the point upon which the force $\mathbf{F}$ acts.

Suppose we denote the unit vector along OL by λ and recall the expressions (3.34) and (3.11) for the projection of a vector on a given axis and for the moment $\mathbf{M}_O$ of a force $\mathbf{F}$. Then we can express M_{OL} as

**Moment about an axis
through the origin**

$$M_{OL} = \lambda \cdot \mathbf{M}_O = \lambda \cdot (\mathbf{r} \times \mathbf{F}) \quad (3.40)$$

This shows that the moment M_{OL} of $\mathbf{F}$ about the axis OL is the scalar obtained by forming the mixed triple product of $\boldsymbol{\lambda}$, $\mathbf{r}$, and $\mathbf{F}$. We can also express M_{OL} in the form of a determinant,

$$M_{OL} = \begin{vmatrix} \lambda_x & \lambda_y & \lambda_z \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix} \quad (3.41)$$

where $\lambda_x, \lambda_y, \lambda_z =$ direction cosines of axis OL

$x, y, z =$ coordinates of point of application of $\mathbf{F}$

$F_x, F_y, F_z =$ components of force $\mathbf{F}$

The physical significance of the moment M_{OL} of a force $\mathbf{F}$ about a fixed axis OL becomes more apparent if we resolve $\mathbf{F}$ into two rectangular components $\mathbf{F}_1$ and $\mathbf{F}_2$, with $\mathbf{F}_1$ parallel to OL and $\mathbf{F}_2$ lying in a plane P perpendicular to OL (Fig. 3.23). Resolving $\mathbf{r}$ similarly into two components $\mathbf{r}_1$ and $\mathbf{r}_2$ and substituting for $\mathbf{F}$ and $\mathbf{r}$ into Eq. (3.40), we get

$$\begin{aligned} M_{OL} &= \boldsymbol{\lambda} \cdot [(\mathbf{r}_1 + \mathbf{r}_2) \times (\mathbf{F}_1 + \mathbf{F}_2)] \\ &= \boldsymbol{\lambda} \cdot (\mathbf{r}_1 \times \mathbf{F}_1) + \boldsymbol{\lambda} \cdot (\mathbf{r}_1 \times \mathbf{F}_2) + \boldsymbol{\lambda} \cdot (\mathbf{r}_2 \times \mathbf{F}_1) + \boldsymbol{\lambda} \cdot (\mathbf{r}_2 \times \mathbf{F}_2) \end{aligned}$$

Note that all of the mixed triple products except the last one are equal to zero because they involve vectors that are coplanar when drawn from a common origin (Sec. 3.2B). Therefore, this expression reduces to

$$M_{OL} = \boldsymbol{\lambda} \cdot (\mathbf{r}_2 \times \mathbf{F}_2) \quad (3.42)$$

The vector product $\mathbf{r}_2 \times \mathbf{F}_2$ is perpendicular to the plane P and represents the moment of the component $\mathbf{F}_2$ of $\mathbf{F}$ about the point Q where OL intersects P . Therefore, the scalar M_{OL} , which is positive if $\mathbf{r}_2 \times \mathbf{F}_2$ and OL have the same sense and is negative otherwise, measures the tendency of $\mathbf{F}_2$ to make the rigid body rotate about the fixed axis OL . The other component $\mathbf{F}_1$ of $\mathbf{F}$ does not tend to make the body rotate about OL , because $\mathbf{F}_1$ and OL are parallel. Therefore, we conclude that

The moment M_{OL} of $\mathbf{F}$ about OL measures the tendency of the force $\mathbf{F}$ to impart to the rigid body a rotation about the fixed axis OL .

From the definition of the moment of a force about an axis, it follows that the moment of $\mathbf{F}$ about a coordinate axis is equal to the component of $\mathbf{M}_O$ along that axis. If we substitute each of the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ for $\boldsymbol{\lambda}$ in Eq. (3.40), we obtain expressions for the *moments of $\mathbf{F}$ about the coordinate axes*. These expressions are respectively equal to those obtained earlier for the components of the moment $\mathbf{M}_O$ of $\mathbf{F}$ about O :

$$\begin{aligned} M_x &= yF_z - zF_y \\ M_y &= zF_x - xF_z \\ M_z &= xF_y - yF_x \end{aligned} \quad (3.18)$$

Just as the components F_x , F_y , and F_z of a force $\mathbf{F}$ acting on a rigid body measure, respectively, the tendency of $\mathbf{F}$ to move the rigid body in the x , y , and z directions, the moments M_x , M_y , and M_z of $\mathbf{F}$ about the coordinate axes measure the tendency of $\mathbf{F}$ to impart to the rigid body a rotation about the x , y , and z axes, respectively.

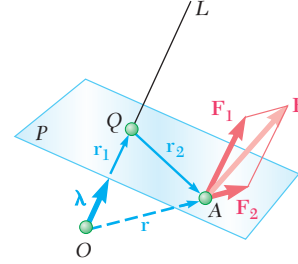


Fig. 3.23 By resolving the force $\mathbf{F}$ into components parallel to the axis OL and in a plane perpendicular to the axis, we can show that the moment M_{OL} of $\mathbf{F}$ about OL measures the tendency of $\mathbf{F}$ to rotate the rigid body about the axis.

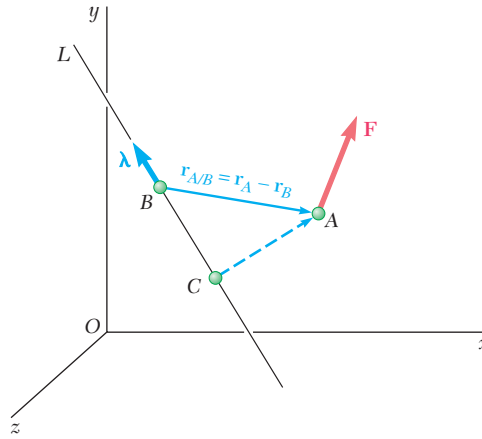


Fig. 3.24 The moment of a force about an axis or line L can be found by evaluating the mixed triple product at a point B on the line. The choice of B is arbitrary, since using any other point on the line, such as C , yields the same result.

More generally, we can obtain the moment of a force $\mathbf{F}$ applied at A about an axis that does not pass through the origin by choosing an arbitrary point B on the axis (Fig. 3.24) and determining the projection on the axis BL of the moment $\mathbf{M}_B$ of $\mathbf{F}$ about B . The equation for this projection is given here.

Moment about an arbitrary axis

$$M_{BL} = \boldsymbol{\lambda} \cdot \mathbf{M}_B = \boldsymbol{\lambda} \cdot (\mathbf{r}_{A/B} \times \mathbf{F}) \quad (3.43)$$

where $\mathbf{r}_{A/B} = \mathbf{r}_A - \mathbf{r}_B$ represents the vector drawn from B to A . Expressing M_{BL} in the form of a determinant, we have

$$M_{BL} = \begin{vmatrix} \lambda_x & \lambda_y & \lambda_z \\ x_{A/B} & y_{A/B} & z_{A/B} \\ F_x & F_y & F_z \end{vmatrix} \quad (3.44)$$

where $\lambda_x, \lambda_y, \lambda_z =$ direction cosines of axis BL

$$\begin{aligned} x_{A/B} &= x_A - x_B & y_{A/B} &= y_A - y_B & z_{A/B} &= z_A - z_B \\ F_x, F_y, F_z &= \text{components of force } \mathbf{F} \end{aligned}$$

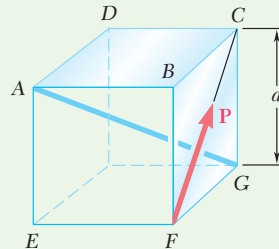
Note that this result is independent of the choice of the point B on the given axis. Indeed, denoting by M_{CL} the moment obtained with a different point C , we have

$$\begin{aligned} M_{CL} &= \boldsymbol{\lambda} \cdot [(\mathbf{r}_A - \mathbf{r}_C) \times \mathbf{F}] \\ &= \boldsymbol{\lambda} \cdot [(\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}] + \boldsymbol{\lambda} \cdot [(\mathbf{r}_B - \mathbf{r}_C) \times \mathbf{F}] \end{aligned}$$

However, since the vectors $\boldsymbol{\lambda}$ and $\mathbf{r}_B - \mathbf{r}_C$ lie along the same line, the volume of the parallelepiped having the vectors $\boldsymbol{\lambda}$, $\mathbf{r}_B - \mathbf{r}_C$, and $\mathbf{F}$ for sides is zero, as is the mixed triple product of these three vectors (Sec. 3.2B). The expression obtained for M_{CL} thus reduces to its first term, which is the expression used earlier to define M_{BL} . In addition, it follows from Sec. 3.1E that, when computing the moment of $\mathbf{F}$ about the given axis, A can be any point on the line of action of $\mathbf{F}$.

Sample Problem 3.5

A cube of side a is acted upon by a force $\mathbf{P}$ along the diagonal of a face, as shown. Determine the moment of $\mathbf{P}$ (a) about A , (b) about the edge AB , (c) about the diagonal AG of the cube. (d) Using the result of part c, determine the perpendicular distance between AG and FC .



STRATEGY: Use the equations presented in this section to compute the moments asked for. You can find the distance between AG and FC from the expression for the moment M_{AG} .

MODELING and ANALYSIS:

a. Moment about A. Choosing x , y , and z axes as shown (Fig. 1), resolve into rectangular components the force $\mathbf{P}$ and the vector $\mathbf{r}_{F/A} = \overrightarrow{AF}$ drawn from A to the point of application F of $\mathbf{P}$.

$$\begin{aligned}\mathbf{r}_{F/A} &= a\mathbf{i} - a\mathbf{j} = a(\mathbf{i} - \mathbf{j}) \\ \mathbf{P} &= (P/\sqrt{2})\mathbf{j} - (P/\sqrt{2})\mathbf{k} = (P/\sqrt{2})(\mathbf{j} - \mathbf{k})\end{aligned}$$

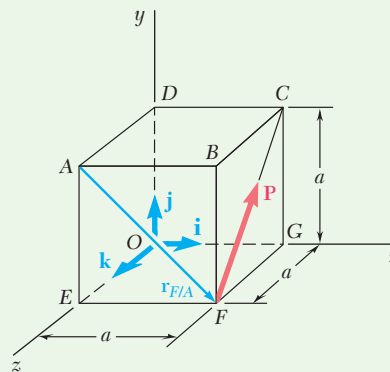


Fig. 1 Position vector $\mathbf{r}_{F/A}$ and force vector $\mathbf{P}$ relative to chosen coordinate system.

The moment of $\mathbf{P}$ about A is the vector product of these two vectors:

$$\begin{aligned}\mathbf{M}_A &= \mathbf{r}_{F/A} \times \mathbf{P} = a(\mathbf{i} - \mathbf{j}) \times (P/\sqrt{2})(\mathbf{j} - \mathbf{k}) \\ \mathbf{M}_A &= (aP/\sqrt{2})(\mathbf{i} + \mathbf{j} + \mathbf{k}) \quad \blacktriangleleft\end{aligned}$$

b. Moment about AB. You want the projection of $\mathbf{M}_A$ on AB :

$$\begin{aligned}M_{AB} &= \mathbf{i} \cdot \mathbf{M}_A = \mathbf{i} \cdot (aP/\sqrt{2})(\mathbf{i} + \mathbf{j} + \mathbf{k}) \\ M_{AB} &= aP/\sqrt{2} \quad \blacktriangleleft\end{aligned}$$

(continued)

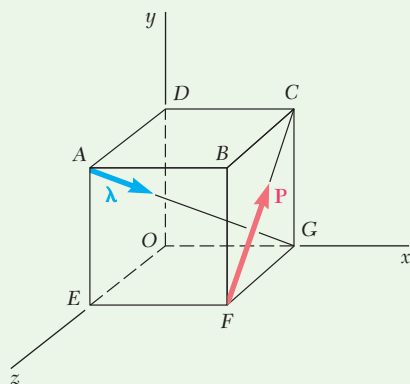


Fig. 2 Unit vector λ used to determine moment of $\mathbf{P}$ about AG .

You can verify that since AB is parallel to the x axis, M_{AB} is also the x component of the moment $\mathbf{M}_A$.

c. Moment about diagonal AG . You obtain the moment of $\mathbf{P}$ about AG by projecting $\mathbf{M}_A$ on AG . If you denote the unit vector along AG by λ (Fig. 2), the calculation looks like this:

$$\lambda = \frac{\overrightarrow{AG}}{AG} = \frac{a\mathbf{i} - a\mathbf{j} - a\mathbf{k}}{a\sqrt{3}} = (1/\sqrt{3})(\mathbf{i} - \mathbf{j} - \mathbf{k})$$

$$M_{AG} = \lambda \cdot \mathbf{M}_A = (1/\sqrt{3})(\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot (aP/\sqrt{2})(\mathbf{i} + \mathbf{j} + \mathbf{k})$$

$$M_{AG} = (aP/\sqrt{6})(1 - 1 - 1) \quad M_{AG} = -aP/\sqrt{6} \quad \blacktriangleleft$$

Alternative Method. You can also calculate the moment of $\mathbf{P}$ about AG from the determinant form:

$$M_{AG} = \begin{vmatrix} \lambda_x & \lambda_y & \lambda_z \\ x_{F/A} & y_{F/A} & z_{F/A} \\ F_x & F_y & F_z \end{vmatrix} = \begin{vmatrix} 1/\sqrt{3} & -1/\sqrt{3} & -1/\sqrt{3} \\ a & -a & 0 \\ 0 & P/\sqrt{2} & -P/\sqrt{2} \end{vmatrix} = -aP/\sqrt{6}$$

d. Perpendicular Distance between AG and FC . First note that $\mathbf{P}$ is perpendicular to the diagonal AG . You can check this by forming the scalar product $\mathbf{P} \cdot \lambda$ and verifying that it is zero:

$$\mathbf{P} \cdot \lambda = (P/\sqrt{2})(\mathbf{j} - \mathbf{k}) \cdot (1/\sqrt{3})(\mathbf{i} - \mathbf{j} - \mathbf{k}) = (P\sqrt{6})(0 - 1 + 1) = 0$$

You can then express the moment M_{AG} as $-Pd$, where d is the perpendicular distance from AG to FC (Fig. 3). (The negative sign is needed because the rotation imparted to the cube by $\mathbf{P}$ appears as clockwise to an observer at G .) Using the value found for M_{AG} in part c,

$$M_{AG} = -Pd = -aP/\sqrt{6} \quad d = a/\sqrt{6} \quad \blacktriangleleft$$

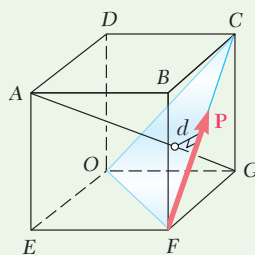


Fig. 3 Perpendicular distance d from AG to FC .

REFLECT and THINK: In a problem like this, it is important to visualize the forces and moments in three dimensions so you can choose the appropriate equations for finding them and also recognize the geometric relationships between them.

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will apply the *scalar product* (or *dot product*) of two vectors to determine the *angle formed by two given vectors* and the *projection of a force on a given axis*. You will also use the *mixed triple product* of three vectors to find the *moment of a force about a given axis* and the *perpendicular distance between two lines*.

1. Calculating the angle formed by two given vectors. First express the vectors in terms of their components and determine the magnitudes of the two vectors. Then find the cosine of the desired angle by dividing the scalar product of the two vectors by the product of their magnitudes [Eq. (3.30)].

2. Computing the projection of a vector $\mathbf{P}$ on a given axis OL . In general, begin by expressing $\mathbf{P}$ and the unit vector $\boldsymbol{\lambda}$, which defines the direction of the axis, in component form. Take care that $\boldsymbol{\lambda}$ has the correct sense (that is, $\boldsymbol{\lambda}$ is directed from O to L). The required projection is then equal to the scalar product $\mathbf{P} \cdot \boldsymbol{\lambda}$. However, if you know the angle θ formed by $\mathbf{P}$ and $\boldsymbol{\lambda}$, the projection is also given by $P \cos \theta$.

3. Determining the moment M_{OL} of a force about a given axis OL . We defined M_{OL} as

$$M_{OL} = \boldsymbol{\lambda} \cdot \mathbf{M}_O = \boldsymbol{\lambda} \cdot (\mathbf{r} \times \mathbf{F}) \quad (3.40)$$

where $\boldsymbol{\lambda}$ is the unit vector along OL and $\mathbf{r}$ is a position vector *from any point* on the line OL to *any point* on the line of action of $\mathbf{F}$. As was the case for the moment of a force about a point, choosing the most convenient position vector will simplify your calculations. Also, recall the warning of the preceding section: The vectors $\mathbf{r}$ and $\mathbf{F}$ must have the correct sense, and they must be placed in the proper order. The procedure you should follow when computing the moment of a force about an axis is illustrated in part *c* of Sample Prob. 3.5. The two essential steps in this procedure are (1) express $\boldsymbol{\lambda}$, $\mathbf{r}$, and $\mathbf{F}$ in terms of their rectangular components and (2) evaluate the mixed triple product $\boldsymbol{\lambda} \cdot (\mathbf{r} \times \mathbf{F})$ to determine the moment about the axis. In most three-dimensional problems, the most convenient way to compute the mixed triple product is by using a determinant.

As noted in the text, when $\boldsymbol{\lambda}$ is directed along one of the coordinate axes, M_{OL} is equal to the scalar component of $\mathbf{M}_O$ along that axis.

4. Determining the perpendicular distance between two lines. Remember that it is the perpendicular component $\mathbf{F}_2$ of the force $\mathbf{F}$ that tends to make a body rotate about a given axis OL (Fig. 3.23). It then follows that

$$M_{OL} = F_2 d$$

(continued)

where M_{OL} is the moment of $\mathbf{F}$ about axis OL and d is the perpendicular distance between OL and the line of action of $\mathbf{F}$. This last equation provides a simple technique for determining d . First assume that a force $\mathbf{F}$ of known magnitude F lies along one of the given lines and that the unit vector $\mathbf{\lambda}$ lies along the other line. Next compute the moment M_{OL} of the force $\mathbf{F}$ about the second line using the method discussed above. The magnitude of the parallel component, F_1 , of $\mathbf{F}$ is obtained using the scalar product:

$$F_1 = \mathbf{F} \cdot \mathbf{\lambda}$$

The value of F_2 is then determined from

$$F_2 = \sqrt{F^2 - F_1^2}$$

Finally, substitute the values of M_{OL} and F_2 into the equation $M_{OL} = F_2 d$ and solve for d .

You should now realize that the calculation of the perpendicular distance in part d of Sample Prob. 3.5 was simplified by $\mathbf{P}$ being perpendicular to the diagonal AG . In general, the two given lines will not be perpendicular, so you will have to use the technique just outlined when determining the perpendicular distance between them.

Problems

3.35 Given the vectors $\mathbf{P} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, $\mathbf{Q} = 5\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$, and $\mathbf{S} = -3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$, compute the scalar products $\mathbf{P} \cdot \mathbf{Q}$, $\mathbf{P} \cdot \mathbf{S}$, and $\mathbf{Q} \cdot \mathbf{S}$.

3.36 Form the scalar product $\mathbf{B} \cdot \mathbf{C}$ and use the result obtained to prove the identity

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

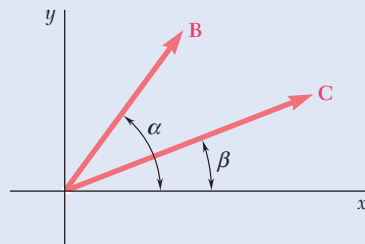


Fig. P3.36

3.37 Three cables are used to support a container as shown. Determine the angle formed by cables AB and AD .

3.38 Three cables are used to support a container as shown. Determine the angle formed by cables AC and AD .

3.39 Knowing that the tension in cable AC is 280 lb, determine (a) the angle between cable AC and the boom AB , (b) the projection on AB of the force exerted by cable AC at point A .

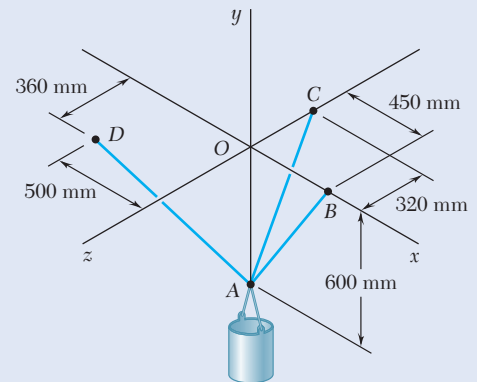


Fig. P3.37 and P3.38

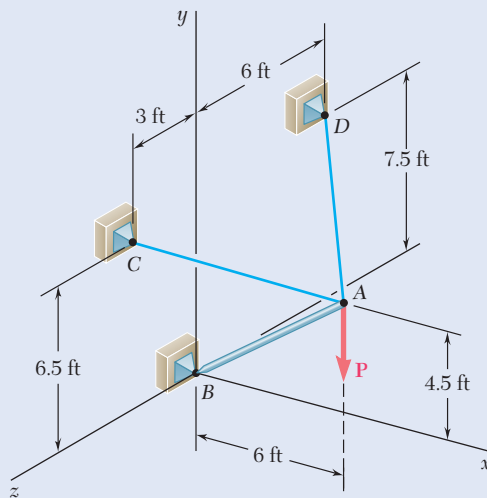


Fig. P3.39 and P3.40

3.40 Knowing that the tension in cable AD is 180 lb, determine (a) the angle between cable AD and the boom AB , (b) the projection on AB of the force exerted by cable AD at point A .

- 3.41** Ropes AB and BC are two of the ropes used to support a tent. The two ropes are attached to a stake at B . If the tension in rope AB is 540 N, determine (a) the angle between rope AB and the stake, (b) the projection on the stake of the force exerted by rope AB at point B .

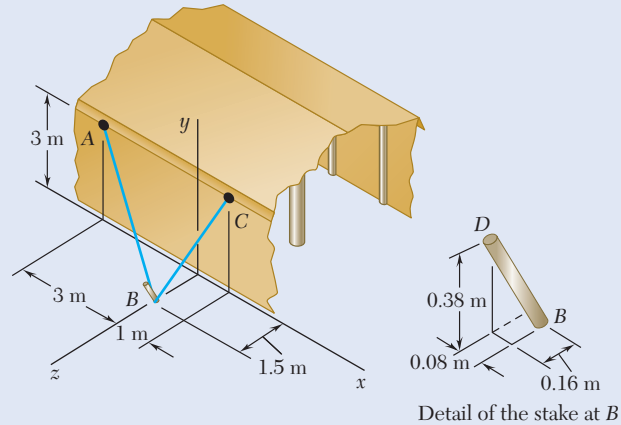


Fig. P3.41 and P3.42

- 3.42** Ropes AB and BC are two of the ropes used to support a tent. The two ropes are attached to a stake at B . If the tension in rope BC is 490 N, determine (a) the angle between rope BC and the stake, (b) the projection on the stake of the force exerted by rope BC at point B .

- 3.43** The 20-in. tube AB can slide along a horizontal rod. The ends A and B of the tube are connected by elastic cords to the fixed point C . For the position corresponding to $x = 11$ in., determine the angle formed by the two cords, (a) using Eq. (3.30), (b) applying the law of cosines to triangle ABC .

- 3.44** Solve Prob. 3.43 for the position corresponding to $x = 4$ in.

- 3.45** Determine the volume of the parallelepiped of Fig. 3.20b when (a) $\mathbf{P} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$, $\mathbf{Q} = -2\mathbf{i} - 5\mathbf{j} + \mathbf{k}$, and $\mathbf{S} = 7\mathbf{i} + \mathbf{j} - \mathbf{k}$, (b) $\mathbf{P} = 5\mathbf{i} - \mathbf{j} + 6\mathbf{k}$, $\mathbf{Q} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$, and $\mathbf{S} = -3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$.

- 3.46** Given the vectors $\mathbf{P} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{Q} = 4\mathbf{i} + Q_y\mathbf{j} - 2\mathbf{k}$, and $\mathbf{S} = 2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, determine the value of Q_y for which the three vectors are coplanar.

- 3.47** A crane is oriented so that the end of the 25-m boom AO lies in the yz plane. At the instant shown, the tension in cable AB is 4 kN. Determine the moment about each of the coordinate axes of the force exerted on A by cable AB .

- 3.48** The 25-m crane boom AO lies in the yz plane. Determine the maximum permissible tension in cable AB if the absolute value of moments about the coordinate axes of the force exerted on A by cable AB must be

$$|M_x| \leq 60 \text{ kN}\cdot\text{m}, |M_y| \leq 12 \text{ kN}\cdot\text{m}, |M_z| \leq 8 \text{ kN}\cdot\text{m}$$

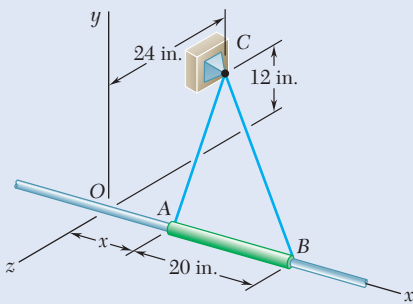


Fig. P3.43

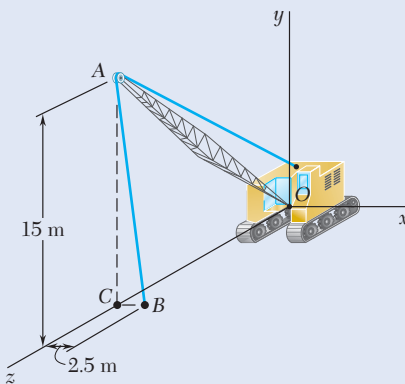


Fig. P3.47 and P3.48

- 3.49** To loosen a frozen valve, a force $\mathbf{F}$ with a magnitude of 70 lb is applied to the handle of the valve. Knowing that $\theta = 25^\circ$, $M_x = -61 \text{ lb}\cdot\text{ft}$, and $M_z = -43 \text{ lb}\cdot\text{ft}$, determine ϕ and d .

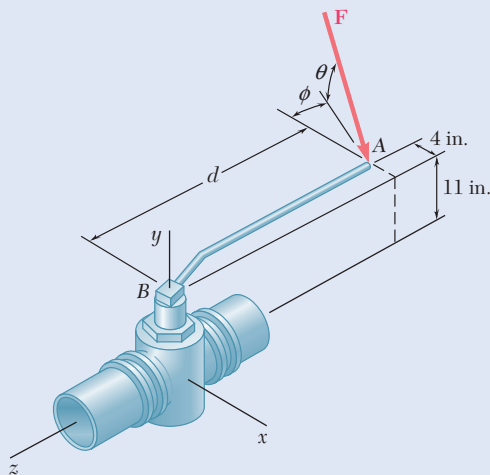


Fig. P3.49 and P3.50

- 3.50** When a force $\mathbf{F}$ is applied to the handle of the valve shown, its moments about the x and z axes are $M_x = -77 \text{ lb}\cdot\text{ft}$ and $M_z = -81 \text{ lb}\cdot\text{ft}$, respectively. For $d = 27 \text{ in.}$, determine the moment M_y of $\mathbf{F}$ about the y axis.

- 3.51** To lift a heavy crate, a man uses a block and tackle attached to the bottom of an I-beam at hook B . Knowing that the moments about the y and the z axes of the force exerted at B by portion AB of the rope are, respectively, $120 \text{ N}\cdot\text{m}$ and $-460 \text{ N}\cdot\text{m}$, determine the distance a .

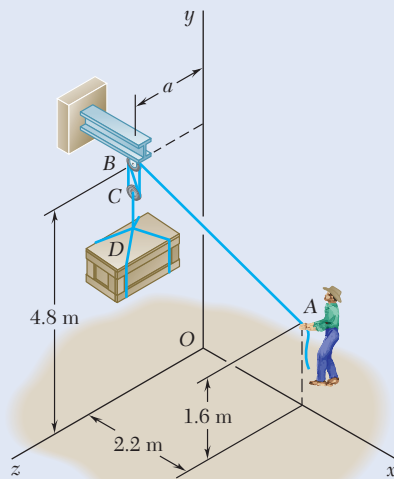


Fig. P3.51 and P3.52

- 3.52** To lift a heavy crate, a man uses a block and tackle attached to the bottom of an I-beam at hook B . Knowing that the man applies a 195-N force to end A of the rope and that the moment of that force about the y axis is $132 \text{ N}\cdot\text{m}$, determine the distance a .

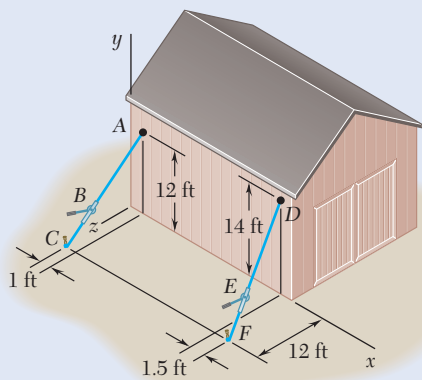


Fig. P3.53

3.53 A farmer uses cables and winch pullers B and E to plumb one side of a small barn. If it is known that the sum of the moments about the x axis of the forces exerted by the cables on the barn at points A and D is equal to 4728 lb·ft, determine the magnitude of T_{DE} when $T_{AB} = 255$ lb.

3.54 Solve Prob. 3.53 when the tension in cable AB is 306 lb.

3.55 The 23-in. vertical rod CD is welded to the midpoint C of the 50-in. rod AB . Determine the moment about AB of the 235-lb force P .

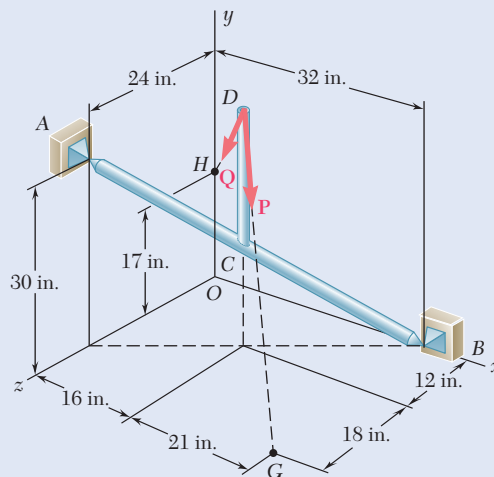


Fig. P3.55 and P3.56

3.56 The 23-in. vertical rod CD is welded to the midpoint C of the 50-in. rod AB . Determine the moment about AB of the 174-lb force Q .

3.57 The frame ACD is hinged at A and D and is supported by a cable that passes through a ring at B and is attached to hooks at G and H . Knowing that the tension in the cable is 450 N, determine the moment about the diagonal AD of the force exerted on the frame by portion BH of the cable.

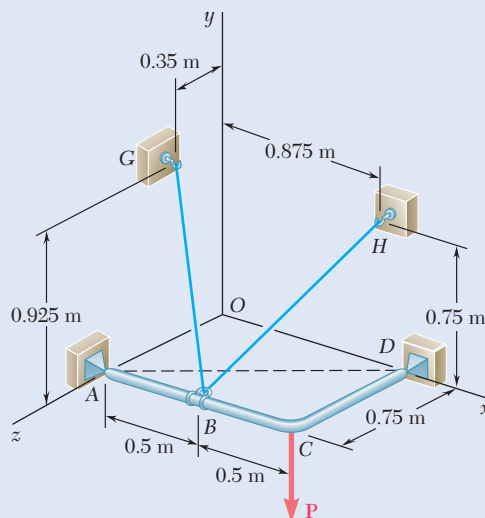


Fig. P3.57

- 3.58** In Prob. 3.57, determine the moment about the diagonal AD of the force exerted on the frame by portion BG of the cable.
- 3.59** The triangular plate ABC is supported by ball-and-socket joints at B and D and is held in the position shown by cables AE and CF . If the force exerted by cable AE at A is 55 N, determine the moment of that force about the line joining points D and B .
- 3.60** The triangular plate ABC is supported by ball-and-socket joints at B and D and is held in the position shown by cables AE and CF . If the force exerted by cable CF at C is 33 N, determine the moment of that force about the line joining points D and B .
- 3.61** A regular tetrahedron has six edges of length a . A force $\mathbf{P}$ is directed as shown along edge BC . Determine the moment of $\mathbf{P}$ about edge OA .

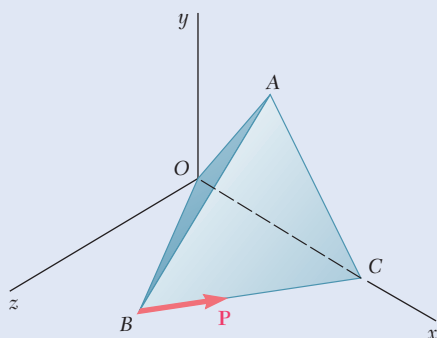


Fig. P3.61 and P3.62

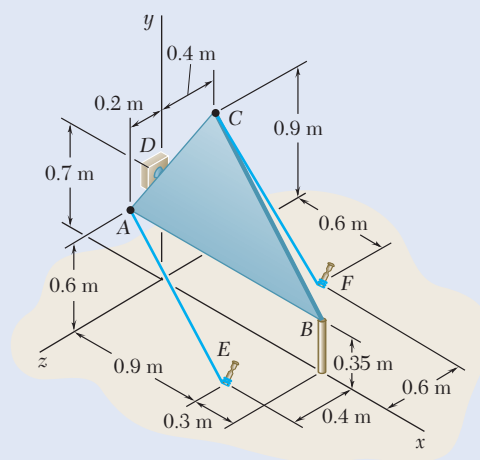


Fig. P3.59 and P3.60

- 3.62** A regular tetrahedron has six edges of length a . (a) Show that two opposite edges, such as OA and BC , are perpendicular to each other. (b) Use this property and the result obtained in Prob. 3.61 to determine the perpendicular distance between edges OA and BC .
- 3.63** Two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ in space have the same magnitude F . Prove that the moment of $\mathbf{F}_1$ about the line of action of $\mathbf{F}_2$ is equal to the moment of $\mathbf{F}_2$ about the line of action of $\mathbf{F}_1$.
- *3.64** In Prob. 3.55, determine the perpendicular distance between rod AB and the line of action of $\mathbf{P}$.
- *3.65** In Prob. 3.56, determine the perpendicular distance between rod AB and the line of action of $\mathbf{Q}$.
- *3.66** In Prob. 3.57, determine the perpendicular distance between portion BH of the cable and the diagonal AD .
- *3.67** In Prob. 3.58, determine the perpendicular distance between portion BG of the cable and the diagonal AD .
- *3.68** In Prob. 3.59, determine the perpendicular distance between cable AE and the line joining points D and B .
- *3.69** In Prob. 3.60, determine the perpendicular distance between cable CF and the line joining points D and B .

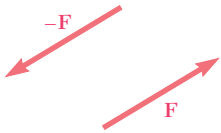


Fig. 3.25 A couple consists of two forces with equal magnitude, parallel lines of action, and opposite sense.

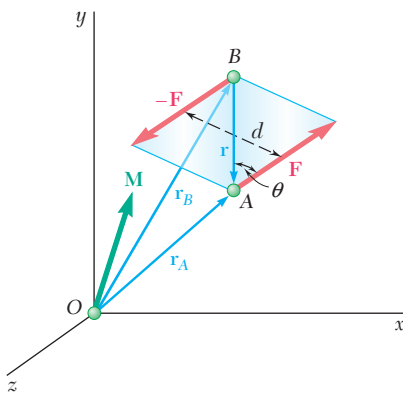


Fig. 3.26 The moment $\mathbf{M}$ of the couple about O is the sum of the moments of $\mathbf{F}$ and of $-\mathbf{F}$ about O .



Photo 3.1 The parallel upward and downward forces of equal magnitude exerted on the arms of the lug nut wrench are an example of a couple.

3.3 COUPLES AND FORCE-COUPLE SYSTEMS

Now that we have studied the effects of forces and moments on a rigid body, we can ask if it is possible to simplify a system of forces and moments without changing these effects. It turns out that we *can* replace a system of forces and moments with a simpler and equivalent system. One of the key ideas used in such a transformation is called a couple.

3.3A Moment of a Couple

Two forces $\mathbf{F}$ and $-\mathbf{F}$, having the same magnitude, parallel lines of action, and opposite sense, are said to form a **couple** (Fig. 3.25). Clearly, the sum of the components of the two forces in any direction is zero. The sum of the moments of the two forces about a given point, however, is not zero. The two forces do not cause the body on which they act to move along a line (translation), but they do tend to make it rotate.

Let us denote the position vectors of the points of application of $\mathbf{F}$ and $-\mathbf{F}$ by $\mathbf{r}_A$ and $\mathbf{r}_B$, respectively (Fig. 3.26). The sum of the moments of the two forces about O is

$$\mathbf{r}_A \times \mathbf{F} + \mathbf{r}_B \times (-\mathbf{F}) = (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}$$

Setting $\mathbf{r}_A - \mathbf{r}_B = \mathbf{r}$, where $\mathbf{r}$ is the vector joining the points of application of the two forces, we conclude that the sum of the moments of $\mathbf{F}$ and $-\mathbf{F}$ about O is represented by the vector

$$\mathbf{M} = \mathbf{r} \times \mathbf{F} \quad (3.45)$$

The vector $\mathbf{M}$ is called the *moment of the couple*. It is perpendicular to the plane containing the two forces, and its magnitude is

$$M = rF \sin \theta = Fd \quad (3.46)$$

where d is the perpendicular distance between the lines of action of $\mathbf{F}$ and $-\mathbf{F}$ and θ is the angle between $\mathbf{F}$ (or $-\mathbf{F}$) and $\mathbf{r}$. The sense of $\mathbf{M}$ is defined by the right-hand rule.

Note that the vector $\mathbf{r}$ in Eq. (3.45) is independent of the choice of the origin O of the coordinate axes. Therefore, we would obtain the same result if the moments of $\mathbf{F}$ and $-\mathbf{F}$ had been computed about a different point O' . Thus, the moment $\mathbf{M}$ of a couple is a *free vector* (Sec. 2.1B), which can be applied at any point (Fig. 3.27).

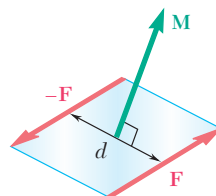


Fig. 3.27 The moment $\mathbf{M}$ of a couple equals the product of $\mathbf{F}$ and d , is perpendicular to the plane of the couple, and may be applied at any point of that plane.

From the definition of the moment of a couple, it also follows that two couples—one consisting of the forces $\mathbf{F}_1$ and $-\mathbf{F}_1$, the other of the forces $\mathbf{F}_2$ and $-\mathbf{F}_2$ (Fig. 3.28)—have equal moments if

$$F_1 d_1 = F_2 d_2 \quad (3.47)$$

provided that the two couples lie in parallel planes (or in the same plane) and have the same sense (i.e., clockwise or counterclockwise).

3.3B Equivalent Couples

Imagine that three couples act successively on the same rectangular box (Fig. 3.29). As we have just seen, the only motion a couple can impart to a rigid body is a rotation. Since each of the three couples shown has the same moment $\mathbf{M}$ (same direction and same magnitude $M = 120 \text{ lb}\cdot\text{in.}$), we can expect each couple to have the same effect on the box.

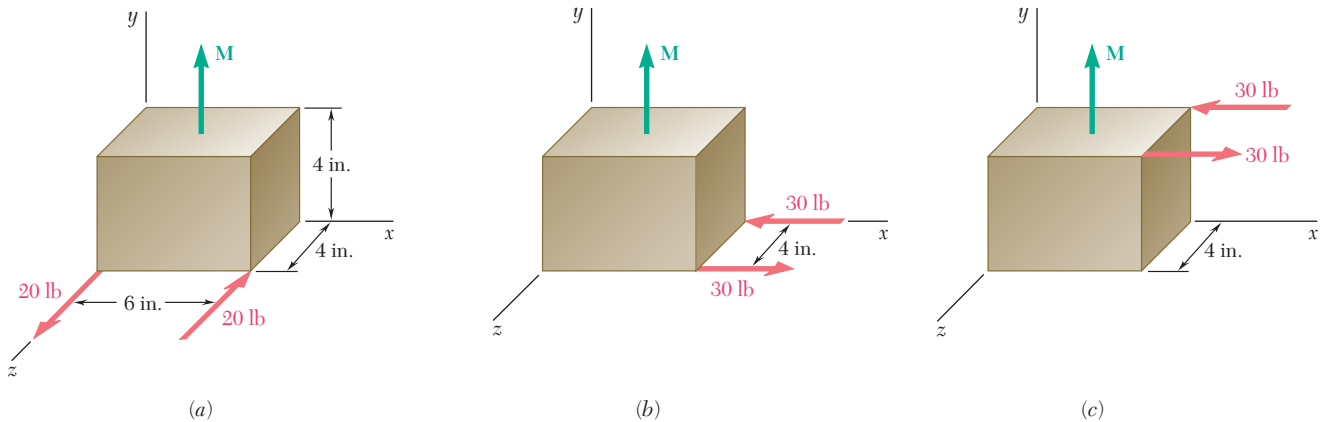


Fig. 3.29 Three equivalent couples. (a) A couple acting on the bottom of the box, acting counterclockwise viewed from above; (b) a couple in the same plane and with the same sense but larger forces than in (a); (c) a couple acting in a different plane but same sense.

As reasonable as this conclusion appears, we should not accept it hastily. Although intuition is of great help in the study of mechanics, it should not be accepted as a substitute for logical reasoning. Before stating that two systems (or groups) of forces have the same effect on a rigid body, we should prove that fact on the basis of the experimental evidence introduced so far. This evidence consists of the parallelogram law for the addition of two forces (Sec. 2.1A) and the principle of transmissibility (Sec. 3.1B). Therefore, we state that **two systems of forces are equivalent** (i.e., they have the same effect on a rigid body) **if we can transform one of them into the other by means of one or several of the following operations:** (1) replacing two forces acting on the same particle by their resultant; (2) resolving a force into two components; (3) canceling two equal and opposite forces acting on the same particle; (4) attaching to the same particle two equal and opposite forces; and (5) moving a force along its line of action. Each of these operations is easily justified on the basis of the parallelogram law or the principle of transmissibility.

Let us now prove that **two couples having the same moment $\mathbf{M}$ are equivalent**. First consider two couples contained in the same plane, and

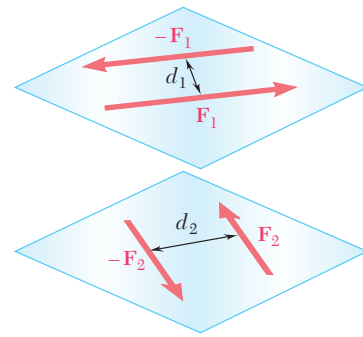


Fig. 3.28 Two couples have the same moment if they lie in parallel planes, have the same sense, and if $F_1 d_1 = F_2 d_2$.

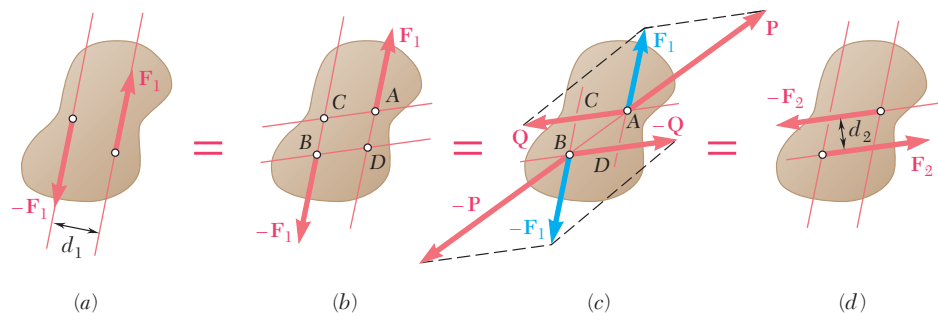


Fig. 3.30 Four steps in transforming one couple to another couple in the same plane by using simple operations. (a) Starting couple; (b) label points of intersection of lines of action of the two couples; (c) resolve forces from first couple into components; (d) final couple.

assume that this plane coincides with the plane of the figure (Fig. 3.30). The first couple consists of the forces $\mathbf{F}_1$ and $-\mathbf{F}_1$ of magnitude F_1 , located at a distance d_1 from each other (Fig. 3.30a). The second couple consists of the forces $\mathbf{F}_2$ and $-\mathbf{F}_2$ of magnitude F_2 , located at a distance d_2 from each other (Fig. 3.30d). Since the two couples have the same moment $\mathbf{M}$, which is perpendicular to the plane of the figure, they must have the same sense (assumed here to be counterclockwise), and the relation

$$F_1 d_1 = F_2 d_2 \quad (3.47)$$

must be satisfied. To prove that they are equivalent, we shall show that the first couple can be transformed into the second by means of the operations listed previously.

Let us denote by A , B , C , and D the points of intersection of the lines of action of the two couples. We first slide the forces $\mathbf{F}_1$ and $-\mathbf{F}_1$ until they are attached, respectively, at A and B , as shown in Fig. 3.30b. We then resolve force $\mathbf{F}_1$ into a component $\mathbf{P}$ along line AB and a component $\mathbf{Q}$ along AC (Fig. 3.30c). Similarly, we resolve force $-\mathbf{F}_1$ into $-\mathbf{P}$ along AB and $-\mathbf{Q}$ along BD . The forces $\mathbf{P}$ and $-\mathbf{P}$ have the same magnitude, the same line of action, and opposite sense; we can move them along their common line of action until they are applied at the same point and may then be canceled. Thus, the couple formed by $\mathbf{F}_1$ and $-\mathbf{F}_1$ reduces to a couple consisting of $\mathbf{Q}$ and $-\mathbf{Q}$.

We now show that the forces $\mathbf{Q}$ and $-\mathbf{Q}$ are respectively equal to the forces $-\mathbf{F}_2$ and $\mathbf{F}_2$. We obtain the moment of the couple formed by $\mathbf{Q}$ and $-\mathbf{Q}$ by computing the moment of $\mathbf{Q}$ about B . Similarly, the moment of the couple formed by $\mathbf{F}_1$ and $-\mathbf{F}_1$ is the moment of $\mathbf{F}_1$ about B . However, by Varignon's theorem, the moment of $\mathbf{F}_1$ is equal to the sum of the moments of its components $\mathbf{P}$ and $\mathbf{Q}$. Since the moment of $\mathbf{P}$ about B is zero, the moment of the couple formed by $\mathbf{Q}$ and $-\mathbf{Q}$ must be equal to the moment of the couple formed by $\mathbf{F}_1$ and $-\mathbf{F}_1$. Recalling Eq. (3.47), we have

$$Q d_2 = F_1 d_1 = F_2 d_2 \quad \text{and} \quad Q = F_2$$

Thus, the forces $\mathbf{Q}$ and $-\mathbf{Q}$ are respectively equal to the forces $-\mathbf{F}_2$ and $\mathbf{F}_2$, and the couple of Fig. 3.30a is equivalent to the couple of Fig. 3.30d.

Now consider two couples contained in parallel planes P_1 and P_2 . We prove that they are equivalent if they have the same moment. In view of the preceding discussion, we can assume that the couples consist of forces of the same magnitude F acting along parallel lines (Fig. 3.31a and d). We propose to show that the couple contained in plane P_1 can be transformed into the couple contained in plane P_2 by means of the standard operations listed previously.

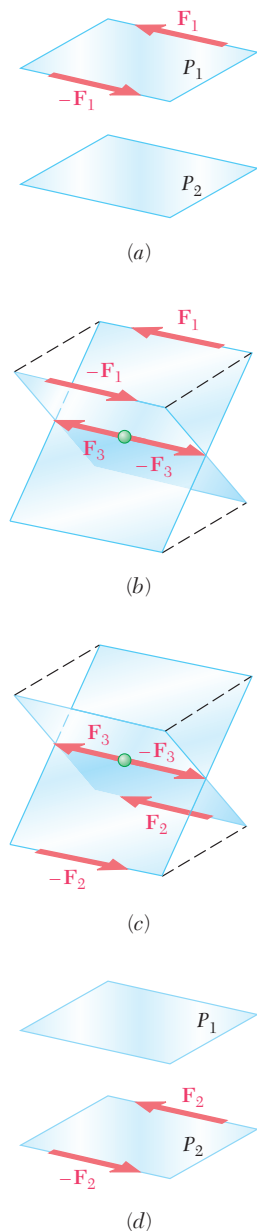


Fig. 3.31 Four steps in transforming one couple to another couple in a parallel plane by using simple operations. (a) Initial couple; (b) add a force pair along the line of intersection of two diagonal planes; (c) replace two couples with equivalent couples in the same planes; (d) final couple.

Let us consider the two diagonal planes defined respectively by the lines of action of $\mathbf{F}_1$ and $-\mathbf{F}_2$ and by those of $-\mathbf{F}_1$ and $\mathbf{F}_2$ (Fig. 3.31*b*). At a point on their line of intersection, we attach two forces $\mathbf{F}_3$ and $-\mathbf{F}_3$, which are respectively equal to $\mathbf{F}_1$ and $-\mathbf{F}_1$. The couple formed by $\mathbf{F}_1$ and $-\mathbf{F}_3$ can be replaced by a couple consisting of $\mathbf{F}_3$ and $-\mathbf{F}_2$ (Fig. 3.31*c*), because both couples clearly have the same moment and are contained in the same diagonal plane. Similarly, the couple formed by $-\mathbf{F}_1$ and $\mathbf{F}_3$ can be replaced by a couple consisting of $-\mathbf{F}_3$ and $\mathbf{F}_2$. Canceling the two equal and opposite forces $\mathbf{F}_3$ and $-\mathbf{F}_3$, we obtain the desired couple in plane P_2 (Fig. 3.31*d*). Thus, we conclude that two couples having the same moment $\mathbf{M}$ are equivalent, whether they are contained in the same plane or in parallel planes.

The property we have just established is very important for the correct understanding of the mechanics of rigid bodies. It indicates that when a couple acts on a rigid body, it does not matter where the two forces forming the couple act or what magnitude and direction they have. The only thing that counts is the *moment* of the couple (magnitude and direction). Couples with the same moment have the same effect on the rigid body.

3.3C Addition of Couples

Consider two intersecting planes P_1 and P_2 and two couples acting respectively in P_1 and P_2 . Recall that each couple is a free vector in its respective plane and can be represented within this plane by any combination of equal, opposite, and parallel forces and of perpendicular distance of separation that provides the same sense and magnitude for this couple. Thus, we can assume, without any loss of generality, that the couple in P_1 consists of two forces $\mathbf{F}_1$ and $-\mathbf{F}_1$ perpendicular to the line of intersection of the two planes and acting respectively at A and B (Fig. 3.32*a*). Similarly, we can assume that the couple in P_2 consists of two forces $\mathbf{F}_2$ and $-\mathbf{F}_2$ perpendicular to AB and acting respectively at A and B . It is clear that the resultant $\mathbf{R}$ of $\mathbf{F}_1$ and $\mathbf{F}_2$ and the resultant $-\mathbf{R}$ of $-\mathbf{F}_1$ and $-\mathbf{F}_2$ form a couple. Denoting the vector joining B to A by $\mathbf{r}$ and recalling the definition of the moment of a couple (Sec. 3.3A), we express the moment $\mathbf{M}$ of the resulting couple as

$$\mathbf{M} = \mathbf{r} \times \mathbf{R} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2)$$

By Varignon's theorem, we can expand this expression as

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2$$

The first term in this expression represents the moment $\mathbf{M}_1$ of the couple in P_1 , and the second term represents the moment $\mathbf{M}_2$ of the couple in P_2 . Therefore, we have

$$\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2 \quad (3.48)$$

We conclude that the sum of two couples of moments $\mathbf{M}_1$ and $\mathbf{M}_2$ is a couple of moment $\mathbf{M}$ equal to the vector sum of $\mathbf{M}_1$ and $\mathbf{M}_2$ (Fig. 3.32*b*). We can extend this conclusion to state that any number of couples can be added to produce one resultant couple, as

$$\mathbf{M} = \Sigma \mathbf{M} = \Sigma(\mathbf{r} \times \mathbf{F})$$

3.3D Couple Vectors

We have seen that couples with the same moment, whether they act in the same plane or in parallel planes, are equivalent. Therefore, we have no need to draw the actual forces forming a given couple in order to define its effect

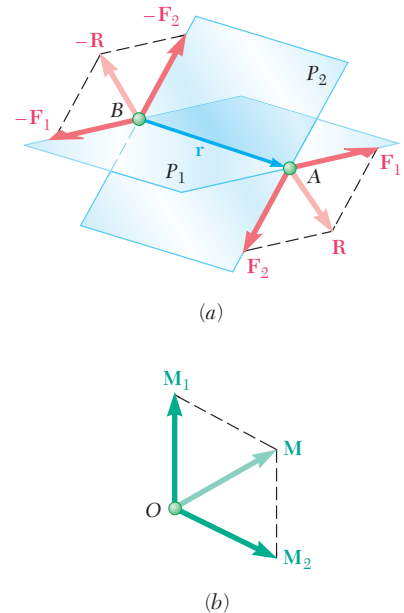


Fig. 3.32 (a) We can add two couples, each acting in one of two intersecting planes, to form a new couple. (b) The moment of the resultant couple is the vector sum of the moments of the component couples.

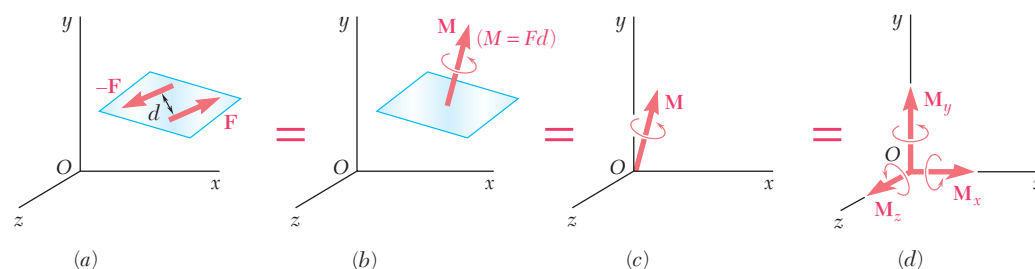


Fig. 3.33 (a) A couple formed by two forces can be represented by (b) a couple vector, oriented perpendicular to the plane of the couple. (c) The couple vector is a free vector and can be moved to other points of application, such as the origin. (d) A couple vector can be resolved into components along the coordinate axes.

on a rigid body (Fig. 3.33a). It is sufficient to draw an arrow equal in magnitude and direction to the moment $\mathbf{M}$ of the couple (Fig. 3.33b). We have also seen that the sum of two couples is itself a couple and that we can obtain the moment $\mathbf{M}$ of the resultant couple by forming the vector sum of the moments $\mathbf{M}_1$ and $\mathbf{M}_2$ of the given couples. Thus, couples obey the law of addition of vectors, so the arrow used in Fig. 3.33b to represent the couple defined in Fig. 3.33a truly can be considered a vector.

The vector representing a couple is called a **couple vector**. Note that, in Fig. 3.33, we use a red arrow to distinguish the couple vector, *which represents the couple itself*, from the *moment* of the couple, which was represented by a green arrow in earlier figures. Also note that we added the symbol $\curvearrowright$ to this red arrow to avoid any confusion with vectors representing forces. A couple vector, like the moment of a couple, is a free vector. Therefore, we can choose its point of application at the origin of the system of coordinates, if so desired (Fig. 3.33c). Furthermore, we can resolve the couple vector $\mathbf{M}$ into component vectors $\mathbf{M}_x$, $\mathbf{M}_y$, and $\mathbf{M}_z$ that are directed along the coordinate axes (Fig. 3.33d). These component vectors represent couples acting, respectively, in the yz , zx , and xy planes.

3.3E Resolution of a Given Force into a Force at O and a Couple

Consider a force $\mathbf{F}$ acting on a rigid body at a point A defined by the position vector $\mathbf{r}$ (Fig. 3.34a). Suppose that for some reason it would simplify the analysis to have the force act at point O instead. Although we can move $\mathbf{F}$ along its line of action (principle of transmissibility), we cannot move it to a point O that does not lie on the original line of action without modifying the action of $\mathbf{F}$ on the rigid body.

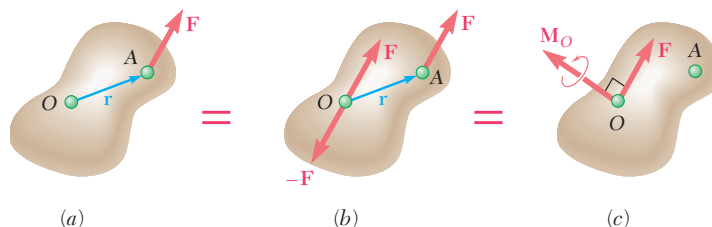


Fig. 3.34 Replacing a force with a force and a couple. (a) Initial force $\mathbf{F}$ acting at point A ; (b) attaching equal and opposite forces at O ; (c) force $\mathbf{F}$ acting at point O and a couple.

We can, however, attach two forces at point O , one equal to $\mathbf{F}$ and the other equal to $-\mathbf{F}$, without modifying the action of the original force on the rigid body (Fig. 3.34b). As a result of this transformation, we now have a force $\mathbf{F}$ applied at O ; the other two forces form a couple of moment $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$. Thus,

Any force $\mathbf{F}$ acting on a rigid body can be moved to an arbitrary point O provided that we add a couple whose moment is equal to the moment of $\mathbf{F}$ about O .

The couple tends to impart to the rigid body the same rotational motion about O that force $\mathbf{F}$ tended to produce before it was transferred to O . We represent the couple by a couple vector $\mathbf{M}_O$ that is perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{F}$. Since $\mathbf{M}_O$ is a free vector, it may be applied anywhere; for convenience, however, the couple vector is usually attached at O together with $\mathbf{F}$. This combination is referred to as a **force-couple system** (Fig. 3.34c).

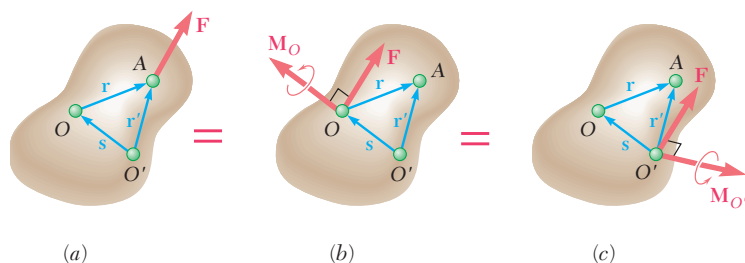


Fig. 3.35 Moving a force to different points. (a) Initial force $\mathbf{F}$ acting at A ; (b) force $\mathbf{F}$ acting at O and a couple; (c) force $\mathbf{F}$ acting at O' and a different couple.

If we move force $\mathbf{F}$ from A to a different point O' (Fig. 3.35a and c), we have to compute the moment $\mathbf{M}_{O'} = \mathbf{r}' \times \mathbf{F}$ of $\mathbf{F}$ about O' and add a new force-couple system consisting of $\mathbf{F}$ and the couple vector $\mathbf{M}_{O'}$ at O' . We can obtain the relation between the moments of $\mathbf{F}$ about O and O' as

$$\mathbf{M}_{O'} = \mathbf{r}' \times \mathbf{F} = (\mathbf{r} + \mathbf{s}) \times \mathbf{F} = \mathbf{r} \times \mathbf{F} + \mathbf{s} \times \mathbf{F}$$

$$\mathbf{M}_{O'} = \mathbf{M}_O + \mathbf{s} \times \mathbf{F} \quad (3.49)$$

where $\mathbf{s}$ is the vector joining O' to O . Thus, we obtain the moment $\mathbf{M}_{O'}$ of $\mathbf{F}$ about O' by adding to the moment $\mathbf{M}_O$ of $\mathbf{F}$ about O the vector product $\mathbf{s} \times \mathbf{F}$, representing the moment about O' of the force $\mathbf{F}$ applied at O .

We also could have established this result by observing that, in order to transfer to O' the force-couple system attached at O (Fig. 3.35b and c), we could freely move the couple vector $\mathbf{M}_O$ to O' . However, to move force $\mathbf{F}$ from O to O' , we need to add to $\mathbf{F}$ a couple vector whose moment is equal to the moment about O' of force $\mathbf{F}$ applied at O . Thus, the couple vector $\mathbf{M}_{O'}$ must be the sum of $\mathbf{M}_O$ and the vector $\mathbf{s} \times \mathbf{F}$.

As noted here, the force-couple system obtained by transferring a force $\mathbf{F}$ from a point A to a point O consists of $\mathbf{F}$ and a couple vector $\mathbf{M}_O$ perpendicular to $\mathbf{F}$. Conversely, any force-couple system consisting of a force $\mathbf{F}$ and a couple vector $\mathbf{M}_O$ that are *mutually perpendicular* can be replaced by a single equivalent force. This is done by moving force $\mathbf{F}$ in the plane perpendicular to $\mathbf{M}_O$ until its moment about O is equal to the moment of the couple being replaced.



Photo 3.2 The force exerted by each hand on the wrench could be replaced with an equivalent force-couple system acting on the nut.

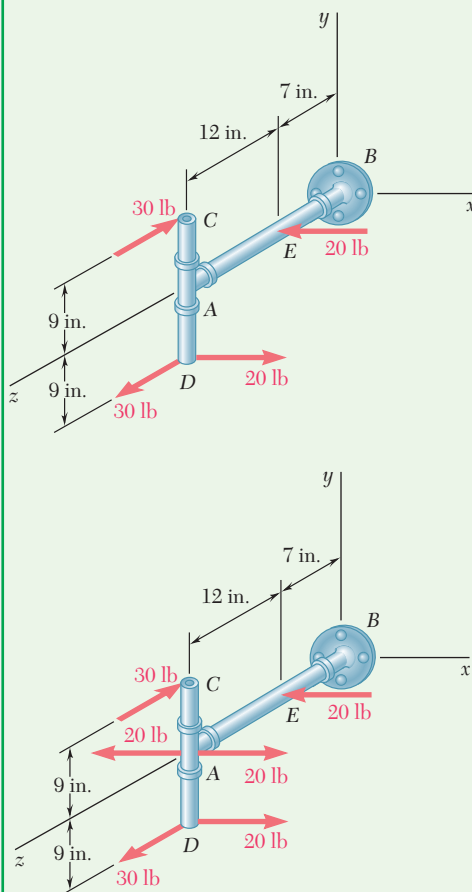


Fig. 1 Placing two equal and opposite 20-lb forces at A to simplify calculations.

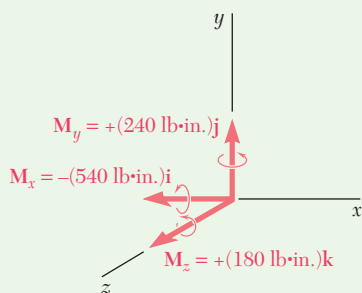


Fig. 2 The three couples represented as couple vectors.

Sample Problem 3.6

Determine the components of the single couple equivalent to the two couples shown.

STRATEGY: Look for ways to add equal and opposite forces to the diagram that, along with already known perpendicular distances, will produce new couples with moments along the coordinate axes. These can be combined into a single equivalent couple.

MODELING: You can simplify the computations by attaching two equal and opposite 20-lb forces at A (Fig. 1). This enables you to replace the original 20-lb-force couple by two new 20-lb-force couples: one lying in the zx plane and the other in a plane parallel to the xy plane.

ANALYSIS: You can represent these three couples by three couple vectors $\mathbf{M}_x$, $\mathbf{M}_y$, and $\mathbf{M}_z$ directed along the coordinate axes (Fig. 2). The corresponding moments are

$$M_x = -(30 \text{ lb})(18 \text{ in.}) = -540 \text{ lb}\cdot\text{in.}$$

$$M_y = +(20 \text{ lb})(12 \text{ in.}) = +240 \text{ lb}\cdot\text{in.}$$

$$M_z = +(20 \text{ lb})(9 \text{ in.}) = +180 \text{ lb}\cdot\text{in.}$$

These three moments represent the components of the single couple $\mathbf{M}$ equivalent to the two given couples. You can write $\mathbf{M}$ as

$$\mathbf{M} = -(540 \text{ lb}\cdot\text{in.})\mathbf{i} + (240 \text{ lb}\cdot\text{in.})\mathbf{j} + (180 \text{ lb}\cdot\text{in.})\mathbf{k} \quad \blacktriangleleft$$

REFLECT and THINK: You can also obtain the components of the equivalent single couple $\mathbf{M}$ by computing the sum of the moments of the four given forces about an arbitrary point. Selecting point D, the moment is (Fig. 3)

$$\mathbf{M} = \mathbf{M}_D = (18 \text{ in.})\mathbf{j} \times (-30 \text{ lb})\mathbf{k} + [(9 \text{ in.})\mathbf{j} - (12 \text{ in.})\mathbf{k}] \times (-20 \text{ lb})\mathbf{i}$$

After computing the various cross products, you get the same result, as

$$\mathbf{M} = -(540 \text{ lb}\cdot\text{in.})\mathbf{i} + (240 \text{ lb}\cdot\text{in.})\mathbf{j} + (180 \text{ lb}\cdot\text{in.})\mathbf{k} \quad \blacktriangleleft$$

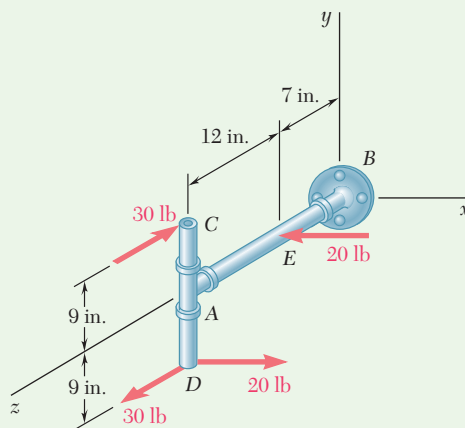
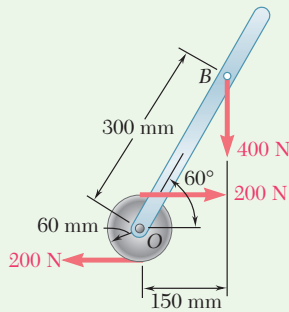


Fig. 3 Using the given force system, the equivalent single couple can also be determined from the sum of moments of the forces about any point, such as point D.



Sample Problem 3.7

Replace the couple and force shown by an equivalent single force applied to the lever. Determine the distance from the shaft to the point of application of this equivalent force.

STRATEGY: First replace the given force and couple by an equivalent force-couple system at O . By moving the force of this force-couple system a distance that creates the same moment as the couple, you can then replace the system with one equivalent force.

MODELING and ANALYSIS: To replace the given force and couple, move the force $\mathbf{F} = -(400 \text{ N})\mathbf{j}$ to O , and at the same time, add a couple of moment $\mathbf{M}_O$ that is equal to the moment about O of the force in its original position (Fig. 1). Thus,

$$\begin{aligned}\mathbf{M}_O &= \overrightarrow{OB} \times \mathbf{F} = [(0.150 \text{ m})\mathbf{i} + (0.260 \text{ m})\mathbf{j}] \times (-400 \text{ N})\mathbf{j} \\ &= -(60 \text{ N}\cdot\text{m})\mathbf{k}\end{aligned}$$

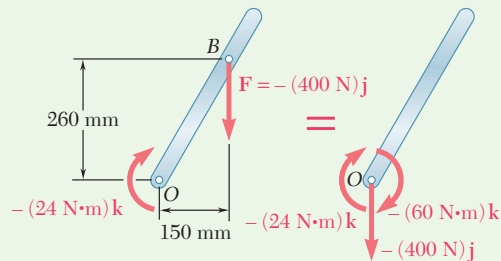


Fig. 1 Replacing given force and couple with an equivalent force-couple at O .

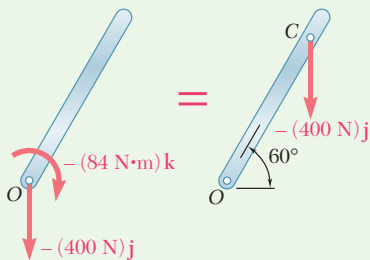


Fig. 2 Resultant couple eliminated by moving force $\mathbf{F}$.

When you add this new couple to the couple of moment $-(24 \text{ N}\cdot\text{m})\mathbf{k}$ formed by the two 200-N forces, you obtain a couple of moment $-(84 \text{ N}\cdot\text{m})\mathbf{k}$ (Fig. 2). You can replace this last couple by applying $\mathbf{F}$ at a point C chosen in such a way that

$$\begin{aligned}-(84 \text{ N}\cdot\text{m})\mathbf{k} &= \overrightarrow{OC} \times \mathbf{F} \\ &= [(OC) \cos 60^\circ \mathbf{i} + (OC) \sin 60^\circ \mathbf{j}] \times (-400 \text{ N})\mathbf{j} \\ &= -(OC) \cos 60^\circ (400 \text{ N})\mathbf{k}\end{aligned}$$

The result is

$$(OC) \cos 60^\circ = 0.210 \text{ m} = 210 \text{ mm} \quad OC = 420 \text{ mm} \quad \blacktriangleleft$$

REFLECT and THINK: Since the effect of a couple does not depend on its location, you can move the couple of moment $-(24 \text{ N}\cdot\text{m})\mathbf{k}$ to B , obtaining a force-couple system at B (Fig. 3). Now you can eliminate this couple by applying $\mathbf{F}$ at a point C chosen in such a way that

$$\begin{aligned}-(24 \text{ N}\cdot\text{m})\mathbf{k} &= \overrightarrow{BC} \times \mathbf{F} \\ &= -(BC) \cos 60^\circ (400 \text{ N})\mathbf{k}\end{aligned}$$

The conclusion is

$$\begin{aligned}(BC) \cos 60^\circ &= 0.060 \text{ m} = 60 \text{ mm} & BC &= 120 \text{ mm} \\ OC &= OB + BC = 300 \text{ mm} + 120 \text{ mm} & OC &= 420 \text{ mm} \quad \blacktriangleleft\end{aligned}$$

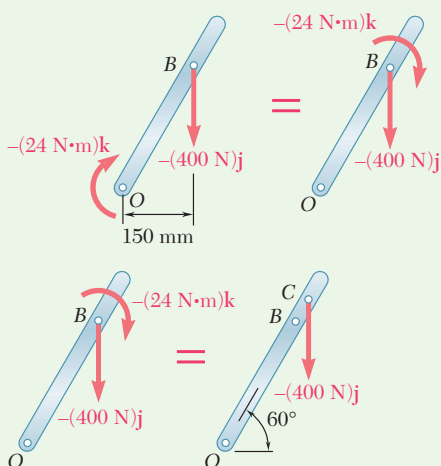


Fig. 3 Couple can be moved to B with no change in effect. This couple can then be eliminated by moving force $\mathbf{F}$.

SOLVING PROBLEMS ON YOUR OWN

In this section, we discussed the properties of *couples*. To solve the following problems, remember that the net effect of a couple is to produce a moment $\mathbf{M}$. Since this moment is independent of the point about which it is computed, $\mathbf{M}$ is a *free vector* and remains unchanged if you move it from point to point. Also, two couples are *equivalent* (that is, they have the same effect on a given rigid body) if they produce the same moment.

When determining the moment of a couple, all previous techniques for computing moments apply. Also, since the moment of a couple is a free vector, you should compute its value relative to the most convenient point.

Because the only effect of a couple is to produce a moment, it is possible to represent a couple with a vector, called the *couple vector*, that is equal to the moment of the couple. The couple vector is a free vector and is represented by a special symbol, $\mathcal{C}$, to distinguish it from force vectors.

In solving the problems in this section, you will be called upon to perform the following operations:

1. Adding two or more couples. This results in a new couple, the moment of which is obtained by adding vectorially the moments of the given couples [Sample Prob. 3.6].

2. Replacing a force with an equivalent force-couple system at a specified point. As explained in Sec. 3.3E, the force of a force-couple system is equal to the original force, whereas the required couple vector is equal to the moment of the original force about the given point. In addition, it is important to note that the force and the couple vector are perpendicular to each other. Conversely, it follows that a force-couple system can be reduced to a single force only if the force and couple vector are mutually perpendicular (see the next paragraph).

3. Replacing a force-couple system (with $\mathbf{F}$ perpendicular to $\mathbf{M}$) with a single equivalent force. The requirement that $\mathbf{F}$ and $\mathbf{M}$ be mutually perpendicular is satisfied in all two-dimensional problems. The single equivalent force is equal to $\mathbf{F}$ and is applied in such a way that its moment about the original point of application is equal to $\mathbf{M}$ [Sample Prob. 3.7].

Problems

- 3.70** Two 80-N forces are applied as shown to the corners B and D of a rectangular plate. (a) Determine the moment of the couple formed by the two forces by resolving each force into horizontal and vertical components and adding the moments of the two resulting couples. (b) Use the result obtained to determine the perpendicular distance between lines BE and DF .
- 3.71** Two parallel 40-N forces are applied to a lever as shown. Determine the moment of the couple formed by the two forces (a) by resolving each force into horizontal and vertical components and adding the moments of the two resulting couples, (b) by using the perpendicular distance between the two forces, (c) by summing the moments of the two forces about point A .

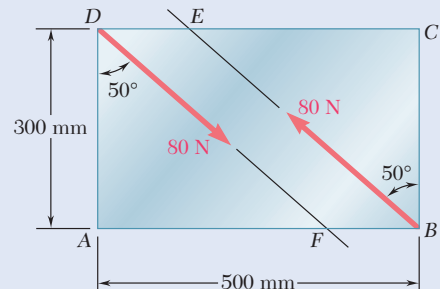


Fig. P3.70

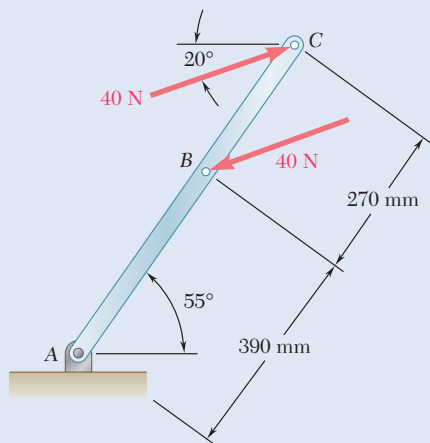


Fig. P3.71

- 3.72** Four $\frac{1}{2}$ -in.-diameter pegs are attached to a board as shown. Two strings are passed around the pegs and pulled with the forces indicated. (a) Determine the resultant couple acting on the board. (b) If only one string is used, around which pegs should it pass and in what directions should it be pulled to create the same couple with the minimum tension in the string? (c) What is the value of that minimum tension?
- 3.73** Four pegs of the same diameter are attached to a board as shown. Two strings are passed around the pegs and pulled with the forces indicated. Determine the diameter of the pegs knowing that the resultant couple applied to the board is 1132.5 lb-in. counterclockwise.

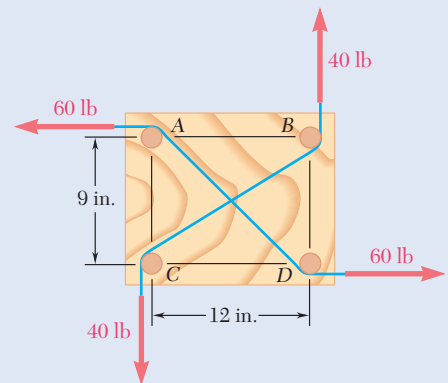


Fig. P3.72 and P3.73

- 3.74** A piece of plywood in which several holes are being drilled successively has been secured to a workbench by means of two nails. Knowing that the drill exerts a 12-N·m couple on the piece of plywood, determine the magnitude of the resulting forces applied to the nails if they are located (a) at A and B, (b) at B and C, (c) at A and C.

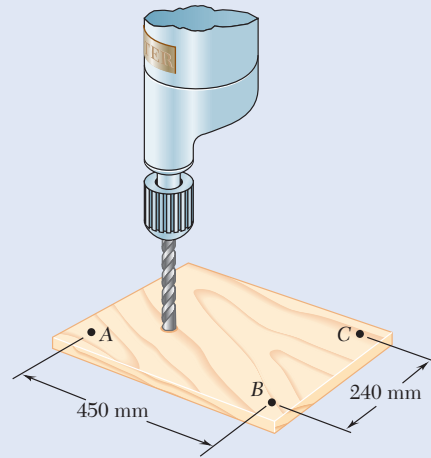


Fig. P3.74

- 3.75** The two shafts of a speed-reducer unit are subjected to couples of magnitude $M_1 = 15$ lb·ft and $M_2 = 3$ lb·ft, respectively. Replace the two couples with a single equivalent couple, specifying its magnitude and the direction of its axis.

- 3.76** If $P = 0$ in the figure, replace the two remaining couples with a single equivalent couple, specifying its magnitude and the direction of its axis.

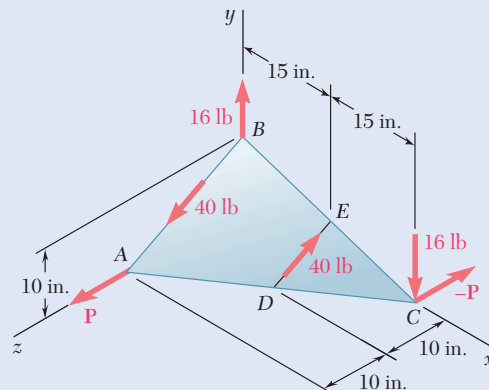


Fig. P3.76 and P3.77

- 3.77** If $P = 20$ lb in the figure, replace the three couples with a single equivalent couple, specifying its magnitude and the direction of its axis.

- 3.78** Replace the two couples shown with a single equivalent couple, specifying its magnitude and the direction of its axis.

- 3.79** Solve Prob. 3.78, assuming that two 10-N vertical forces have been added, one acting upward at C and the other downward at B.

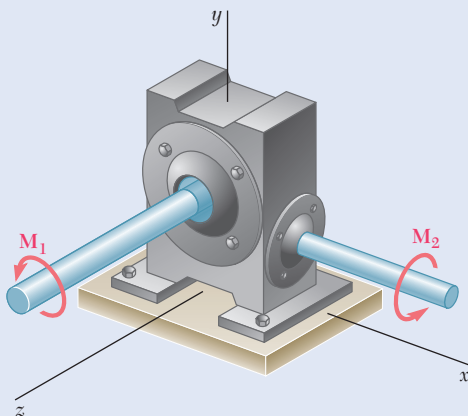


Fig. P3.75

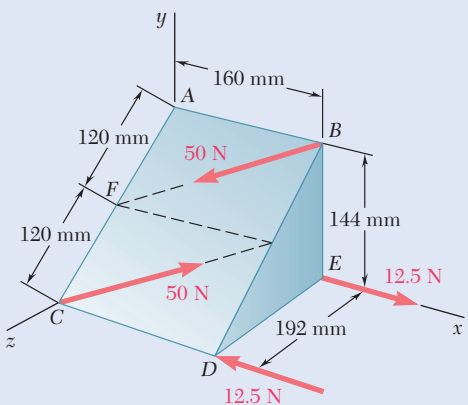


Fig. P3.78

- 3.80** Shafts A and B connect the gear box to the wheel assemblies of a tractor, and shaft C connects it to the engine. Shafts A and B lie in the vertical yz plane, while shaft C is directed along the x axis. Replace the couples applied to the shafts by a single equivalent couple, specifying its magnitude and the direction of its axis.

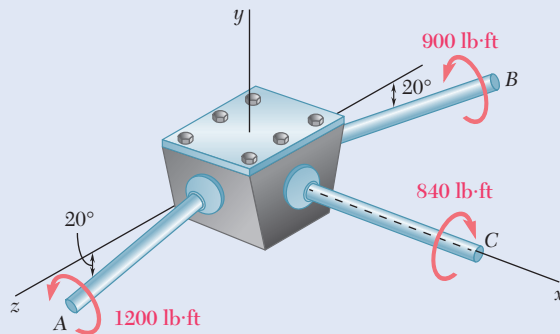


Fig. P3.80

- 3.81** A 500-N force is applied to a bent plate as shown. Determine (a) an equivalent force-couple system at B , (b) an equivalent system formed by a vertical force at A and a force at B .

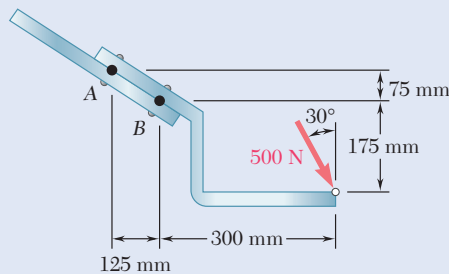


Fig. P3.81

- 3.82** The tension in the cable attached to the end C of an adjustable boom ABC is 560 lb. Replace the force exerted by the cable at C with an equivalent force-couple system (a) at A , (b) at B .

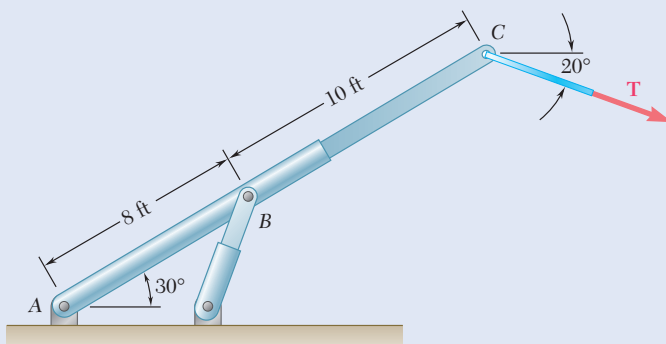


Fig. P3.82

- 3.83** A dirigible is tethered by a cable attached to its cabin at B . If the tension in the cable is 1040 N, replace the force exerted by the cable at B with an equivalent system formed by two parallel forces applied at A and C .

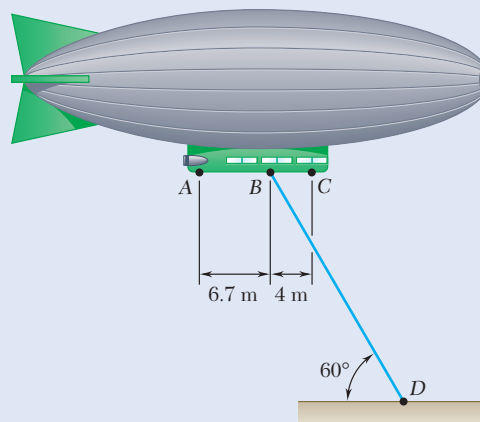


Fig. P3.83

- 3.84** A 30-lb vertical force $\mathbf{P}$ is applied at A to the bracket shown, which is held by screws at B and C . (a) Replace $\mathbf{P}$ with an equivalent force-couple system at B . (b) Find the two horizontal forces at B and C that are equivalent to the couple obtained in part a .

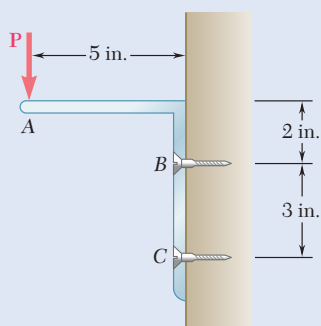


Fig. P3.84

- 3.85** A worker tries to move a rock by applying a 360-N force to a steel bar as shown. (a) Replace that force with an equivalent force-couple system at D . (b) Two workers attempt to move the same rock by applying a vertical force at A and another force at D . Determine these two forces if they are to be equivalent to the single force of part a .

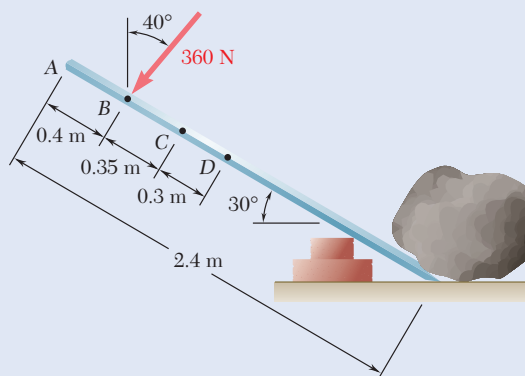


Fig. P3.85 and P3.86

- 3.86** A worker tries to move a rock by applying a 360-N force to a steel bar as shown. If two workers attempt to move the same rock by applying a force at A and a parallel force at C , determine these two forces so that they will be equivalent to the single 360-N force shown in the figure.

- 3.87** The shearing forces exerted on the cross section of a steel channel can be represented by a 900-N vertical force and two 250-N horizontal forces as shown. Replace this force and couple with a single force $\mathbf{F}$ applied at point C , and determine the distance x from C to line BD . (Point C is defined as the *shear center* of the section.)

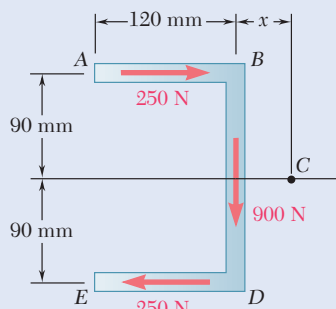


Fig. P3.87

3.88 A force and a couple are applied as shown to the end of a cantilever beam. (a) Replace this system with a single force $\mathbf{F}$ applied at point C , and determine the distance d from C to a line drawn through points D and E . (b) Solve part a if the directions of the two 360-N forces are reversed.

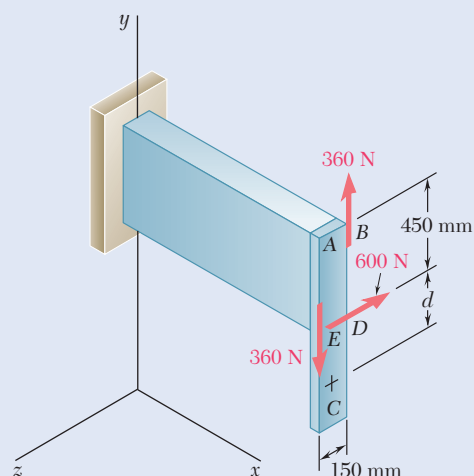


Fig. P3.88

3.89 Three control rods attached to a lever ABC exert on it the forces shown. (a) Replace the three forces with an equivalent force-couple system at B . (b) Determine the single force that is equivalent to the force-couple system obtained in part a , and specify its point of application on the lever.

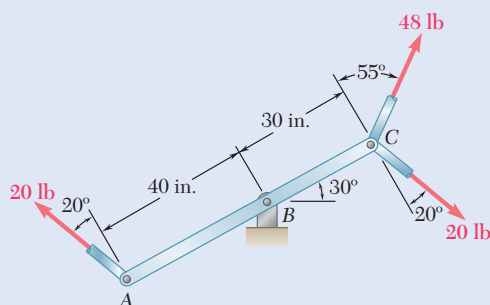


Fig. P3.89

3.90 A rectangular plate is acted upon by the force and couple shown. This system is to be replaced with a single equivalent force. (a) For $\alpha = 40^\circ$, specify the magnitude and line of action of the equivalent force. (b) Specify the value of α if the line of action of the equivalent force is to intersect line CD 300 mm to the right of D .

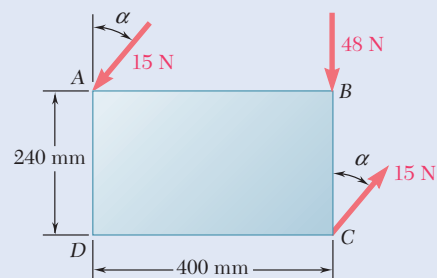


Fig. P3.90

3.91 While tapping a hole, a machinist applies the horizontal forces shown to the handle of the tap wrench. Show that these forces are equivalent to a single force, and specify, if possible, the point of application of the single force on the handle.

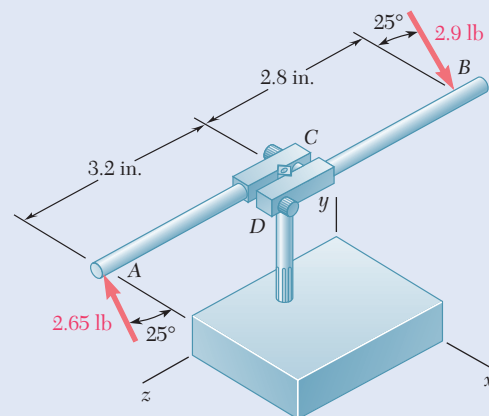


Fig. P3.91

3.92 A hexagonal plate is acted upon by the force $\mathbf{P}$ and the couple shown. Determine the magnitude and the direction of the smallest force $\mathbf{P}$ for which this system can be replaced with a single force at E .

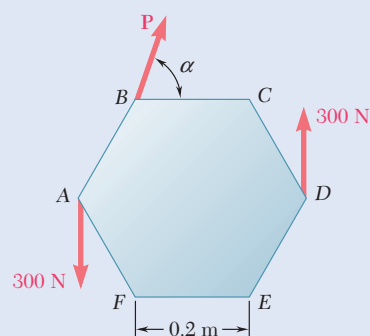


Fig. P3.92

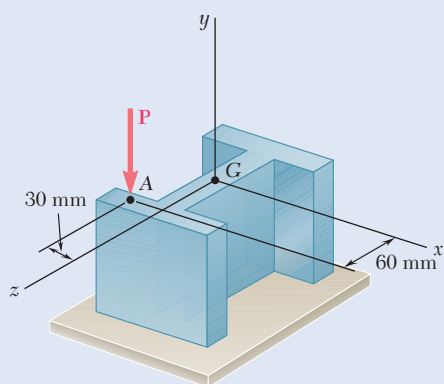


Fig. P3.93

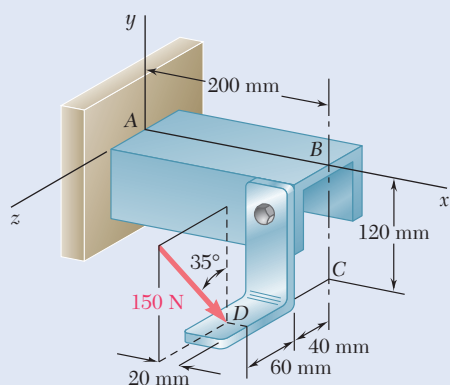


Fig. P3.95

3.93 Replace the 250-kN force $\mathbf{P}$ with an equivalent force-couple system at G .

3.94 A 2.6-kip force is applied at point D of the cast-iron post shown. Replace that force with an equivalent force-couple system at the center A of the base section.

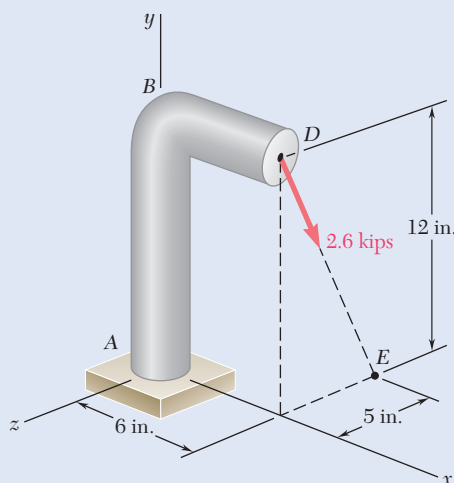


Fig. P3.94

3.95 Replace the 150-N force with an equivalent force-couple system at A .

3.96 To keep a door closed, a wooden stick is wedged between the floor and the doorknob. The stick exerts at B a 175-N force directed along line AB . Replace that force with an equivalent force-couple system at C .

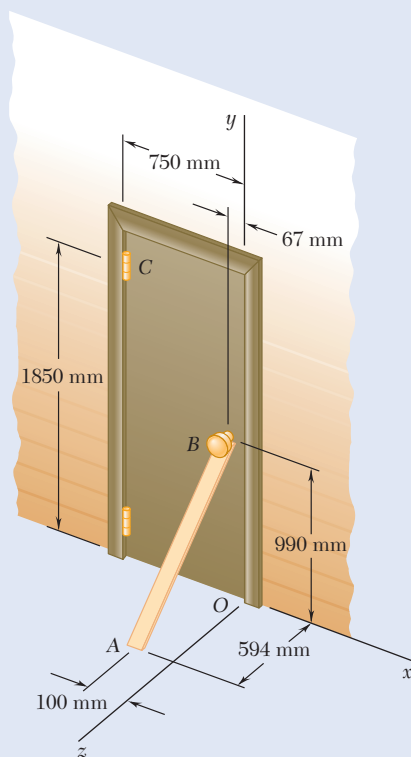


Fig. P3.96

- 3.97** A 46-lb force $\mathbf{F}$ and a 2120-lb-in. couple $\mathbf{M}$ are applied to corner A of the block shown. Replace the given force-couple system with an equivalent force-couple system at corner H .

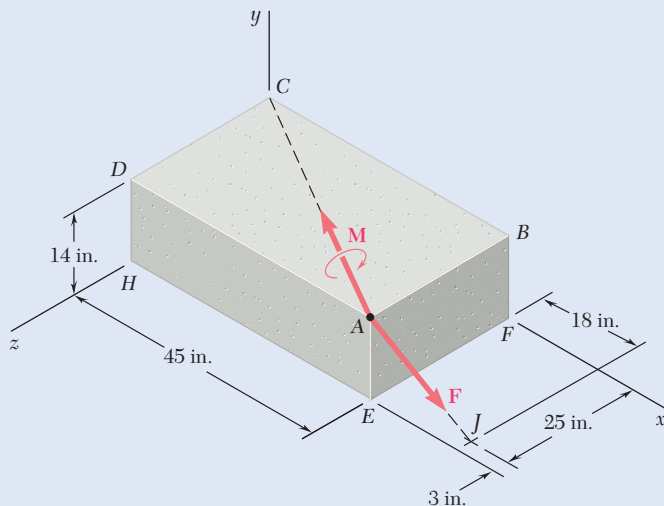


Fig. P3.97

- 3.98** A 110-N force acting in a vertical plane parallel to the yz plane is applied to the 220-mm-long horizontal handle AB of a socket wrench. Replace the force with an equivalent force-couple system at the origin O of the coordinate system.
- 3.99** An antenna is guyed by three cables as shown. Knowing that the tension in cable AB is 288 lb, replace the force exerted at A by cable AB with an equivalent force-couple system at the center O of the base of the antenna.

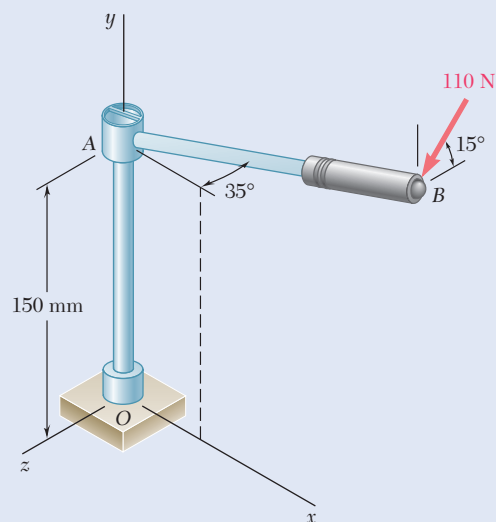


Fig. P3.98

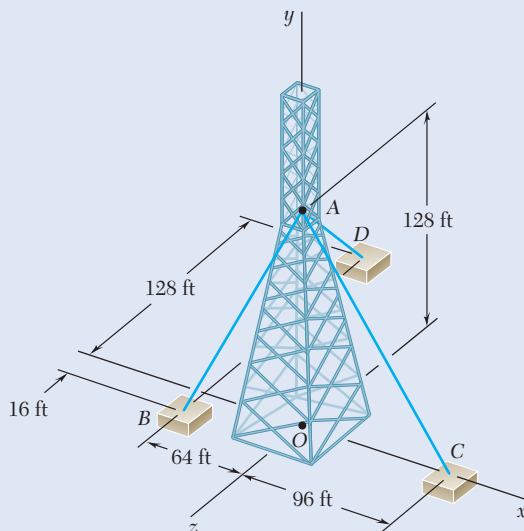


Fig. P3.99 and P3.100

- 3.100** An antenna is guyed by three cables as shown. Knowing that the tension in cable AD is 270 lb, replace the force exerted at A by cable AD with an equivalent force-couple system at the center O of the base of the antenna.

3.4 SIMPLIFYING SYSTEMS OF FORCES

We saw in the preceding section that we can replace a force acting on a rigid body with a force-couple system that may be easier to analyze. However, the true value of a force-couple system is that we can use it to replace not just one force but a system of forces to simplify analysis and calculations.

3.4A Reducing a System of Forces to a Force-Couple System

Consider a system of forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$, acting on a rigid body at the points $A_1, A_2, A_3, \dots$, defined by the position vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$, etc. (Fig. 3.36a). As seen in the preceding section, we can move $\mathbf{F}_1$ from A_1 to a given point O if we add a couple of moment $\mathbf{M}_1$ equal to the moment $\mathbf{r}_1 \times \mathbf{F}_1$ of $\mathbf{F}_1$ about O . Repeating this procedure with $\mathbf{F}_2, \mathbf{F}_3, \dots$, we obtain the system shown in Fig. 3.36b, which consists of the original forces, now acting at O , and the added couple vectors. Since the forces are now concurrent, they can be added vectorially and replaced by their resultant $\mathbf{R}$. Similarly, the couple vectors $\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3, \dots$, can be added vectorially and replaced by a single couple vector $\mathbf{M}_O^R$. Thus,

We can reduce any system of forces, however complex, to an equivalent force-couple system acting at a given point O .

Note that, although each of the couple vectors $\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3, \dots$ in Fig. 3.36b is perpendicular to its corresponding force, the resultant force $\mathbf{R}$ and the resultant couple vector $\mathbf{M}_O^R$ shown in Fig. 3.36c are not, in general, perpendicular to each other.

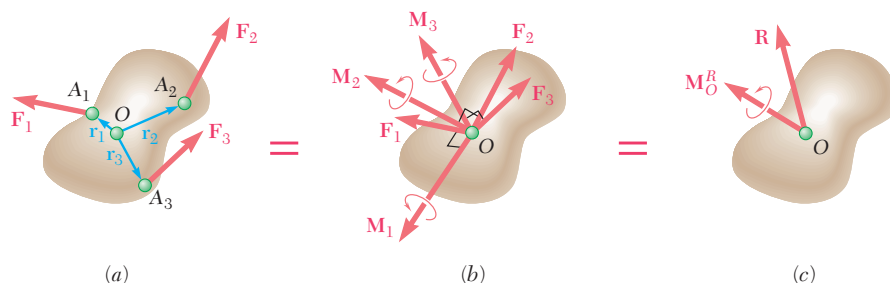


Fig. 3.36 Reducing a system of forces to a force-couple system. (a) Initial system of forces; (b) all the forces moved to act at point O , with couple vectors added; (c) all the forces reduced to a resultant force vector and all the couple vectors reduced to a resultant couple vector.

The equivalent force-couple system is defined by

Force-couple system

$$\mathbf{R} = \Sigma \mathbf{F} \quad \mathbf{M}_O^R = \Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) \quad (3.50)$$

These equations state that we obtain force $\mathbf{R}$ by adding all of the forces of the system, whereas we obtain the moment of the resultant couple vector $\mathbf{M}_O^R$, called the **moment resultant** of the system, by adding the moments about O of all the forces of the system.

Once we have reduced a given system of forces to a force and a couple at a point O , we can replace it with a force and a couple at another point O' . The resultant force $\mathbf{R}$ will remain unchanged, whereas the new moment resultant $\mathbf{M}_{O'}^R$ will be equal to the sum of $\mathbf{M}_O^R$ and the moment about O' of force $\mathbf{R}$ attached at O (Fig. 3.37). We have

$$\mathbf{M}_{O'}^R = \mathbf{M}_O^R + \mathbf{s} \times \mathbf{R} \quad (3.51)$$

In practice, the reduction of a given system of forces to a single force $\mathbf{R}$ at O and a couple vector $\mathbf{M}_O^R$ is carried out in terms of components. Resolving each position vector $\mathbf{r}$ and each force $\mathbf{F}$ of the system into rectangular components, we have

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (3.52)$$

$$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k} \quad (3.53)$$

Substituting for $\mathbf{r}$ and $\mathbf{F}$ in Eq. (3.50) and factoring out the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, we obtain $\mathbf{R}$ and $\mathbf{M}_O^R$ in the form

$$\mathbf{R} = R_x\mathbf{i} + R_y\mathbf{j} + R_z\mathbf{k} \quad \mathbf{M}_O^R = M_x^R\mathbf{i} + M_y^R\mathbf{j} + M_z^R\mathbf{k} \quad (3.54)$$

The components R_x , R_y , and R_z represent, respectively, the sums of the x , y , and z components of the given forces and measure the tendency of the system to impart to the rigid body a translation in the x , y , or z direction. Similarly, the components M_x^R , M_y^R , and M_z^R represent, respectively, the sum of the moments of the given forces about the x , y , and z axes and measure the tendency of the system to impart to the rigid body a rotation about the x , y , or z axis.

If we need to know the magnitude and direction of force $\mathbf{R}$, we can obtain them from the components R_x , R_y , and R_z by means of the relations in Eqs. (2.18) and (2.19) of Sec. 2.4A. Similar computations yield the magnitude and direction of the couple vector $\mathbf{M}_O^R$.

3.4B Equivalent and Equipollent Systems of Forces

We have just seen that any system of forces acting on a rigid body can be reduced to a force-couple system at a given point O . This equivalent force-couple system characterizes completely the effect of the given force system on the rigid body.

Two systems of forces are equivalent if they can be reduced to the same force-couple system at a given point O .

Recall that the force-couple system at O is defined by the relations in Eq. (3.50). Therefore, we can state that

Two systems of forces, $F_1, F_2, F_3, \dots$, and $F'_1, F'_2, F'_3, \dots$, that act on the same rigid body are equivalent if, and only if, the sums of the forces and the sums of the moments about a given point O of the forces of the two systems are, respectively, equal.

Mathematically, the necessary and sufficient conditions for the two systems of forces to be equivalent are

Conditions for equivalent systems of forces

$$\Sigma \mathbf{F} = \Sigma \mathbf{F}' \quad \text{and} \quad \Sigma \mathbf{M}_O = \Sigma \mathbf{M}'_O \quad (3.55)$$

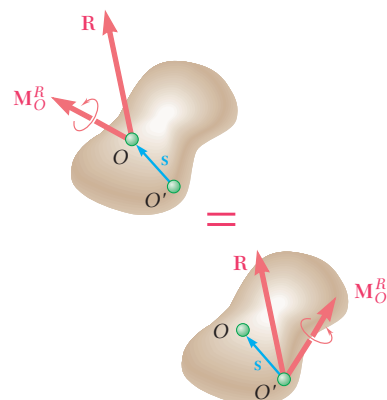


Fig. 3.37 Once a system of forces has been reduced to a force-couple system at one point, we can replace it with an equivalent force-couple system at another point. The force resultant stays the same, but we have to add the moment of the resultant force about the new point to the resultant couple vector.

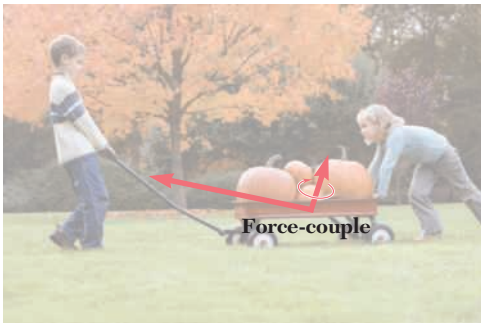
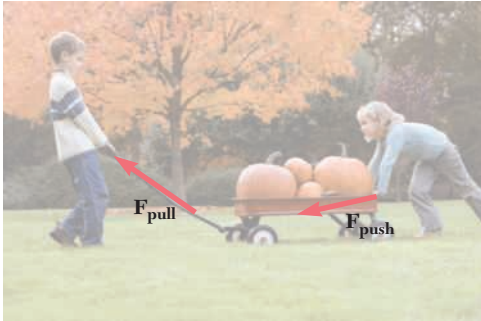


Photo 3.3 The forces exerted by the children upon the wagon can be replaced with an equivalent force-couple system when analyzing the motion of the wagon.

Note that to prove that two systems of forces are equivalent, we must establish the second of the relations in Eq. (3.55) with respect to *only one point* O . It will hold, however, with respect to *any point* if the two systems are equivalent.

Resolving the forces and moments in Eqs. (3.55) into their rectangular components, we can express the necessary and sufficient conditions for the equivalence of two systems of forces acting on a rigid body as

$$\begin{array}{lll} \Sigma F_x = \Sigma F'_x & \Sigma F_y = \Sigma F'_y & \Sigma F_z = \Sigma F'_z \\ \Sigma M_x = \Sigma M'_x & \Sigma M_y = \Sigma M'_y & \Sigma M_z = \Sigma M'_z \end{array} \quad (3.56)$$

These equations have a simple physical significance. They express that

Two systems of forces are equivalent if they tend to impart to the rigid body (1) the same translation in the x , y , and z directions, respectively, and (2) the same rotation about the x , y , and z axes, respectively.

In general, when two systems of vectors satisfy Eqs. (3.55) or (3.56), i.e., when their resultants and their moment resultants about an arbitrary point O are respectively equal, the two systems are said to be **equipollent**. The result just established can thus be restated as

If two systems of forces acting on a rigid body are equipollent, they are also equivalent.

It is important to note that this statement does not apply to *any* system of vectors. Consider, for example, a system of forces acting on a set of independent particles that do *not* form a rigid body. A different system of forces acting on the same particles may happen to be equipollent to the first one; i.e., it may have the same resultant and the same moment resultant. Yet, since different forces now act on the various particles, their effects on these particles are different; the two systems of forces, while equipollent, are *not equivalent*.

3.4C Further Reduction of a System of Forces

We have now seen that any given system of forces acting on a rigid body can be reduced to an equivalent force-couple system at O , consisting of a force $\mathbf{R}$ equal to the sum of the forces of the system, and a couple vector $\mathbf{M}_O^R$ of moment equal to the moment resultant of the system.

When $\mathbf{R} = 0$, the force-couple system reduces to the couple vector $\mathbf{M}_O^R$. The given system of forces then can be reduced to a single couple called the **resultant couple** of the system.

What are the conditions under which a given system of forces can be reduced to a single force? It follows from the preceding section that we can replace the force-couple system at O by a single force $\mathbf{R}$ acting along a new line of action if $\mathbf{R}$ and $\mathbf{M}_O^R$ are mutually perpendicular. The systems of forces that can be reduced to a single force, or *resultant*, are therefore the systems for which force $\mathbf{R}$ and the couple vector $\mathbf{M}_O^R$ are mutually perpendicular. This condition is *generally not satisfied* by systems of forces in space, but it is *satisfied* by systems consisting of (1) concurrent forces, (2) coplanar forces, or (3) parallel forces. Let's look at each case separately.

1. **Concurrent forces** act at the same point; therefore, we can add them directly to obtain their resultant $\mathbf{R}$. Thus, they always reduce to a single force. Concurrent forces were discussed in detail in Chap. 2.

2. Coplanar forces act in the same plane, which we assume to be the plane of the figure (Fig. 3.38a). The sum $\mathbf{R}$ of the forces of the system also lies in the plane of the figure, whereas the moment of each force about O and thus the moment resultant $\mathbf{M}_O^R$ are perpendicular to that plane. The force-couple system at O consists, therefore, of a force $\mathbf{R}$ and a couple vector $\mathbf{M}_O^R$ that are mutually perpendicular (Fig. 3.38b).[†] We can reduce them to a single force $\mathbf{R}$ by moving $\mathbf{R}$ in the plane of the figure until its moment about O becomes equal to $\mathbf{M}_O^R$. The distance from O to the line of action of $\mathbf{R}$ is $d = M_O^R/R$ (Fig. 3.38c).

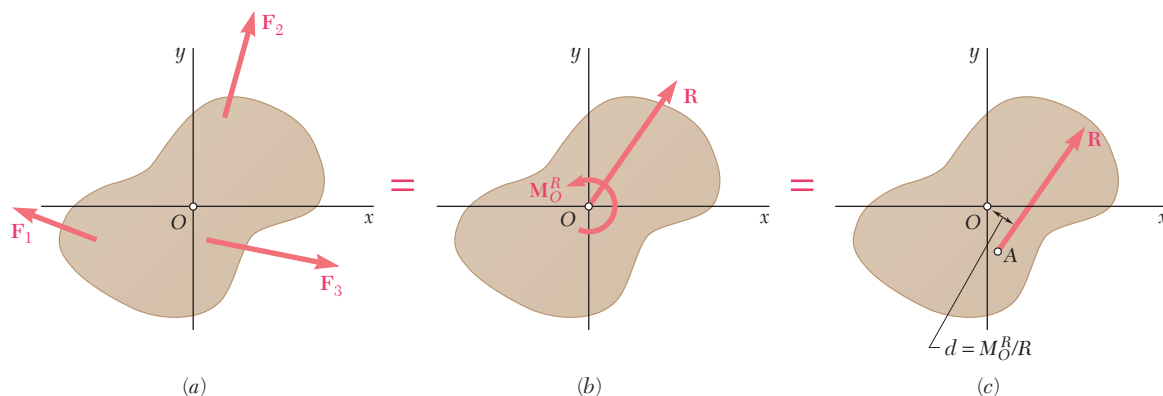


Fig. 3.38 Reducing a system of coplanar forces. (a) Initial system of forces; (b) equivalent force-couple system at O ; (c) moving the resultant force to a point A such that the moment of $\mathbf{R}$ about O equals the couple vector.

As noted earlier, the reduction of a system of forces is considerably simplified if we resolve the forces into rectangular components. The force-couple system at O is then characterized by the components (Fig. 3.39a)

$$R_x = \sum F_x \quad R_y = \sum F_y \quad M_z^R = M_O^R = \sum M_O \quad (3.57)$$

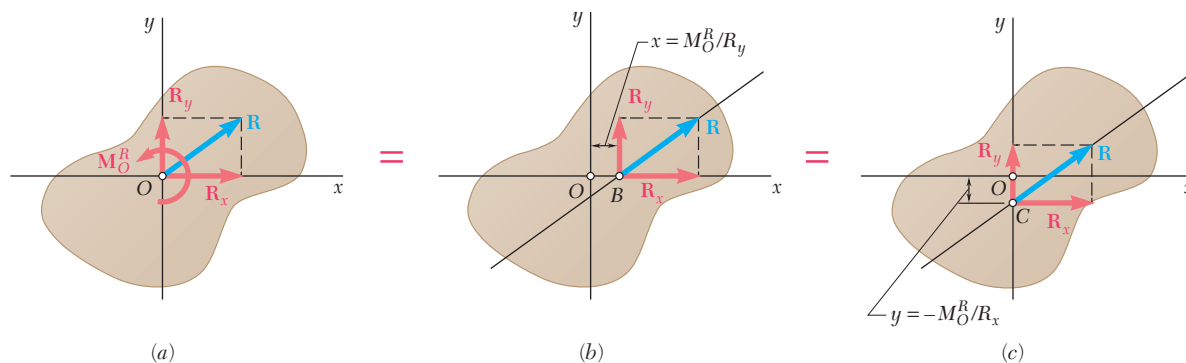


Fig. 3.39 Reducing a system of coplanar forces by using rectangular components. (a) From Fig. 3.38(b), resolve the resultant into components along the x and y axes; (b) determining the x intercept of the final line of action of the resultant; (c) determining the y intercept of the final line of action of the resultant.

[†]Because the couple vector $\mathbf{M}_O^R$ is perpendicular to the plane of the figure, we represent it by the symbol $\curvearrowright$. A counterclockwise couple $\curvearrowright$ represents a vector pointing out of the page and a clockwise couple $\curvearrowleft$ represents a vector pointing into the page.

To reduce the system to a single force $\mathbf{R}$, the moment of $\mathbf{R}$ about O must be equal to $\mathbf{M}_O^R$. If we denote the coordinates of the point of application of the resultant by x and y and apply equation (3.22) of Sec. 3.1F, we have

$$xR_y - yR_x = M_O^R$$

This represents the equation of the line of action of $\mathbf{R}$. We can also determine the x and y intercepts of the line of action of the resultant directly by noting that $\mathbf{M}_O^R$ must be equal to the moment about O of the y component of $\mathbf{R}$ when $\mathbf{R}$ is attached at B (Fig. 3.39b) and to the moment of its x component when $\mathbf{R}$ is attached at C (Fig. 3.39c).

3. **Parallel forces** have parallel lines of action and may or may not have the same sense. Assuming here that the forces are parallel to the y axis (Fig. 3.40a), we note that their sum $\mathbf{R}$ is also parallel to the y axis.

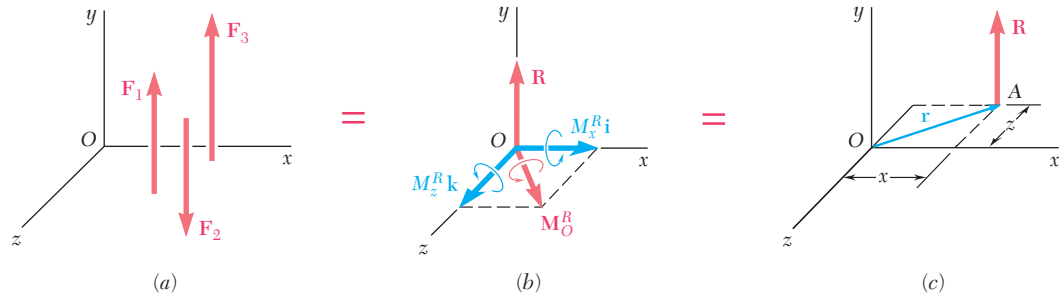


Fig. 3.40 Reducing a system of parallel forces. (a) Initial system of forces; (b) equivalent force-couple system at O , resolved into components; (c) moving $\mathbf{R}$ to point A , chosen so that the moment of $\mathbf{R}$ about O equals the resultant moment about O .

On the other hand, since the moment of a given force must be perpendicular to that force, the moment about O of each force of the system and thus the moment resultant $\mathbf{M}_O^R$ lie in the xz plane. The force-couple system at O consists, therefore, of a force $\mathbf{R}$ and a couple vector $\mathbf{M}_O^R$ that are mutually perpendicular (Fig. 3.40b). We can reduce them to a single force $\mathbf{R}$ (Fig. 3.40c) or, if $\mathbf{R} = 0$, to a single couple of moment $\mathbf{M}_O^R$.

In practice, the force-couple system at O is characterized by the components

$$R_y = \sum F_y \quad M_x^R = \sum M_x \quad M_z^R = \sum M_z \quad (3.58)$$

The reduction of the system to a single force can be carried out by moving $\mathbf{R}$ to a new point of application $A(x, 0, z)$, which is chosen so that the moment of $\mathbf{R}$ about O is equal to $\mathbf{M}_O^R$.

$$\mathbf{r} \times \mathbf{R} = \mathbf{M}_O^R$$

$$(x\mathbf{i} + z\mathbf{k}) \times R_y\mathbf{j} = M_x^R\mathbf{i} + M_z^R\mathbf{k}$$



Photo 3.4 The parallel wind forces acting on the highway signs can be reduced to a single equivalent force. Determining this force can simplify the calculation of the forces acting on the supports of the frame to which the signs are attached.

By computing the vector products and equating the coefficients of the corresponding unit vectors in both sides of the equation, we obtain two scalar equations that define the coordinates of A :

$$-zR_y = M_x^R \quad \text{and} \quad xR_y = M_z^R$$

These equations express the fact that the moments of $\mathbf{R}$ about the x and z axes must be equal, respectively, to M_x^R and M_z^R .

*3.4D Reduction of a System of Forces to a Wrench

In the general case of a system of forces in space, the equivalent force-couple system at O consists of a force $\mathbf{R}$ and a couple vector $\mathbf{M}_O^R$ that are not perpendicular and where neither is zero (Fig. 3.41a). This system of forces *cannot* be reduced to a single force or to a single couple. However, we still have a way of simplifying this system further.

The simplification method consists of first replacing the couple vector by two other couple vectors that are obtained by resolving $\mathbf{M}_O^R$ into a component $\mathbf{M}_1$ along $\mathbf{R}$ and a component $\mathbf{M}_2$ in a plane perpendicular to $\mathbf{R}$ (Fig. 3.41b). Then we can replace the couple vector $\mathbf{M}_2$ and force $\mathbf{R}$ by a single force $\mathbf{R}$ acting along a new line of action. The original system of forces thus reduces to $\mathbf{R}$ and to the couple vector $\mathbf{M}_1$ (Fig. 3.41c), i.e., to $\mathbf{R}$ and a couple acting in the plane perpendicular to $\mathbf{R}$.

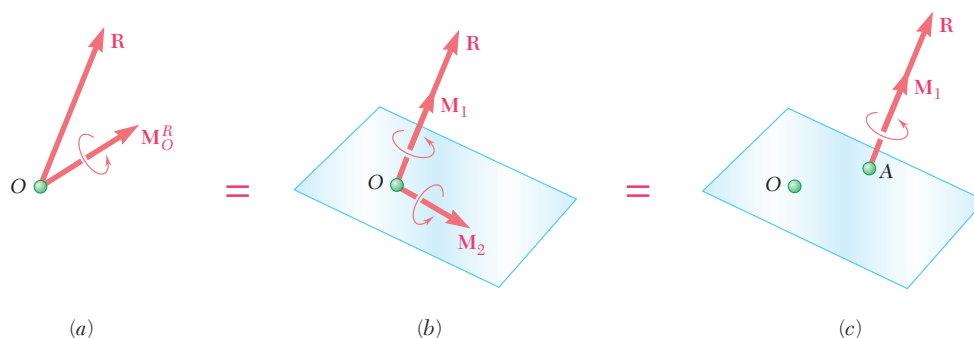


Fig. 3.41 Reducing a system of forces to a wrench. (a) General force system reduced to a single force and a couple vector, not perpendicular to each other; (b) resolving the couple vector into components along the line of action of the force and perpendicular to it; (c) moving the force and collinear couple vector (the wrench) to eliminate the couple vector perpendicular to the force.

This particular force-couple system is called a **wrench** because the resulting combination of push and twist is the same as that caused by an actual wrench. The line of action of $\mathbf{R}$ is known as the *axis of the wrench*, and the ratio $p = M_1/R$ is called the *pitch of the wrench*. A wrench therefore consists of two collinear vectors: a force $\mathbf{R}$ and a couple vector

$$\mathbf{M}_1 = p\mathbf{R} \quad (3.59)$$

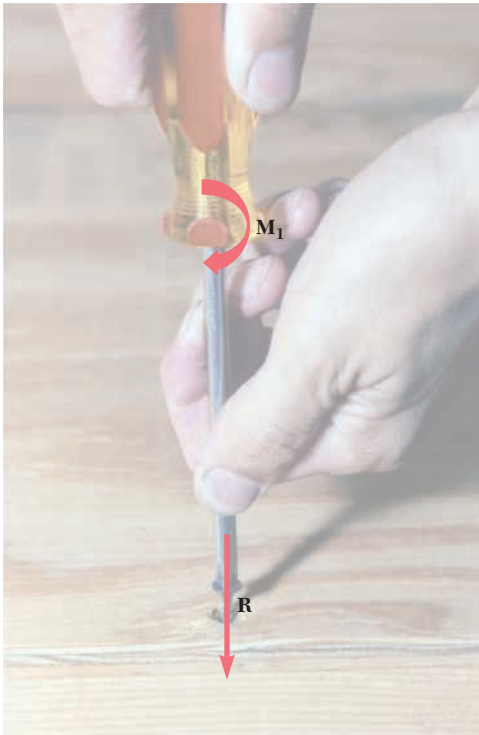


Photo 3.5 The pushing-turning action associated with the tightening of a screw illustrates the collinear lines of action of the force and couple vector that constitute a wrench.

Recall the expression in Eq. (3.33) for the projection of a vector on the line of action of another vector. Using this equation, we note that the projection of $\mathbf{M}_O^R$ on the line of action of $\mathbf{R}$ is

$$M_1 = \frac{\mathbf{R} \cdot \mathbf{M}_O^R}{R}$$

Thus, we can express the pitch of the wrench as[†]

$$p = \frac{M_1}{R} = \frac{\mathbf{R} \cdot \mathbf{M}_O^R}{R^2} \quad (3.60)$$

To define the axis of the wrench, we can write a relation involving the position vector $\mathbf{r}$ of an arbitrary point P located on that axis. We first attach the resultant force $\mathbf{R}$ and couple vector $\mathbf{M}_1$ at P (Fig. 3.42). Then, since the moment about O of this force-couple system must be equal to the moment resultant $\mathbf{M}_O^R$ of the original force system, we have

$$\mathbf{M}_1 + \mathbf{r} \times \mathbf{R} = \mathbf{M}_O^R \quad (3.61)$$

Alternatively, using Eq. (3.59), we have

$$p\mathbf{R} + \mathbf{r} \times \mathbf{R} = \mathbf{M}_O^R \quad (3.62)$$

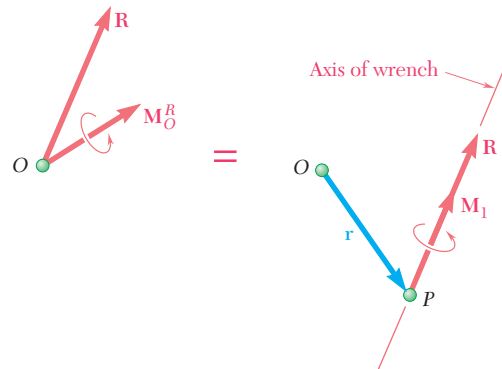


Fig. 3.42 By finding the position vector $\mathbf{r}$ that locates any arbitrary point on the axis of the wrench, you can define the axis.

[†]The expressions obtained for the projection of the couple vector on the line of action of $\mathbf{R}$ and for the pitch of the wrench are independent of the choice of point O . Using the relation (3.51) of Sec. 3.4A, we note that if a different point O' had been used, the numerator in (3.60) would have been

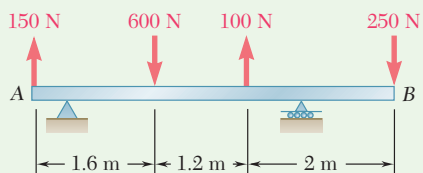
$$\mathbf{R} \cdot \mathbf{M}_{O'}^R = \mathbf{R} \cdot (\mathbf{M}_O^R + \mathbf{s} \times \mathbf{R}) = \mathbf{R} \cdot \mathbf{M}_O^R + \mathbf{R} \cdot (\mathbf{s} \times \mathbf{R})$$

Since the mixed triple product $\mathbf{R} \cdot (\mathbf{s} \times \mathbf{R})$ is identically equal to zero, we have

$$\mathbf{R} \cdot \mathbf{M}_{O'}^R = \mathbf{R} \cdot \mathbf{M}_O^R$$

Thus, the scalar product $\mathbf{R} \cdot \mathbf{M}_O^R$ is independent of the choice of point O .

Sample Problem 3.8



A 4.80-m-long beam is subjected to the forces shown. Reduce the given system of forces to (a) an equivalent force-couple system at A, (b) an equivalent force-couple system at B, (c) a single force or resultant. *Note:* Since the reactions at the supports are not included in the given system of forces, the given system will not maintain the beam in equilibrium.

STRATEGY: The *force* part of an equivalent force-couple system is simply the sum of the forces involved. The *couple* part is the sum of the moments caused by each force relative to the point of interest. Once you find the equivalent force-couple at one point, you can transfer it to any other point by a moment calculation.

MODELING and ANALYSIS:

a. Force-Couple System at A. The force-couple system at A equivalent to the given system of forces consists of a force $\mathbf{R}$ and a couple $\mathbf{M}_A^R$ defined as (Fig. 1):

$$\begin{aligned}\mathbf{R} &= \Sigma \mathbf{F} \\ &= (150 \text{ N})\mathbf{j} - (600 \text{ N})\mathbf{j} + (100 \text{ N})\mathbf{j} - (250 \text{ N})\mathbf{j} = -(600 \text{ N})\mathbf{j} \\ \mathbf{M}_A^R &= \Sigma (\mathbf{r} \times \mathbf{F}) \\ &= (1.6\mathbf{i}) \times (-600\mathbf{j}) + (2.8\mathbf{i}) \times (100\mathbf{j}) + (4.8\mathbf{i}) \times (-250\mathbf{j}) \\ &= -(1880 \text{ N}\cdot\text{m})\mathbf{k}\end{aligned}$$

The equivalent force-couple system at A is thus

$$\mathbf{R} = 600 \text{ N} \downarrow \quad \mathbf{M}_A^R = 1880 \text{ N}\cdot\text{m} \downarrow \quad \blacktriangleleft$$

b. Force-Couple System at B. You want to find a force-couple system at B equivalent to the force-couple system at A determined in part a. The force $\mathbf{R}$ is unchanged, but you must determine a new couple $\mathbf{M}_B^R$, the moment of which is equal to the moment about B of the force-couple system determined in part a (Fig. 2). You have

$$\begin{aligned}\mathbf{M}_B^R &= \mathbf{M}_A^R + \overrightarrow{BA} \times \mathbf{R} \\ &= -(1880 \text{ N}\cdot\text{m})\mathbf{k} + (-4.8 \text{ m})\mathbf{i} \times (-600 \text{ N})\mathbf{j} \\ &= -(1880 \text{ N}\cdot\text{m})\mathbf{k} + (2880 \text{ N}\cdot\text{m})\mathbf{k} = +(1000 \text{ N}\cdot\text{m})\mathbf{k}\end{aligned}$$

The equivalent force-couple system at B is thus

$$\mathbf{R} = 600 \text{ N} \downarrow \quad \mathbf{M}_B^R = 1000 \text{ N}\cdot\text{m} \uparrow \quad \blacktriangleleft$$

c. Single Force or Resultant. The resultant of the given system of forces is equal to $\mathbf{R}$, and its point of application must be such that the moment of $\mathbf{R}$ about A is equal to $\mathbf{M}_A^R$ (Fig. 3). This equality of moments leads to

$$\begin{aligned}\mathbf{r} \times \mathbf{R} &= \mathbf{M}_A^R \\ x\mathbf{i} \times (-600 \text{ N})\mathbf{j} &= -(1880 \text{ N}\cdot\text{m})\mathbf{k} \\ -x(600 \text{ N})\mathbf{k} &= -(1880 \text{ N}\cdot\text{m})\mathbf{k}\end{aligned}$$

(continued)

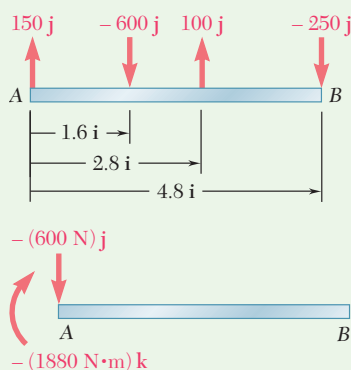


Fig. 1 Force-couple system at A that is equivalent to given system of forces.

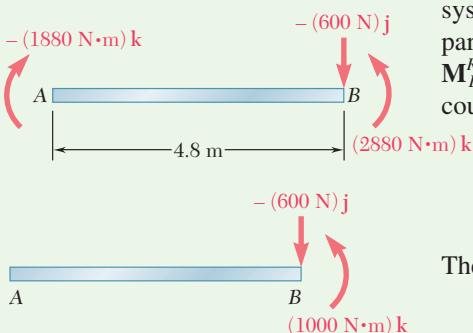


Fig. 2 Finding force-couple system at B equivalent to that determined in part a.

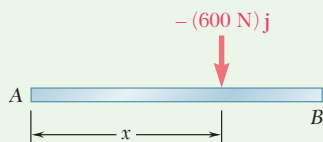


Fig. 3 Single force that is equivalent to given system of forces.

Solving for x , you get $x = 3.13$ m. Thus, the single force equivalent to the given system is defined as

$$\mathbf{R} = 600 \text{ N} \downarrow \quad x = 3.13 \text{ m} \quad \blacktriangleleft$$

REFLECT and THINK: This reduction of a given system of forces to a single equivalent force uses the same principles that you will use later for finding centers of gravity and centers of mass, which are important parameters in engineering mechanics.

Sample Problem 3.9

Four tugboats are bringing an ocean liner to its pier. Each tugboat exerts a 5000-lb force in the direction shown. Determine (a) the equivalent force-couple system at the foremast O , (b) the point on the hull where a single, more powerful tugboat should push to produce the same effect as the original four tugboats.

STRATEGY: The equivalent force-couple system is defined by the sum of the given forces and the sum of the moments of those forces at a particular point. A single tugboat could produce this system by exerting the resultant force at a point of application that produces an equivalent moment.

MODELING and ANALYSIS:

a. Force-Couple System at O . Resolve each of the given forces into components, as in Fig. 1 (kip units are used). The force-couple system at O equivalent to the given system of forces consists of a force $\mathbf{R}$ and a couple $\mathbf{M}_O^R$ defined as

$$\begin{aligned} \mathbf{R} &= \sum \mathbf{F} \\ &= (2.50\mathbf{i} - 4.33\mathbf{j}) + (3.00\mathbf{i} - 4.00\mathbf{j}) + (-5.00\mathbf{j}) + (3.54\mathbf{i} + 3.54\mathbf{j}) \\ &= 9.04\mathbf{i} - 9.79\mathbf{j} \end{aligned}$$

$$\begin{aligned} \mathbf{M}_O^R &= \sum (\mathbf{r} \times \mathbf{F}) \\ &= (-90\mathbf{i} + 50\mathbf{j}) \times (2.50\mathbf{i} - 4.33\mathbf{j}) \\ &\quad + (100\mathbf{i} + 70\mathbf{j}) \times (3.00\mathbf{i} - 4.00\mathbf{j}) \\ &\quad + (400\mathbf{i} + 70\mathbf{j}) \times (-5.00\mathbf{j}) \\ &\quad + (300\mathbf{i} - 70\mathbf{j}) \times (3.54\mathbf{i} + 3.54\mathbf{j}) \\ &= (390 - 125 - 400 - 210 - 2000 + 1062 + 248)\mathbf{k} \\ &= -1035\mathbf{k} \end{aligned}$$

The equivalent force-couple system at O is thus (Fig. 2)

$$\mathbf{R} = (9.04 \text{ kips})\mathbf{i} - (9.79 \text{ kips})\mathbf{j} \quad \mathbf{M}_O^R = -(1035 \text{ kip}\cdot\text{ft})\mathbf{k}$$

or

$$\mathbf{R} = 13.33 \text{ kips} \searrow 47.3^\circ \quad \mathbf{M}_O^R = 1035 \text{ kip}\cdot\text{ft} \downarrow \quad \blacktriangleleft$$

(continued)

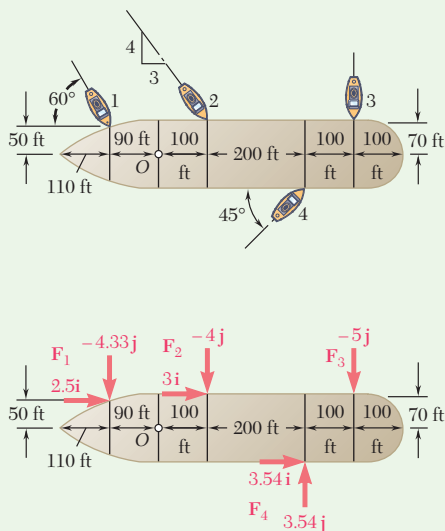


Fig. 1 Given forces resolved into components.

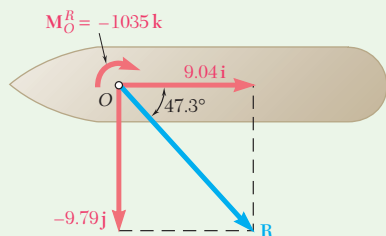


Fig. 2 Equivalent force-couple system at O .

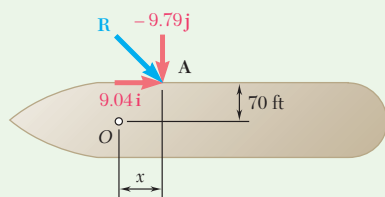


Fig. 3 Point of application of single tugboat to create same effect as given force system.

Remark: Since all the forces are contained in the plane of the figure, you would expect the sum of their moments to be perpendicular to that plane. Note that you could obtain the moment of each force component directly from the diagram by first forming the product of its magnitude and perpendicular distance to O and then assigning to this product a positive or a negative sign, depending upon the sense of the moment.

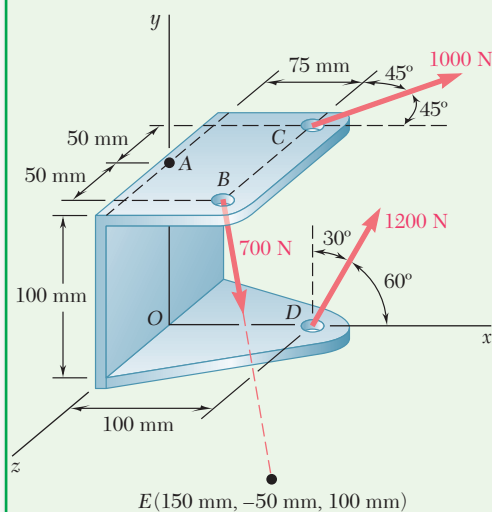
b. Single Tugboat. The force exerted by a single tugboat must be equal to $\mathbf{R}$, and its point of application A must be such that the moment of $\mathbf{R}$ about O is equal to $\mathbf{M}_O^R$ (Fig. 3). Observing that the position vector of A is

$$\mathbf{r} = x\mathbf{i} + 70\mathbf{j}$$

you have

$$\begin{aligned}\mathbf{r} \times \mathbf{R} &= \mathbf{M}_O^R \\ (x\mathbf{i} + 70\mathbf{j}) \times (9.04\mathbf{i} - 9.79\mathbf{j}) &= -1035\mathbf{k} \\ -x(9.79)\mathbf{k} - 633\mathbf{k} &= -1035\mathbf{k} \quad x = 41.1 \text{ ft} \quad \blacktriangleleft\end{aligned}$$

REFLECT and THINK: Reducing the given situation to that of a single force makes it easier to visualize the overall effect of the tugboats in maneuvering the ocean liner. But in practical terms, having four boats applying force allows for greater control in slowing and turning a large ship in a crowded harbor.



Sample Problem 3.10

Three cables are attached to a bracket as shown. Replace the forces exerted by the cables with an equivalent force-couple system at A .

STRATEGY: First determine the relative position vectors drawn from point A to the points of application of the various forces and resolve the forces into rectangular components. Then sum the forces and moments.

MODELING and ANALYSIS: Note that $\mathbf{F}_B = (700 \text{ N})\boldsymbol{\lambda}_{BE}$ where

$$\boldsymbol{\lambda}_{BE} = \frac{\overrightarrow{BE}}{BE} = \frac{75\mathbf{i} - 150\mathbf{j} + 50\mathbf{k}}{175}$$

Using meters and newtons, the position and force vectors are

$$\begin{aligned}\mathbf{r}_{B/A} = \overrightarrow{AB} &= 0.075\mathbf{i} + 0.050\mathbf{k} & \mathbf{F}_B &= 300\mathbf{i} - 600\mathbf{j} + 200\mathbf{k} \\ \mathbf{r}_{C/A} = \overrightarrow{AC} &= 0.075\mathbf{i} - 0.050\mathbf{k} & \mathbf{F}_C &= 707\mathbf{i} - 707\mathbf{k} \\ \mathbf{r}_{D/A} = \overrightarrow{AD} &= 0.100\mathbf{i} - 0.100\mathbf{j} & \mathbf{F}_D &= 600\mathbf{i} + 1039\mathbf{j}\end{aligned}$$

The force-couple system at A equivalent to the given forces consists of a force $\mathbf{R} = \Sigma \mathbf{F}$ and a couple $\mathbf{M}_A^R = \Sigma (\mathbf{r} \times \mathbf{F})$. Obtain the force $\mathbf{R}$ by adding respectively the x , y , and z components of the forces:

$$\mathbf{R} = \Sigma \mathbf{F} = (1607 \text{ N})\mathbf{i} + (439 \text{ N})\mathbf{j} - (507 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

(continued)

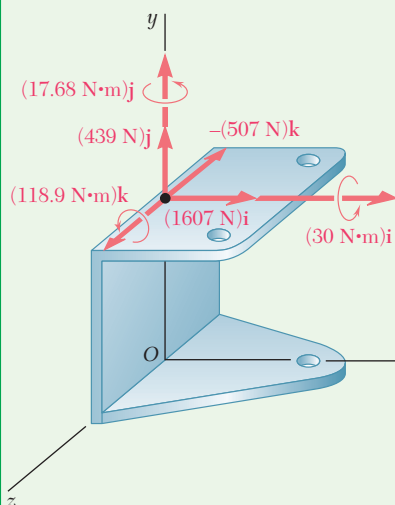


Fig. 1 Rectangular components of equivalent force-couple system at A.

The computation of $\mathbf{M}_A^R$ is facilitated by expressing the moments of the forces in the form of determinants (Sec. 3.1F). Thus,

$$\begin{aligned}\mathbf{r}_{B/A} \times \mathbf{F}_B &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.075 & 0 & 0.050 \\ 300 & -600 & 200 \end{vmatrix} = 30\mathbf{i} & -45\mathbf{k} \\ \mathbf{r}_{C/A} \times \mathbf{F}_C &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.075 & 0 & -0.050 \\ 707 & 0 & -707 \end{vmatrix} = & 17.68\mathbf{j} \\ \mathbf{r}_{D/A} \times \mathbf{F}_D &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.100 & -0.100 & 0 \\ 600 & 1039 & 0 \end{vmatrix} = & 163.9\mathbf{k}\end{aligned}$$

Adding these expressions, you have

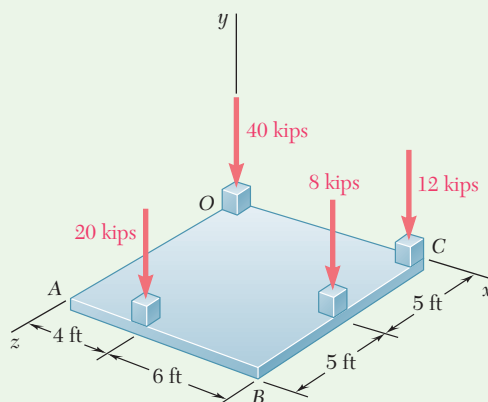
$$\mathbf{M}_A^R = \Sigma(\mathbf{r} \times \mathbf{F}) = (30 \text{ N}\cdot\text{m})\mathbf{i} + (17.68 \text{ N}\cdot\text{m})\mathbf{j} + (118.9 \text{ N}\cdot\text{m})\mathbf{k} \quad \blacktriangleleft$$

Figure 1 shows the rectangular components of the force $\mathbf{R}$ and the couple $\mathbf{M}_A^R$.

REFLECT and THINK: The determinant approach to calculating moments shows its advantages in a general three-dimensional problem such as this.

Sample Problem 3.11

A square foundation mat supports the four columns shown. Determine the magnitude and point of application of the resultant of the four loads.



STRATEGY: Start by reducing the given system of forces to a force-couple system at the origin O of the coordinate system. Then reduce the system further to a single force applied at a point with coordinates x , z .

MODELING: The force-couple system consists of a force $\mathbf{R}$ and a couple vector $\mathbf{M}_O^R$ defined as

$$\mathbf{R} = \Sigma \mathbf{F} \qquad \mathbf{M}_O^R = \Sigma(\mathbf{r} \times \mathbf{F})$$

(continued)

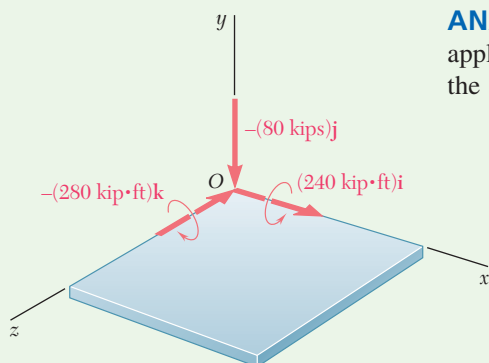


Fig. 1 Force-couple system at O that is equivalent to given force system.

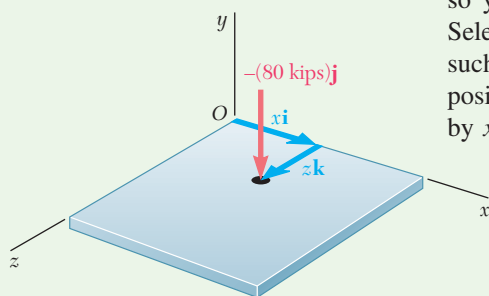


Fig. 2 Single force that is equivalent to given force system.

ANALYSIS: After determining the position vectors of the points of application of the various forces, you may find it convenient to arrange the computations in tabular form. The results are shown in Fig. 1.

$\mathbf{r}, \text{ft}$	$\mathbf{F}, \text{kips}$	$\mathbf{r} \times \mathbf{F}, \text{kip}\cdot\text{ft}$
0	$-40\mathbf{j}$	0
$10\mathbf{i}$	$-12\mathbf{j}$	$-120\mathbf{k}$
$10\mathbf{i} + 5\mathbf{k}$	$-8\mathbf{j}$	$40\mathbf{i} - 80\mathbf{k}$
$4\mathbf{i} + 10\mathbf{k}$	$-20\mathbf{j}$	$200\mathbf{i} - 80\mathbf{k}$
	$\mathbf{R} = -80\mathbf{j}$	$\mathbf{M}_O^R = 240\mathbf{i} - 280\mathbf{k}$

The force $\mathbf{R}$ and the couple vector $\mathbf{M}_O^R$ are mutually perpendicular, so you can reduce the force-couple system further to a single force $\mathbf{R}$. Select the new point of application of $\mathbf{R}$ in the plane of the mat and in such a way that the moment of $\mathbf{R}$ about O is equal to $\mathbf{M}_O^R$. Denote the position vector of the desired point of application by $\mathbf{r}$ and its coordinates by x and z (Fig. 2). Then

$$\begin{aligned}\mathbf{r} \times \mathbf{R} &= \mathbf{M}_O^R \\ (x\mathbf{i} + z\mathbf{k}) \times (-80\mathbf{j}) &= 240\mathbf{i} - 280\mathbf{k} \\ -80x\mathbf{k} + 80z\mathbf{i} &= 240\mathbf{i} - 280\mathbf{k}\end{aligned}$$

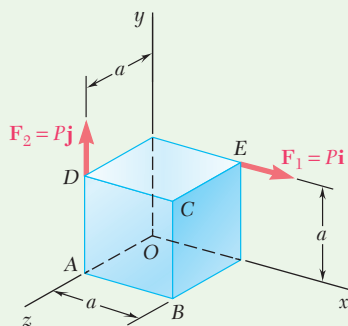
It follows that

$$\begin{aligned}-80x &= -280 & 80z &= 240 \\ x &= 3.50 \text{ ft} & z &= 3.00 \text{ ft}\end{aligned}$$

The resultant of the given system of forces is

$$\mathbf{R} = 80 \text{ kips} \downarrow \quad \text{at } x = 3.50 \text{ ft}, z = 3.00 \text{ ft} \quad \blacktriangleleft$$

REFLECT and THINK: The fact that the given forces are all parallel simplifies the calculations, so the final step becomes just a two-dimensional analysis.



Sample Problem 3.12

Two forces of the same magnitude P act on a cube of side a as shown. Replace the two forces by an equivalent wrench, and determine (a) the magnitude and direction of the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the yz plane.

STRATEGY: The first step is to determine the equivalent force-couple system at the origin O . Then you can reduce this system to a wrench and determine its properties.

(continued)

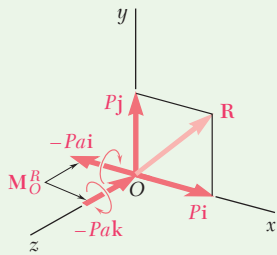


Fig. 1 Force-couple system at O that is equivalent to the given force system.

MODELING and ANALYSIS:

Equivalent Force-Couple System at O . The position vectors of the points of application E and D of the two given forces are $\mathbf{r}_E = a\mathbf{i} + a\mathbf{j}$ and $\mathbf{r}_D = a\mathbf{j} + a\mathbf{k}$. The resultant $\mathbf{R}$ of the two forces and their moment resultant $\mathbf{M}_O^R$ about O are (Fig. 1)

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 = P\mathbf{i} + P\mathbf{j} = P(\mathbf{i} + \mathbf{j}) \quad (1)$$

$$\begin{aligned} \mathbf{M}_O^R &= \mathbf{r}_E \times \mathbf{F}_1 + \mathbf{r}_D \times \mathbf{F}_2 = (a\mathbf{i} + a\mathbf{j}) \times P\mathbf{i} + (a\mathbf{j} + a\mathbf{k}) \times P\mathbf{j} \\ &= -Pa\mathbf{k} - Pa\mathbf{i} = -Pa(\mathbf{i} + \mathbf{k}) \end{aligned} \quad (2)$$

a. Resultant Force $\mathbf{R}$. It follows from Eq. (1) and Fig. 1 that the resultant force $\mathbf{R}$ has a magnitude of $R = P\sqrt{2}$, lies in the xy plane, and forms angles of 45° with the x and y axes. Thus

$$R = P\sqrt{2} \quad \theta_x = \theta_y = 45^\circ \quad \theta_z = 90^\circ \quad \blacktriangleleft$$

b. Pitch of the Wrench. Using equation (3.60) of Sec. 3.4D and Eqs. (1) and (2) above, the pitch p of the wrench is

$$p = \frac{\mathbf{R} \cdot \mathbf{M}_O^R}{R^2} = \frac{P(\mathbf{i} + \mathbf{j}) \cdot (-Pa)(\mathbf{i} + \mathbf{k})}{(P\sqrt{2})^2} = \frac{-P^2a(1 + 0 + 0)}{2P^2} \quad p = -\frac{a}{2} \quad \blacktriangleleft$$

c. Axis of the Wrench. From the pitch and from Eq. (3.59), the wrench consists of the force $\mathbf{R}$ found in Eq. (1) and the couple vector

$$\mathbf{M}_1 = p\mathbf{R} = -\frac{a}{2}P(\mathbf{i} + \mathbf{j}) = -\frac{Pa}{2}(\mathbf{i} + \mathbf{j}) \quad (3)$$

To find the point where the axis of the wrench intersects the yz plane, set the moment of the wrench about O equal to the moment resultant $\mathbf{M}_O^R$ of the original system:

$$\mathbf{M}_1 + \mathbf{r} \times \mathbf{R} = \mathbf{M}_O^R$$

Alternatively, noting that $\mathbf{r} = y\mathbf{j} + z\mathbf{k}$ (Fig. 2) and substituting for $\mathbf{R}$, $\mathbf{M}_O^R$, and $\mathbf{M}_1$ from Eqs. (1), (2), and (3), we have

$$-\frac{Pa}{2}(\mathbf{i} + \mathbf{j}) + (y\mathbf{j} + z\mathbf{k}) \times P(\mathbf{i} + \mathbf{j}) = -Pa(\mathbf{i} + \mathbf{k})$$

$$-\frac{Pa}{2}\mathbf{i} - \frac{Pa}{2}\mathbf{j} - Py\mathbf{k} + Pz\mathbf{j} - Pz\mathbf{i} = -Pa\mathbf{i} - Pa\mathbf{k}$$

Equating the coefficients of $\mathbf{k}$ and then the coefficients of $\mathbf{j}$, the final result is

$$y = a \quad z = a/2 \quad \blacktriangleleft$$

REFLECT and THINK: Conceptually, reducing a system of forces to a wrench is simply an additional application of finding an equivalent force-couple system.

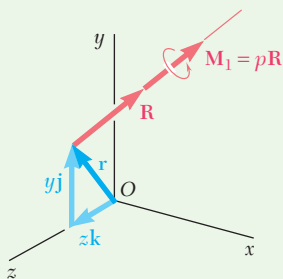


Fig. 2 Wrench that is equivalent to the given force system.

SOLVING PROBLEMS ON YOUR OWN

In this section you studied the reduction and simplification of force systems. In solving the problems that follow, you will be asked to perform the following operations.

1. Reducing a force system to a force and a couple at a given point A .

The force is the *resultant* $\mathbf{R}$ of the system that is obtained by adding the various forces. The moment of the couple is the *moment resultant* of the system that is obtained by adding the moments about A of the various forces. We have

$$\mathbf{R} = \Sigma \mathbf{F} \qquad \mathbf{M}_A^R = \Sigma (\mathbf{r} \times \mathbf{F})$$

where the position vector $\mathbf{r}$ is drawn from A to *any point* on the line of action of $\mathbf{F}$.

2. Moving a force-couple system from point A to point B . If you wish to reduce a given force system to a force-couple system at point B , you need not recompute the moments of the forces about B after you have reduced it to a force-couple system at point A . The resultant $\mathbf{R}$ remains unchanged, and you can obtain the new moment resultant $\mathbf{M}_B^R$ by adding the moment about B of the force $\mathbf{R}$ applied at A to $\mathbf{M}_A^R$ [Sample Prob. 3.8]. Denoting the vector drawn from B to A as $\mathbf{s}$, you have

$$\mathbf{M}_B^R = \mathbf{M}_A^R + \mathbf{s} \times \mathbf{R}$$

3. Checking whether two force systems are equivalent. First reduce each force system to a force-couple system *at the same, but arbitrary, point A* (as explained in the first operation). The two force systems are equivalent (that is, they have the same effect on the given rigid body) if the two reduced force-couple systems are identical; that is, if

$$\Sigma \mathbf{F} = \Sigma \mathbf{F}' \qquad \text{and} \qquad \Sigma \mathbf{M}_A = \Sigma \mathbf{M}'_A$$

You should recognize that if the first of these equations is not satisfied—that is, if the two systems do not have the same resultant $\mathbf{R}$ —the two systems cannot be equivalent, and there is no need to check whether or not the second equation is satisfied.

4. Reducing a given force system to a single force. First reduce the given system to a force-couple system consisting of the resultant $\mathbf{R}$ and the couple vector $\mathbf{M}_A^R$ at

(continued)

some convenient point A (as explained in the first operation). Recall from Section 3.4 that further reduction to a single force is possible *only if the force $\mathbf{R}$ and the couple vector $\mathbf{M}_A^R$ are mutually perpendicular*. This will certainly be the case for systems of forces that are either *concurrent*, *coplanar*, or *parallel*. You can then obtain the required single force by moving $\mathbf{R}$ until its moment about A is equal to $\mathbf{M}_A^R$, as you did in several problems in Section 3.4. More formally, the position vector $\mathbf{r}$ drawn from A to any point on the line of action of the single force $\mathbf{R}$ must satisfy the equation

$$\mathbf{r} \times \mathbf{R} = \mathbf{M}_A^R$$

This procedure was illustrated in Sample Probs. 3.8, 3.9, and 3.11.

5. Reducing a given force system to a wrench. If the given system includes forces that are not concurrent, coplanar, or parallel, the equivalent force-couple system at a point A will consist of a force $\mathbf{R}$ and a couple vector $\mathbf{M}_A^R$ that, in general, *are not mutually perpendicular*. (To check whether $\mathbf{R}$ and $\mathbf{M}_A^R$ are mutually perpendicular, form their scalar product. If this product is zero, they are mutually perpendicular; otherwise, they are not.) If $\mathbf{R}$ and $\mathbf{M}_A^R$ are not mutually perpendicular, the force-couple system (and thus the given system of forces) *cannot be reduced to a single force*. However, the system can be reduced to a *wrench*—the combination of a force $\mathbf{R}$ and a couple vector $\mathbf{M}_1$ directed along a common line of action called the *axis of the wrench* (Fig. 3.42). The ratio $p = M_1/R$ is called the *pitch* of the wrench.

To reduce a given force system to a wrench, you should follow these steps,

- a. Reduce the given system to an equivalent force-couple system $(\mathbf{R}, \mathbf{M}_O^R)$, typically located at the origin O .
- b. Determine the pitch p from Eq. (3.60),

$$p = \frac{M_1}{R} = \frac{\mathbf{R} \cdot \mathbf{M}_O^R}{R^2}$$

and the couple vector from $\mathbf{M}_1 = p\mathbf{R}$.

- c. Set the moment about O of the wrench equal to the moment resultant $\mathbf{M}_O^R$ of the force-couple system at O :

$$\mathbf{M}_1 + \mathbf{r} \times \mathbf{R} = \mathbf{M}_O^R \quad (3.61)$$

This equation allows you to determine the point where the line of action of the wrench intersects a specified plane, since the position vector $\mathbf{r}$ is directed from O to that point. These steps are illustrated in Sample Prob. 3.12. Although determining a wrench and the point where its axis intersects a plane may appear difficult, the process is simply the application of several of the ideas and techniques developed in this chapter. Once you have mastered the wrench, you can feel confident that you understand much of Chap. 3.

Problems

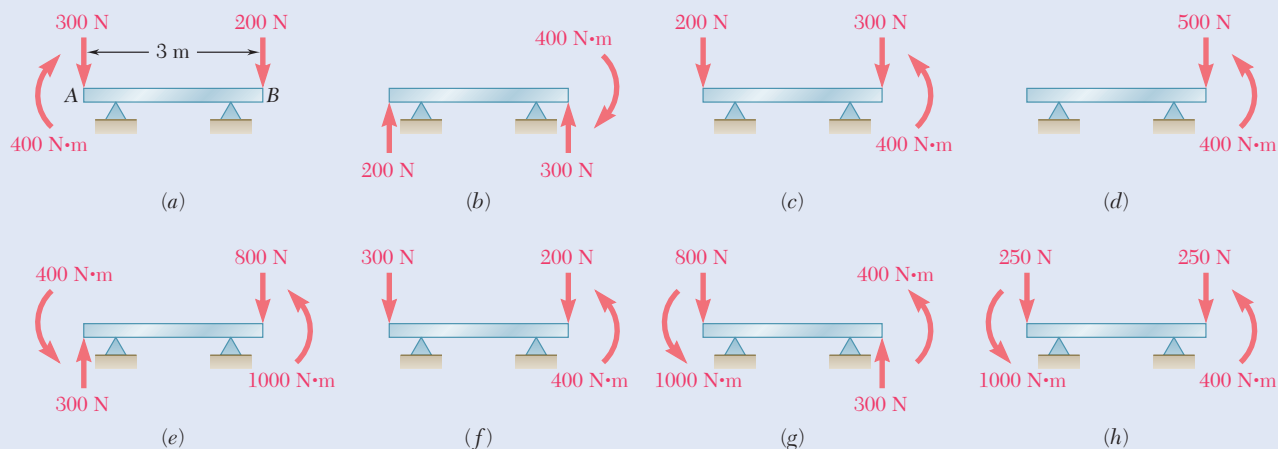


Fig. P3.101

3.101 A 3-m-long beam is subjected to a variety of loadings. (a) Replace each loading with an equivalent force-couple system at end A of the beam. (b) Which of the loadings are equivalent?

3.102 A 3-m-long beam is loaded as shown. Determine the loading of Prob. 3.101 that is equivalent to this loading.

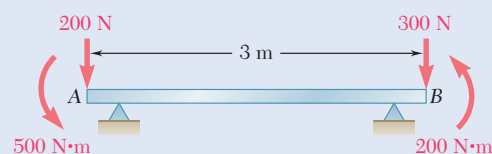


Fig. P3.102

3.103 Determine the single equivalent force and the distance from point A to its line of action for the beam and loading of (a) Prob. 3.101a, (b) Prob. 3.101b, (c) Prob. 3.102.

3.104 Five separate force-couple systems act at the corners of a piece of sheet metal that has been bent into the shape shown. Determine which of these systems is equivalent to a force $\mathbf{F} = (10 \text{ lb})\mathbf{i}$ and a couple of moment $\mathbf{M} = (15 \text{ lb}\cdot\text{ft})\mathbf{j} + (15 \text{ lb}\cdot\text{ft})\mathbf{k}$ located at the origin.

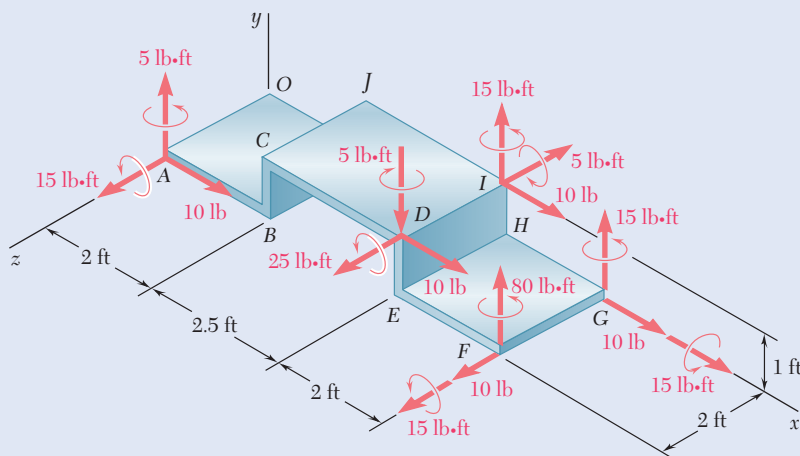


Fig. P3.104

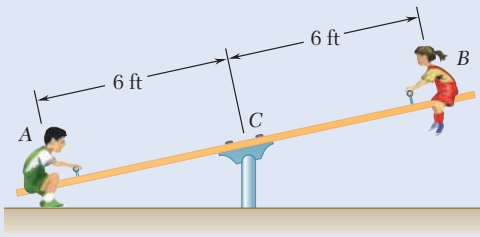


Fig. P3.105

3.105 The weights of two children sitting at ends A and B of a seesaw are 84 lb and 64 lb, respectively. Where should a third child sit so that the resultant of the weights of the three children will pass through C if she weighs (a) 60 lb, (b) 52 lb?

3.106 Three stage lights are mounted on a pipe as shown. The lights at A and B each weigh 4.1 lb, while the one at C weighs 3.5 lb. (a) If $d = 25$ in., determine the distance from D to the line of action of the resultant of the weights of the three lights. (b) Determine the value of d so that the resultant of the weights passes through the midpoint of the pipe.

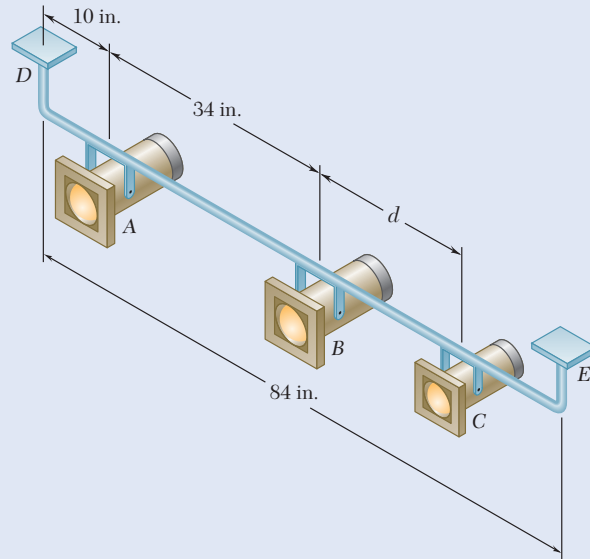


Fig. P3.106

3.107 A beam supports three loads of given magnitude and a fourth load whose magnitude is a function of position. If $b = 1.5$ m and the loads are to be replaced with a single equivalent force, determine (a) the value of a so that the distance from support A to the line of action of the equivalent force is maximum, (b) the magnitude of the equivalent force and its point of application on the beam.

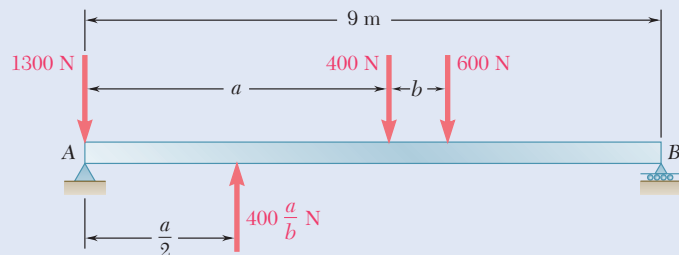


Fig. P3.107

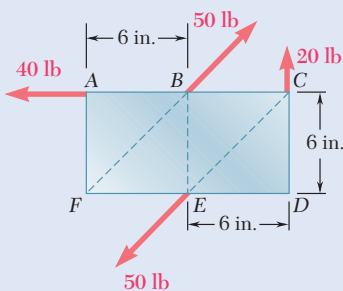


Fig. P3.108

3.108 A 6×12 -in. plate is subjected to four loads as shown. Find the resultant of the four loads and the two points at which the line of action of the resultant intersects the edge of the plate.

3.109 A 32-lb motor is mounted on the floor. Find the resultant of the weight and the forces exerted on the belt, and determine where the line of action of the resultant intersects the floor.

3.110 To test the strength of a 625×500 -mm suitcase, forces are applied as shown. If $P = 88$ N, (a) determine the resultant of the applied forces, (b) locate the two points where the line of action of the resultant intersects the edge of the suitcase.

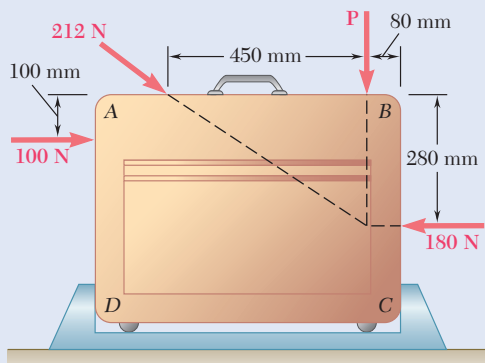


Fig. P3.110

3.111 Solve Prob. 3.110, assuming that $P = 138$ N.

3.112 Pulleys A and B are mounted on bracket CDEF. The tension on each side of the two belts is as shown. Replace the four forces with a single equivalent force, and determine where its line of action intersects the bottom edge of the bracket.

3.113 A truss supports the loading shown. Determine the equivalent force acting on the truss and the point of intersection of its line of action with a line drawn through points A and G.

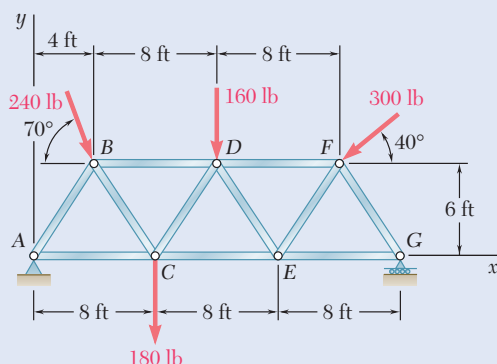


Fig. P3.113

3.114 A couple of magnitude $M = 80$ lb·in. and the three forces shown are applied to an angle bracket. (a) Find the resultant of this system of forces. (b) Locate the points where the line of action of the resultant intersects line AB and line BC.

3.115 A couple $\mathbf{M}$ and the three forces shown are applied to an angle bracket. Find the moment of the couple if the line of action of the resultant of the force system is to pass through (a) point A, (b) point B, (c) point C.

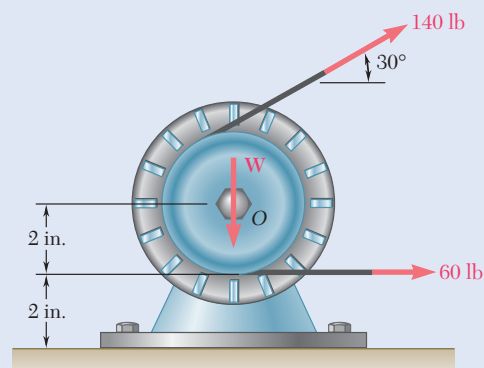


Fig. P3.109

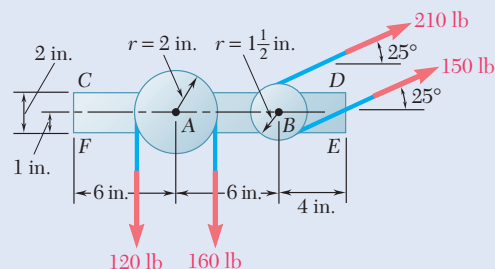


Fig. P3.112

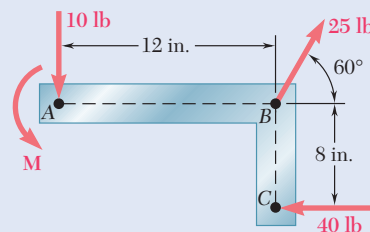


Fig. P3.114 and P3.115

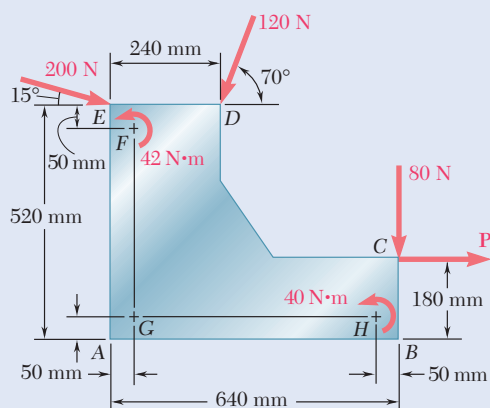


Fig. P3.116

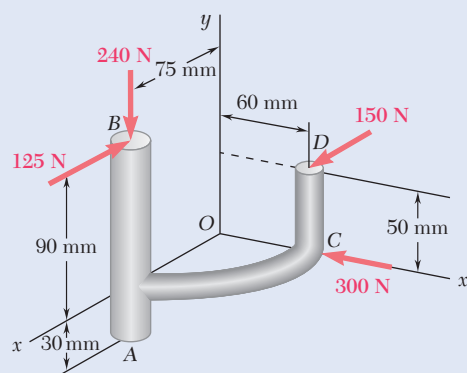


Fig. P3.119

3.116 A machine component is subjected to the forces and couples shown. The component is to be held in place by a single rivet that can resist a force but not a couple. For $P = 0$, determine the location of the rivet hole if it is to be located (a) on line FG , (b) on line GH .

3.117 Solve Prob. 3.116, assuming that $P = 60$ N.

3.118 As follower AB rolls along the surface of member C , it exerts a constant force F perpendicular to the surface. (a) Replace F with an equivalent force-couple system at the point D obtained by drawing the perpendicular from the point of contact to the x axis. (b) For $a = 1$ m and $b = 2$ m, determine the value of x for which the moment of the equivalent force-couple system at D is maximum.

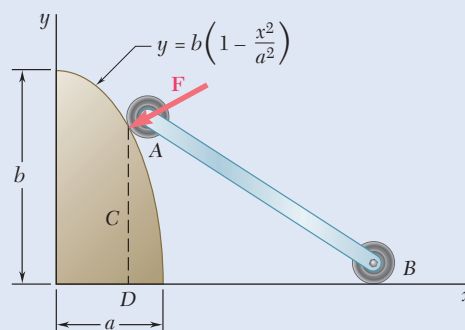


Fig. P3.118

3.119 A machine component is subjected to the forces shown, each of which is parallel to one of the coordinate axes. Replace these forces with an equivalent force-couple system at A .

3.120 Two 150-mm-diameter pulleys are mounted on line shaft AD . The belts at B and C lie in vertical planes parallel to the yz plane. Replace the belt forces shown with an equivalent force-couple system at A .

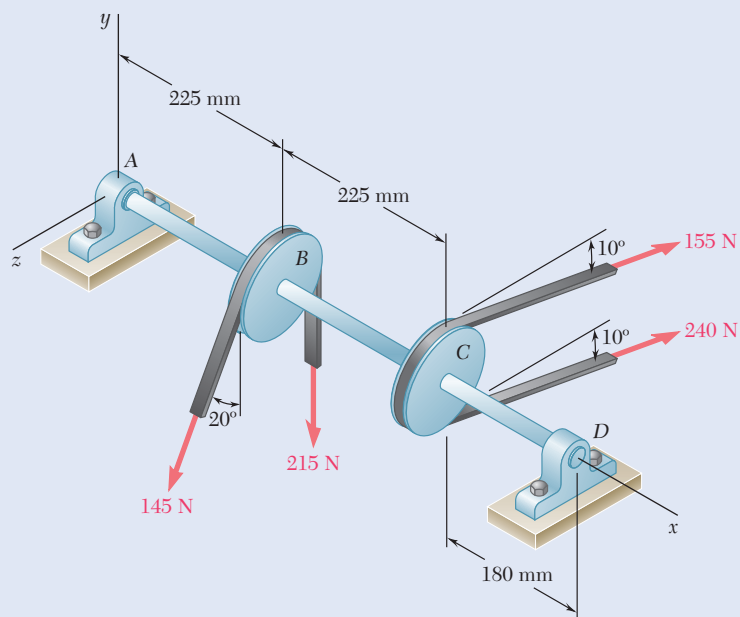


Fig. P3.120

- 3.121** As an adjustable brace BC is used to bring a wall into plumb, the force-couple system shown is exerted on the wall. Replace this force-couple system with an equivalent force-couple system at A if $R = 21.2$ lb and $M = 13.25$ lb·ft.

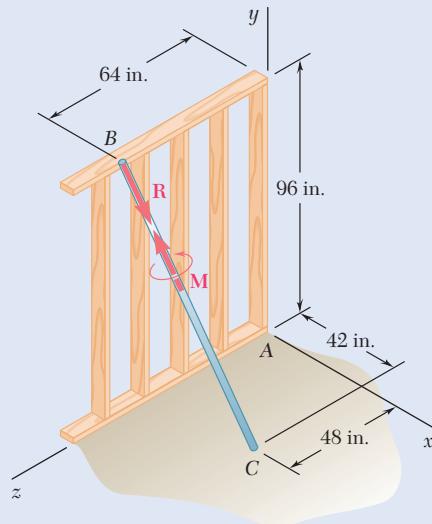


Fig. P3.121

- 3.122** In order to unscrew the tapped faucet A , a plumber uses two pipe wrenches as shown. By exerting a 40-lb force on each wrench at a distance of 10 in. from the axis of the pipe and in a direction perpendicular to the pipe and to the wrench, he prevents the pipe from rotating, and thus he avoids loosening or further tightening the joint between the pipe and the tapped elbow C . Determine (a) the angle θ that the wrench at A should form with the vertical if elbow C is not to rotate about the vertical, (b) the force-couple system at C equivalent to the two 40-lb forces when this condition is satisfied.

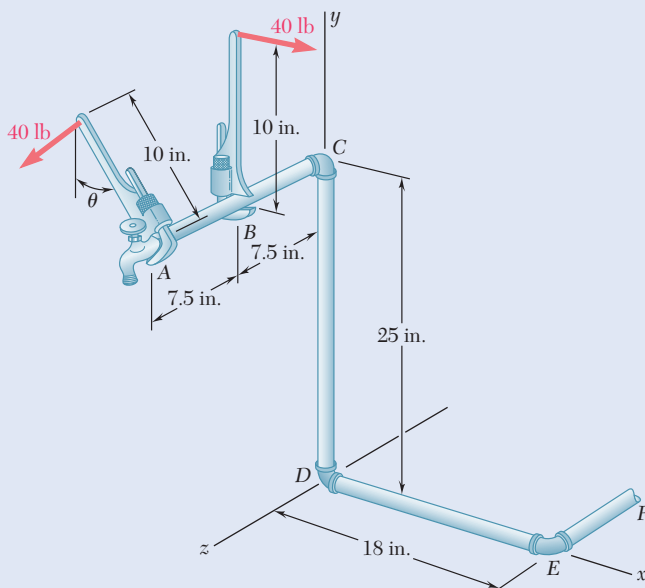


Fig. P3.122

- 3.123** Assuming $\theta = 60^\circ$ in Prob. 3.122, replace the two 40-lb forces with an equivalent force-couple system at D and determine whether the plumber's action tends to tighten or loosen the joint between (a) pipe CD and elbow D , (b) elbow D and pipe DE . Assume all threads to be right-handed.

- 3.124** Four forces are applied to the machine component $ABDE$ as shown. Replace these forces with an equivalent force-couple system at A .

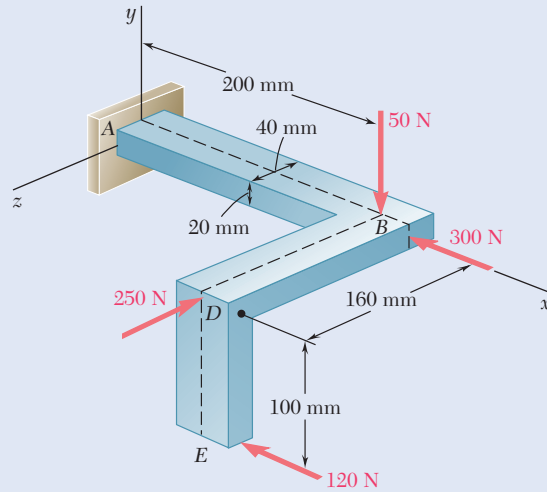


Fig. P3.124

- 3.125** A blade held in a brace is used to tighten a screw at A . (a) Determine the forces exerted at B and C , knowing that these forces are equivalent to a force-couple system at A consisting of $\mathbf{R} = -(25 \text{ N})\mathbf{i} + R_y\mathbf{j} + R_z\mathbf{k}$ and $\mathbf{M}_A^R = -(13.5 \text{ N}\cdot\text{m})\mathbf{i}$. (b) Find the corresponding values of R_y and R_z . (c) What is the orientation of the slot in the head of the screw for which the blade is least likely to slip when the brace is in the position shown?

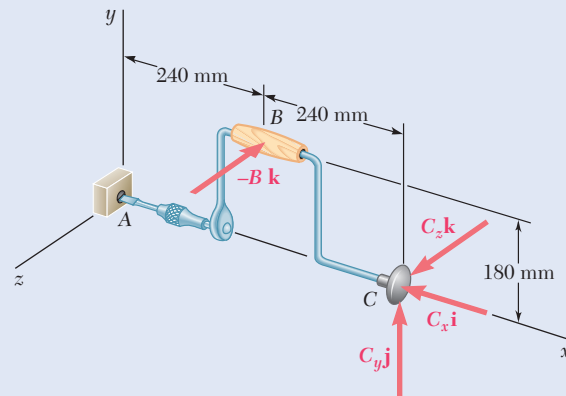


Fig. P3.125

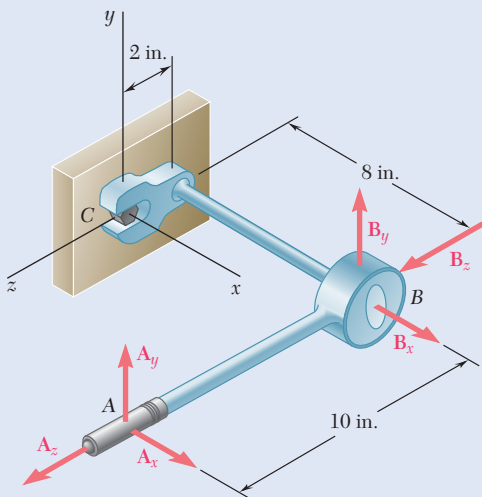


Fig. P3.126

- 3.126** A mechanic uses a crowfoot wrench to loosen a bolt at C . The mechanic holds the socket wrench handle at points A and B and applies forces at these points. Knowing that these forces are equivalent to a force-couple system at C consisting of the force $\mathbf{C} = -(8 \text{ lb})\mathbf{i} + (4 \text{ lb})\mathbf{k}$ and the couple $\mathbf{M}_C = (360 \text{ lb}\cdot\text{in.})\mathbf{i}$, determine the forces applied at A and at B when $A_z = 2 \text{ lb}$.

3.127 Three children are standing on a 5×5 -m raft. If the weights of the children at points A , B , and C are 375 N, 260 N, and 400 N, respectively, determine the magnitude and the point of application of the resultant of the three weights.

3.128 Three children are standing on a 5×5 -m raft. The weights of the children at points A , B , and C are 375 N, 260 N, and 400 N, respectively. If a fourth child weighing 425 N climbs onto the raft, determine where she should stand if the other children remain in the positions shown and if the line of action of the resultant of the four weights is to pass through the center of the raft.

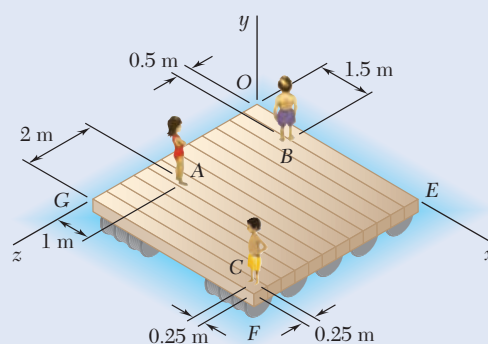


Fig. P3.127 and P3.128

3.129 Four signs are mounted on a frame spanning a highway, and the magnitudes of the horizontal wind forces acting on the signs are as shown. Determine the magnitude and the point of application of the resultant of the four wind forces when $a = 1$ ft and $b = 12$ ft.

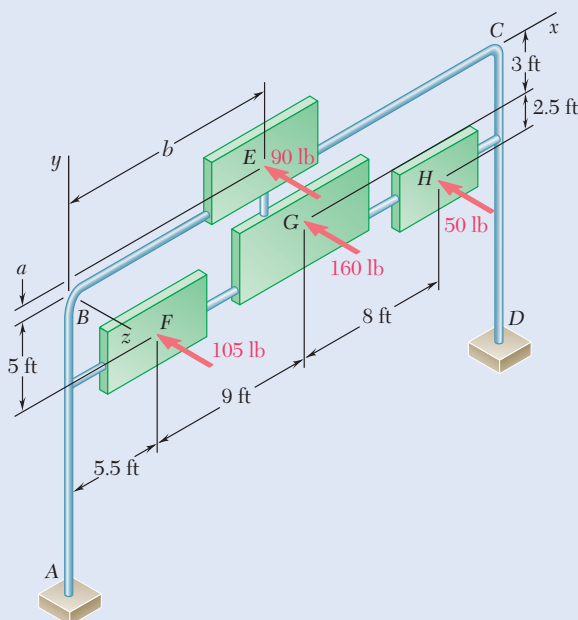


Fig. P3.129 and P3.130

3.130 Four signs are mounted on a frame spanning a highway, and the magnitudes of the horizontal wind forces acting on the signs are as shown. Determine a and b so that the point of application of the resultant of the four forces is at G .

3.131 A concrete foundation mat of 5-m radius supports four equally spaced columns, each of which is located 4 m from the center of the mat. Determine the magnitude and the point of application of the resultant of the four loads.

3.132 Determine the magnitude and the point of application of the smallest additional load that must be applied to the foundation mat of Prob. 3.131 if the resultant of the five loads is to pass through the center of the mat.

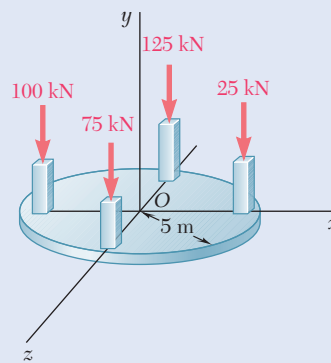


Fig. P3.131

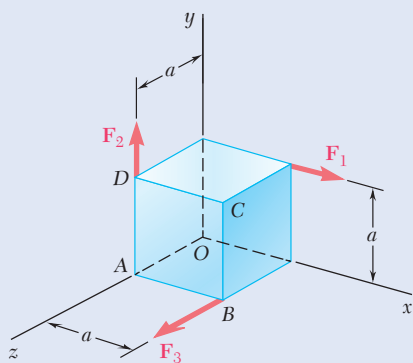


Fig. P3.133

***3.133** Three forces of the same magnitude P act on a cube of side a as shown. Replace the three forces with an equivalent wrench and determine (a) the magnitude and direction of the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the axis of the wrench.

***3.134** A piece of sheet metal is bent into the shape shown and is acted upon by three forces. If the forces have the same magnitude P , replace them with an equivalent wrench and determine (a) the magnitude and the direction of the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the axis of the wrench.

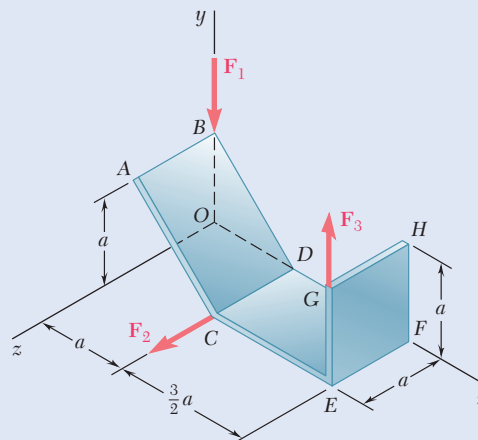


Fig. P3.134

***3.135 and *3.136** The forces and couples shown are applied to two screws as a piece of sheet metal is fastened to a block of wood. Reduce the forces and the couples to an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the xz plane.

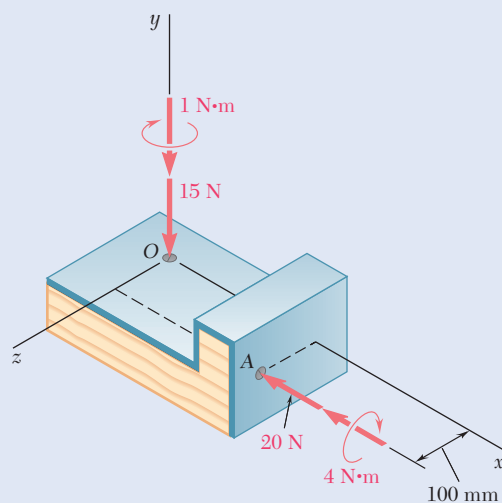


Fig. P3.135

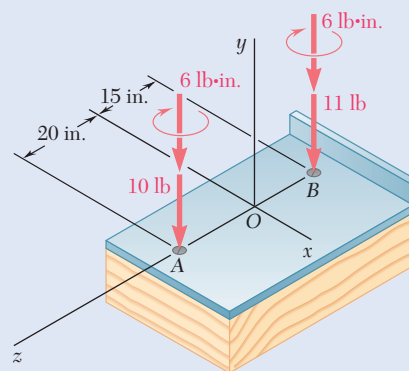


Fig. P3.136

***3.137 and *3.138** Two bolts at A and B are tightened by applying the forces and couples shown. Replace the two wrenches with a single equivalent wrench and determine (a) the resultant $\mathbf{R}$, (b) the pitch of the single equivalent wrench, (c) the point where the axis of the wrench intersects the xz plane.

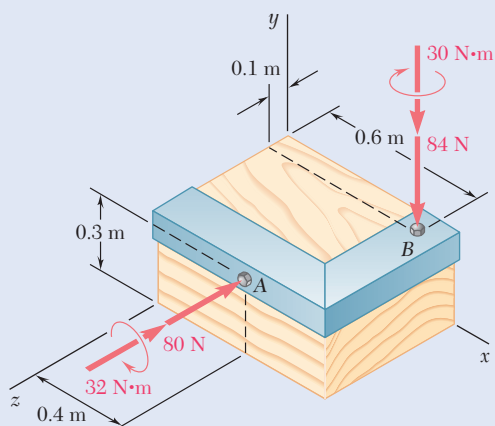


Fig. P3.137

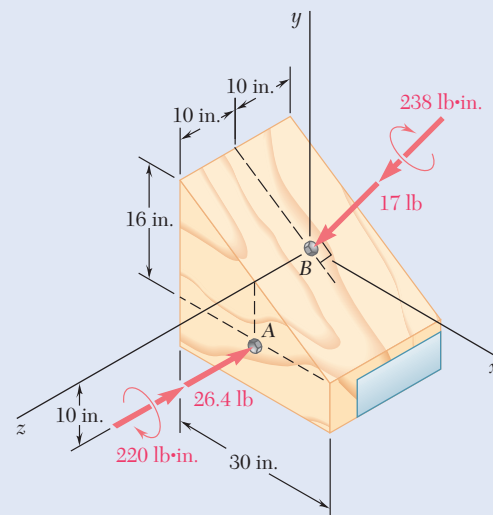


Fig. P3.138

***3.139** Two ropes attached at A and B are used to move the trunk of a fallen tree. Replace the forces exerted by the ropes with an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the yz plane.

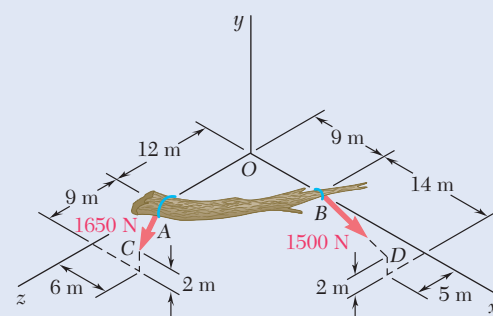


Fig. P3.139

***3.140** A flagpole is guyed by three cables. If the tensions in the cables have the same magnitude P , replace the forces exerted on the pole with an equivalent wrench and determine (a) the resultant force $\mathbf{R}$, (b) the pitch of the wrench, (c) the point where the axis of the wrench intersects the xz plane.

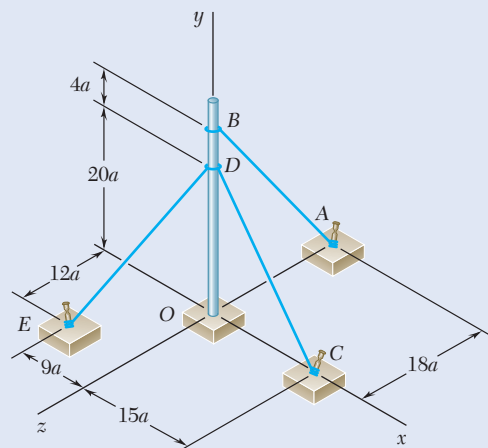


Fig. P3.140

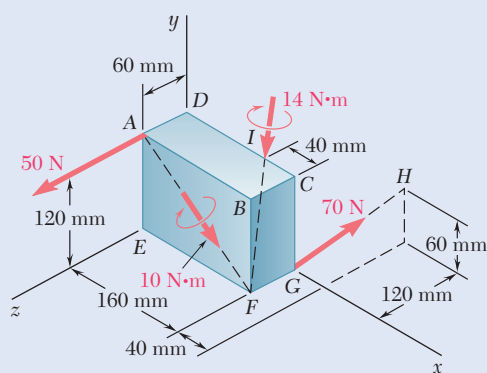


Fig. P3.141

***3.141 and *3.142** Determine whether the force-and-couple system shown can be reduced to a single equivalent force $\mathbf{R}$. If it can, determine $\mathbf{R}$ and the point where the line of action of $\mathbf{R}$ intersects the yz plane. If it cannot be reduced, replace the given system with an equivalent wrench and determine its resultant, its pitch, and the point where its axis intersects the yz plane.

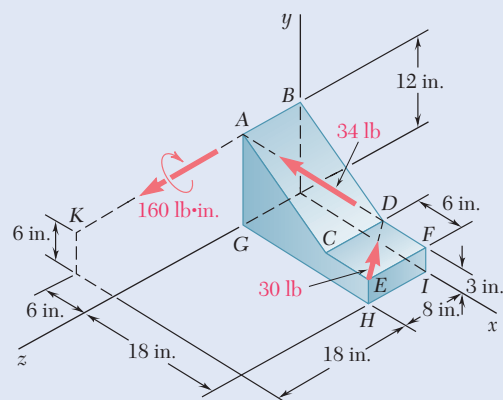


Fig. P3.142

***3.143** Replace the wrench shown with an equivalent system consisting of two forces perpendicular to the y axis and applied respectively at A and B .

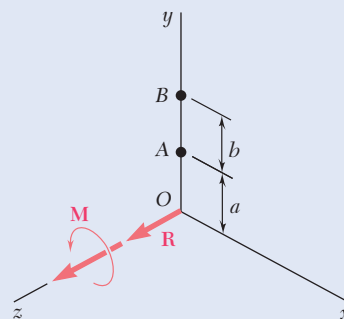


Fig. P3.143

***3.144** Show that, in general, a wrench can be replaced with two forces chosen in such a way that one force passes through a given point while the other force lies in a given plane.

***3.145** Show that a wrench can be replaced with two perpendicular forces, one of which is applied at a given point.

***3.146** Show that a wrench can be replaced with two forces, one of which has a prescribed line of action.

Review and Summary

Principle of Transmissibility

In this chapter, we presented the effects of forces exerted on a rigid body. We began by distinguishing between **external** and **internal** forces [Sec. 3.1A]. We then explained that, according to the **principle of transmissibility**, the effect of an external force on a rigid body remains unchanged if we move that force along its line of action [Sec. 3.1B]. In other words, two forces $\mathbf{F}$ and $\mathbf{F}'$ acting on a rigid body at two different points have the same effect on that body if they have the same magnitude, same direction, and same line of action (Fig. 3.43). Two such forces are said to be **equivalent**.

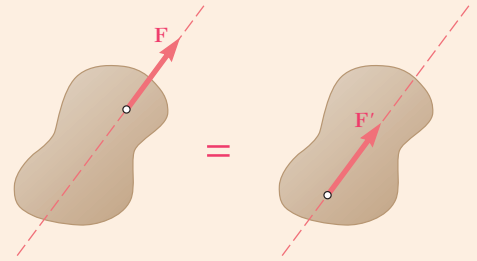


Fig. 3.43

Vector Product

Before proceeding with the discussion of **equivalent systems of forces**, we introduced the concept of the **vector product of two vectors** [Sec. 3.1C]. We defined the vector product

$$\mathbf{V} = \mathbf{P} \times \mathbf{Q}$$

of the vectors $\mathbf{P}$ and $\mathbf{Q}$ as a vector perpendicular to the plane containing $\mathbf{P}$ and $\mathbf{Q}$ (Fig. 3.44) with a magnitude of

$$V = PQ \sin \theta \quad (3.1)$$

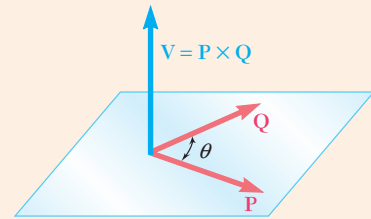
and directed in such a way that a person located at the tip of $\mathbf{V}$ will observe the rotation to be counterclockwise through θ , bringing the vector $\mathbf{P}$ in line with the vector $\mathbf{Q}$. The three vectors $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{V}$ —taken in that order—are said to form a *right-handed triad*. It follows that the vector products $\mathbf{Q} \times \mathbf{P}$ and $\mathbf{P} \times \mathbf{Q}$ are represented by equal and opposite vectors:

$$\mathbf{Q} \times \mathbf{P} = -(\mathbf{P} \times \mathbf{Q}) \quad (3.4)$$

It also follows from the definition of the vector product of two vectors that the vector products of the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are

$$\mathbf{i} \times \mathbf{i} = 0 \quad \mathbf{i} \times \mathbf{j} = \mathbf{k} \quad \mathbf{j} \times \mathbf{i} = -\mathbf{k}$$

and so on. You can determine the sign of the vector product of two unit vectors by arranging in a circle and in counterclockwise order the three letters representing the unit vectors (Fig. 3.45): The vector product of two unit vectors is positive if they follow each other in counterclockwise order and negative if they follow each other in clockwise order.



(a)



(b)

Fig. 3.44

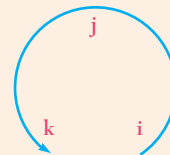


Fig. 3.45

Rectangular Components of Vector Product

The **rectangular components of the vector product** $\mathbf{V}$ of two vectors $\mathbf{P}$ and $\mathbf{Q}$ are expressed [Sec. 3.1D] as

$$\begin{aligned} V_x &= P_y Q_z - P_z Q_y \\ V_y &= P_z Q_x - P_x Q_z \\ V_z &= P_x Q_y - P_y Q_x \end{aligned} \quad (3.9)$$

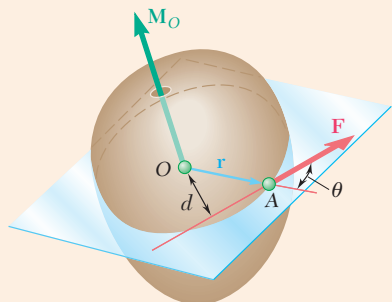


Fig. 3.46

We can also express the components of a vector product as a determinant:

$$\mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix} \quad (3.10)$$

Moment of a Force about a Point

We defined the **moment of a force $\mathbf{F}$ about a point O** [Sec. 3.1E] as the vector product

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (3.11)$$

where $\mathbf{r}$ is the *position vector* drawn from O to the point of application A of the force $\mathbf{F}$ (Fig. 3.46). Denoting the angle between the lines of action of $\mathbf{r}$ and $\mathbf{F}$ as θ , we found that the magnitude of the moment of $\mathbf{F}$ about O is

$$M_O = rF \sin \theta = Fd \quad (3.12)$$

where d represents the perpendicular distance from O to the line of action of $\mathbf{F}$.

Rectangular Components of Moment

The **rectangular components of the moment $\mathbf{M}_O$ of a force $\mathbf{F}$** [Sec. 3.1F] are

$$\begin{aligned} M_x &= yF_z - zF_y \\ M_y &= zF_x - xF_z \\ M_z &= xF_y - yF_x \end{aligned} \quad (3.18)$$

where x , y , and z are the components of the position vector $\mathbf{r}$ (Fig. 3.47). Using a determinant form, we also wrote

$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix} \quad (3.19)$$

In the more general case of the moment about an arbitrary point B of a force $\mathbf{F}$ applied at A , we had

$$\mathbf{M}_B = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_{A/B} & y_{A/B} & z_{A/B} \\ F_x & F_y & F_z \end{vmatrix} \quad (3.21)$$

where $x_{A/B}$, $y_{A/B}$, and $z_{A/B}$ denote the components of the vector $\mathbf{r}_{A/B}$:

$$x_{A/B} = x_A - x_B \quad y_{A/B} = y_A - y_B \quad z_{A/B} = z_A - z_B$$

In the case of *problems involving only two dimensions*, we can assume the force $\mathbf{F}$ lies in the xy plane. Its moment $\mathbf{M}_B$ about a point B in the same plane is perpendicular to that plane (Fig. 3.48) and is completely defined by the scalar

$$M_B = (x_A - x_B)F_y - (y_A - y_B)F_x \quad (3.23)$$

Various methods for computing the moment of a force about a point were illustrated in Sample Probs. 3.1 through 3.4.

Scalar Product of Two Vectors

The **scalar product** of two vectors $\mathbf{P}$ and $\mathbf{Q}$ [Sec. 3.2A], denoted by $\mathbf{P} \cdot \mathbf{Q}$, is defined as the scalar quantity

$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \theta \quad (3.24)$$

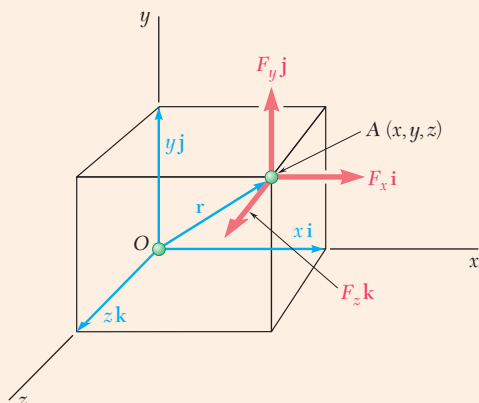


Fig. 3.47

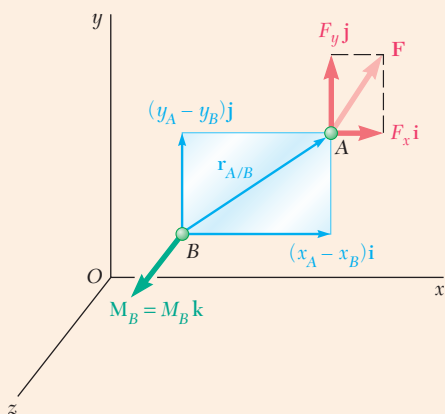


Fig. 3.48

where θ is the angle between $\mathbf{P}$ and $\mathbf{Q}$ (Fig. 3.49). By expressing the scalar product of $\mathbf{P}$ and $\mathbf{Q}$ in terms of the rectangular components of the two vectors, we determined that

$$\mathbf{P} \cdot \mathbf{Q} = P_x Q_x + P_y Q_y + P_z Q_z \quad (3.28)$$

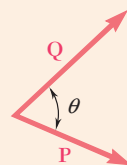


Fig. 3.49

Projection of a Vector on an Axis

We obtain the **projection of a vector $\mathbf{P}$ on an axis OL** (Fig. 3.50) by forming the scalar product of $\mathbf{P}$ and the unit vector $\boldsymbol{\lambda}$ along OL . We have

$$P_{OL} = \mathbf{P} \cdot \boldsymbol{\lambda} \quad (3.34)$$

Using rectangular components, this becomes

$$P_{OL} = P_x \cos \theta_x + P_y \cos \theta_y + P_z \cos \theta_z \quad (3.35)$$

where θ_x , θ_y , and θ_z denote the angles that the axis OL forms with the coordinate axes.

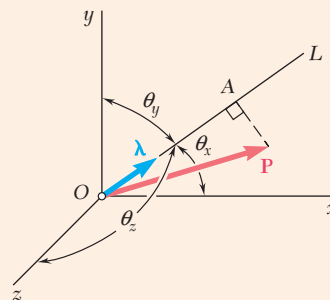


Fig. 3.50

Mixed Triple Product of Three Vectors

We defined the **mixed triple product** of the three vectors $\mathbf{S}$, $\mathbf{P}$, and $\mathbf{Q}$ as the scalar expression

$$\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) \quad (3.36)$$

obtained by forming the scalar product of $\mathbf{S}$ with the vector product of $\mathbf{P}$ and $\mathbf{Q}$ [Sec. 3.2B]. We showed that

$$\mathbf{S} \cdot (\mathbf{P} \times \mathbf{Q}) = \begin{vmatrix} S_x & S_y & S_z \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix} \quad (3.39)$$

where the elements of the determinant are the rectangular components of the three vectors.

Moment of a Force about an Axis

We defined the **moment of a force $\mathbf{F}$ about an axis OL** [Sec. 3.2C] as the projection OC on OL of the moment $\mathbf{M}_O$ of the force $\mathbf{F}$ (Fig. 3.51), i.e., as the mixed triple product of the unit vector $\boldsymbol{\lambda}$, the position vector $\mathbf{r}$, and the force $\mathbf{F}$:

$$M_{OL} = \boldsymbol{\lambda} \cdot \mathbf{M}_O = \boldsymbol{\lambda} \cdot (\mathbf{r} \times \mathbf{F}) \quad (3.40)$$

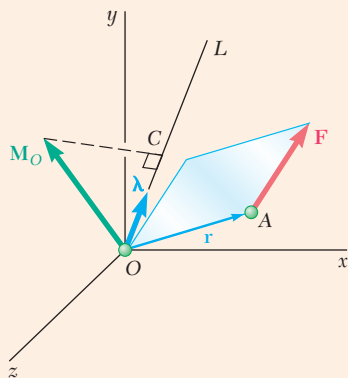


Fig. 3.51

The determinant form for the mixed triple product is

$$M_{OL} = \begin{vmatrix} \lambda_x & \lambda_y & \lambda_z \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix} \quad (3.41)$$

where

$\lambda_x, \lambda_y, \lambda_z$ = direction cosines of axis OL

x, y, z = components of $\mathbf{r}$

F_x, F_y, F_z = components of $\mathbf{F}$

An example of determining the moment of a force about a skew axis appears in Sample Prob. 3.5.

Couples

Two forces $\mathbf{F}$ and $-\mathbf{F}$ having the same magnitude, parallel lines of action, and opposite sense are said to form a **couple** [Sec. 3.3A]. The moment of a couple is independent of the point about which it is computed; it is a vector $\mathbf{M}$ perpendicular to the plane of the couple and equal in magnitude to the product of the common magnitude F of the forces and the perpendicular distance d between their lines of action (Fig. 3.52).

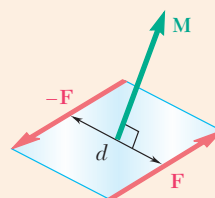


Fig. 3.52

Two couples having the same moment $\mathbf{M}$ are *equivalent*, i.e., they have the same effect on a given rigid body [Sec. 3.3B]. The sum of two couples is itself a couple [Sec. 3.3C], and we can obtain the moment $\mathbf{M}$ of the resultant couple by adding vectorially the moments $\mathbf{M}_1$ and $\mathbf{M}_2$ of the original couples [Sample Prob. 3.6]. It follows that we can represent a couple by a vector, called a **couple vector**, equal in magnitude and direction to the moment $\mathbf{M}$ of the couple [Sec. 3.3D]. A couple vector is a *free vector* that can be attached to the origin O if so desired and resolved into components (Fig. 3.53).

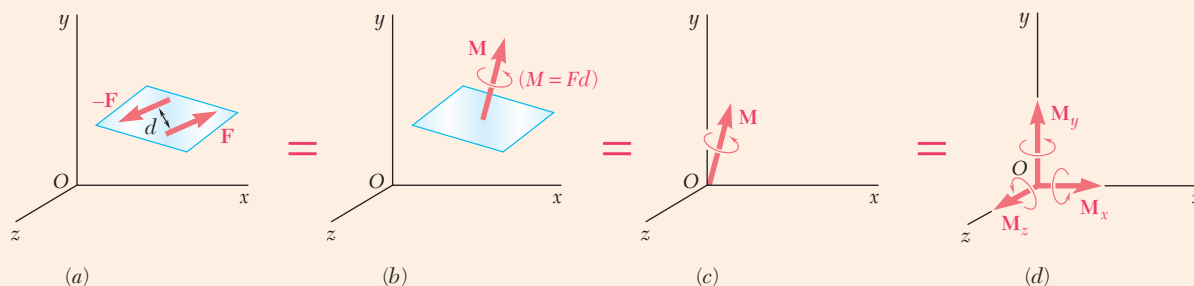


Fig. 3.53

Force-Couple System

Any force $\mathbf{F}$ acting at a point A of a rigid body can be replaced by a **force-couple system** at an arbitrary point O consisting of the force $\mathbf{F}$ applied at O

and a couple of moment $\mathbf{M}_O$, which is equal to the moment about O of the force $\mathbf{F}$ in its original position [Sec. 3.3E]. Note that the force $\mathbf{F}$ and the couple vector $\mathbf{M}_O$ are always perpendicular to each other (Fig. 3.54).

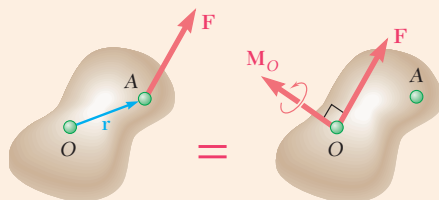


Fig. 3.54

Reduction of a System of Forces to a Force-Couple System

It follows [Sec. 3.4A] that *any system of forces can be reduced to a force-couple system at a given point O* by first replacing each of the forces of the system by an equivalent force-couple system at O (Fig. 3.55) and then adding all of the forces and all of the couples to obtain a resultant force $\mathbf{R}$ and a resultant couple vector $\mathbf{M}_O^R$ [Sample Probs. 3.8 through 3.11]. In general, the resultant $\mathbf{R}$ and the couple vector $\mathbf{M}_O^R$ will not be perpendicular to each other.

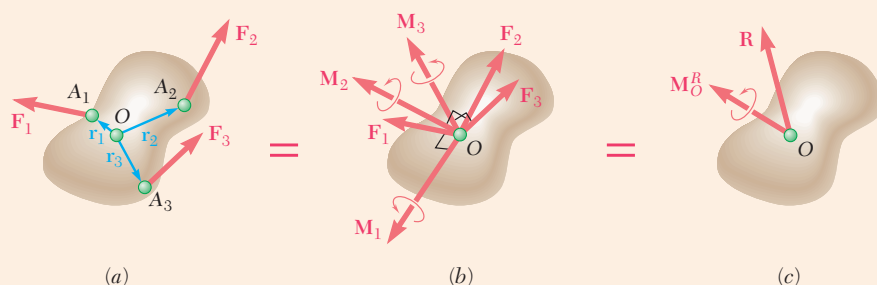


Fig. 3.55

Equivalent Systems of Forces

We concluded [Sec. 3.4B] that, as far as rigid bodies are concerned, *two systems of forces, $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$ and $\mathbf{F}'_1, \mathbf{F}'_2, \mathbf{F}'_3, \dots$, are equivalent if, and only if,*

$$\Sigma \mathbf{F} = \Sigma \mathbf{F}' \quad \text{and} \quad \Sigma \mathbf{M}_O = \Sigma \mathbf{M}'_O \quad (3.55)$$

Further Reduction of a System of Forces

If the resultant force $\mathbf{R}$ and the resultant couple vector $\mathbf{M}_O^R$ are perpendicular to each other, we can further reduce the force-couple system at O to a single resultant force [Sec. 3.4C]. This is the case for systems consisting of (a) concurrent forces (cf. Chap. 2), (b) coplanar forces [Sample Probs. 3.8 and 3.9], or (c) parallel forces [Sample Prob. 3.11]. If the resultant $\mathbf{R}$ and the couple vector $\mathbf{M}_O^R$ are *not* perpendicular to each other, the system *cannot* be reduced to a single force. We can, however, reduce it to a special type of force-couple system called a *wrench*, consisting of the resultant $\mathbf{R}$ and a couple vector $\mathbf{M}_1$ directed along $\mathbf{R}$ [Sec. 3.4D and Sample Prob. 3.12].

Review Problems

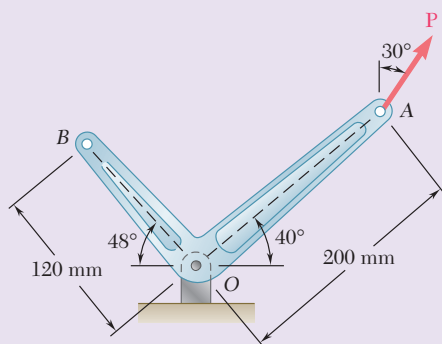


Fig. P3.147

3.147 A 300-N force $\mathbf{P}$ is applied at point A of the bell crank shown. (a) Compute the moment of the force $\mathbf{P}$ about O by resolving it into horizontal and vertical components. (b) Using the result of part a , determine the perpendicular distance from O to the line of action of $\mathbf{P}$.

3.148 A winch puller AB is used to straighten a fence post. Knowing that the tension in cable BC is 1040 N and length d is 1.90 m, determine the moment about D of the force exerted by the cable at C by resolving that force into horizontal and vertical components applied (a) at point C , (b) at point E .

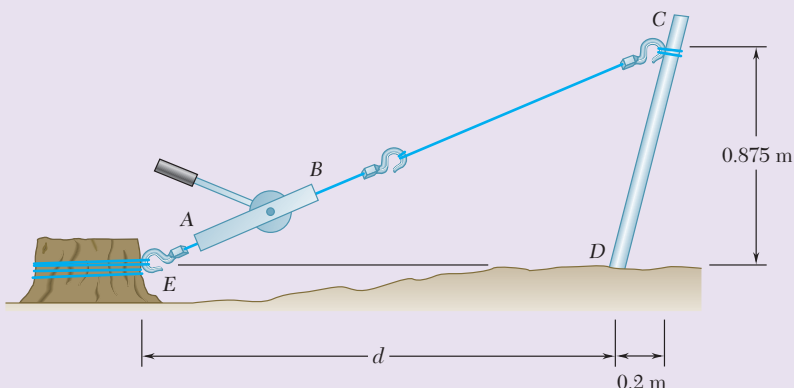


Fig. P3.148

3.149 A small boat hangs from two davits, one of which is shown in the figure. The tension in line $ABAD$ is 82 lb. Determine the moment about C of the resultant force $\mathbf{R}_A$ exerted on the davit at A .

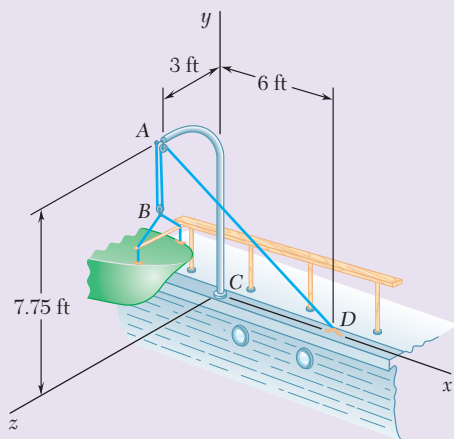


Fig. P3.149

3.150 Consider the volleyball net shown. Determine the angle formed by guy wires AB and AC .

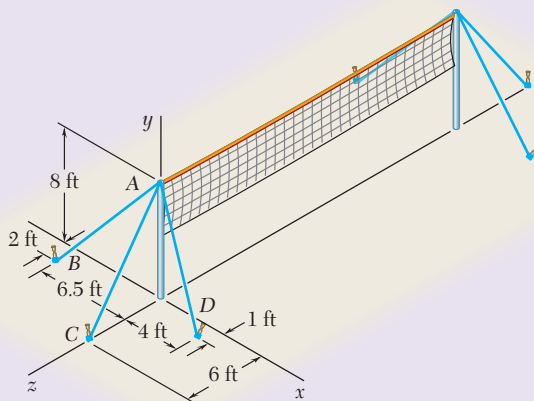


Fig. P3.150

3.151 A single force $\mathbf{P}$ acts at C in a direction perpendicular to the handle BC of the crank shown. Determine the moment M_x of $\mathbf{P}$ about the x axis when $\theta = 65^\circ$, knowing that $M_y = -15 \text{ N}\cdot\text{m}$ and $M_z = -36 \text{ N}\cdot\text{m}$.

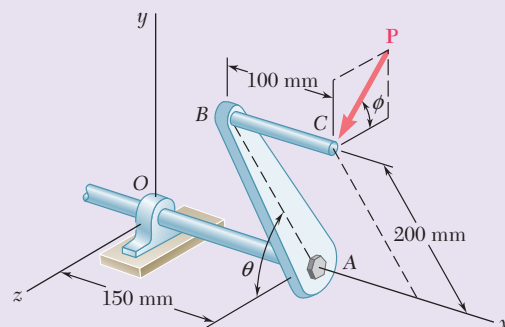


Fig. P3.151

3.152 A small boat hangs from two davits, one of which is shown in the figure. It is known that the moment about the z axis of the resultant force $\mathbf{R}_A$ exerted on the davit at A must not exceed $279 \text{ lb}\cdot\text{ft}$ in absolute value. Determine the largest allowable tension in line $ABAD$ when $x = 6 \text{ ft}$.

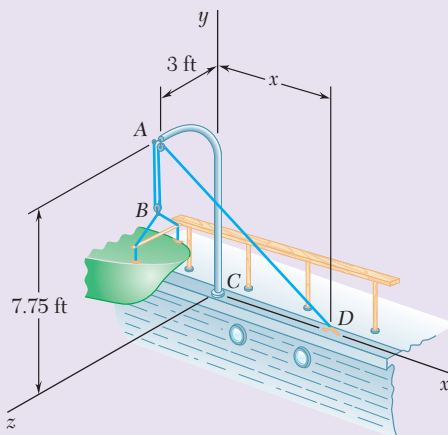


Fig. P3.152

3.153 In a manufacturing operation, three holes are drilled simultaneously in a workpiece. If the holes are perpendicular to the surfaces of the workpiece, replace the couples applied to the drills with a single equivalent couple, specifying its magnitude and the direction of its axis.

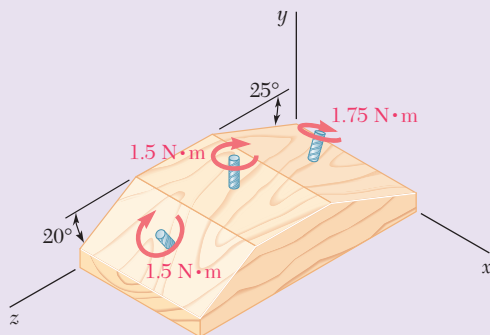


Fig. P3.153

3.154 A 260-lb force is applied at A to the rolled-steel section shown. Replace that force with an equivalent force-couple system at the center C of the section.

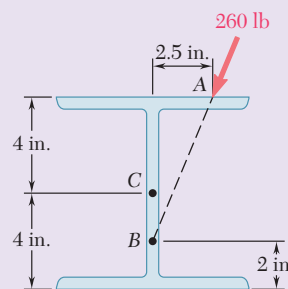


Fig. P3.154

3.155 The force and couple shown are to be replaced by an equivalent single force. Knowing that $P = 2Q$, determine the required value of α if the line of action of the single equivalent force is to pass through (a) point A , (b) point C .

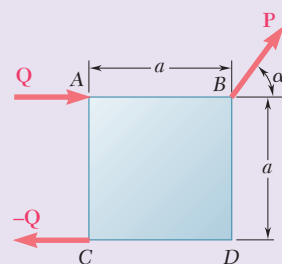


Fig. P3.155

- 3.156** A 77-N force $\mathbf{F}_1$ and a 31-N·m couple $\mathbf{M}_1$ are applied to corner E of the bent plate shown. If $\mathbf{F}_1$ and $\mathbf{M}_1$ are to be replaced with an equivalent force-couple system $(\mathbf{F}_2, \mathbf{M}_2)$ at corner B and if $(M_2)_z = 0$, determine (a) the distance d , (b) $\mathbf{F}_2$ and $\mathbf{M}_2$.

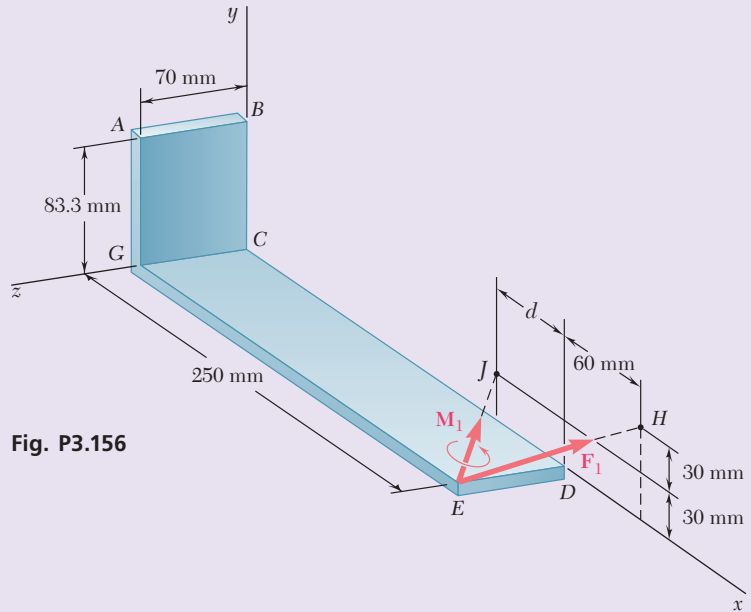


Fig. P3.156

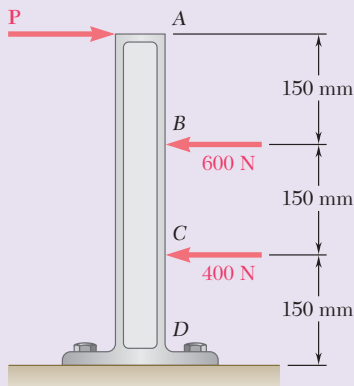


Fig. P3.157

- 3.157** Three horizontal forces are applied as shown to a vertical cast-iron arm. Determine the resultant of the forces and the distance from the ground to its line of action when (a) $P = 200$ N, (b) $P = 2400$ N, (c) $P = 1000$ N.

- 3.158** While using a pencil sharpener, a student applies the forces and couple shown. (a) Determine the forces exerted at B and C knowing that these forces and the couple are equivalent to a force-couple system at A consisting of the force $\mathbf{R} = (2.6 \text{ lb})\mathbf{i} + R_y\mathbf{j} - (0.7 \text{ lb})\mathbf{k}$ and the couple $\mathbf{M}_A^R = M_x\mathbf{i} + (1.0 \text{ lb}\cdot\text{ft})\mathbf{j} - (0.72 \text{ lb}\cdot\text{ft})\mathbf{k}$. (b) Find the corresponding values of R_y and M_x .

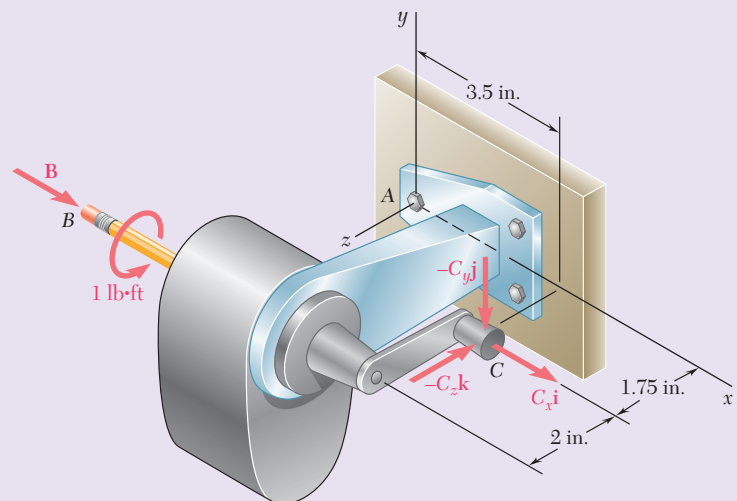


Fig. P3.158



4

Equilibrium of Rigid Bodies

The Tianjin Eye is a Ferris wheel that straddles a bridge over the Hai River in China. The structure is designed so that the support reactions at the wheel bearings as well as those at the base of the frame maintain equilibrium under the effects of vertical gravity and horizontal wind forces.

Introduction

Free-Body Diagrams

4.1 EQUILIBRIUM IN TWO DIMENSIONS

4.1A Reactions for a Two-Dimensional Structure

4.1B Rigid-Body Equilibrium in Two Dimensions

4.1C Statically Indeterminate Reactions and Partial Constraints

4.2 TWO SPECIAL CASES

4.2A Equilibrium of a Two-Force Body

4.2B Equilibrium of a Three-Force Body

4.3 EQUILIBRIUM IN THREE DIMENSIONS

4.3A Rigid-Body Equilibrium in Three Dimensions

4.3B Reactions for a Three-Dimensional Structure

Objectives

- **Analyze** the static equilibrium of rigid bodies in two and three dimensions.
- **Consider** the attributes of a properly drawn free-body diagram, an essential tool for the equilibrium analysis of rigid bodies.
- **Examine** rigid bodies supported by statically indeterminate reactions and partial constraints.
- **Study** two cases of particular interest: the equilibrium of two-force and three-force bodies.

Introduction

We saw in Chapter 3 how to reduce the external forces acting on a rigid body to a force-couple system at some arbitrary point O . When the force and the couple are both equal to zero, the external forces form a system equivalent to zero, and the rigid body is said to be in **equilibrium**.

We can obtain the necessary and sufficient conditions for the equilibrium of a rigid body by setting $\mathbf{R}$ and $\mathbf{M}_O^R$ equal to zero in the relations of Eq. (3.50) of Sec. 3.4A:

$$\Sigma \mathbf{F} = 0 \quad \Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) = 0 \quad (4.1)$$

Resolving each force and each moment into its rectangular components, we can replace these vector equations for the equilibrium of a rigid body with the following six scalar equations:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma F_z = 0 \quad (4.2)$$

$$\Sigma M_x = 0 \quad \Sigma M_y = 0 \quad \Sigma M_z = 0 \quad (4.3)$$

We can use these equations to determine unknown forces applied to the rigid body or unknown reactions exerted on it by its supports. Note that Eqs. (4.2) express the fact that the components of the external forces in the x , y , and z directions are balanced; Eqs. (4.3) express the fact that the moments of the external forces about the x , y , and z axes are balanced. Therefore, for a rigid body in equilibrium, the system of external forces imparts no translational or rotational motion to the body.

In order to write the equations of equilibrium for a rigid body, we must first identify all of the forces acting on that body and then draw the corresponding **free-body diagram**. In this chapter, we first consider the equilibrium of *two-dimensional structures* subjected to forces contained in their planes and study how to draw their free-body diagrams. In addition to the forces *applied* to a structure, we must also consider the *reactions* exerted on the structure by its supports. A specific reaction is associated with each type of support. You will see how to determine whether the structure is properly supported, so that you can know in advance whether you can solve the equations of equilibrium for the unknown forces and reactions.

Later in this chapter, we consider the equilibrium of three-dimensional structures, and we provide the same kind of analysis to these structures and their supports.

Free-Body Diagrams

In solving a problem concerning a rigid body in equilibrium, it is essential to consider *all* of the forces acting on the body. It is equally important to exclude any force that is *not* directly applied to the body. Omitting a force or adding an extraneous one would destroy the conditions of equilibrium. Therefore, the first step in solving the problem is to draw a **free-body diagram** of the rigid body under consideration.

We have already used free-body diagrams on many occasions in Chap. 2. However, in view of their importance to the solution of equilibrium problems, we summarize here the steps you must follow in drawing a correct free-body diagram.

1. Start with a clear decision regarding the choice of the free body to be analyzed. Mentally, you need to detach this body from the ground and separate it from all other bodies. Then you can sketch the contour of this isolated body.
2. Indicate all external forces on the free-body diagram. These forces represent the actions exerted *on* the free body *by* the ground and *by* the bodies that have been detached. In the diagram, apply these forces at the various points where the free body was supported by the ground or

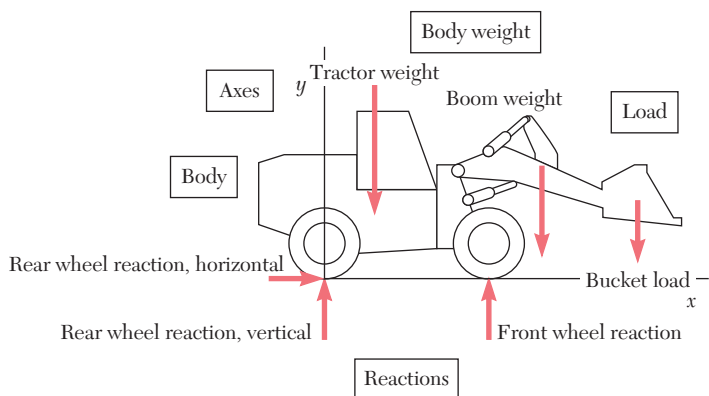


Photo 4.1 A tractor supporting a bucket load. As shown, its free-body diagram should include all external forces acting on the tractor.

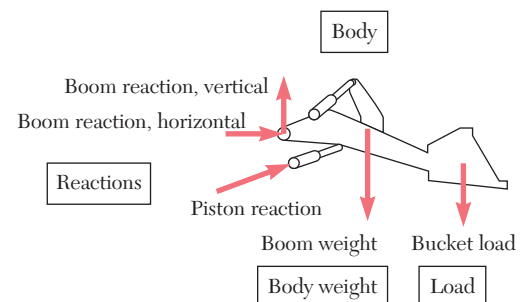


Photo 4.2 Tractor bucket and boom. In Chap. 6, we will see how to determine the internal forces associated with interconnected members such as these using free-body diagrams like the one shown.

was connected to the other bodies. Generally, you should include the *weight* of the free body among the external forces, since it represents the attraction exerted by the earth on the various particles forming the free body. You will see in Chapter 5 that you should draw the weight so it acts at the center of gravity of the body. If the free body is made of several parts, do *not* include the forces the various parts exert on each other among the external forces. These forces are internal forces as far as the free body is concerned.

3. Clearly mark the magnitudes and directions of the *known external forces* on the free-body diagram. Recall that when indicating the directions of these forces, the forces are those exerted *on*, and not *by*, the free body. Known external forces generally include the *weight* of the free body and *forces applied* for a given purpose.
4. *Unknown external forces* usually consist of the **reactions** through which the ground and other bodies oppose a possible motion of the free body. The reactions constrain the free body to remain in the same position; for that reason, they are sometimes called *constraining forces*. Reactions are exerted at the points where the free body is *supported by* or *connected to* other bodies; you should clearly indicate these points. Reactions are discussed in detail in Secs. 4.1 and 4.3.
5. The free-body diagram should also include dimensions, since these may be needed for computing moments of forces. Any other detail, however, should be omitted.

4.1 EQUILIBRIUM IN TWO DIMENSIONS

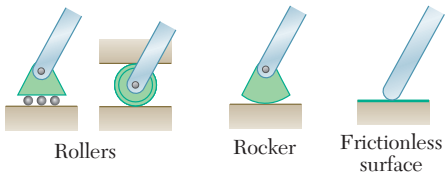
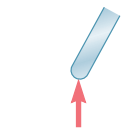
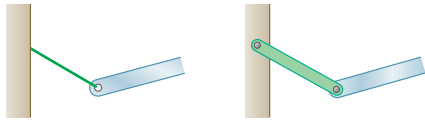
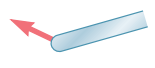
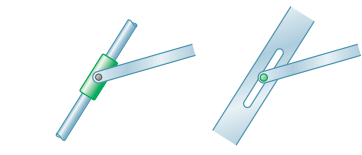
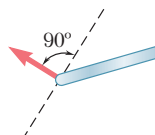
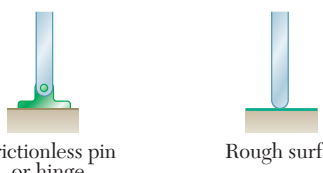
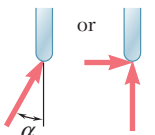
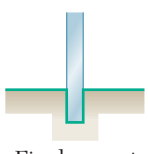
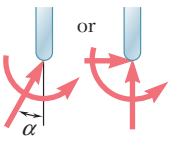
In the first part of this chapter, we consider the equilibrium of two-dimensional structures; i.e., we assume that the structure being analyzed and the forces applied to it are contained in the same plane. Clearly, the reactions needed to maintain the structure in the same position are also contained in this plane.

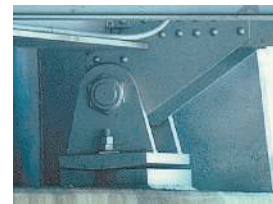
4.1A Reactions for a Two-Dimensional Structure

The reactions exerted on a two-dimensional structure fall into three categories that correspond to three types of **supports** or **connections**.

1. **Reactions Equivalent to a Force with a Known Line of Action.** Supports and connections causing reactions of this type include *rollers*, *rockers*, *frictionless surfaces*, *short links and cables*, *collars on frictionless rods*, and *frictionless pins in slots*. Each of these supports and connections can prevent motion in one direction only. Figure 4.1 shows these supports and connections together with the reactions they produce. Each reaction involves *one unknown*—specifically, the magnitude of the reaction. In problem solving, you should denote this magnitude by an appropriate letter. The line of action of the reaction is known and should be indicated clearly in the free-body diagram.

The sense of the reaction must be as shown in Fig. 4.1 for cases of a frictionless surface (toward the free body) or a cable (away from the free body). The reaction can be directed either way in the cases of double-track rollers, links, collars on rods, or pins in slots. Generally, we

Support or Connection	Reaction	Number of Unknowns
 Rollers Rocker Frictionless surface	 Force with known line of action perpendicular to surface	1
 Short cable Short link	 Force with known line of action along cable or link	1
 Collar on frictionless rod Frictionless pin in slot	 Force with known line of action perpendicular to rod or slot	1
 Frictionless pin or hinge Rough surface	 Force of unknown direction	2
 Fixed support	 Force and couple	3



This rocker bearing supports the weight of a bridge. The convex surface of the rocker allows the bridge to move slightly horizontally.



Links are often used to support suspended spans of highway bridges.



Force applied to the slider exerts a normal force on the rod, causing the window to open.



Pin supports are common on bridges and overpasses.



This cantilever support is fixed at one end and extends out into space at the other end.

Fig. 4.1 Reactions of supports and connections in two dimensions.

assume that single-track rollers and rockers are reversible, so the corresponding reactions can be directed either way.

- Reactions Equivalent to a Force of Unknown Direction and Magnitude.** Supports and connections causing reactions of this type include *frictionless pins in fitted holes*, *hinges*, and *rough surfaces*. They can prevent translation of the free body in all directions, but they cannot prevent the body from rotating about the connection. Reactions of this group involve *two unknowns* and are usually represented by their x and

y components. In the case of a rough surface, the component normal to the surface must be directed away from the surface.

- 3. Reactions Equivalent to a Force and a Couple.** These reactions are caused by *fixed supports* that oppose any motion of the free body and thus constrain it completely. Fixed supports actually produce forces over the entire surface of contact; these forces, however, form a system that can be reduced to a force and a couple. Reactions of this group involve *three unknowns* usually consisting of the two components of the force and the moment of the couple.

When the sense of an unknown force or couple is not readily apparent, do not attempt to determine it. Instead, arbitrarily assume the sense of the force or couple; the sign of the answer will indicate whether the assumption is correct or not. (A positive answer means the assumption is correct, while a negative answer means the assumption is incorrect.)

4.1B Rigid-Body Equilibrium in Two Dimensions

The conditions stated in Sec. 4.1A for the equilibrium of a rigid body become considerably simpler for the case of a two-dimensional structure. Choosing the x and y axes to be in the plane of the structure, we have

$$F_z = 0 \quad M_x = M_y = 0 \quad M_z = M_O$$

for each of the forces applied to the structure. Thus, the six equations of equilibrium stated in Sec. 4.1 reduce to three equations:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_O = 0 \quad (4.4)$$

Since $\Sigma M_O = 0$ must be satisfied regardless of the choice of the origin O , we can write the equations of equilibrium for a two-dimensional structure in the more general form

Equations of equilibrium in two dimensions

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_A = 0 \quad (4.5)$$

where A is any point in the plane of the structure. These three equations can be solved for no more than *three unknowns*.

You have just seen that unknown forces include reactions and that the number of unknowns corresponding to a given reaction depends upon the type of support or connection causing that reaction. Referring to Fig. 4.1, note that you can use the equilibrium equations (4.5) to determine the reactions associated with two rollers and one cable, or one fixed support, or one roller and one pin in a fitted hole, etc.

For example, consider Fig. 4.2a, in which the truss shown is in equilibrium and is subjected to the given forces $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$. The truss is held in place by a pin at A and a roller at B . The pin prevents point A from moving by exerting a force on the truss that can be resolved into the components A_x and A_y . The roller keeps the truss from rotating about A by exerting the vertical force $\mathbf{B}$. The free-body diagram of the truss is shown in Fig. 4.2b; it includes the reactions A_x , A_y , and $\mathbf{B}$ as well as the applied forces $\mathbf{P}$, $\mathbf{Q}$, and $\mathbf{S}$ (in x and y component form) and the weight $\mathbf{W}$ of the truss.

Since the truss is in equilibrium, the sum of the moments about A of all of the forces shown in Fig. 4.2b is zero, or $\Sigma M_A = 0$. We can use

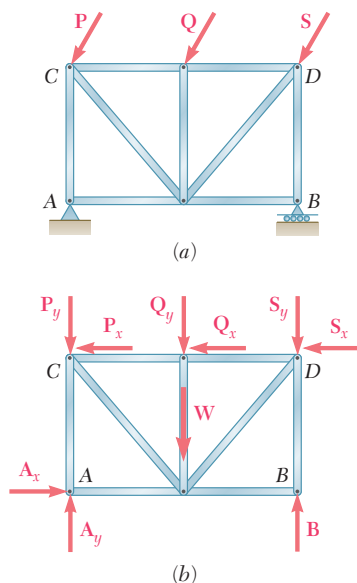


Fig. 4.2 (a) A truss supported by a pin and a roller; (b) free-body diagram of the truss.

this equation to determine the magnitude B because the equation does not contain A_x or A_y . Then, since the sum of the x components and the sum of the y components of the forces are zero, we write the equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$. From these equations, we can obtain the components A_x and A_y , respectively.

We could obtain an additional equation by noting that the sum of the moments of the external forces about a point other than A is zero. We could write, for instance, $\Sigma M_B = 0$. This equation, however, does not contain any new information, because we have already established that the system of forces shown in Fig. 4.2b is equivalent to zero. The additional equation *is not independent* and cannot be used to determine a fourth unknown. It can be useful, however, for checking the solution obtained from the original three equations of equilibrium.

Although the three equations of equilibrium cannot be *augmented* by additional equations, any of them can be *replaced* by another equation. Properly chosen, the new system of equations still describes the equilibrium conditions but may be easier to work with. For example, an alternative system of equations for equilibrium is

$$\Sigma F_x = 0 \quad \Sigma M_A = 0 \quad \Sigma M_B = 0 \quad (4.6)$$

Here the second point about which the moments are summed (in this case, point B) cannot lie on the line parallel to the y axis that passes through point A (Fig. 4.2b). These equations are sufficient conditions for the equilibrium of the truss. The first two equations indicate that the external forces must reduce to a single vertical force at A . Since the third equation requires that the moment of this force be zero about a point B that is not on its line of action, the force must be zero, and the rigid body is in equilibrium.

A third possible set of equilibrium equations is

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad \Sigma M_C = 0 \quad (4.7)$$

where the points A , B , and C do not lie in a straight line (Fig. 4.2b). The first equation requires that the external forces reduce to a single force at A ; the second equation requires that this force pass through B ; and the third equation requires that it pass through C . Since the points A , B , C do not lie in a straight line, the force must be zero, and the rigid body is in equilibrium.

Notice that the equation $\Sigma M_A = 0$, stating that the sum of the moments of the forces about pin A is zero, possesses a more definite physical meaning than either of the other two equations (4.7). These two equations express a similar idea of balance but with respect to points about which the rigid body is not actually hinged. They are, however, as useful as the first equation. The choice of equilibrium equations should not be unduly influenced by their physical meaning. Indeed, in practice, it is desirable to choose equations of equilibrium containing only one unknown, since this eliminates the necessity of solving simultaneous equations. You can obtain equations containing only one unknown by summing moments about the point of intersection of the lines of action of two unknown forces or, if these forces are parallel, by summing force components in a direction perpendicular to their common direction.

For example, in Fig. 4.3, in which the truss shown is held by rollers at A and B and a short link at D , we can eliminate the reactions at A and B by summing x components. We can eliminate the reactions at A and D

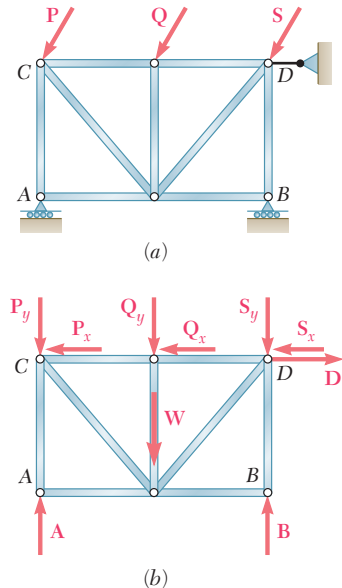


Fig. 4.3 (a) A truss supported by two rollers and a short link; (b) free-body diagram of the truss.

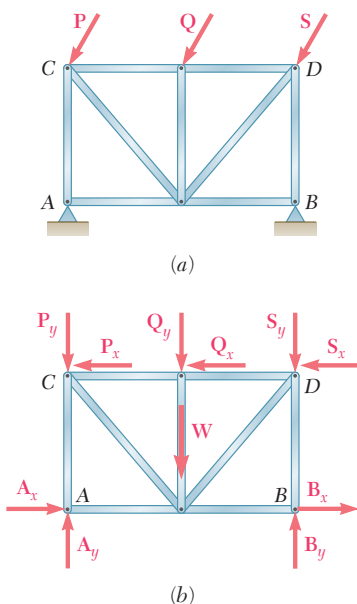


Fig. 4.4 (a) Truss with statically indeterminate reactions; (b) free-body diagram.

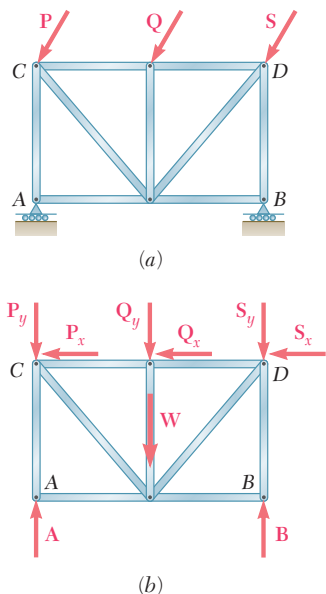


Fig. 4.5 (a) Truss with partial constraints; (b) free-body diagram.

by summing moments about C and the reactions at B and D by summing moments about D . The resulting equations are

$$\Sigma F_x = 0 \quad \Sigma M_C = 0 \quad \Sigma M_D = 0$$

Each of these equations contains only one unknown.

4.1C Statically Indeterminate Reactions and Partial Constraints

In the two examples considered in Figs. 4.2 and 4.3, the types of supports used were such that the rigid body could not possibly move under the given loads or under any other loading conditions. In such cases, the rigid body is said to be **completely constrained**. Recall that the reactions corresponding to these supports involved *three unknowns* and could be determined by solving the three equations of equilibrium. When such a situation exists, the reactions are said to be **statically determinate**.

Consider Fig. 4.4a, in which the truss shown is held by pins at A and B . These supports provide more constraints than are necessary to keep the truss from moving under the given loads or under any other loading conditions. Note from the free-body diagram of Fig. 4.4b that the corresponding reactions involve *four unknowns*. We pointed out in Sec. 4.1D that only three independent equilibrium equations are available; therefore, in this case, we have *more unknowns than equations*. As a result, we cannot determine all of the unknowns. The equations $\Sigma M_A = 0$ and $\Sigma M_B = 0$ yield the vertical components B_y and A_y , respectively, but the equation $\Sigma F_x = 0$ gives only the sum $A_x + B_x$ of the horizontal components of the reactions at A and B . The components A_x and B_x are **statically indeterminate**. We could determine their magnitudes by considering the deformations produced in the truss by the given loading, but this method is beyond the scope of statics and belongs to the study of mechanics of materials.

Let's consider the opposite situation. The supports holding the truss shown in Fig. 4.5a consist of rollers at A and B . Clearly, the constraints provided by these supports are not sufficient to keep the truss from moving. Although they prevent any vertical motion, the truss is free to move horizontally. The truss is said to be **partially constrained**.[†] From the free-body diagram in Fig. 4.5b, note that the reactions at A and B involve only *two unknowns*. Since three equations of equilibrium must still be satisfied, we have *fewer unknowns than equations*. In such a case, one of the equilibrium equations will not be satisfied in general. The equations $\Sigma M_A = 0$ and $\Sigma M_B = 0$ can be satisfied by a proper choice of reactions at A and B , but the equation $\Sigma F_x = 0$ is not satisfied unless the sum of the horizontal components of the applied forces happens to be zero. We thus observe that the equilibrium of the truss of Fig. 4.5 cannot be maintained under general loading conditions.

From these examples, it would appear that, if a rigid body is to be completely constrained and if the reactions at its supports are to be statically determinate, **there must be as many unknowns as there are equations of equilibrium**. When this condition is *not* satisfied, we can be certain that either the rigid body is not completely constrained or that the reactions at its supports

[†]Partially constrained bodies are often referred to as *unstable*. However, to avoid confusion between this type of instability, due to insufficient constraints, and the type of instability considered in Chap. 10, which relates to the behavior of a rigid body when its equilibrium is disturbed, we shall restrict the use of the words *stable* and *unstable* to the latter case.

are not statically determinate. It is also possible that the rigid body is not completely constrained *and* that the reactions are statically indeterminate.

You should note, however, that, although this condition is *necessary*, it is *not sufficient*. In other words, the fact that the number of unknowns is equal to the number of equations is no guarantee that a body is completely constrained or that the reactions at its supports are statically determinate. Consider Fig. 4.6a, which shows a truss held by rollers at *A*, *B*, and *E*. We have three unknown reactions of *A*, *B*, and *E* (Fig. 4.6b), but the equation $\Sigma F_x = 0$ is not satisfied unless the sum of the horizontal components of the applied forces happens to be zero. Although there are a sufficient number of constraints, these constraints are not properly arranged, so the truss is free to move horizontally. We say that the truss is **improperly constrained**. Since only two equilibrium equations are left for determining three unknowns, the reactions are statically indeterminate. Thus, improper constraints also produce static indeterminacy.

The truss shown in Fig. 4.7 is another example of improper constraints—and of static indeterminacy. This truss is held by a pin at *A* and by rollers at *B* and *C*, which altogether involve four unknowns. Since only three independent equilibrium equations are available, the reactions at the supports are statically indeterminate. On the other hand, we note that the equation $\Sigma M_A = 0$ cannot be satisfied under general loading conditions, since the lines of action of the reactions *B* and *C* pass through *A*. We conclude that the truss can rotate about *A* and that it is improperly constrained.[†]

The examples of Figs. 4.6 and 4.7 lead us to conclude that

A rigid body is improperly constrained whenever the supports (even though they may provide a sufficient number of reactions) are arranged in such a way that the reactions must be either concurrent or parallel.[‡]

In summary, to be sure that a two-dimensional rigid body is completely constrained and that the reactions at its supports are statically determinate, you should verify that the reactions involve three—and only three—unknowns and that the supports are arranged in such a way that they do not require the reactions to be either concurrent or parallel.

Supports involving statically indeterminate reactions should be used with care in the design of structures and only with a full knowledge of the problems they may cause. On the other hand, the analysis of structures possessing statically indeterminate reactions often can be partially carried out by the methods of statics. In the case of the truss of Fig. 4.4, for example, we can determine the vertical components of the reactions at *A* and *B* from the equilibrium equations.

For obvious reasons, supports producing partial or improper constraints should be avoided in the design of stationary structures. However, a partially or improperly constrained structure will not necessarily collapse; under particular loading conditions, equilibrium can be maintained. For example, the trusses of Figs. 4.5 and 4.6 will be in equilibrium if the applied forces *P*, *Q*, and *S* are vertical. Besides, structures designed to move *should* be only partially constrained. A railroad car, for instance, would be of little use if it were completely constrained by having its brakes applied permanently.

[†]Rotation of the truss about *A* requires some “play” in the supports at *B* and *C*. In practice such play will always exist. In addition, we note that if the play is kept small, the displacements of the rollers *B* and *C* and, thus, the distances from *A* to the lines of action of the reactions *B* and *C* will also be small. The equation $\Sigma M_A = 0$ then requires that *B* and *C* be very large, a situation which can result in the failure of the supports at *B* and *C*.

[‡]Because this situation arises from an inadequate arrangement or *geometry* of the supports, it is often referred to as *geometric instability*.

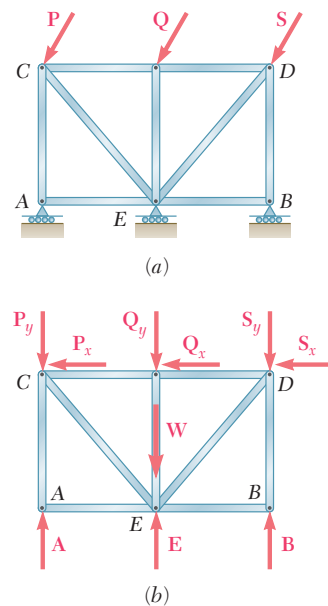


Fig. 4.6 (a) Truss with improper constraints; (b) free-body diagram.

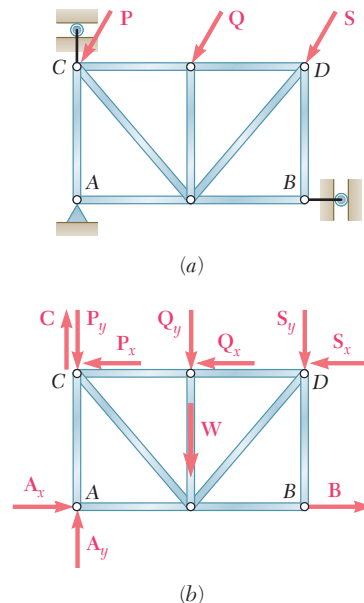


Fig. 4.7 (a) Truss with improper constraints; (b) free-body diagram.

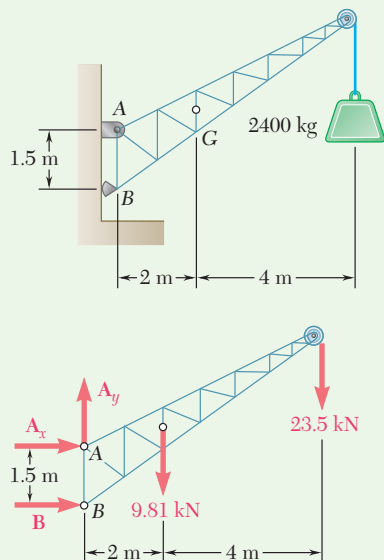


Fig. 1 Free-body diagram of crane.

Sample Problem 4.1

A fixed crane has a mass of 1000 kg and is used to lift a 2400-kg crate. It is held in place by a pin at *A* and a rocker at *B*. The center of gravity of the crane is located at *G*. Determine the components of the reactions at *A* and *B*.

STRATEGY: Draw a free-body diagram to show all of the forces acting on the crane, then use the equilibrium equations to calculate the values of the unknown forces.

MODELING:

Free-Body Diagram. By multiplying the masses of the crane and of the crate by $g = 9.81 \text{ m/s}^2$, you obtain the corresponding weights—that is, 9810 N or 9.81 kN, and 23 500 N or 23.5 kN (Fig. 1). The reaction at pin *A* is a force of unknown direction; you can represent it by components A_x and A_y . The reaction at the rocker *B* is perpendicular to the rocker surface; thus, it is horizontal. Assume that A_x , A_y , and B act in the directions shown.

ANALYSIS:

Determination of B . The sum of the moments of all external forces about point *A* is zero. The equation for this sum contains neither A_x nor A_y , since the moments of A_x and A_y about *A* are zero. Multiplying the magnitude of each force by its perpendicular distance from *A*, you have

$$+\circlearrowleft \sum M_A = 0: \quad +B(1.5 \text{ m}) - (9.81 \text{ kN})(2 \text{ m}) - (23.5 \text{ kN})(6 \text{ m}) = 0$$

$$B = +107.1 \text{ kN} \quad \mathbf{B = 107.1 \text{ kN} \rightarrow}$$

Since the result is positive, the reaction is directed as assumed.

Determination of A_x . Determine the magnitude of A_x by setting the sum of the horizontal components of all external forces to zero.

$$+\rightarrow \sum F_x = 0: \quad A_x + B = 0$$

$$A_x + 107.1 \text{ kN} = 0$$

$$A_x = -107.1 \text{ kN} \quad \mathbf{A_x = 107.1 \text{ kN} \leftarrow}$$

Since the result is negative, the sense of A_x is opposite to that assumed originally.

Determination of A_y . The sum of the vertical components must also equal zero. Therefore,

$$+\uparrow \sum F_y = 0: \quad A_y - 9.81 \text{ kN} - 23.5 \text{ kN} = 0$$

$$A_y = +33.3 \text{ kN} \quad \mathbf{A_y = 33.3 \text{ kN} \uparrow}$$

Adding the components A_x and A_y vectorially, you can find that the reaction at *A* is 112.2 kN $\searrow 17.3^\circ$.

REFLECT and THINK: You can check the values obtained for the reactions by recalling that the sum of the moments of all the external forces about any point must be zero. For example, considering point *B* (Fig. 2), you can show

$$+\circlearrowleft \sum M_B = -(9.81 \text{ kN})(2 \text{ m}) - (23.5 \text{ kN})(6 \text{ m}) + (107.1 \text{ kN})(1.5 \text{ m}) = 0$$

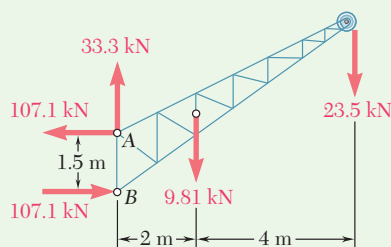


Fig. 2 Free-body diagram of crane with solved reactions.

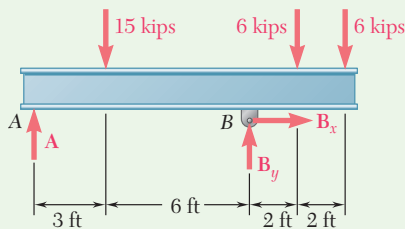
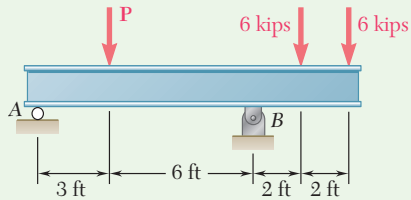


Fig. 1 Free-body diagram of beam.

Sample Problem 4.2

Three loads are applied to a beam as shown. The beam is supported by a roller at A and by a pin at B . Neglecting the weight of the beam, determine the reactions at A and B when $P = 15$ kips.

STRATEGY: Draw a free-body diagram of the beam, then write the equilibrium equations, first summing forces in the x direction and then summing moments at A and at B .

MODELING:

Free-Body Diagram. The reaction at A is vertical and is denoted by A (Fig. 1). Represent the reaction at B by components B_x and B_y . Assume that each component acts in the direction shown.

ANALYSIS:

Equilibrium Equations. Write the three equilibrium equations and solve for the reactions indicated:

$$+\rightarrow \Sigma F_x = 0: \quad B_x = 0 \quad B_x = 0 \quad \blacktriangleleft$$

$$+\uparrow \Sigma M_A = 0: \quad -(15 \text{ kips})(3 \text{ ft}) + B_y(9 \text{ ft}) - (6 \text{ kips})(11 \text{ ft}) - (6 \text{ kips})(13 \text{ ft}) = 0$$

$$B_y = +21.0 \text{ kips} \quad B_y = 21.0 \text{ kips} \uparrow \quad \blacktriangleleft$$

$$+\uparrow \Sigma M_B = 0: \quad -A(9 \text{ ft}) + (15 \text{ kips})(6 \text{ ft}) - (6 \text{ kips})(2 \text{ ft}) - (6 \text{ kips})(4 \text{ ft}) = 0$$

$$A = +6.00 \text{ kips} \quad A = 6.00 \text{ kips} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: Check the results by adding the vertical components of all of the external forces:

$$+\uparrow \Sigma F_y = +6.00 \text{ kips} - 15 \text{ kips} + 21.0 \text{ kips} - 6 \text{ kips} - 6 \text{ kips} = 0$$

Remark. In this problem, the reactions at both A and B are vertical; however, these reactions are vertical for different reasons. At A , the beam is supported by a roller; hence, the reaction cannot have any horizontal component. At B , the horizontal component of the reaction is zero because it must satisfy the equilibrium equation $\Sigma F_x = 0$ and none of the other forces acting on the beam has a horizontal component.

You might have noticed at first glance that the reaction at B was vertical and dispensed with the horizontal component B_x . This, however, is bad practice. In following it, you run the risk of forgetting the component B_x when the loading conditions require such a component (i.e., when a horizontal load is included). Also, you found the component B_x to be zero by using and solving an equilibrium equation, $\Sigma F_x = 0$. By setting B_x equal to zero immediately, you might not realize that you actually made use of this equation. Thus, you might lose track of the number of equations available for solving the problem.

Sample Problem 4.3

A loading car is at rest on a track forming an angle of 25° with the vertical. The gross weight of the car and its load is 5500 lb, and it acts at a point 30 in. from the track, halfway between the two axles. The car is held by a cable attached 24 in. from the track. Determine the tension in the cable and the reaction at each pair of wheels.

STRATEGY: Draw a free-body diagram of the car to determine the unknown forces, and write equilibrium equations to find their values, summing moments at A and B and then summing forces.

MODELING:

Free-Body Diagram. The reaction at each wheel is perpendicular to the track, and the tension force $\mathbf{T}$ is parallel to the track. Therefore, for convenience, choose the x axis parallel to the track and the y axis perpendicular to the track (Fig. 1). Then resolve the 5500-lb weight into x and y components.

$$W_x = +(5500 \text{ lb}) \cos 25^\circ = +4980 \text{ lb}$$

$$W_y = -(5500 \text{ lb}) \sin 25^\circ = -2320 \text{ lb}$$

ANALYSIS:

Equilibrium Equations. Take moments about A to eliminate $\mathbf{T}$ and $\mathbf{R}_1$ from the computation.

$$+\circlearrowleft \Sigma M_A = 0: \quad -(2320 \text{ lb})(25 \text{ in.}) - (4980 \text{ lb})(6 \text{ in.}) + R_2(50 \text{ in.}) = 0$$

$$R_2 = +1758 \text{ lb} \quad \nearrow \quad \blacktriangleleft$$

Then take moments about B to eliminate $\mathbf{T}$ and $\mathbf{R}_2$ from the computation.

$$+\circlearrowleft \Sigma M_B = 0: \quad (2320 \text{ lb})(25 \text{ in.}) - (4980 \text{ lb})(6 \text{ in.}) - R_1(50 \text{ in.}) = 0$$

$$R_1 = +562 \text{ lb} \quad \nearrow \quad \blacktriangleleft$$

Determine the value of T by summing forces in the x direction.

$$\searrow + \Sigma F_x = 0: \quad +4980 \text{ lb} - T = 0$$

$$T = +4980 \text{ lb} \quad \nwarrow \quad \blacktriangleleft$$

Figure 2 shows the computed values of the reactions.

REFLECT and THINK: You can verify the computations by summing forces in the y direction.

$$\nearrow + \Sigma F_y = +562 \text{ lb} + 1758 \text{ lb} - 2320 \text{ lb} = 0$$

You could also check the solution by computing moments about any point other than A or B .

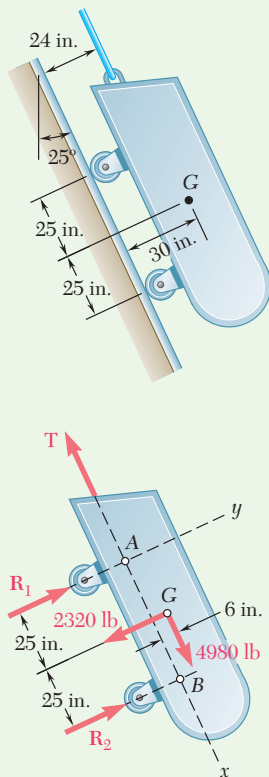


Fig. 1 Free-body diagram of car.

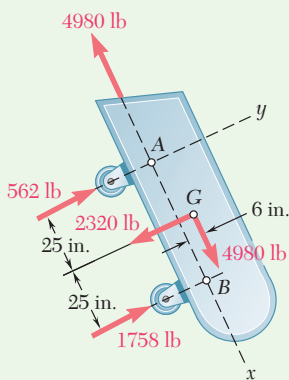
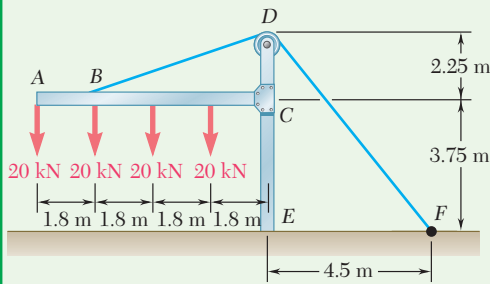


Fig. 2 Free-body diagram of car with solved reactions.



Sample Problem 4.4

The frame shown supports part of the roof of a small building. Knowing that the tension in the cable is 150 kN, determine the reaction at the fixed end E .

STRATEGY: Draw a free-body diagram of the frame and of the cable BDF . The support at E is fixed, so the reactions here include a moment; to determine its value, sum moments about point E .

MODELING:

Free-Body Diagram. Represent the reaction at the fixed end E by the force components E_x and E_y and the couple M_E (Fig. 1). The other forces acting on the free body are the four 20-kN loads and the 150-kN force exerted at end F of the cable.

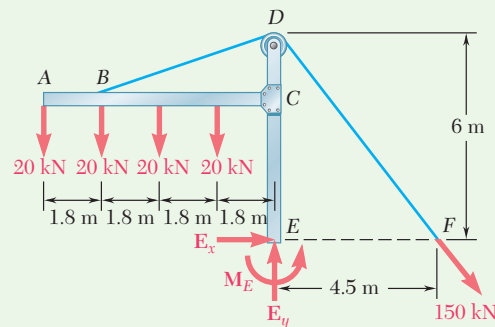


Fig. 1 Free-body diagram of frame.

ANALYSIS:

Equilibrium Equations. First note that

$$DF = \sqrt{(4.5 \text{ m})^2 + (6 \text{ m})^2} = 7.5 \text{ m}$$

Then you can write the three equilibrium equations and solve for the reactions at E .

$$\begin{aligned} \rightarrow \Sigma F_x = 0: \quad E_x + \frac{4.5}{7.5}(150 \text{ kN}) &= 0 \\ E_x &= -90.0 \text{ kN} \quad \mathbf{E_x = 90.0 \text{ kN} \leftarrow} \end{aligned}$$

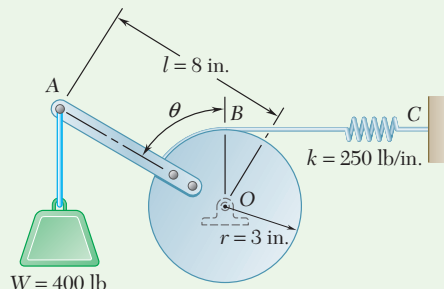
$$\begin{aligned} +\uparrow \Sigma F_y = 0: \quad E_y - 4(20 \text{ kN}) - \frac{6}{7.5}(150 \text{ kN}) &= 0 \\ E_y &= +200 \text{ kN} \quad \mathbf{E_y = 200 \text{ kN} \uparrow} \end{aligned}$$

$$\begin{aligned} +\curvearrowright \Sigma M_E = 0: \quad (20 \text{ kN})(7.2 \text{ m}) + (20 \text{ kN})(5.4 \text{ m}) + (20 \text{ kN})(3.6 \text{ m}) \\ + (20 \text{ kN})(1.8 \text{ m}) - \frac{6}{7.5}(150 \text{ kN})(4.5 \text{ m}) + M_E &= 0 \\ M_E &= +180.0 \text{ kN}\cdot\text{m} \quad \mathbf{M_E = 180.0 \text{ kN}\cdot\text{m} \curvearrowright} \end{aligned}$$

REFLECT and THINK: The cable provides a fourth constraint, making this situation statically indeterminate. This problem therefore gave us the value of the cable tension, which would have been determined by means other than statics. We could then use the three available independent static equilibrium equations to solve for the remaining three reactions.

Sample Problem 4.5

A 400-lb weight is attached at A to the lever shown. The constant of the spring BC is $k = 250$ lb/in., and the spring is unstretched when $\theta = 0$. Determine the position of equilibrium.



STRATEGY: Draw a free-body diagram of the lever and cylinder to show all forces acting on the body (Fig. 1), then sum moments about O . Your final answer should be the angle θ .

MODELING:

Free-Body Diagram. Denote by s the deflection of the spring from its unstretched position and note that $s = r\theta$. Then $F = ks = kr\theta$.

ANALYSIS:

Equilibrium Equation. Sum the moments of $\mathbf{W}$ and $\mathbf{F}$ about O to eliminate the reactions supporting the cylinder. The result is

$$+\circlearrowleft \Sigma M_O = 0: \quad Wl \sin \theta - r(kr\theta) = 0 \quad \sin \theta = \frac{kr^2}{Wl} \theta$$

Substituting the given data yields

$$\sin \theta = \frac{(250 \text{ lb/in.})(3 \text{ in.})^2}{(400 \text{ lb})(8 \text{ in.})} \theta \quad \sin \theta = 0.703 \theta$$

Solving by trial and error, the angle is $\theta = 0$ $\theta = 80.3^\circ$ ◀

REFLECT and THINK: The weight could represent any vertical force acting on the lever. The key to the problem is to express the spring force as a function of the angle θ .

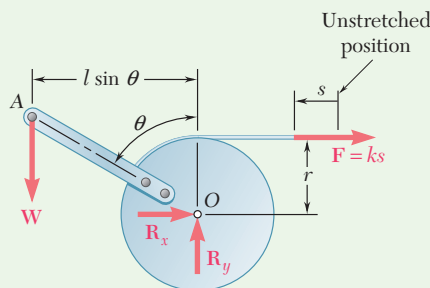


Fig. 1 Free-body diagram of the lever and cylinder.

SOLVING PROBLEMS ON YOUR OWN

You saw that, for a rigid body in equilibrium, the system of external forces is equivalent to zero. To solve an equilibrium problem, your first task is to draw a neat, reasonably large **free-body diagram** on which you show all external forces. You should include both known and unknown forces.

For a two-dimensional rigid body, the reactions at the supports can involve one, two, or three unknowns, depending on the type of support (Fig. 4.1). A correct free-body diagram is essential for the successful solution of a problem. Never proceed with the solution of a problem until you are sure that your free-body diagram includes all loads, all reactions, and the weight of the body (if appropriate).

1. You can write three equilibrium equations and solve them for *three unknowns*. The three equations might be

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_O = 0$$

However, usually several alternative sets of equations are possible, such as

$$\Sigma F_x = 0 \quad \Sigma M_A = 0 \quad \Sigma M_B = 0$$

where point *B* is chosen in such a way that the line *AB* is not parallel to the *y* axis, or

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad \Sigma M_C = 0$$

where the points *A*, *B*, and *C* do not lie along a straight line.

2. To simplify your solution, it may be helpful to use one of the following solution techniques.

a. By summing moments about the point of intersection of the lines of action of two unknown forces, you obtain an equation in a single unknown.

b. By summing components in a direction perpendicular to two unknown parallel forces, you also obtain an equation in a single unknown.

3. After drawing your free-body diagram, you may find that one of the following special situations arises.

a. The reactions involve fewer than three unknowns. The body is said to be **partially constrained** and motion of the body is possible.

b. The reactions involve more than three unknowns. The reactions are said to be **statically indeterminate**. Although you may be able to calculate one or two reactions, you cannot determine all of them.

c. The reactions pass through a single point or are parallel. The body is said to be **improperly constrained** and motion can occur under a general loading condition.

Problems

FREE-BODY PRACTICE PROBLEMS

4.F1 Two crates, each of mass 350 kg, are placed as shown in the bed of a 1400-kg pick-up truck. Draw the free-body diagram needed to determine the reactions at each of the two rear wheels A and front wheels B .

4.F2 A lever AB is hinged at C and attached to a control cable at A . If the lever is subjected to a 75-lb vertical force at B , draw the free-body diagram needed to determine the tension in the cable and the reaction at C .

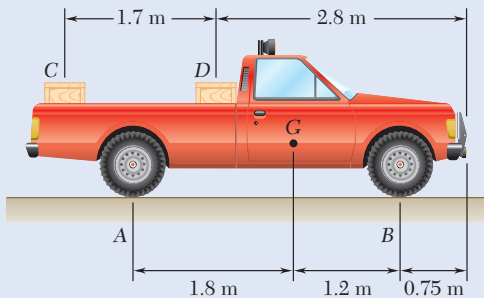


Fig. P4.F1

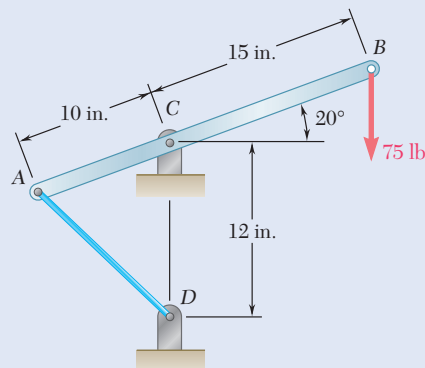


Fig. P4.F2

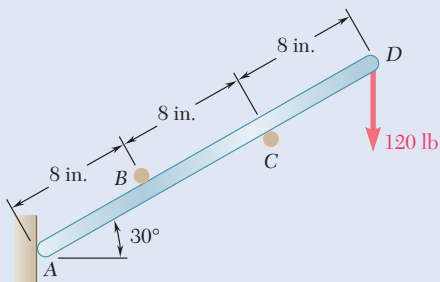


Fig. P4.F3

4.F3 A light rod AD is supported by frictionless pegs at B and C and rests against a frictionless wall at A . A vertical 120-lb force is applied at D . Draw the free-body diagram needed to determine the reactions at A , B , and C .

4.F4 A tension of 20 N is maintained in a tape as it passes through the support system shown. Knowing that the radius of each pulley is 10 mm, draw the free-body diagram needed to determine the reaction at C .

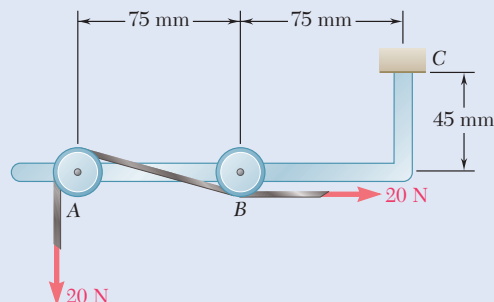


Fig. P4.F4

END-OF-SECTION PROBLEMS

- 4.1** A gardener uses a 60-N wheelbarrow to transport a 250-N bag of fertilizer. What force must she exert on each handle?
- 4.2** The gardener of Prob. 4.1 wishes to transport a second 250-N bag of fertilizer at the same time as the first one. Determine the maximum allowable horizontal distance from the axle A of the wheelbarrow to the center of gravity of the second bag if she can hold only 75 N with each arm.
- 4.3** A 2100-lb tractor is used to lift 900 lb of gravel. Determine the reaction at each of the two (a) rear wheels A , (b) front wheels B .

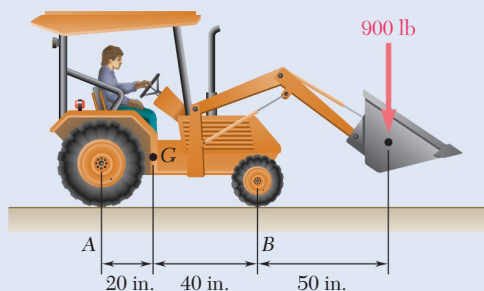


Fig. P4.3

- 4.4** For the beam and loading shown, determine (a) the reaction at A , (b) the tension in cable BC .
- 4.5** A load of lumber of weight $W = 25$ kN is being raised by a mobile crane. The weight of boom ABC and the combined weight of the truck and driver are as shown. Determine the reaction at each of the two (a) front wheels H , (b) rear wheels K .
- 4.6** A load of lumber of weight $W = 25$ kN is being raised by a mobile crane. Knowing that the tension is 25 kN in all portions of cable AEF and that the weight of boom ABC is 3 kN, determine (a) the tension in rod CD , (b) the reaction at pin B .
- 4.7** A T-shaped bracket supports the four loads shown. Determine the reactions at A and B (a) if $a = 10$ in., (b) if $a = 7$ in.

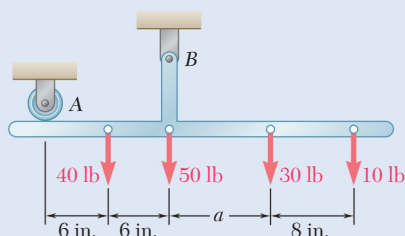


Fig. P4.7

- 4.8** For the bracket and loading of Prob. 4.7, determine the smallest distance a if the bracket is not to move.

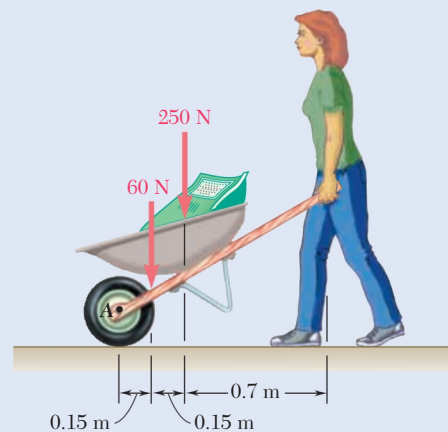


Fig. P4.1

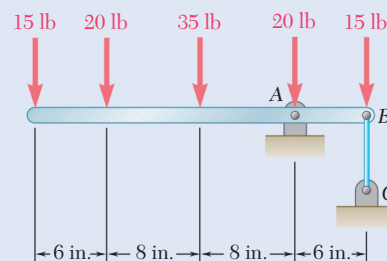


Fig. P4.4

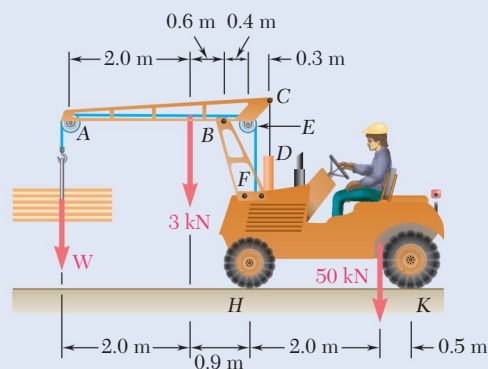


Fig. P4.5 and P4.6

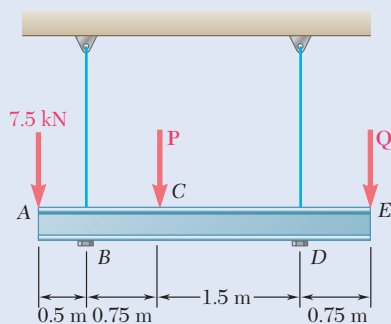


Fig. P4.9 and P4.10

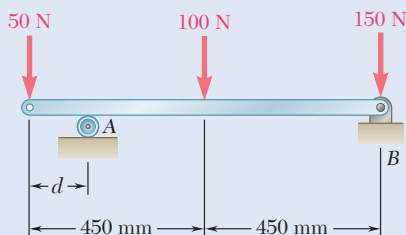


Fig. P4.13

4.9 Three loads are applied as shown to a light beam supported by cables attached at B and D . Neglecting the weight of the beam, determine the range of values of Q for which neither cable becomes slack when $P = 0$.

4.10 Three loads are applied as shown to a light beam supported by cables attached at B and D . Knowing that the maximum allowable tension in each cable is 12 kN and neglecting the weight of the beam, determine the range of values of Q for which the loading is safe when $P = 0$.

4.11 For the beam of Prob. 4.10, determine the range of values of Q for which the loading is safe when $P = 5$ kN.

4.12 For the beam of Sample Prob. 4.2, determine the range of values of P for which the beam will be safe, knowing that the maximum allowable value of each of the reactions is 25 kips and that the reaction at A must be directed upward.

4.13 The maximum allowable value of each of the reactions is 180 N. Neglecting the weight of the beam, determine the range of the distance d for which the beam is safe.

4.14 For the beam and loading shown, determine the range of the distance a for which the reaction at B does not exceed 100 lb downward or 200 lb upward.

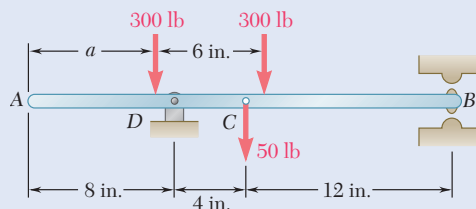


Fig. P4.14

4.15 Two links AB and DE are connected by a bell crank as shown. Knowing that the tension in link AB is 720 N, determine (a) the tension in link DE , (b) the reaction at C .

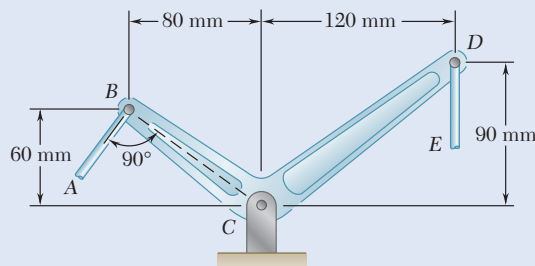


Fig. P4.15 and P4.16

4.16 Two links AB and DE are connected by a bell crank as shown. Determine the maximum force that can be safely exerted by link AB on the bell crank if the maximum allowable value for the reaction at C is 1600 N.

- 4.17** The required tension in cable AB is 200 lb. Determine (a) the vertical force P that must be applied to the pedal, (b) the corresponding reaction at C .
- 4.18** Determine the maximum tension that can be developed in cable AB if the maximum allowable value of the reaction at C is 250 lb.
- 4.19** The bracket BCD is hinged at C and attached to a control cable at B . For the loading shown, determine (a) the tension in the cable, (b) the reaction at C .

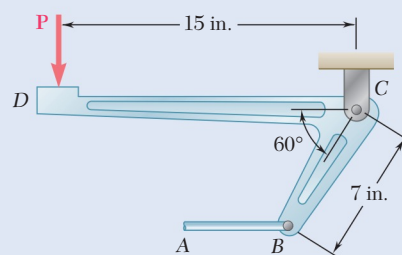


Fig. P4.17 and P4.18

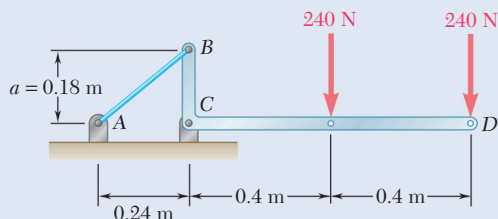


Fig. P4.19

- 4.20** Solve Prob. 4.19, assuming that $a = 0.32$ m.
- 4.21** The 40-ft boom AB weighs 2 kips; the distance from the axle A to the center of gravity G of the boom is 20 ft. For the position shown, determine (a) the tension T in the cable, (b) the reaction at A .

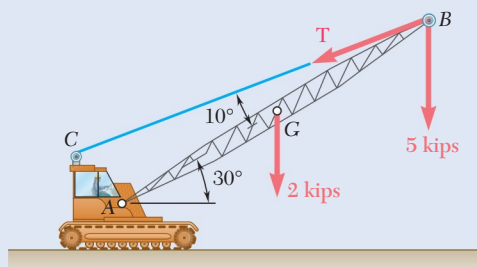


Fig. P4.21

- 4.22** A lever AB is hinged at C and attached to a control cable at A . If the lever is subjected to a 500-N horizontal force at B , determine (a) the tension in the cable, (b) the reaction at C .

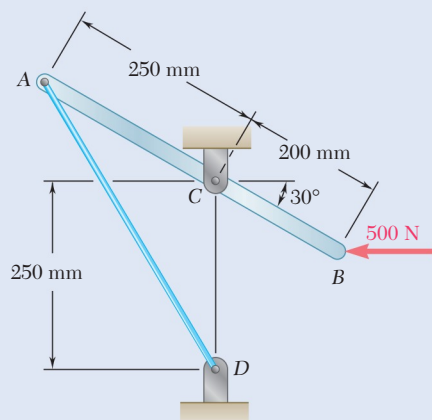


Fig. P4.22

4.23 and 4.24 For each of the plates and loadings shown, determine the reactions at A and B.

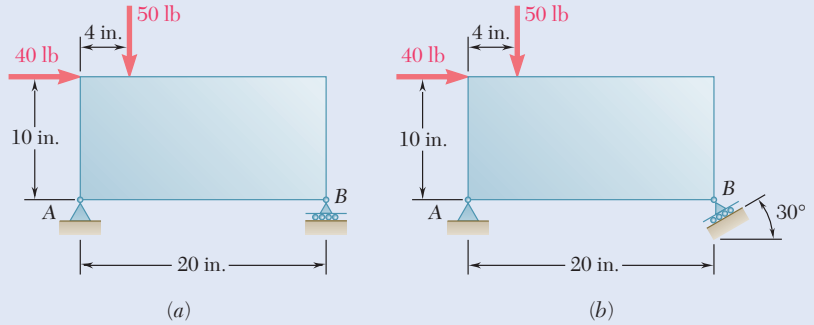


Fig. P4.23

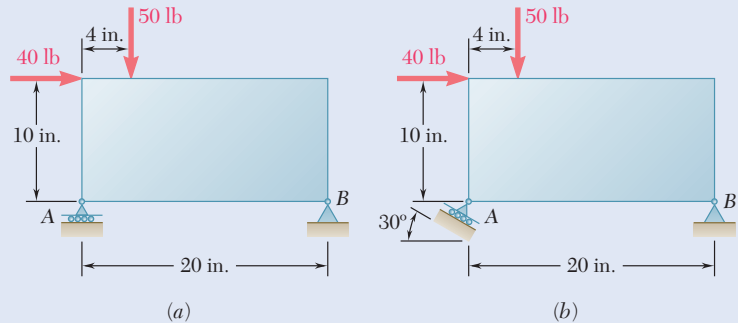


Fig. P4.24

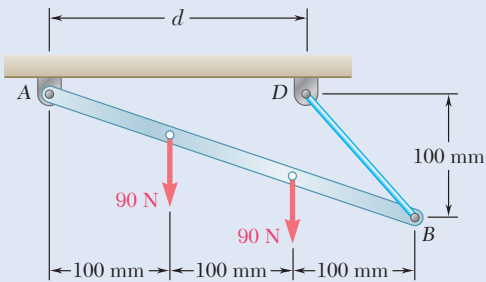


Fig. P4.25 and P4.26

4.25 A rod AB, hinged at A and attached at B to cable BD, supports the loads shown. Knowing that $d = 200$ mm, determine (a) the tension in cable BD, (b) the reaction at A.

4.26 A rod AB, hinged at A and attached at B to cable BD, supports the loads shown. Knowing that $d = 150$ mm, determine (a) the tension in cable BD, (b) the reaction at A.

4.27 Determine the reactions at A and B when (a) $\alpha = 0$, (b) $\alpha = 90^\circ$, (c) $\alpha = 30^\circ$.

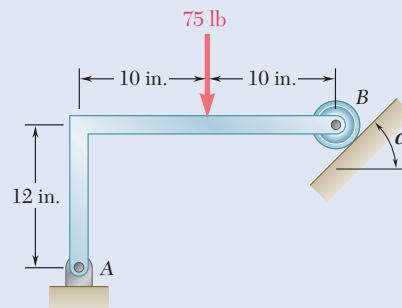


Fig. P4.27

4.28 Determine the reactions at A and C when (a) $\alpha = 0$, (b) $\alpha = 30^\circ$.

4.29 Rod ABC is bent in the shape of an arc of circle of radius R . Knowing that $\theta = 30^\circ$, determine the reaction (a) at B, (b) at C.

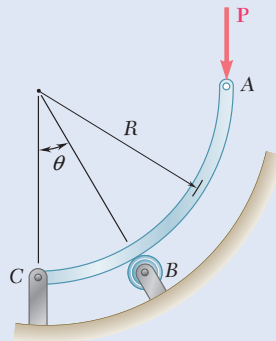


Fig. P4.29 and P4.30

4.30 Rod ABC is bent in the shape of an arc of circle of radius R . Knowing that $\theta = 60^\circ$, determine the reaction (a) at B, (b) at C.

4.31 Neglecting friction, determine the tension in cable ABD and the reaction at C when $\theta = 60^\circ$.

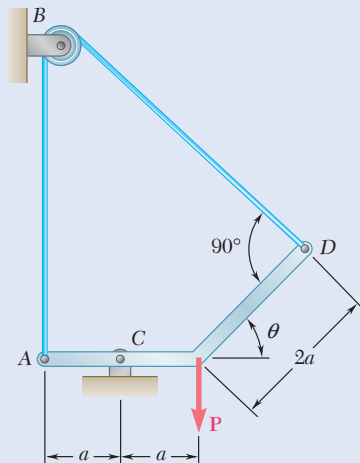


Fig. P4.31 and P4.32

4.32 Neglecting friction, determine the tension in cable ABD and the reaction at C when $\theta = 45^\circ$.

4.33 A force P of magnitude 90 lb is applied to member ACDE that is supported by a frictionless pin at D and by the cable ABE. Since the cable passes over a small pulley at B, the tension may be assumed to be the same in portions AB and BE of the cable. For the case when $a = 3$ in., determine (a) the tension in the cable, (b) the reaction at D.

4.34 Solve Prob. 4.33 for $a = 6$ in.

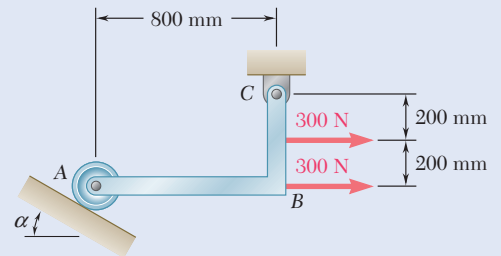


Fig. P4.28

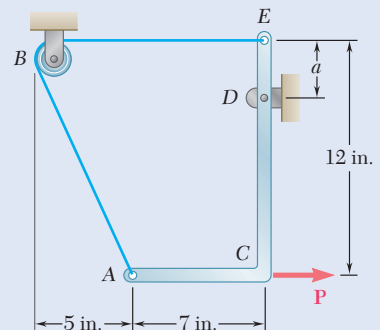


Fig. P4.33

- 4.35** Bar AC supports two 400-N loads as shown. Rollers at A and C rest against frictionless surfaces and a cable BD is attached at B . Determine (a) the tension in cable BD , (b) the reaction at A , (c) the reaction at C .

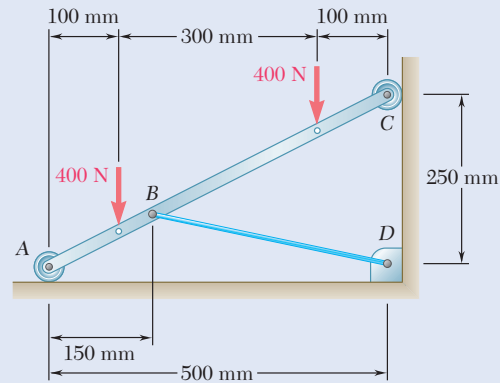


Fig. P4.35

- 4.36** A light bar AD is suspended from a cable BE and supports a 20-kg block at C . The ends A and D of the bar are in contact with frictionless vertical walls. Determine the tension in cable BE and the reactions at A and D .

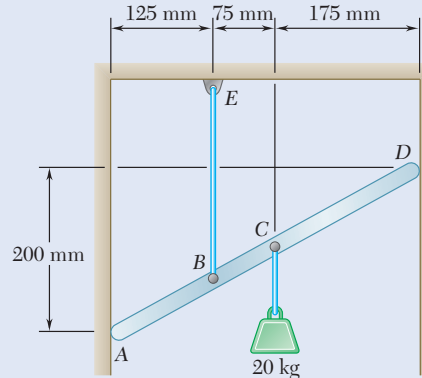


Fig. P4.36

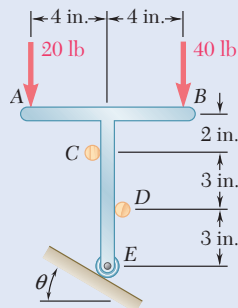


Fig. P4.37 and P4.38

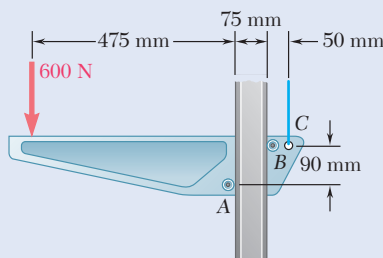


Fig. P4.39

- 4.37** The T-shaped bracket shown is supported by a small wheel at E and pegs at C and D . Neglecting the effect of friction, determine the reactions at C , D , and E when $\theta = 30^\circ$.
- 4.38** The T-shaped bracket shown is supported by a small wheel at E and pegs at C and D . Neglecting the effect of friction, determine (a) the smallest value of θ for which the equilibrium of the bracket is maintained, (b) the corresponding reactions at C , D , and E .
- 4.39** A movable bracket is held at rest by a cable attached at C and by frictionless rollers at A and B . For the loading shown, determine (a) the tension in the cable, (b) the reactions at A and B .

- 4.40** A light bar AB supports a 15-kg block at its midpoint C . Rollers at A and B rest against frictionless surfaces, and a horizontal cable AD is attached at A . Determine (a) the tension in cable AD , (b) the reactions at A and B .

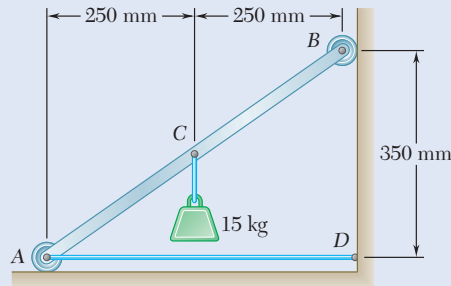


Fig. P4.40

- 4.41** Two slots have been cut in plate DEF , and the plate has been placed so that the slots fit two fixed, frictionless pins A and B . Knowing that $P = 15$ lb, determine (a) the force each pin exerts on the plate, (b) the reaction at F .

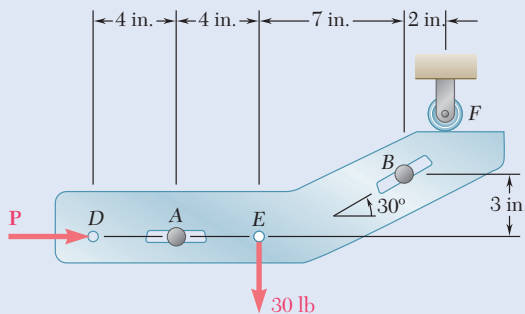


Fig. P4.41

- 4.42** For the plate of Prob. 4.41, the reaction at F must be directed downward, and its maximum value is 20 lb. Neglecting friction at the pins, determine the required range of values of P .
- 4.43** The rig shown consists of a 1200-lb horizontal member ABC and a vertical member DBE welded together at B . The rig is being used to raise a 3600-lb crate at a distance $x = 12$ ft from the vertical member DBE . If the tension in the cable is 4 kips, determine the reaction at E , assuming that the cable is (a) anchored at F as shown in the figure, (b) attached to the vertical member at a point located 1 ft above E .

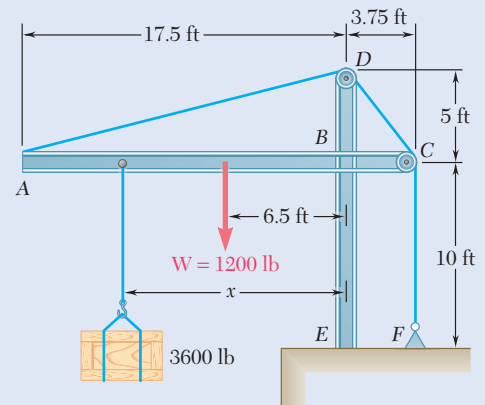


Fig. P4.43

- 4.44** For the rig and crate of Prob. 4.43 and assuming that cable is anchored at F as shown, determine (a) the required tension in cable $ADCF$ if the maximum value of the couple at E as x varies from 1.5 to 17.5 ft is to be as small as possible, (b) the corresponding maximum value of the couple.

- 4.45** A 175-kg utility pole is used to support at C the end of an electric wire. The tension in the wire is 600 N, and the wire forms an angle of 15° with the horizontal at C . Determine the largest and smallest allowable tensions in the guy cable BD if the magnitude of the couple at A may not exceed $500 \text{ N}\cdot\text{m}$.

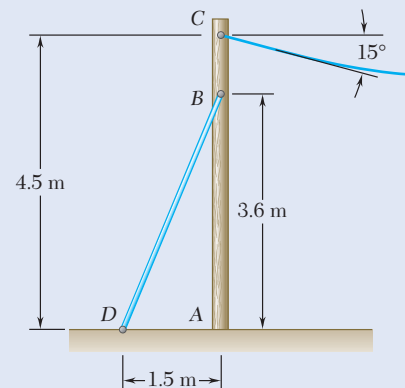


Fig. P4.45

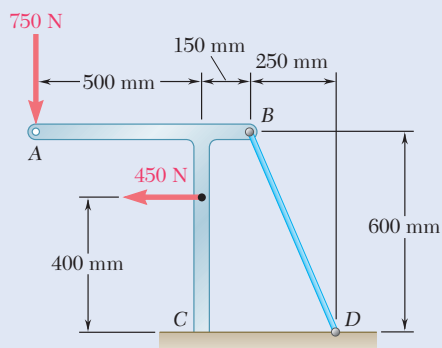


Fig. P4.46 and P4.47

4.46 Knowing that the tension in wire BD is 1300 N, determine the reaction at the fixed support C of the frame shown.

4.47 Determine the range of allowable values of the tension in wire BD if the magnitude of the couple at the fixed support C is not to exceed $100 \text{ N}\cdot\text{m}$.

4.48 Beam AD carries the two 40-lb loads shown. The beam is held by a fixed support at D and by the cable BE that is attached to the counterweight W . Determine the reaction at D when (a) $W = 100 \text{ lb}$, (b) $W = 90 \text{ lb}$.

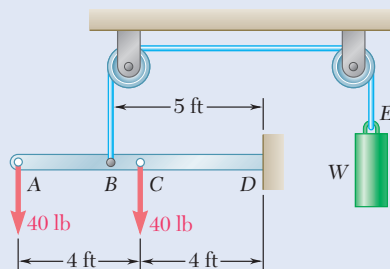


Fig. P4.48 and P4.49

4.49 For the beam and loading shown, determine the range of values of W for which the magnitude of the couple at D does not exceed $40 \text{ lb}\cdot\text{ft}$.

4.50 An 8-kg mass can be supported in the three different ways shown. Knowing that the pulleys have a 100-mm radius, determine the reaction at A in each case.

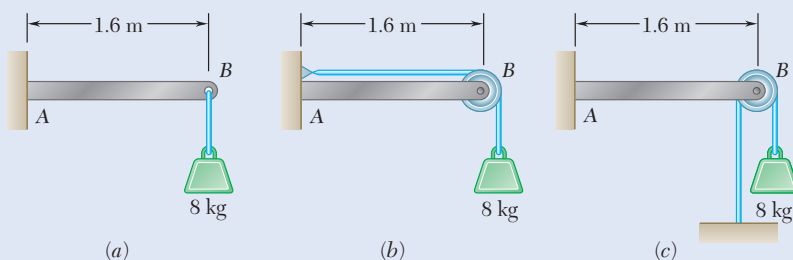


Fig. P4.50

4.51 A uniform rod AB with a length of l and weight of W is suspended from two cords AC and BC of equal length. Determine the angle θ corresponding to the equilibrium position when a couple M is applied to the rod.

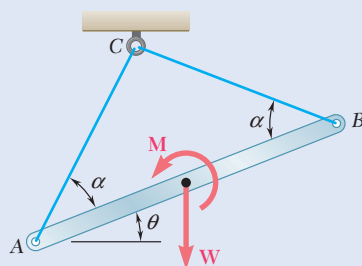


Fig. P4.51